

Mood alternations

A synchronic and diachronic study of
negated complement clauses in
Spanish

Raquel Montero Estebaranz

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Barcelona, winter 2025

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1 Introduction

1.1 Mood and modality

The term *modality* is used in linguistics to describe the phenomenon whereby speakers can talk about and describe situations which need not be real (Portner 2009).¹ For instance, sentence (1a) describes what the speaker considers obligatory/necessary according to a set of principles (deontic modality), sentence (1b) indicates what the speaker considers best according to their desires/preferences (bouletic modality), and sentences (1c) and (1d) indicate the degree of certainty of the speaker about the proposition (epistemic modality).

- | | | |
|-----|--|----------------------|
| (1) | a. Celeste must go to school. | (deontic modality) |
| | b. Celeste should try this chocolate. | (bouletic modality) |
| | c. Celeste might be at school. | (epistemic modality) |
| | d. Celeste must be at school. | (epistemic modality) |

Modality, thus, covers a broad range of semantic distinctions – epistemic, bouletic, teleological, alethic, etc. – see von Stechow (2006). In English, such distinctions can be expressed by means of modal auxiliaries (such as *must* and *may*, see (1) above), but they can be expressed in many different ways. For example, through modal adverbs, see (2).

- (2) I will **probably** be home by midnight.

Cross-linguistic work on modality has shown that there is a great deal of variation in how these modal distinctions are expressed across the world's languages (see de Haan 2006 and Nauze 2008 for an overview of different strategies). For example, in Tamil, modal affixes² such as *-laam* can be optionally attached to the verb to express permission, see (3) (de Haan 2006). In Japanese, periphrastic

¹For other definitions of the term *modality* see Bybee & Fleischman (1995), Ridruejo Alonso (1999) and Palmer (2001).

²The key difference between modal affixes and verbal mood is that only the latter is compulsory (de Haan 2006).

1 Introduction

expressions such as *nitigainai* (lit. ‘there is no mistake in’) and *kamosirenai* (lit. ‘can’t know whether’) are used to express the degree of certainty about a proposition, see (4) (Larm 2014, Kaufmann & Tamura 2020); and in languages such as German, modal particles (PRT) can be used with a similar function, see (5) (Coniglio 2008).

(3) **Modal affixes:**

Avan peeca-laam. (Tamil)

3SG speak-PERM

‘He is allowed to speak.’ (de Haan 2006: 36)

(4) **Periphrastic expressions:**

Asita wa ame ga huru nitigaina-i/kamosirena-i. (Japanese)

tomorrow TOP rain NOM fall.NPST must-NPST/may-NPST

‘It must/may rain tomorrow.’ (Kaufmann & Tamura 2020: 3)

(5) **Modal particles:**

Hans ist wohl auf See. (German)

Hans is PRT at Sea

‘Hans is probably at sea.’ (Coniglio 2008: 92)

In Spanish, which will be the focus of this book, and in Romance more generally³, modal concepts can be expressed by means of what is called *mood morphology*. These are affixes that obligatorily attach to the verb and reflect the modal properties of the context in which the verb occurs (Portner 2018). Given the functional nature of verbal morphology in Spanish, however, the semantic categories of Tense, Aspect and Mood (known as TAM) are combined/fused into one inflectional morpheme (i.e., there is no specific morph indicating mood distinctions alone).

Following Alcoba (1999), the verb in Spanish can be decomposed into four components: (i) the root, (ii) the theme vowel (ThV), which indicates the conjugation class of the verb⁴, (iii) the Tense, Aspect and Mood (TAM) morpheme, and (iv) the Number and Person (NP) morpheme, see (6):

(6) Root + ThV + TAM + NP

³Within the Romance family, Romanian differs in this respect, as it marks the subjunctive mood by means of the particle *să* followed by the verb (Farkas 1981).

⁴In Spanish, there are three main conjugation classes: *-ar*, *-er* and *-ir*. These conjugations are also known as the 1st, 2nd and 3rd conjugations respectively.

The form *cantábamos* of the verb *cantar* ‘to sing’, for example, can be decomposed into the four units in (7). First, the root *cant*, which carries the lexical input. Second, the theme vowel *-á*, which indicates that the verb belongs to the 1st conjugation class. Third, the morpheme *-ba*, which indicates that the verb is in the past tense, carries imperfective aspect, and that its mood is indicative. Fourth, the morpheme *-mos* which indicates that the word is first person plural.

- (7) cant- á- ba- mos (Spanish)
sing- THV- PST.IPFV.IND- 1PL

Regarding the TAM morpheme, traditional grammars of Spanish distinguish four non-periphrastic tenses – present, future, past, and conditional –, two aspects – perfect and imperfect –, and three moods – the indicative, the subjunctive and the imperative – (Real Academia Española 2010). Not all possible combinations of these categories exist. For instance, there is no synthetic form for the past perfect subjunctive; and contrary to other languages, for example, Latin (Azofra Sierra 2010) or Cheyenne (see Murray 2016 and references cited therein), the imperative in Spanish does not show temporal oppositions. These characteristics are illustrated in Table 1.1, where the paradigm of the verb *cantar* ‘to sing’ in the 2nd person plural is exemplified.

Table 1.1: Conjugation table of the verb *cantar* ‘to sing’. The table only shows the second person plural forms and non-periphrastic tenses. For the full paradigms of Spanish verbs, the interested reader is referred to the Real Academia Española (2010: 49–93).

	Indicative	Subjunctive	Imperative
Present	cant-á + Ø + is	cant-Ø + é + is	–
Past Perfect	cant-a + ste + is	–	–
Past Imperfect	cant-a + ba + is	cant-a + ra/se + is	–
Future	cant-a + ré + is	cant-a + re + is	–
Conditional	cant-a + ría + is	–	–
–	–	–	cant-á + Ø + d

The three mood distinctions present in Spanish are used to express different types of modalities. For instance, the imperative (IMP) can be used in contexts which express orders (deontic modality), see sentence (8a); the subjunctive (SBJV) can be used in contexts which express desires (bouletic modality), see (8b); and the indicative can be used in contexts which express certainty (epistemic modality), see (8c).

(8) Mood morphology

(Spanish)

- a. **Canta** alguna canción.
sing.IMP.2SG some song
'Sing a song!'
- b. Quiero que **cantes** alguna canción.
want that sing.PRS.SBJV.2SG some song
'I want you to sing a song.'
- c. Estoy segura de que **cantarás** alguna canción.
am sure of that sing.FUT.IND.2SG some song
'I am sure you will sing a song.'

It is important to highlight, however, that there is not a one-to-one correspondence between modal notions and how they are grammaticalized in language (Ridruejo Alonso 1999). One linguistic strategy may be used to express several different modal distinctions. The subjunctive mood, for instance, can be used to express desire (deontic modality), as in (8b), or probability (epistemic modality), as in (9). Conversely, a given modal distinction can be expressed via different linguistic means. The notion of probability can be conveyed by means of the subjunctive or the future tense, see (10).

- (9) Es posible que **cantes** una canción.
is possible that sing.PRS.SBJV.2SG a song
'It's possible that you will sing a song.'
- (10) a. Quizás **sean** las ocho. (probability)
maybe be.PRS.SBJV the eight
'Maybe it's eight o'clock.'
- b. **Serán** las ocho. (probability)
be.FUT.IND the eight
'Maybe it's eight o'clock.'

Moreover, the mapping between meanings and forms is in constant flux. Some forms become obsolete, new forms emerge, and a given form can extend some of its possible contexts of usage or change them altogether. For example, in the evolution of preterit forms from Latin to Modern Spanish (see Figure 1.1), the forms of the preterit imperfect subjunctive (e.g., *cantarem*) and preterit perfect subjunctive (e.g., *cantaverim*) became obsolete, and were substituted by new periphrastic forms, e.g., *haya cantado* (Veiga 2006). The Modern Spanish preterit pluperfect

subjunctive (e.g., *cantara/cantase*), on the other hand, emerged from the Latin indicative and subjunctive preterit pluperfect forms (*cantaveram* and *cantavissem*, respectively). That is, an indicative form in Latin was reanalyzed as a subjunctive form (Veiga 2006). For a complete overview of the changes in verbal morphology from Latin to Romance the interested reader is referred to Maiden (2018), and for the changes from Latin to Spanish in particular see Azofra Sierra (2019).

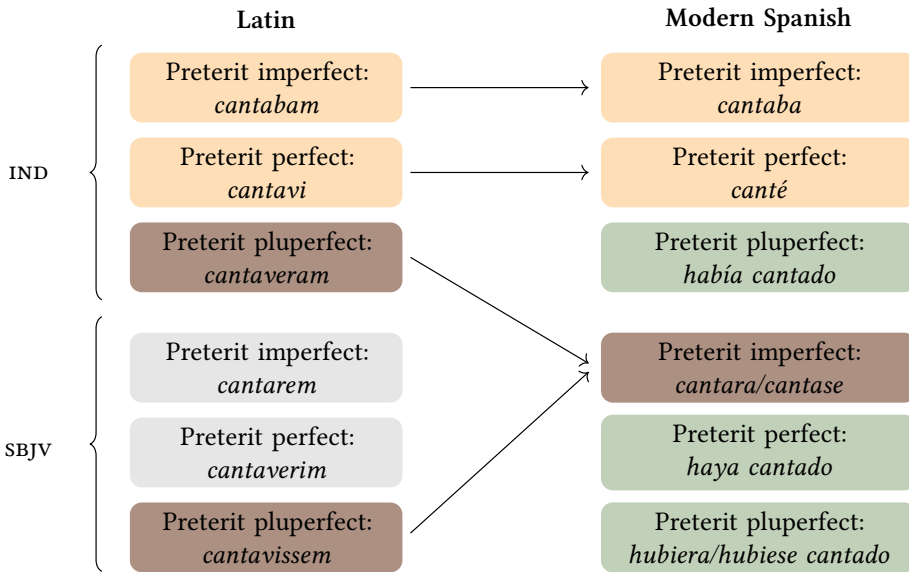


Figure 1.1: Changes in the preterit forms from Latin to Modern Spanish. For an in-depth description of the changes that the Spanish verbal system underwent, see Veiga (2006).

These semasiological and onomasiological changes add a layer of complexity when analyzing the matching between mood and modality; an aspect that has sparked a huge amount of research on the topic, both from a synchronic and a diachronic perspective. Moreover, as the following sections explain in more detail, the choice of mood morphology is constrained by the presence/absence of other linguistic phenomena such as negation, NPIs, tense, aspect, type of matrix predicate etc., but the way these phenomena interact with mood has also not been constant in the Spanish language.

This book aims to contribute to this existing literature by investigating how mood in Spanish interacts with negation, and how the different meanings attached to these expressions have changed over time and could be formally represented and explained. Section 1.2 presents the notions of intensional and polarity

mood selection. Sections 1.3 and 1.4 describe the two types of mood in Iberian Spanish and the variability and constraints they exhibit. After this descriptive overview, Section 1.5 presents the research questions and aims of the book, and Section 1.6 explains the main proposal of the book. The chapter concludes by presenting the structure of the rest of the book.

1.2 Mood selection: Intensional and polarity mood

An important aspect of mood morphology is that the choice of one mood over another is determined by elements (overt or covert) that are structurally higher than the verb in the clause. This phenomenon is known in the literature as mood selection, and can be triggered by many different types of elements: verbs, adjectives, complementizers, prepositions, etc. (Bosque 2012). For instance, the verb *lamentar* ‘regret’, the adjective *raro* ‘weird’, the complementizer *como* ‘how/if’, and the preposition *sin* ‘without’, all select for a verb with subjunctive morphology (see (11a), (11b), (11c), and (11d), respectively).

- (11) a. **Lamento** que [estés / *estás] triste.
 regret that be.PRS.SBJV be.PRS.IND sad
 ‘I regret that you are sad.’
- b. **Es raro** que [estés / *estás] triste.
 Is weird that be.PRS.SBJV be.PRS.IND sad
 ‘It’s weird that you are sad.’
- c. **Como** Carlos [continue / *continua] así,
 if Carlos continue.PRS.SBJV continue.PRS.IND this.way,
 suspenderá.
 fail.FUT.IND.3SG
 ‘If Carlos continues behaving this way, he will fail.’
- d. **Sin** que tu lo [sepas / *sabes].
 Without that you it know.PRS.SBJV know.PRS.IND
 ‘Without you knowing.’

Depending on the type of mood selector, it is commonly agreed in the literature that there are two types of mood: intensional mood, which is licensed by a lexical predicate, and polarity mood, which is licensed by a polarity operator (Quer 1998). In modern Spanish, for example, the verb *querer* ‘want’ forces the embedded verb to appear with the subjunctive morphology (what is referred to

as ‘intensional subjunctive’), and a verb such as *saber* ‘know’ forces the embedded verb to appear in the indicative, see (12). However, if a verb that takes the indicative in affirmative clauses (e.g., *know*) is negated, the embedded verb can display the indicative or the subjunctive mood (‘polarity subjunctive’), see (13).⁵

(12) Intensional mood:

- a. Edna **quería** que la [**dejaban/dejasen*] ser libre.
Edna wanted that her let.IND let.SBJV be free
‘Edna wanted to be allowed to be free.’
- b. Edna **sabía** que un amigo [*iba* /**fuese*] a venir.
Edna knew that a friend was.IND was.SBJV to come
‘Edna knew that a friend was going to come.’

(13) Polarity mood:

- Calixta **no** sabía que un amigo [*iba* /**fuese*] a venir.
Calixta not knew that a friend was.IND was.SBJV to come
‘Calixta didn’t know that a friend was going to come.’

While the choice of mood of individual elements is language-specific, the linguistic literature has identified certain patterns that tend to re-occur cross-linguistically (Rothstein & Thieroff 2010). For example, within the Romance family, the verb *want* tends to select the subjunctive mood (14)⁶. Other predicates, however, are more unstable and display more variation (Portner & Rubinstein 2020). For instance, the verb *hope*, see (15), selects the subjunctive in Spanish and Galician, the indicative in French, and both moods in Portuguese, Italian, Romanian and to some extent Catalan (Portner & Rubinstein 2020, Dindelegan & Maiden 2013, Badia i Margarit 1994).

(14) Volitional *want*

- a. Vull que acabi la tesi. (Catalan)
want that finish.SBJV the dissertation
‘I want her/him to finish the dissertation.’ (Quer 2010: 164)
- b. Quero que fale. (Galician)
want that speak.SBJV
‘I want him/her to speak.’ (Frías Conde 2015: 16)

⁵The difference between intensional and polarity mood has been illustrated with Spanish, but the distinction is present in most of the Romance languages (see Quer 1998 for Catalan, Marques 2024 for Portuguese and Homer 2008 for French and Italian).

⁶Portner & Rubinstein (2020) reports that for some speakers of Italian, allow the indicative in certain contexts.

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- c. O João quer que o pai brinque com ele. (Portuguese)
The John wants that the father play.SBJV with him
'John wants his father to play with him.' (Becker 2010a: 184)
 - d. Voglio che A. parta oggi. (Italian)
want that A. leave.SBJV today
'I want A. to leave today.' (Squartini 2010: 245)
 - e. Vrea să plec. (Romanian)
wants sĂ_{sbjv} leave.SBJV
'(S)he wants me to leave.' (Dindelegan & Maiden 2013: 44)
 - f. Je veux qu' il vienne. (French)
I want that he come.SBJV
'I want him to come.' (Soutet 2000: 32)
 - g. Victoria quiere que Marcela venga al picnic. (Spanish)
Victoria wants that Marcela come.SBJV to.the picnic
'Victoria wants Marcelona to come to the picnic.' (Villalta 2008: 470)
- (15) Volitional *hope*
- a. Esperem que tot [acabarà /acabi⁷] bé. (Catalan)
Hope that all end.FUT.IND end.PRS.SBJV well
'We hope everything will end up well.'
(Badia i Margarit 1994: 635)
 - b. Xoán espera que María [*estará /estea] contenta. (Galician)
Xoán hopes that María be.FUT.IND be.PRS.SBJV happy
'Xoán hopes that Mary is/will be happy.'
(p.c. Natalia Jardón)

⁷Badia i Margarit (1994: 635) states that the verb *esperar* has two interpretations: when it means to *wait* it tends to take the subjunctive, when it means to *hope* it can take both the indicative or the subjunctive, although the subjunctive in this context may be due to the influence from Spanish. The Institut d'Estudis Catalans (2016: 936), in turn, proposes that when the indicative is used the verb is closer to English *expect*, and when the subjunctive is used the verb is interpreted as *hope*: "l'indicatiu s'associa a la creença que alguna cosa s'esdevindrà [...] i el subjuntiu a la idea de desig" (Institut d'Estudis Catalans 2016: 936). The *Gran Diccionari de la llengua catalana* (GDLC) (Enciclopèdia Catalana 1998), however, gives examples with both the indicative (*espero que hi reeixiràs*) and the subjunctive (*espero que t'hagi anat bé*) when the verb has a desiderative interpretation. In contrast, in Portner & Rubinstein (2020) it is reported that the native consultants they asked only allowed for the subjunctive with *esperar*. Given the variability across sources, there might be some variation across speakers.

1.2 Mood selection: *Intensional and polarity mood*

- c. J. espera que a M. [estará /esteja] feliz. (Portuguese)
 J. hopes that the M. be.FUT.IND be.PRS.SBJV happy
 ‘J. hopes that M. is/will be happy.’
 (Portner & Rubinstein 2020: 381)
- d. Gianni spera che Maria [sarà /sia] contenta. (Italian)
 Gianni hopes that Maria be.FUT.IND be.PRS.SBJV happy
 ‘Gianni hopes that Mary is/will be happy.’
 (Portner & Rubinstein 2020: 382)
- e. Ion speră ca Maria [va fi /să fie] fericită. (Romanian)
 Ion hopes that Maria FUT be /SBJV be happy
 ‘Ion hopes that Maria is/will be happy’
 (Portner & Rubinstein 2020: 384)
- f. Pierre espère que Marie [sera /*soit] heureuse. (French)
 Pierre hopes that Marie be.FUT.IND be.PRS.SBJV happy
 ‘Pierre hopes that Maria is/will be happy.’
 (Portner & Rubinstein 2020: 344)
- g. Juan espera que María [*estará /está] contenta. (Spanish)
 Juan hopes that María be.FUT.IND be.PRS.SBJV happy
 ‘Juan hopes that María is/will be happy.’
 (Portner & Rubinstein 2020: 347)

Across the Romance languages, there is also variation in mood selection with emotive-factive, doxastic and dubitative predicates, to name a few (Farkas 1992, Portner & Rubinstein 2020, Mari & Portner 2021, Dindelegan & Maiden 2013). A summary of the key differences and similarities reported in the literature can be found in Table 1.2. The data for Spanish and Portuguese is from the European varieties. The superscript numbers indicate the exact source where the information can be found. In some cases, the linguistic literature was further supplemented with native speakers’ intuitions and corpus data. This was done either because different grammars had different rules, or because the specific verb was not mentioned. Regarding causative verbs (e.g., *make*) in Spanish and other Romance languages, these verbs can take a finite complement clause (16a), but this option is precluded in Italian and French (16b) (Vivanco 2019). Given that Table 1.2 is only intended to provide information regarding the competition between the indicative and the subjunctive, the corresponding cells have been left empty (–).

- (16) a. Haré que él venga en tren. (Spanish)
 make.FUT that he come.SBJV on train.

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- b. * Je ferais qu' il vienne en train. (French)
I make.FUT that he come.SBJV on train.
'I will make him come by train.' (Vivanco 2019: 78)

Overall, the data illustrates that in addition to *want*, directive verbs like *order*, causative verbs such as *make*, and weak epistemic predicates such as *be possible* tend to select the subjunctive mood across the Romance languages. Interestingly, no predicate consistently selects the indicative mood across the selected languages. The prototypical indicative selector is the verb *know*, but this verb can take the subjunctive mood in Romanian to convey ability readings (Dindelegan & Maiden 2013: 44). The other indicative-selecting verbs are highly consistent across of the chosen languages, but in Italian these predicates allow for both mood options (Giannakidou & Mari 2021). A general theory of mood selection should aim to explain these differences as well as the similarities that emerge cross-linguistically. Investigating how mood selection changes diachronically, thus, provides a first step in this direction, as it can shed light on the reasons behind why the languages might start to diverge from one another.

Looking at the Latin mood system (Pinkster 1995, Magni 2010, Pinkster 2021), one observes that those predicates that select the subjunctive across the Romance languages also selected the subjunctive in Latin. For example, in Latin, verbs of volition (*uolo* 'want'), see (17), and directives (*impero* 'order') took subjunctive complement clauses (Magni 2010) and they still do across Romance.

- (17) hodie uxorem **ducas**, ut dixi, **uolo** (Latin)
today wife marry.SBJV that say want.1SG
'I want you to get married today, as I have said'
(*apud* Magni 2010: 245)

However, in some cases, similarities across the Romance languages can be due to convergent trajectories rather than feature inheritance. For example, while emotive-factive verbs tend to take the subjunctive across most of the Romance languages, in Latin they could take the indicative.

- (18) sane **gaudeo** quod te **interpellaui** (Latin)
certainly be.glad.1SG that you interrupted.IND
'I am truly glad that I interrupted you.' (López Fonseca 2009: 465)

It is also important to highlight the fact that Latin had two main means of complementation which have evolved differently across the Romance languages: (i)

1.2 Mood selection: Intensional and polarity mood

Table 1.2: Mood selection in declarative affirmative clauses across several Romance languages

Verb	Catalan ¹	Galician	E. Portuguese	Italian	Romanian	French	E. Spanish ¹⁹
Epistemic (<i>know</i>)	ind	ind ²	ind ⁵	ind ⁹	ind/(sbjv) ¹³	ind ¹⁶	ind
Doxastic (<i>believe</i>)	ind	ind(sbjv) ⁴	ind/sbjv ⁵	sbjv(ind) ⁹	ind ¹⁴	ind ¹⁶	ind
Certainty (<i>be certain/sure</i>)	ind	ind ²	ind ⁶	ind(sbjv) ⁹	ind ¹⁴	ind ¹⁴	ind
Perception (<i>see</i>)	ind	ind ²	ind ⁷	ind ¹²	ind ¹⁵	ind ¹⁸	ind
Communication ^a (<i>tell</i>)	ind	ind ^{3,12}	ind ⁵	ind(sbjv) ⁹	ind ¹³	ind ¹⁶	ind
Commissive (<i>promise</i>)	ind	ind ¹²	ind ⁵	ind ¹²	ind/sbjv ¹⁴	ind ¹⁶	ind
Fiction (<i>dream</i>)	ind	ind/(sbjv) ¹²	ind ⁵	ind(sbjv) ^{b9}	ind ¹⁴	ind/(sbjv) ²⁰	ind/(sbjv) ²¹
Volition I (<i>want</i>)	sbjv	sbjv ²	sbjv ⁵	sbjv ⁹	sbjv ¹³	sbjv ¹⁶	sbjv
Volition II (<i>hope</i>)	sbjv/(ind)	sbjv ¹²	ind/sbjv ⁸	ind/sbjv ⁸	ind/ sbjv ⁸	ind ⁸	sbjv
Directive (<i>order</i>)	sbjv	sbjv ²	sbjv ⁵	sbjv ⁹	sbjv ¹⁴	sbjv ¹⁶	sbjv
Causative (<i>make</i>)	sbjv	–	sbjv ⁵	–	sbjv ¹³	–	sbjv
Dubitative (<i>doubt</i>)	sbjv	sbjv ¹²	sbjv ⁵	sbjv ¹⁰	ind ¹³	sbjv ¹⁴	sbjv
Emotive-factive (<i>regret</i>)	sbjv	sbjv ¹²	sbjv ⁵	sbjv ¹¹	ind ¹⁴	sbjv/(ind) ¹⁷	sbjv
Weak epistemic (<i>be possible</i>)	sbjv	sbjv ¹²	sbjv ⁶	sbjv ⁹	sbjv ¹⁴	sbjv ¹⁴	sbjv

References: 1. Institut d'Estudis Catalans (2016: 933–939;1006–1008) 2. Álvarez & Xove (2002: 272–274) 3. Frías Conde (2015) 4. Jardón & Björnsdóttir (2026) 5. Marques (2010) 6. Becker (2010a) 7. Martins (2008) 8. Portner & Rubinstein (2020) 9. Giannakidou & Mari (2021) 10. Manzini (2008) 11. Giorgi (2009) 12. Native consultant 13. Dindelegan & Maiden (2013) 14. Farkas (1992) 15. Becker (2010b) 16. Silk (2018) 17. Schlenker (2005) 18. Achard (2002) 19. Villalta (2008) 20. Baunaz & Puskás (2022: 23) 21. Delbecque & Lamiroy (1999)

^aWithin communication verbs there is some variation. For example, *say* allows the subjunctive with an order interpretation, but others like *tell* do not.

^bWhile Giannakidou & Mari (2021: 29) argue it is possible to have both the indicative and the subjunctive, Giorgi & Pianesi (2001: 66) state that the subjunctive is ruled out. The difference may stem from variations across verbs within the category. Giannakidou & Mari (2021) use *immaginare* and Giorgi (2009) *sognare*.

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accusative and infinitive (AcI) constructions (19a), and (ii) complementizer constructions (19b) (Ledgeway 2012: 235–258). While AcI constructions with cognitive verbs (e.g., doxastic, epistemic) have been lost in some Romance languages, complementizer constructions have remained the norm (Pons Rodríguez 2008). However, the way in which language reorganized the internal complementation system differed, potentially leading to cross-linguistically variability in the complementation patterns as a whole.

- (19) a. *tacitum te dicere credo* (Latin)
 silent you.ACC say.INF believe.1SG
 ‘I fancy you say to yourself’
 (Ledgeway 2012: 240)
- b. *scis enim quod epulum dedi* (Latin)
 know.2SG for that feast gave.1SG.IND
 ‘For you remember that I gave a public banquet once.’
 (Ledgeway 2012: 240)

Given that the present-day mood patterns as well as their historical evolution are language-specific, the rest of the book will focus on Spanish, drawing parallels to other systems when necessary. The following sections will describe the Spanish mood selection system and its diachronic development in more detail, starting with intensional mood (Section 1.3), and then polarity mood (Section 1.4). The main goal is to determine which aspects of the phenomenon have remained constant diachronically, which ones have changed, and which ones we do not have enough information about and should be further investigated.

1.3 Intensional mood selection in Iberian Spanish

As was shown in the previous section, mood selection is subject to a considerable amount of cross-linguistic variability, sometimes displaying different selection patterns even within the same language family and language (see, for example, Rothstein & Thieroff 2010 for an overview of mood selection in the languages of Europe, Portner & Rubinstein 2020 and Quer 2010 for a discussion about the Romance family in particular, and Busch 2017 for the internal variability within Spanish).

Regarding Spanish, although the vast majority of predicates have a similar selectional pattern across the different varieties, there are certain predicates which

do show differences. For example, the verb *creer* can only take an indicative embedded verb in Iberian Spanish⁸, but allows the subjunctive in Venezuelan Spanish (Quer 2010) and Yucatec Spanish (Bove 2018), see (20).

- (20) a. Creo que Luis [ama /*ame] a María. (Iberian Sp.)
 believe.1SG that Luis love.PRS.IND love.PRS.SBJV to Mary
 ‘I believe that Luis loves Mary.’
- b. Creo que Luis ame a María. (Venezuelan Sp.)
 believe.1SG that Luis love.PRS.SBJV to Mary
 ‘I believe that Luis loves Mary.’ (Quer 2010: 169)
- c. Cree que su amigo sea tramposo. (Yucatec Sp.)
 believe.1SG that his friend be.PRS.SBJV liar
 ‘He believes that his friend is a liar.’ (Bove 2018: 128)

Given the complex pattern of variation, the present work will concentrate exclusively on Iberian Spanish and its diachronic development.

Starting first with the Modern Iberian Spanish mood selection, Villalta (2008) provides one of the most extensive lists of predicate classes and their selectional patterns. Among the predicates that select the indicative are epistemic (e.g., *saber* ‘know’), doxastic (e.g., *creer* ‘believe’), certainty (e.g., *estar seguro* ‘be sure’), communication (e.g., *contar* ‘tell’), commissive (e.g., *prometer* ‘promise’), and perception (e.g., *ver* ‘see’) predicates. Villalta (2008) also includes in this list fiction verbs (e.g., *soñar*), but as Delbecque & Lamiroy (1999) discuss, this is a slight simplification as these verbs can take the subjunctive (in addition to the indicative) if they are interpreted as volitional (21). Usually this interpretation requires the preposition *con* ‘with’. It must be acknowledged that this second use is more infrequent. For instance, a quick search in the CREA for [*soñar*] + *que* + *V_{ind}* outputted 247 results, whereas the same search with an embedded subjunctive verb, i.e. [*soñar*] + *que* + *V_{sbjv}*, did not find any results. This frequency difference is important because for Old Spanish only the textual record is available, which provides a partial view of the real underlying system. If one finds both moods in the corpora, it can be taken as evidence that the two meanings might have always existed in the language. However, the absence of one of the forms (specially given that in the present-day language one is less frequent) cannot be used as evidence that the subjunctive is a recent development.

⁸The terms Iberian Spanish, Peninsular Spanish and European Spanish are all often used in the literature interchangeably to refer to those varieties spoken in Spain.

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- (21) a. Los niños sueñan que los Reyes les **traen** muchos juguetes.
 the kids dream that the Kings them bring.IND many toys
 ‘The kids dream that the Three Wise Men bring them many toys.’
 b. Los niños sueñan con que los Reyes les **traigan** muchos
 the kids dream with that the Kings them bring.SBJV many
 juguetes.
 toys
 ‘The kids dream/wish that the Three Wise Men will bring them many
 toys.’ (Delbecque & Lamiroy 1999: 2051)

Among the predicates that select the subjunctive mood are desire/volitional (e.g., *querer* ‘want’, *esperar* ‘hope’), emotive-factive (e.g., *lamentar* ‘regret’), modal (e.g., *ser posible* ‘be possible’), directive/request (e.g., *ordenar* ‘command’, *suplicar* ‘beg’), causative (*conseguir* ‘achieve’) and dubitative (e.g., *dudar* ‘doubt’) predicates.

This behavior was similar in Old Spanish⁹ for most of the predicates mentioned above. Jensen & Lathrop (1973) state that epistemic predicates, certainty predicates, communication verbs, and fiction verbs all appeared with an indicative complement clause. This has been corroborated by searching for examples in the *Diccionario histórico de la lengua española* (CDH) (Real Academia Española 2013). For the verb *saber* ‘know’ (epistemic), *ser cierto* ‘be true’ (certainty), *contar* ‘tell’ (communication), *soñar* ‘dream’ (fiction), and *adivinar* ‘guess’ (mental judgment), all examples that could be found took the indicative. It is important to note, however, that for the predicate *adivinar* only 11 sentences could be found, and the conclusions might not be fully representative of the language as a whole. Moreover, as was mentioned previously, the fact that the subjunctive is not found with *soñar* need not be indicative of the fact that the subjunctive use after this predicate is a new development. It could be due to the subjunctive option being less frequent, as in the present-day language. Some of the examples can be found in (22)–(25).

- (22) **sabré** que es verdat. (epistemic)
 know.FUT.IND that be.PRS.IND true
 ‘I will know that it is true.’
 (*La fazienda de Ultra Mar*, 13th century, CDH)

⁹Contrary to other languages which have three clearly delimited historical periods (e.g., Old English, Middle English, Modern English), most Spanish historical manuals identify two key periods: Old Spanish (9th–15th centuries) and Modern Spanish (15th century onward) (Eberenz 1991). There is a lot of ongoing discussion regarding different periodization alternatives (for an overview see Eberenz 2023). For simplicity, this book will follow the standard periodization unless explicitly indicated.

- (23) **es cierto** que nos **podremos** defender (certainty)
 is true that us can.FUT.IND defend
 ‘It’s true that we will be able to protect ourselves.’
 (*El Conde Lucanor*, 1325–1335, CDH)
- (24) **contava** que **avié** grant imperio ganado (communication)
 told that have.PST.IND big empire win
 ‘He told that he had conquered a great empire.’
 (*Libro de Alexandre*, 1240–1250, CDH)
- (25) **soñaua** que el **andaua** enel mo[n]te (fiction)
 dreamt that he walk.PST.IND in.the mountain
 ‘He dreamt that he walked in the mountain.’
 (*Cuento de don Tristán de Leonís*, 1240–1250, CDH)

Similarly, some volitional predicates such as *want* (not all volitional predicates behaved the same, see more on this below), modal, directive/request, causative and doubt¹⁰ verbs seem to have selected subjunctive embedded verbs in Old Spanish as well. The relevant examples(26)-(29) have been extracted from Jensen & Lathrop (1973) and the CDH corpus. With the exception of the verb *want*, the number of sentences that could be found in the corpus was rather small.

- (26) **quiero** que esto **sea** (volitional)
 want that this be.PRS.SBJV
 ‘I want this to be...’
 (Poema de Fernán González, 1260, apud Jensen & Lathrop 1973)
- (27) **es posible** que el agua **venga** (modal)
 is possible that the water come.PRS.SBJV
 ‘It’s possible that the water comes.’
 (*Libro de Palladio*, 1380–1385, CDH)
- (28) **suplícote** que **perdones** mi lengua (request)
 beg=you that forgive.PRS.SBJV my tongue
 ‘I beg you will forgive my tongue.’
 (Manrique 155, 1047 apud Jensen & Lathrop 1973)

¹⁰Jensen & Lathrop (1973: 61) note that *dudar* ‘doubt’ requires the subjunctive, unless it is negated, in which case it can also take the indicative. This same behavior is encountered in the present-day language.

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- (29) Joan **hizo** que los **acogiessen** en el castillo. (causative)
 Joan made that them host.PST.SBJV.3PL in the castle
 ‘Joan had them welcome in the castle.’
 (*Gran crónica de Alfonso XI*, 1348–1379, CDH)
- (30) **dubdamos** que non nos lo **ouiesse** fecho saber (doubt)
 doubt that not us it had.SBJV done know
 ‘We doubt that he wouldn’t have let us know.’
 (*Respuesta de Jaime II a una carta de Don Juan Manuel (CCCXXI)*, 1327, CDH)

There are, however, several predicates that had a different behavior in Old Spanish in comparison to Modern Spanish. Doxastic, commissive, and, to a certain extent, perception and some desire verbs could take both indicative and subjunctive complements. Moreover, emotive-factive predicates took mostly indicative complements, in stark contrast to the modern language which mostly takes the subjunctive. The rest of this section will illustrate these differences with examples.

The doxastic verb *creer* could take both the indicative and the subjunctive in the past, but, as was mentioned previously, this is not possible in Modern Iberian Spanish (Quer 2010).

- (31) **Creo** que [es /*sea] Nazaret. (Modern Iberian Sp.)
 believe that be.PRS.IND be.PRS.SBJV Nazaret
 ‘I believe it’s Nazaret.’
- (32) a. Nazaret **creo** que **sea**. (Old Sp.)
 Nazaret believe that be.PRS.SBJV
 ‘I believe that it is Nazaret.’
 (Arcipreste de Hita, 14th century, apud Quer 2010)
- b. e **creo** que **es** verdat lo que escriuio Ouidio
 and believe that be.PRS.IND true it that wrote Ovid
 ‘And I think it is true what Ovid wrote.’
 (*Historia troyana en prosa y verso*, 1270, CDH)

Similarly, commissive verbs such as *promise* could take both the indicative and the subjunctive in Old Spanish, but this is no longer the case. ‘Verbs that indicate guarantees and promises, such as *dar fianza*, *poner fiador*, *asegurar*, *poner*, *prometer*, etc., very often contain a strong volitive element and therefore

take a subjunctive. [...] The verb *prometer*, if void of any element of volition, takes the indicative.” (Jensen & Lathrop 1973: 51). The relevant examples are provided in (33)-(34).¹¹

- (33) **Prometo** que [viviré] /[*viva]. (Modern Iberian Sp.)
 promise that live.FUT.IND live.PRS.SBJV
 ‘I promise I’ll live/survive.’
- (34) a. Çient quintales **promete** que **dará** de su auer. (Old Sp.)
 hundred quintal promise that give.FUT.IND of his have
 ‘He promises he will give one hundred quintals of what he has.’
 (Anonymous, *Libro de Apolonio*, 1250, apud Jensen & Lathrop 1973)
- b. E **prometo** que **viva** en todas mis cosas segund la
 and promise that live.PRS.SBJV in all my thing according to
 costumbre.
 tradition
 ‘and I promise I’ll live in all my affairs according to tradition.’
 (Fernando III, *Fuero Juzgo*, 1241, apud Jensen & Lathrop 1973)

Regarding perception verbs, Jensen & Lathrop (1973) have stated that these verbs took the indicative in Old Spanish, but looking at corpus data, one does find some sporadic cases in which the verb appeared with the subjunctive as well, see (36b). It must be highlighted though that these examples are very infrequent, so it is not completely clear to what extent this was a productive pattern in the language. For the time being, I assume that some variability might have been present, though a more quantitative study would be needed to corroborate it.

- (35) **Vio** que [**tardaba** /***tardase**]. (Modern Ib. Sp.)
 see.PST.IND that took.time.PST.IND took.time.PST.SBJV
 ‘He saw that it took a long time.’
- (36) a. **vyo** que **crecia** Seila (Old Sp.)
 see.PST.IND that grow.PST.IND Seila
 ‘He saw that Seila grew.’
 (*La fazienda de Ultra Mar*, 1200, CDH)

¹¹Although the examples provided in this case are in the future tense, note this verb can select the indicative even in non-future tenses.

(i) Te prometo que estoy bien.
to.you promise that be.PRS.IND.1SG well
'I promise you I am doing well.'

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- b. **vio** que se **tardasse**
 see.PST.IND that CLT take.time.PST.SBJV
 ‘He saw that it took time’
 (*Libro de Alexandre*, 1300, Mark David *Corpus del Español*)

Regarding volitive predicates such as *esperar* ‘hope’, the literature states that “[t]he indicative is used almost exclusively after *esperar*” (Jensen & Lathrop 1973), but nonetheless some uses of the subjunctive can also be found (Blake & Sánchez Lancis 2018). Again, this is in contrast to Modern Spanish which only allows the subjunctive after these predicates.

- (37) **Espero** que me [**des** /*das]. (Modern Iberian Sp.)
 Hope that me give.PRS.SBJV give.PRS.IND
 ‘I hope you (will) give me.’
- (38) a. **espera** que **faga** viento (Old Sp.)
 hope that make.PRS.SBJV wind
 ‘He hopes it will be windy.’
 (Ferrer Sayol, 1380–1385, apud Blake & Sánchez Lancis 2018)
- b. **esperamos** que **daredes**
 hope that give.FUT.IND
 ‘We hope you will give...’
 (Anonymous, 1290, apud Blake & Sánchez Lancis 2018)

Lastly, Jensen & Lathrop (1973) state that ‘[v]erbs of high emotional content, such as *pesar*, *ser alegre*, *espantar*, etc., occur generally with the indicative in Old Spanish.’ (Jensen & Lathrop 1973: 56). This is in stark contrast to Modern Iberian Spanish, where these predicates take for the most part a subjunctive complement:

- (39) **Lamento** que no [**seamos** /*somos] de aquí. (Modern Ib. Sp.)
 regret that not be.PRS.SBJV be.PRS.IND from here
 ‘I regret we are not from here.’
- (40) **Pesa** me que no **somos** çerteros del lugar (Old Sp.)
 grieve me that not be.PRS.IND sure of.the place
 ‘I regret that we’re sure about the place.’
 (Gonzalo de Berceo, early 13th century, apud Quer 2010)

A summary of the mood selectional patterns presented in this section can be found in Table 1.3. As can be seen, although a considerable number of predicates

have retained their original behavior, several predicates have either reduced the possible moods they can select (e.g., doxastic, commissive, perception and desire predicates such as ‘hope’) or they have changed them altogether (e.g., emotive-factive predicates). It is also interesting to note that no verb that took exclusively the subjunctive in Old Spanish has changed its selection since; an aspect which will play an important role when trying to explain what might have changed in the system.

Table 1.3: Intensional mood selectional patterns in Modern and Old Spanish in declarative affirmative clauses. Predicates in bold indicate that they have undergone a change from Old Spanish to Modern Spanish. Parentheses indicate that the option is possible but not frequent. For the predicate *soñar* it is not possible to determine whether the language did not allow the subjunctive or whether it is an infrequent option (as in Modern Spanish). For this reason, it has been marked in gray.

Verb Class	Example	Old Spanish	Modern Spanish
Epistemic	<i>saber</i> ‘know’	ind	ind
Doxastic	<i>creer</i> ‘believe’	ind/sbjv	ind
Certainty	<i>ser seguro</i> ‘be sure’	ind	ind
Perception	<i>ver</i> ‘see’	ind/(sbjv)	ind
Communication	<i>contar</i> ‘tell’	ind	ind
Commissive	<i>prometer</i> ‘promise’	ind/sbjv	ind
Fiction	<i>soñar</i> ‘dream’	ind(?sbjv)	ind/(sbjv)
Volition I	<i>querer</i> ‘want’	sbjv	sbjv
Volition II	<i>esperar</i> ‘hope’	ind/(sbjv)	sbjv
Directive/Request	<i>suplicar</i> ‘beg’	sbjv	sbjv
Causative	<i>hacer</i> ‘make/force’	sbjv	sbjv
Dubitative	<i>dudar</i> ‘doubt’	sbjv	sbjv
Emotive-factive	<i>lamentar</i> ‘regret’	ind	sbjv/(ind)
Weak Epistemic	<i>es posible</i> ‘be possible’	sbjv	sbjv

Overall, this data shows that besides the morphological changes presented in the previous section, mood has also undergone a substantial amount of change in terms of selection. This raises interesting questions as to how to formalize this change (as most current semantic theories are mostly synchronic and hence highly static) and why this change might have occurred, questions which will be discussed more in depth in the following chapters. Before that, in order to gain a more comprehensive picture of how mood selection works as a whole, it is first

necessary to describe polarity mood and the linguistic phenomena that constrain its behavior. This will be tackled in the next section.

1.4 Polarity mood selection in Iberian Spanish

As was introduced in the previous sections, polarity mood is a phenomenon whereby a verb that selects the indicative mood in affirmative contexts, can select both indicative and subjunctive morphology when negated. This is illustrated with the verb *creer* ‘believe’ in (41).

- (41) a. Creía que iban a venir.
believed that go.PST.IND to come
‘I believed that they were coming.’
b. No creía que [iban /fueran] a venir.
Not believed that go.PST.IND go.PST.SBJV to come
‘I didn’t believe that they were coming.’

In Modern Iberian Spanish, however, the structure’s overall behavior can be affected by several linguistic phenomena, to wit: (i) the type of predicate class the matrix verb belongs to, (ii) the person, number, and tense of the matrix predicate, (iii) the type of matrix clause, (iv) the information status of the embedded clause, and (v) the presence/absence of strong NPIs. The following paragraphs will describe more in depth how these five phenomena affect polarity mood in the modern language.

Starting with the verb class, the meaning that the alternation conveys seems to be different depending on the type of matrix predicate. If the matrix verb is semi-factive, the indicative option presupposes speaker commitment towards *p*, while the subjunctive does not carry such a presupposition, see (42) (Quer 1998). With other verb classes (e.g., non-factives) this presuppositional difference is not as clear-cut, see (43), and recent experimental evidence, see Montero & Romero (2023), could only replicate the effect of mood on speaker commitment when the verb was semi-factive.

- (42) a. María no sabía que Ana estaba enferma. $\leadsto$ Ana was sick.
María not knew that Ana be.PST.IND sick
‘María didn’t know that Ana was sick.’
a. María no sabía que Ana estuviese enferma.
María not knew that Ana be.PST.SBJV sick
‘María didn’t know whether Ana was sick.’

- (43) a. *María no pensaba que Juana estaba enferma.*
 María not thought that Juana be.PST.IND sick
 ‘María didn’t think that Juana was sick.’
 a. *María no pensaba que Juana estuviese enferma.*
 María not thought that Juana be.PST.SBJV sick
 ‘María didn’t think that Juana was sick.’

Second, the presence of a first person subject and a present tense matrix verb reduces the acceptability of the indicative, a phenomenon that will be referred to as ‘The First Person Effect’ (Quer 1998). For instance, in sentence (44), despite having the verb *creer* ‘believe’ negated, the use of the indicative is considered to be only marginally grammatical to native speakers.

- (44) *Yo no creo que [??está /esté] frío.*
 I not believe.PRS.IND.1SG that be.PRS.IND be.PRS.SBJV cold
 ‘I don’t think it is cold.’

Third, if the matrix clause is imperative, the only available option is the indicative (Borgonovo 2003). In example (45), even if the verb *creer* ‘believe’ is negated, the use of the subjunctive is ungrammatical, because the matrix clause is imperative.

- (45) *¡No creas que el dinero [crece /*crezca] de los árboles!*
 not believe that the money grow.PRS.IND grow.PRS.SBJV of the trees
 ‘Don’t believe money grows on trees!’

Fourth, there is an effect of information status, whereby the indicative option is preferred if the information presented in the subordinate clause is new in the context (Faulkner 2021). It must be mentioned, though, that this effect is rather small, and experimentally the mean difference between using the indicative and the subjunctive was less than 0.5 on a 7-point Likert Scale (Faulkner 2021).

Fifth, the presence of a Strong Negative Polarity Item (NPI) such as *hasta las siete* ‘until seven’ in the embedded clause, see (46), degrades the use of the indicative (see Rivero 1971 and for a recent experimental study Montero et al. 2025).

- (46) *No pensaba que [??iban /fueran] a llegar hasta las siete.*
 Not think that go.PST.IND go.PST.SBJV to arrive until the seven
 ‘I didn’t think they were going to arrive until seven.’

1 Introduction

It is interesting to note that this effect only occurs in embedded clauses, and the presence of an NPI in an indicative matrix clause is fully grammatical, see (47).

- (47) No iban a llegar hasta las siete.
Not go.PST.IND to arrive until the seven
'They weren't going to arrive until seven.'

In summary, polarity mood is an intricate phenomenon and the availability of different moods and their interpretations is constrained by multiple factors, not just the presence/absence of negation. As will be shown in Chapter 2, this complexity is difficult to explain in theoretical terms, and no current proposal is able to capture all the different effects presented above. Moreover, it raises the question of whether this complex pattern was already present at earlier stages of the language or has developed over time.

According to B-Violette (2019), polarity mood was already present in Vulgar Latin when the matrix predicate was a semi-factive verb (e.g., *know*), but the intricacies regarding how it spread to other verb classes and contexts are still poorly understood. In the case of French, B-Violette (2019) argues that the process was slow but by the end of the Middle French period, the pattern had spread to most predicates. Regarding Spanish, there is no study, that I am aware of, analyzing the spread of polarity mood, the frequencies or the linguistic factors that constrained the phenomenon. Searching in the CDH corpus, one can observe that indeed some of the predicates that exclusively selected the indicative (epistemic, certainty, and communication predicates) could take the subjunctive in negated clauses by 1200–1400 (for fictional and mental judgment predicates there were no examples with negation in the relevant period).

- (48) a. non **sabe** que la yglesia **ouiesse** dos juizes
not know that the church have.PST.SBJV two judges
'He doesn't know that there were two judges in the church.'
(«Sentencia» (*Documentos de la Catedral de León*), 1266, CDH)
- b. non era **çierto** que don Joan Manuel e don Joan Nuñez le
not be true that don Joan Manuel and don Joan Nuñez him
quisiessen servir
want.PST.SBJV serve
'It was not true that Don Joan Manuel and Don Joan Nuñez wanted to serve him.'
(*Gran crónica de Alfonso XI*, 1348–1379, CDH)

c. no cuenta que fallase

not tell that fail.PST.SBJV

‘He didn’t tell that he failed.’ (*Cuento de don Tristán de Leonís*, 1313–c1410)

However, one single example is not representative of the language as a whole, as it might represent a transcription error, a stylistic convention, etc. Hence, a more nuanced study regarding the frequencies would be needed to determine whether polarity mood was possible outside of cognitive-factive verbs.

Lastly, regarding the five phenomena that affect polarity mood in Modern Spanish (the First Person Effect, the type of matrix predicate, the type of matrix clause, the information status of the embedded clause, and the presence/absence of NPIs), there is no literature that I could find on the topic. A preliminary search indicates that the constraints might have been different in Old Spanish as one finds examples with the indicative even in those cases where the present-day language rejects them.

- (49) a. No creo que codicias la vida. (Old Sp.)
not believe that lust.PRS.IND the life

‘I don’t think you covet life.’

(1496, *Tratado que hizo Nicolás Núñez sobre el que Sant Pedro compuso de Leriano y Laureola llamado Cárcel de amor*)

- b. No creo que [*codicias/ codicies] la vida.
not believe that lust.PRS.IND lust.PRS.SBJV the life

(Modern Iberian Sp.)

‘I don’t think you covet life.’

There is, thus, an important gap regarding the development of polarity mood in the history of Spanish. The next section will describe how the present book aims to fill this and other gaps in our understanding of mood selection.

1.5 Research aims

As was shown in the previous sections, mood in Spanish has been subject to a substantial amount of variation, both in terms of its morphology and in terms of its selectional pattern. Although there is a considerable amount of research analyzing the morphological changes in the mood system (see Veiga 2006 for an overview), less work has been devoted to investigating how and why mood selection has changed over time (two notable exceptions being Jensen & Lathrop

1973 and Blake & Sánchez Lancis 2018). This gap is especially salient when looking at polarity mood, a phenomenon which, to my knowledge, has never been extensively investigated in the history of Spanish.

Moreover, mood in Spanish interacts with many other linguistic phenomena, such as verb class, negation, information status, etc. creating a complex interwoven pattern of connections. In the last 30 years, there has been a huge amount of research focusing on how to formalize the interactions of mood with the matrix predicate's verb class, i.e., intensional mood (see Portner 2018 for an overview), but the other effects are less well studied.

This book aims to complement the existing literature on the topic, by providing an in-depth analysis of polarity mood and how it has developed over time from both a theoretical and an empirical perspective. It must be highlighted that although the focus is polarity mood, the overarching intention is to provide an analysis that could explain mood selection more generally. For this reason, intensional mood will be occasionally mentioned to draw ideas and possible connections. The following sections summarize the research aims of the book.

1.5.1 Empirical goals: Diachronic development of polarity mood

From an empirical point of view, the main goal is to provide new corpus data regarding how polarity mood developed in the history of Spanish. The phenomenon will be explored in terms of frequencies, and how the frequencies might change based on the linguistic context. Based on this information, it will be explored how the phenomenon was affected by: the verb class of the matrix predicate, effect (i); the tense and person on the matrix predicate, effect (ii); and the type of matrix clause, effect (iii). The other two effects (the effect of information status and NPIs) are harder to investigate using corpora given the low frequency of NPIs in speech, and the subjectivity and time-consuming nature of annotating for information status. The effects will, thus, not be studied using a corpus-based methodology, but it will be discussed how the theoretical proposal could explain how these effects might have developed.

1.5.2 Theoretical goals: A new theory of polarity mood

From a theoretical point of view, the book aims to achieve two main goals. First, provide a theory of polarity mood that will explain how it can interact with the five phenomena mentioned in the previous sections: (i) the verb class effect, (ii) the first person effect, (iii) the type of matrix clause, (iv) the information status effect, and (v) the NPIs effect. Second, provide a possible explanation as to how

these effects might have emerged in the history of Spanish. The next section will describe in a nutshell the theoretical proposal that this book puts forth.

1.6 Main proposal in a nutshell

Drawing inspiration from the competing grammars framework (Kroch 1989, Yang 2000) and the theory of Variational Specialization (Wallenberg 2019), it is proposed that there might have been more than one mood selectional mechanism in the history of the language. Mood was initially used as a hedging device – similar to mood in present-day Bulgarian (Smirnova 2011) – to express the degree of commitment (grammar G_0). Then, a new meaning emerged, whereby the indicative was used to express that the world of evaluation lies in some accessible Context Set (grammar G_1) (see Schlenker 2005). At the same time, a grammaticalization process occurred, whereby mood selection started to be used as a purely syntactic marker (grammar G_2) – similar to grammaticalized Genitive case in Lithuanian (Arkadiev 2016). These three grammars were in competition, and while grammar G_1 won the competition in negated contexts in which the verb was semi-factive already by the year 1500, in those contexts where the matrix predicate was non-factive the competition took longer to be resolved. As a consequence, in the present-day language, there might still be a mixture of grammars G_1 and G_2 when the verb is negated and non-factive.

Under this account, the first person effect can be seen as a remnant of the initial grammar G_0 showing a preference for the subjunctive mood if the matrix predicate was first person and present tense, as hedging devices are more frequent in these constructions (Koev et al. 2021). The effect of NPIs can be seen as a remnant of the G_1 grammar, as in this grammar the indicative carries a presupposition which blocks the licensing of NPIs (as proposed in Montero et al. 2025). The effect of information status can be understood as a result of the competing grammars trying to specialize across the information dimension, but failing to do so as specialization cannot occur along a continuous variable (see Wallenberg 2019). The different behavior across verb classes can be explained as a result of different grammars winning in different contexts: with semi-factive verbs G_1 won over (and hence the alternation is semantically driven), and with non-factive verbs grammar G_2 (syntactic mood) is winning over (which would explain why it is extremely hard to determine if the alternation conveys any meaning in these environments). Lastly, the effect of imperative clauses can be derived if one assumes that grammar G_2 , syntactic mood, behaves similarly to other dependent cases (as in Marantz 1991). That is, polarity subjunctive mood depends on the

presence of another unvalued verbal element with which to agree. If this analysis is on the right track, then both G_1 and G_2 would only produce sentences with the indicative mood in imperative clauses. Given that with non-factive predicates the argument is that only these two grammars remain in competition, the fact that these clauses do not show mood alternations is accounted for.

The next section explains in more detail the structure of the book.

1.7 Structure of the book

The book is divided into six core chapters. The first three (chapters 2, 3, and 4) are more theoretical in nature and lay down the theoretical basis for the following chapters.

Chapter 2 starts by reviewing previous accounts on polarity mood in Spanish. The chapter shows that although current theories are able to describe several of the linguistic phenomena affecting the phenomenon in the present-day language, none, at least in their current formulations, is capable of explaining all the effects.

Chapter 3, then, provides an overview of some of the most influential semantic and syntactic theories of intensional mood. The aim is to determine if any of the existing accounts could be extended to also explain polarity mood, and to highlight key insights from previous literature.

Chapter 4 presents a new theoretical proposal for polarity mood. Drawing inspiration from previous research, this new theory combines insights from the semantic and syntactic literature on mood, and proposes, using the competing grammars framework (Kroch 1989, Yang 2000) as the foundational base, that there might have been several grammars in competition in the history of the language. Two grammars in which mood distinctions carry an interpretable difference (i.e., as a marker of hedging and as a marker indicating that the world of evaluation lies in some accessible Context Set), and one in which mood is becoming grammaticalized and lacking any meaning. The competition, however, has not been fully resolved yet, and as a consequence, even Modern Spanish might represent a transitory stage between grammars.

The following three chapters in the book are more empirical in nature and attempt to understand polarity mood from a diachronic perspective, focusing on how the approach proposed in Chapter 4 could explain its development.

Chapter 5 looks at the effect of verb class showing how the phenomenon of polarity subjunctive has evolved differently in the case of semi-factive and non-factive verbs. With non-factives, the use of the indicative started to decrease across all constructions during the 18th century (a decrease that is still ongoing

today). With semi-factives, the use of the indicative only decreased after first person constructions, and the decrease started much earlier, during the 15th century. The chapter then explores the possibility of explaining this behavior as a result of Variational Specialization (Wallenberg 2019). The proposed analysis not only accounts for the different diachronic behavior of first person constructions across verb types in diachrony, but could also explain the experimental data in Montero & Romero (2023) where a significant difference in terms of speaker commitment was found with semi-factive verbs but not with non-factive verbs.

Chapter 6 tackles the issue of the First Person Effect in more detail. It proposes that although the effect is attested with both semi-factive and non-factive verbs, the source of the effect is different. In the case of semi-factive verbs, the effect emerged as a result of grammar G_1 winning over the other grammars during the 15th century, as the use of the indicative with a first person construction would lead to a semantic contradiction in this grammar (e.g., the speaker knows p and the speaker doesn't know p). In the case of non-factive verbs, the effect emerged as a combination of G_0 taking more subjunctive complements when the matrix verb was first person and present, and the subsequent drop of the indicative during the 18th century caused by G_2 winning the other grammars. Given that the use of the indicative in the two constructions decreased at similar speed, the original difference created by grammar G_0 across constructions was preserved in the language in these environments. The first part of the chapter shows how indeed the speed of change across constructions was the same in the case of the verb *say*, and that the differences found with the verb *believe* can be explained by means of a further change (where the complement clause was merged in the structure) which indirectly affected the proportions of mood. The second part of the chapter describes the parallelisms between the Old Spanish and the present-day Bulgarian mood systems and explores the possibility of extending the analysis proposed by Smirnova (2011) for Bulgarian to Old Spanish. This third grammar G_0 (mood as a hedging device) is the one responsible, under this account, for the first person effect observed with non-factive verbs. The chapter then concludes by showing how the proposed three-grammar competition is feasible in theoretical terms.

Chapter 7 explores the remaining three effects: the effect of imperative clauses, the effect of information status and the effect of NPIs. Starting with the effect of imperative clauses, the chapter argues that if syntactic mood G_2 is analyzed as a dependent mood – similar to dependent cases (Marantz 1991) – then both grammar G_1 and G_2 predict the indicative mood to be used in negative imperatives. The emergence of the effect during the 18th century may be a result of grammar

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G_0 losing to the other grammars. The chapter then shows how information status seems to play a relevant factor whenever grammars compete in diachrony. Therefore, the fact that there is also an effect of information status on mood is not unexpected. Moreover, the chapter shows how not only can its effect be accounted for under the proposed approach, but also the size of this effect. Lastly, the chapter concludes with a brief exploration of the interaction between strong NPIs and mood morphology. Following the work by Montero et al. (2025), the section proposes that the effect can be explained via grammar G_1 blocking the licensing of NPIs. Given that it has been shown that non-truth-conditional content can block the licensing of these items, the proposal is that the presupposition that the indicative carries is also capable of blocking the licensing of NPIs.

Chapter 8 concludes the book with a summary of the main empirical observations and how the new proposal explains them. It then discusses future lines of research that could be explored.

2 Theories of polarity mood

This chapter reviews previous accounts of polarity mood (PM) in Spanish.¹ The different proposals will be divided into three major groups: the Presuppositional approach (Section 2.1), the Neg-Raising/Focus approach (Section 2.2), and the information status approach (Section 2.3). As will be shown, there does not seem to be a general consensus either on the meaning that PM conveys or on the best theoretical analysis, all accounts being able to explain the data only partially. Despite these problems, Section 2.4 shows that previous accounts provide a fruitful and interesting testing ground for understanding PM, and they will provide the basis of the following chapters of the book.

2.1 The presuppositional approach

The key idea underlying most work taking this approach (see, for example, Rivero 1971 and Quer 1998) is that the presence of the indicative and the subjunctive after negation conveys a presuppositional difference: the construction with the indicative carries the presupposition that the speaker believes the embedded proposition to be true, whereas the subjunctive does not carry such presupposition. For instance, in example (1) the construction with the verb in the indicative presupposes that the speaker who utters the sentence believes that it is raining, but the matrix subject (Dani) doesn't. In contrast, in the construction with the subjunctive the speaker may or may not be committed to it being raining.

- (1) a. Dani no creía que estaba lloviendo. $\leadsto$ It was raining.
Dani not believe that be.PRS.IND raining
b. Dani no creía que estuviera lloviendo.
Dani not believe that be.PST.SBJV raining
'Dani didn't believe it was raining.'

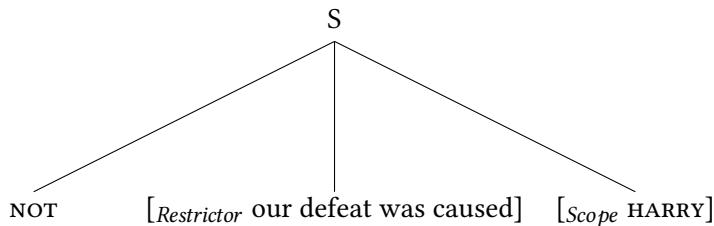
¹Other accounts have been proposed for languages like Greek, see for example Giannakidou (2009), but given the different behavior across languages, this chapter will solely focus on Spanish.

This account has been very influential because it could explain what will be referred to in this book as the First Person Effect: when the subject is first person and the matrix verb appears in present tense the only available embedded mood is the subjunctive, see (2). In these contexts, the sentence with the indicative would lead to a semantic contradiction: the matrix subject (i.e., the speaker) does not believe p and the speaker is committed to p . Thus, the sentence is ruled out on semantic grounds (Quer 1998: 61).²

- (2) # No creo que los estudiantes se merecen un premio.
 not believe that the students CLT deserve.IND a prize
 ‘I don’t believe the students deserve a prize.’ (based on Quer 1998: 61)

This presuppositional idea has been formalized in Quer (1998), where, following Partee (1995)’s analysis of negation, it is proposed that the presuppositional difference can be derived by means of two different types of negation. On the one hand, it has been argued that in sentences such as (3) negation can give rise to tripartite structures, whereby the focused part *Harry* lies within the scope of negation, but the non-focalized content is mapped onto the restrictor of negation triggering a presupposition. Exporting this idea to the realm of mood, Quer (1998) proposes that when the indicative is used in the embedded clause, negation would similarly give rise to a tripartite structure: the embedded clause would be in the scope of negation and the presupposition mapped onto the restrictor, see (4) (Quer 1998: 70).

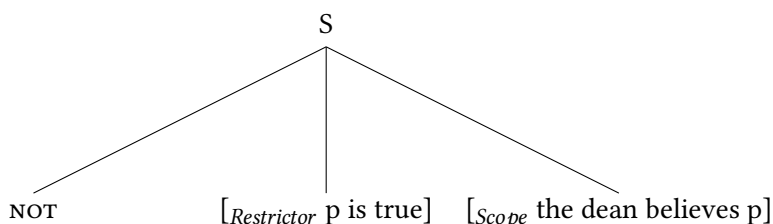
- (3) This time our defeat was not caused by HARRY $\leadsto$ we were defeated



²As it will be later shown, these examples improve in correction contexts, see (i). See also Ridruejo Alonso (1999: 3223) for similar observations.

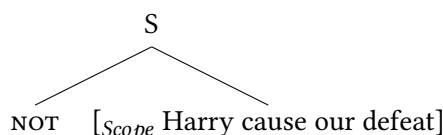
- (i) Context: A says that B believes they will win.
 B: No, yo no creo que van a ganar.
 B: no I not believe that go-IND to win
 ‘No, I don’t believe they are going to win.’

- (4) El decano no cree que los estudiantes se merecen un premio.
The dean doesn't believe that the students deserve.IND a prize $\rightsquigarrow$ p

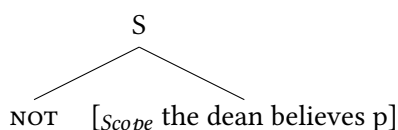


On the other hand, negation has been argued not to give rise to these tripartite structures in other contexts. For instance, a sentence such as (5) does not give rise to a presupposition and it would only requirethetic negation (Quer 1998: 71). Similarly, in the cases when the subjunctive mood is used in the embedded clause, negation would be interpreted asthetic. Since nothing is mapped onto the restrictor there would be no presupposition.

- (5) This time Harry didn't cause our DEFEAT $\rightsquigarrow$ we were defeated



- (6) El decano no cree que los estudiantes se merezcan un premio.
The dean doesn't believe that the students deserve.SBJV a prize $\rightsquigarrow$ p



This account not only derives the correct presuppositions, but it could also derive the effect of mood on the licensing of a Negative Polarity Item (NPI); a possibility which has been explored in Homer (2008) for Italian (which could possibly be extended to Spanish). In these languages the presence of the indicative in an embedded clause blocks the licensing of NPIs. For instance, in Italian the weak NPI *mai* 'ever' is only allowed when the embedded verb is in the subjunctive, see (7).

(7) Italian weak NPIs: (Homer 2008: 431)

- a. * Gianni non pensa che Maria è **mai** andata a Parigi. $\sim p$ true.
Gianni NEG think that Maria be.IND ever gone to Paris
'Gianni doesn't think that Maria has ever been to Paris.'
- b. Gianni non pensa che Maria sia **mai** andata a Parigi.
Gianni NEG think that Maria be.SBJV ever gone to Paris
'Gianni doesn't think that Maria has ever been to Paris.'

Following the line of research that proposes that NPI licensing may be disrupted by non-truth-conditional content, Homer (2008) argues that the presupposition that the indicative carries, namely that p is true, blocks the licensing of NPIs.

A similar effect has been argued to occur in Spanish with strong NPIs, see (8). According to the presuppositional approach, only the indicative version would carry the presupposition that p is true, and hence like in the case of Italian, this presupposition would block the licensing of strong NPIs.

(8) Spanish strong NPIs: (Rivero 1971: 310)

- a. * No creen que probaste **gota de vino**. $\sim p$ is true.
not believe that try.PST.IND drop of wine
'They don't believe you tried a single drop of wine.'
- b. No creen que probaras **gota de vino**.
not believe that try.PST.SBJV drop of wine
'They don't believe you tried a single drop of wine.'

Some potential problems. Although the account could derive the correct presuppositions and potentially also explain the distribution of (strong) NPIs, it does not explain why the alternation in mood only emerges with verbs that select the indicative in affirmative clauses, but not if the matrix verb is a subjunctive selecting verb. For instance, a verb such as *want* does not allow the alternation between the indicative and the subjunctive, even in the presence of negation:

- (9) a. Ana quería que Marcos viniese.
Ana wanted that Marcos come.PST.SBJV
'Ana wanted Marcos to come.'
- b. Ana no quería que Marcos [**venía*/ viniese].
Ana not wanted that Marcos come.PST.IND come.PST.SBJV
'Ana didn't want Marcos to come.'

If negation could generally take these two interpretations (categorical vs.thetic), why is only the second option only available with verbs such as *want*? The analysis would need to be expanded to explain when the different types of negation can be used.

Moreover, looking at the broader empirical picture, there are several usages of the indicative mood which the analysis does not predict. For example, under this account it is not clear why if preceded by an adverb such as *ya* ‘no longer’ the sentence with a first person pronoun and present tense becomes grammatical. According to this account, given that the indicative is used in the embedded clause, the whole sentence should presuppose that the speaker is committed to *p*; but at the same time, the sentence asserts that the speaker does not believe *p*. The construction, thus, should be ungrammatical, contrary to fact.

- (10) Ya no creo que soy una carga para ti.
 No.longer not believe that be.PRS.IND a burden to you
 ‘I not longer believe I’m a burden to you.’

Similarly, there are other contexts for which the analysis does not make the correct predictions. For example, in negative imperatives, where the speaker is clearly not committed to *p*, see (11), the only available option is the indicative mood, but the presuppositional account would predict the subjunctive.

- (11) No creas que es mentira. ✗ it is a lie.
 not believe that be.PRS.IND lie
 ‘Don’t believe that it is a lie!’

Finally and to conclude, the presuppositional difference has been recently tested experimentally in Spanish, but could only be replicated with semi-factive predicates (Montero & Romero 2023). This work tested the amount of speaker commitment towards the embedded proposition after three different verb classes, each including five different predicates: V1 semi-factive verbs (*saber* ‘know’, *ver* ‘see’, *notar* ‘notice’, *recordar* ‘remember’ and *darse cuenta* ‘find out’), V2 non-factive/non-fiction verbs (*creer* ‘believe’, *pensar* ‘think’, *decir* ‘say’, *contar* ‘tell’ and *sospechar* ‘suspect’) and V3 fiction verbs (*soñar* ‘dream’, *fantasear* ‘fantasize’, *inventarse* ‘invent’, *fingir* ‘fake’ and *hacer creer* ‘make believe’). Participants (50 speakers of Iberian Spanish) were given a sentence, see (12) for a template, and then they were asked to rate on a scale from 1 to 5 to what extent they thought the embedded proposition *p* was true. A score of 1 indicated the participant considered the proposition false, 3 that the participant was not sure, and 5 that they considered that the proposition was true.

- (12) SUBJECT didn't VERB [that p.IND/SBJV]. (Critical Items)
Is p true?

Figure 2.1 shows the results of that investigation. The only verb class for which there was a significant and large size effect of Speaker Commitment was for V1 verbs but not for V2 or V3 verbs.

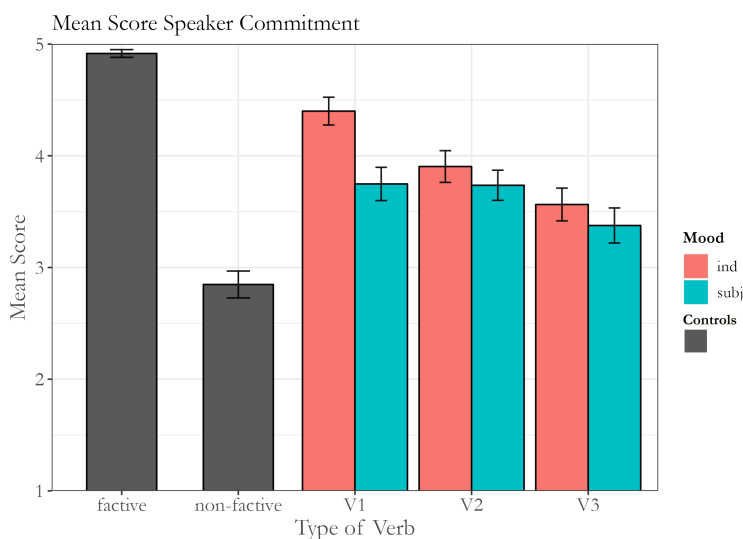


Figure 2.1: Results of the experiment testing the degree of speaker commitment from Montero & Romero (2023). A score of 1 indicates the embedded proposition was rated as false, a 5 that the speaker thought it was true, and a 3 that they didn't know.

This raises the question of how to explain the effect of verbal class on the interpretation of the embedded clause. Current accounts such as Quer (1998) put the weight of the explanation on negation itself and thus do not predict differences across predicates.

2.2 The neg-raising/focus approach

Another influential account is the neg-raising/focus account (see Reider 1990 and Borgonovo 2003, among others). Verbs like *believe* have been categorized in the literature as Neg-raising predicates because they give rise to an inference whereby negation appears to be interpreted below the matrix verb (Horn 1978). For instance, sentence (13) implies (14).

(13) I don't believe you are coming to the party.

(14) I believe you are not coming to the party.

The Neg-raising account puts forth the idea that when the indicative is used the negation is interpreted high, whereas if the subjunctive is used negation is interpreted low. This would explain why, as was illustrated before, strong NPIs such as *gota de vino* 'a drop of wine' are only licensed with subjunctive verbs. Given that negation needs to scope locally over the strong NPI, only in those contexts in which negation is interpreted low such requirement is satisfied.

(15) (Rivero 1971: 310)

- a. No creen que probaras gota de vino.
not believe that try.PST.SBJV drop of wine
'They don't believe you tried a single drop of wine.'
- b. *No creen que probaste gota de vino.
not believe that try.PST.IND drop of wine
'They don't believe you tried a single drop of wine.'

Strong NPIs have been argued to be licensed only in anti-additive environments (see Zwarts 1998, among others).³ An anti-additive environment is one where the following entailments are satisfied:

1. $f(X) \vee f(Y) \implies f(X \wedge Y)$
2. $f(X \vee Y) \implies f(X) \wedge f(Y)$
3. $f(X) \wedge f(Y) \implies f(X \vee Y)$

The first two characterize downward-entailing environments. If, in addition to these two, an environment also satisfies entailment 3, then the environment is said to be anti-additive. For example, the expression *no student* would create an anti-additive environment as illustrated by the entailment relationship in (16). However, an expression such as *not every student* would not create an anti-additive environment, see (17) (Gajewski 2005).

(16) No student smokes and no student drinks $\implies$ No student smokes or drinks

³Gajewski (2011) has proposed that the notion of downward entailing might be sufficient to explain both weak and strong NPIs, but for simplicity this section follows more conventional literature on the topic.

2 Theories of polarity mood

- (17) Not every student smokes and not every student drinks $\nRightarrow$ Not every student smokes or drinks.

Neg-raising predicates, in contrast to non-Neg-raising predicates, are anti-additive (Neg-raising behaves like $ALL \neg$ and non-Neg-raising are like $\neg ALL$) (Gajewski 2005). If the subjunctive allows the Neg-raising inference, as has been argued for, this would make the context anti-additive, hence allowing NPIs:

- (18) John believes the book is not blue and John believes that the book is not black $\Rightarrow$ John believes the book is neither blue nor black.

If the indicative blocks the Neg-raising inference, then the context would no longer be anti-additive and thus it would block strong NPIs. That is, similar to other non-Neg-raising verbs like *say*, the NPI would not be in the correct environment to be licensed.

- (19) Juan no creía que María [**había/ hubiese*] leído un libro
John not believe that Mary have.PST.IND have.PST.SBJV read a book
en años.
in years
'John didn't believe that Mary had read a book in years.'
- (20) * Juan no dijo que María [*había/ hubiese*] leído un libro en
John not say that Mary have.PST.IND have.PST.SBJV read a book in
años.
years
'John didn't say that Mary had read a book in years.'

Some potential problems. Although this proposal could also explain the incompatibility of strong NPIs with indicative embedded verbs, similar to the previous account this approach runs into problems when trying to explain why other verbs like *want* do not trigger the mood alternation, see (9a) above. The verb *want* like *believe* is a Neg-raising predicate: a sentence such as (21) implicates (22). Thus, if mood were used to mark where negation is to be interpreted, one would expect the alternation also to emerge with verbs like *want*, contrary to fact.

- (21) I don't want you to come.
(22) I want you not to come.

Negative imperatives also pose problems to this account, as the presence of the subjunctive is ruled out in these contexts, see (23), but Neg-raising inferences (as well as strong NPIs) are triggered in these environments. For example, sentence (24) shows that a negative imperative can trigger a Neg-raising inference – which is necessary to make the environment anti-veridical and license the strong NPI.

- (23) No creas que es mentira.
not believe that be.PRS.IND lie
'Don't believe that it is a lie!'
- (24) No esperes que él sepa una palabra de francés.
not expect that he know.PRS.SBJV a word of French
'Don't expect him to know a word of French.'
Neg-raising Inference: (you should) expect they won't know a word of French.

More generally, the core generalization that there is a one-to-one correlation between the subjunctive mood and the Neg-raising inference does not seem to be supported by the empirical data.⁴ Looking at present-day corpora there are multiple examples in which a Neg-raising inference is encountered with an indicative verb. The following is an example taken from the *Corpus de Referencia del Español Actual* (CREA) (Real Academia Española 2008). In sentence (25) the Neg-raising inference can be triggered (the reason why he had depression was that he thought that he was not able to handle the position), and nonetheless the indicative is used.

- (25) Tuvo una depresión de dos semanas porque no creía que
had a depression of two weeks because not believed that
iba a poder con el cargo.
go.PST.IND to can with the position
'He had a depression of two weeks because he didn't think he could
handle the position.' (CREA, El País, 02/02/1989)

Similarly, there are cases in which the subjunctive can be used in contexts in which the Neg-raising inference is not triggered. For example, in (26) the focus of negation is on the matrix verb, and given that the sentence is followed by the statement *I know it will break*, there cannot be a Neg-raising inference in this context as it would lead to a contradiction (*I know that it will break* and *I believe that it won't break*). Nonetheless, the use of the subjunctive mood is grammatical.

⁴See also Montero et al. (2025) for recent experimental evidence showing that the correlation could not be obtained experimentally.

- (26) No, no CREO que el puente se vaya a romper, sé que se
no, not believe that the bridge CLT go.PRS.SBJV to break, know that CLT
va a romper.
go to break
'No, I don't BELIEVE that the bridge will break, I KNOW it will break.'

Finally and to conclude, the observed presuppositional difference for semi-factives in Montero & Romero (2023) cannot be derived under this account. There is no formal link between a sentence triggering a speaker commitment presupposition and a predicate allowing neg-raising inferences. Thus, the account would have to postulate two independent meanings for mood morphology which are in competition: one that marks speaker commitment and another that indicates where negation ought to be interpreted. This of course need not be fatal for the analysis as the availability of two competing grammars has been widely demonstrated (Yang 2000, Kroch 1989, Santorini 1992, Pintzuk 2003), but it does highlight that the theory on its own cannot explain the full empirical picture and might need to be expanded with other existing approaches.

2.3 The information status approach

The idea that mood conveys a difference in information status has been discussed in the literature at least since Lavandera (1983) and has been further investigated in Lunn (1989), Mejías-Bikandi (1998) and Faulkner (2021), among others. This last work in particular argues that there are two types of subjunctive mood: one which is compulsory and licensed after predicates such as *want* and *order*, and another in which the subjunctive is optional and alternates with the indicative. In this last case, the indicative conveys that the information presented is new/contrastive. Interesting for current purposes is that among the predicates the work argues to behave in terms of the information status of the embedded clause are negated epistemic/doxastic verbs (e.g., *not know*, *not believe*, *not be sure*, etc.).⁵ For instance, a sentence with the predicate *be true* in which the information is presented as unknown (and hence new) would allow the use of the indicative as in (27), but one with the predicate *believe* in which the information is old the indicative would sound more marked, see (28).

⁵The author uses the term *epistemic verb* to refer to what has been referred to in this book as epistemic and doxastic predicates.

- (27) **Informative** [Comment taken from a blog about the exploitation of gas. The readers are seeing for the first time the president’s response regarding the issue.]
Hay gente que intenta politizar este asunto. No es verdad que tiene que ver con la explotación petrolera, sostuvo el presidente.
 There are people that try to politicize this issue. It is not true that it has.IND to do with the gas exploitation, maintained the president.
 (apud Faulkner 2022: 90)
- (28) **Uninformative** [Comment extracted from a blog about religion. The author of the comment is reiterating the fact that all of the blog’s participants are friends, even if they have different points of view. The blog’s readers are already well-acquainted with this perspective.]
 ?? *No creo que los que estamos en protesta somos enemigos de los que están de acuerdo.*
 I don’t believe that the ones who are in disagreement are.IND enemies of the ones that are in agreement.
 (apud Faulkner 2022: 91)

In order to corroborate these intuitions, Faulkner (2021) carried out an experiment testing several predicate types. In the case of negated epistemics, the results showed no significant effect of information status for the subjunctive clauses, but there was a significant effect of information status for the indicative ones. The following table shows the obtained means and medians for the different constructions on a 4-point Likert Scale:

Table 2.1: Acceptability ratings from Faulkner (2021) for negated epistemics.

Mood	Uninformative	Informative	Context-Free
Ind	2.15 (2 median)	2.54 (3 median)	2.55 (4 median)
Sbjv	3.44 (4 median)	3.44 (4 median)	3.53 (4 median)

The work is interesting insofar as it could explain why there is mood variation in some contexts (after negated predicates), but not in others, for instance, after verbs like *want*. This state of affairs could not be explained by the other two approaches. However, there are other potential problems with the analysis.

Some Potential Problems. Firstly, although there is an effect of information status, the observed effect is rather small (less than 0.5 difference in means), and

importantly, the indicative, although more acceptable, is not really considered fully acceptable (the mean score is 2.5 and never reaches above 3 in the scale), at least not to the same extent that the subjunctive clauses are grammatical in the same contexts (they always obtained scores above 3). That is, one cannot maintain based on these data that the indicative is used when the information is new and the subjunctive when old, as the subjunctive is always acceptable and the indicative never really reaches the same level of acceptance irrespective of context. A possibility is that the intuition was more salient in previous stages of the language and that it is slowly getting lost. This would explain the effect while at the same time accounting for its size. Chapter 7 of this book will explore this idea further.

Another problem with the proposal is that it cannot account for the First Person Effect, the blocking of NPIs by the indicative, the effect of clause type⁶, or the effect of verb class. Starting with the First Person Effect, there is no *a priori* reason why first person constructions would present first person information always as old information; hence, the reason why the subjunctive is mostly used in these constructions is unexplained. The same logic applies to NPIs and negative imperatives which, to my knowledge, are not affected by information status. For example, one could utter a negative imperative with information already mentioned before, as (29) illustrates, and nonetheless the indicative is still the only grammatical option.

- (29) Ya te lo he dicho muchas veces. No creas que el dinero
 already you it have told many times not believe that the money
 [crece/ *crezca] de los árboles.
 grow.PRS.IND grow.PRS.SBJV from the trees
 ‘I have already told you many times. Don’t you think money grows from
 trees.’

Finally, the work does not take into account differences across predicate types. The study combined the results of both semi-factives (*know* and *be true*) and non-factives (*believe* and *think*). It would be interesting to see if this had an impact on the results of information status.

⁶Here, and in the rest of this book, the term ‘clause type’ (which relates to the form of a sentence) has been used rather than ‘sentential force’ (which is related to meaning) (Chierchia & McConnell-Ginet 1990). The reason is two-fold: first, for carrying out the diachronic corpus analysis of the effect in Chapter 7 it was much more straight forward to look at the form rather than the meaning of a given sentence; second, the analysis in Chapter 7 argues that the effect is actually at the syntactic level rather than semantically driven.

All in all, it seems based on experimental evidence that the effect of information status is small and the account could not derive the other observed patterns in the linguistic literature. This is not to say that the effect is less important because it was not very large: ideally, a theory of PM would be able to derive the observed effect of information status as well as its size, while at the same time accounting for the other effects.

2.4 Potential limitations of current theories

The Presuppositional approach was developed to better understand the First Person Effect, here repeated as (30). If one uses the indicative with a first person matrix subject, the sentence would convey two contradictory pieces of meaning: the speaker does not believe *p*, and the speaker is committed to *p*. Hence, the sentence is semantically odd in these contexts.

- (30) * No creo que los estudiantes se merecen un premio.
 not believe that the students CLT deserve.PRS.IND a prize
 ‘I don’t believe the students deserve a prize.’ (Based on Quer 1998: 61)

The Neg-raising/focus approach emerged as a result of investigations on (strong) NPIs, and their interaction with mood morphology, here repeated as (31). If the indicative mood were to block the Neg-raising inference, one would obtain a natural way of deriving the ungrammaticality of having a strong NPI with the indicative: strong NPIs are only licensed in anti-additive environments.

- (31) Rivero (1971: 310)
- a. No creen que probaras gota de vino.
 not believe that try.PST.SBJV drop of wine
 ‘They don’t believe you tried a single drop of wine.’
 - b. * No creen que probaste gota de vino.
 not believe that try.PST.IND drop of wine
 ‘They don’t believe you tried a single drop of wine.’

Finally, the information status approach emerged out of the observation that the indicative in certain contexts seems to convey contrastive (new) information. This idea could explain why in informative contexts the use of the indicative sounds more grammatical than in contexts in which the information is old.

This chapter has shown that, although each individual approach could account for the effect it was originally intended to explain, none of them is able to account

for all the observed effects in the literature at the same time. Moreover, broadening the empirical domain of investigation, current accounts are not compatible with two other linguistic effects: the effect of verb class (Montero & Romero 2023) and the effect of clause type (negative imperatives). Table 2.2 provides a summary of the effects and which of these current approaches can explain them.

Table 2.2: Summary of the effects and approaches of polarity mood.

Approaches/effect	1st Person	NPI	Contrast	Verb Class	Clause type
Presuppositional	✓	✓	–	–	–
Neg-raising	–	✓	–	–	–
Information status	–	–	✓	–	–

Given these problems, the following chapter will discuss current analyses of intensional mood. The intention is to explore to what extent they could be extended to account for polarity mood as well, and to highlight key insights that could prove useful in the formation of a new theory of polarity mood.

3 Insights from Intensional Mood

Most work on mood has focused on explaining either intensional mood or polarity mood, but few works have attempted to provide an analysis of both. As a consequence, the majority of accounts present in the literature are not well suited to explain mood selection in both of these environments. This chapter reviews some of the most influential analyses of intensional mood, in order to determine whether they could be extended to account for polarity mood. The intention is not to provide an in-depth overview of all theories of mood,¹ but to point to major ideas of interest that could be used to account for polarity mood as well.

The chapter starts with an overview of the main syntactic (Section 3.1) and semantic (Section 3.2) theories of intensional mood, and concludes with a discussion of key observations and how they could be deployed when building a new theory of mood.

3.1 Syntactic theories

Although the vast majority of the recent literature devoted to explaining intensional mood takes a semantic approach, there are some works that explore the possibility of analyzing mood selection in syntactic terms. The proposals can be divided into two groups: those that use the notion of **subordination**, and those that explore the parallelism with **case marking** at the nominal level.

3.1.1 Subordination marker

Traditionally, the subjunctive mood was considered a subordination marker, whereas the indicative mood was associated with parataxis (Arias Bedoya 2021). Although this characterization has proven to be too coarse-grained and is obso-

¹There are many book-length overviews on this topic, for example, Zamorano Aguilar (2001), Quer (2017), Portner (2018), among others; for an overview on the topic from a Romance perspective see Mutet (2022) and Lindschouw (2017)

lete nowadays, at least for present-day Spanish,² several accounts still maintain that there might be some core of truth to this generalization.

For example, some authors have argued that the key difference is not between subordination vs. non-subordination but between the type of subordination: complement vs. adjunct subordination (Torrego & Uriagereka 1992, San Martín 2007).

Kempchinsky (2009) explains that in Torrego & Uriagereka (1992) it is suggested that dependent clauses appear in the subjunctive mood in Spanish, see (1). Those complement clauses that appear with the indicative are adjunct-like rather than complement subordinated clauses (either because they are in an appositive relation or have been dislocated), see (2).

- (1) V [COMPLEMENT that ... V_{sbjv}]
- (2) V [pro_i] [ADJUNCT that ... V_{ind}]

As evidence for this, they use the availability of Main Clause Phenomena (e.g., ‘as for’-Topicalization) in indicative complements but not in subjunctive complements:

- (3) (Torrego 2018: 105)
 - a. Aristóteles creía que, *en cuanto a la tragedia*, debía haber tres unidades.
‘Aristotle thought that, *as far as Tragedy was concerned*, there should.IND be three units.’
 - b. *Aristóteles quería que, *en cuanto a la tragedia*, hubiera tres unidades.
‘Aristotle wanted that, *as far as Tragedy was concerned*, there should.SBJV be three units.’

The analysis in a nutshell states that predicates that allow for Main Clause Phenomena (MCP) take adjunct-like subordinated clauses which should have the verb in the indicative; predicates that disallow MCP take complement subordinated clauses and should have the subjunctive morphology.

Following Hooper & Thompson (1973)’s generalization that MCP operations are available with doxastic (non-)factives (Classes B and E) and speech act predicates (Class A), but not with non-asserted and non-presupposed verbs (Class C)

²This is not the case in other languages. For example, in German *Konjunktiv I* is still analyzed by some authors as a pure marker of subordination that has lost its modal meaning (Fábregas 2014: 59; Thieroff 2010: 139–140). Similarly, in several Old Germanic languages (e.g., Old High German) it has been argued that mood had predominantly syntactic functions and only later started gaining more transparent semantico-pragmatic functions (see Coniglio & Hinterhölzl 2020 for an in-depth discussion). See also Giorgi (2009) for a similar proposal for Italian.

and emotive-factives (Class D), the prediction is that the first set of predicates but not the latter will select the indicative. This prediction seems, at least superficially, to be supported by the data:

Table 3.1: Predicates that allow MCP (Hooper & Thompson 1973) and the predictions (prd.) by Torrego & Uriagereka (1992) regarding mood in Spanish (Sp.).

Predicate Type	Example	MCP	Prd. Mood	Sp. Mood
speech act (A)	<i>decir</i> 'say'	yes	ind	ind
doxastic non-factive (B)	<i>pensar</i> 'think'	yes	ind	ind
non-asserted/non-presupposed (C)	<i>negar</i> 'deny'	no	sbjv	sbjv
emotive factives (D)	<i>lamentar</i> 'regret'	no	sbjv	sbjv
doxastic semi-factives (E)	<i>saber</i> 'know'	yes	ind	ind

There are, however, a few potential problems with this analysis. Firstly, the correlation between having the indicative and the structure in (2) – V[pro][Adjunct Clause] – is not supported by the empirical data, nor is it theoretically motivated. In present-day Spanish, there is a construction with the adverb *ya* ('already') that is used for emphasis and which allows the *pro* element in structure (2) to be overtly pronounced. That is, structures with *ya* and the pronoun *lo* have unambiguously the structure in (2). Nonetheless, examples in (4) show that this construction can take both the indicative or the subjunctive, depending on the matrix predicate.

- (4) a. Ya **lo** creo que va a venir.
 PRT it believe that go.PRS.IND to come
 'I believe it, that he/she/it is going to come ($\approx$ It's bound to come).'
- b. Ya **lo** lamento que vaya a venir.
 PRT it regret that go.PRS.SBJV to come
 'I regret it, that he/she/it is going to come.'

Secondly, it has been observed in the literature that not all MCPs (topicalization, dislocation, VP-preposing, speech act adverbs, V2, etc.) pattern the same

across predicate types (see Djärv 2019, Hooper & Thompson 1973, among others, for a more detailed argumentation of the behavior of different MCPs). For example, while epistemic predicates have been argued to allow for VP-preposing, some have argued that these predicates do not allow topicalization, as in (5).³

- (5) a. **Topicalization, epistemic:**
 *Mary **realizes** that [this book], John read.
 (Djärv 2019: 395 and references cited therein)
- b. **VP-preposing, epistemic:**
 Mary plans for John to marry her, and I **know** that [marry her] he will.
 (Caplan & Djärv 2019: 4 and references cited therein)

Thus, it is unclear under their account which is the relevant MCP, if any, that correlates with mood morphology. Meinunger (2017), Truckenbrodt (2019) and Gärtner & Eyþórsson (2020) have explored the possibility that the relevant phenomenon might be V2. These works argue that there is a strong parallelism between the verbs that ban embedded V2 in Germanic (volition, causative, directives, weak epistemic, emotive-factives, and non-assertive), and those that block the indicative in Romance, see (6).

- (6) a. German (no V2): (Truckenbrodt 2019: 392)
 Es ist möglich, dass Lisa da ist/sei / Lisa
 it is possible that Lisa there be.PRS.IND/PRS.KONJ / Lisa
 *ist/*sei da.
 be.PRS.IND/PRS.KONJ there
 ‘It is possible that Lisa is there.’
- b. Spanish (no indicative)
 Es posible que Lisa [esté/ *está] allí.
 is possible that Lisa be.PRS.SBJV be.PRS.IND there
 ‘It is possible that Lisa is there.’

However, these two phenomena have overlapping but not perfectly matching distributions. Negated predicates, for instance, disallow V2 in German but can take indicative complements in Spanish, see (7).

³There is a lot of intra-speaker variability for these judgments. Some native speakers (p.c.) do not notice a contrast between these examples. Similar problems are also discussed later in this chapter. Note, moreover, that there is also cross-linguistic variability in this respect as the judgments in Spanish and English differ.

- (7) a. German (no V2): (Truckenbrodt 2006: 295)
 *Hans glaubt nicht, Peter **geht** nach Hause.
 Han thinks not Peter goes to home.
 ‘Hans doesn’t think that Peter is going home.’
- b. Spanish (both the indicative and the subjunctive)
 Hans no creía que Peter se [estuviese/ estaba] yendo a
 Hans not believed that Peter CLT be.PST.SBJV be.PST.IND going to
 casa.
 home
 ‘Hans didn’t think that Peter was going home.’

Finally, the picture is also complicated by the fact that there is substantial intra-speaker and cross-linguistic variability regarding MCP (Djävrv 2019, Bianchi & Frascarelli 2010, Camacho-Taboada & Jiménez-Fernández 2013). For instance, emotive-factive predicates that usually block topicalization have been claimed to be grammatical in certain contexts: 12 out of 15 speakers reported sentence (8) to be acceptable (Bianchi & Frascarelli 2010).

- (8) ? I am glad that this unrewarding job, she has finally decided to give up.

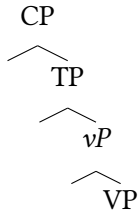
In conclusion, the proposal has both theoretical and empirical problems. Despite these, it will be shown later in the book that a weaker version of this idea might hold true, and that combined with a theory of competing grammars, it could prove fruitful in explaining some of the observed tendencies.

3.1.2 Case marker

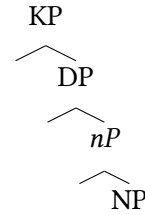
Research of the last 40 years in the generative tradition has revealed that there are strong parallelisms between the internal architecture at the clausal and nominal domains (Szabolcsi 1983, Abney 1987, Ritter 1991, Travis & Lamontagne 1992, Hiraiwa 2005, Barrie et al. 2021). Although the exact characterization of the parallelisms is still widely discussed,⁴ many authors have proposed that the Kase Phrase, Determiner Phrase and Nominal Phrase are the equivalent of the Complementizer Phrase, the Tense Phrase and the Verbal Phrase, respectively, see (9)-(10).

⁴There seems to be disagreement regarding whether the CP corresponds to the DP or KP layer. Some authors have argued that the DP is the equivalent of the CP layer, and NumP is the equivalent of TP (CP>DP; TP>NumP) (Hiraiwa 2005); others have argued that the KP is the equivalent of CP and the DP is the equivalent of the TP layer (CP>KP; TP>DP) (Barrie et al. 2021).

(9) Clausal domain



(10) Nominal domain



Building on this line of work, another idea that has been mentioned in the literature for the characterization of mood is that the subjunctive morphology might be the clausal counterpart of genitive/partitive case at the nominal level (Uriagereka 1988, Kagan 2012).

Although, to the best of my knowledge, there is not a full-fledged syntactic analysis aiming to formalize this idea, the parallelism between the two phenomena has been observed at least since Dahl (1971)'s paper *The Genitive and the Subjunctive in Russian* where it was observed that contexts which can assign genitive case to their nominal complement tend to select the subjunctive mood. In particular, there are two contexts in which there is a strong parallelism.

(I) After intensional predicates like *ask for*: these predicates can assign genitive case to their complement in languages like Russian, see (11a) (a phenomenon known as Intensional Genitive), and the subjunctive mood to their clausal complement in languages like Spanish, see (11b) (a phenomenon termed Intensional Subjunctive).

(11) a. **Intensional Genitive:** (Dahl 1971: 135)

On prosit proščenija. (Russian)
 He asks pardon.GEN
 'He asks for pardon.'

b. **Intensional Subjunctive:**

Él pide que le perdonen. (Spanish)
 He asks that him forgive.PRS.SBJV
 'He asks to be pardoned.'

A list of some of the predicates that have been mentioned in the literature to trigger both Intensional Subjunctive and Intensional Genitive is displayed below (Kagan 2012, Dahl 1971):

- volition verbs: want (*xotet'*, *querer*), wish (*želat'*, *desear*).
- emotive-factive: be afraid of (*bojat'sja*, *temer*).

- directive verbs: demand (*trebovat'*, *demandar*), ask for (*prosit'*, *pedir*).
- causative verbs: achieve (*dobivat'sja*, *conseguir*).

(II) After negation: in the Slavic and Baltic languages, a phenomenon known as Genitive of Negation triggers the use of genitive in a non-oblique internal argument of a verb, which in the affirmative is assigned accusative case, see (12). This phenomenon, although optional in Russian, has been grammaticalized in several languages such as Lithuanian, where the use of genitive is obligatory in these contexts (Kagan 2012, Ambrazas et al. 1997). Similarly, in the Romance languages the phenomenon known as Polarity Subjunctive triggers the subjunctive mood after a predicate that in affirmative clauses requires the indicative, see (13):

(12) **Genitive of Negation:**

- a. Ja ne vižu dna. (Russian)
 I not see day.GEN
 'I can't see the bottom.'
 (Bailyn 1997)
- b. Nežino tiesos. (Lithuanian)
 not.know truth.GEN
 'He doesn't know the truth.'
 (Ambrazas et al. 1997: 503)

(13) **Polarity Subjunctive**

- No veo que haya un fondo. (Spanish)
 not see that there.be.PRS.SBJV a bottom
 'I don't see that there is a bottom.'

There are, however, a few differences which are worth mentioning. First, the adverbs *possible*, *maybe*, etc. trigger mood alternations in Romance but do not license Genitive of Negation in Russian (p.c. Irina Burukina).

- (14) Mara **quizás** [venga/ viene] mañana.
 Mara maybe come.PRS.SBJV come.PRS.IND tomorrow
 'Mara maybe comes tomorrow.'

Second, as pointed out by Kagan (2012), counterfactual conditionals license the subjunctive mood in Spanish but not genitive:

3 *Insights from Intensional Mood*

- (15) a. Esli by Dima sotvoril čudo/*čuda, Maša byla by rada.
 if SBJV Dima created miracle.ACC/GEN Masha was SBJV glad.
 ‘if Dima worked a wonder, Masha would be glad.’ (Kagan 2012: 115)
- b. Si tuviese sueño, estaría bostezando.
 if have.PST.SBJV sleep, be.COND yawning
 ‘If I/he/she were sleepy, I/he/she would be yawning.’

Lastly, while in Romance languages there is a tendency for Polarity Subjunctive to be optional and Intensional Subjunctive to be compulsory, the pattern in which Genitive of Negation is optional and Intensional Genitive is obligatory seems to be absent, at least in the Slavic and Baltic languages, where all other combinatorial patterns exist, see Table 3.2.

Table 3.2: Availability of Intensional Subjunctive/Genitive and Polarity Subjunctive/ Genitive of Negation. Crosslinguistic data on Slavic and Baltic languages based on Kagan (2012: 23)

Pattern	Language	Polarity subjunctive	Intensional subjunctive
OP-OB	Spanish	Optional	Obligatory
	French	Optional	Obligatory
	Italian	Optional	Obligatory
	Portuguese	Optional	Obligatory
Pattern	Language	Genitive of negation	Intensional genitive
OB-OB	Old Church Slavonic	Obligatory	Obligatory
	Lithuanian	Obligatory	Obligatory
OP-OP	Russian	Optional	Optional
	Ukrainian	Optional	Optional
	Belarusian	Optional	Optional
	Latvian	Optional	Optional
OB-OP	Slovenian	Obligatory	Optional
	Polish	Obligatory	? Optional
Absent	Serbo-Croatian	Essentially absent	Essentially absent

Extending the analysis of the genitive case in these languages to the subjunctive mood in Spanish is complicated due to these differences. On the one hand,

if one takes the pattern OP-OP as in languages like Russian, the account would have to explain why in Spanish there is no variation in mood with intensional predicates (volitional, causatives, directives and emotive-factives), see Table 3.3 for the similarities and differences between the two phenomena. On the other hand, extending an analysis of a fully grammaticalized genitive (OBL-OBL) as in Lithuanian has the problem of explaining why there is variation after negated predicates in Spanish. If they were grammaticalized, there should be no alternation. One could potentially claim that in Spanish, Intensional Subjunctive is grammaticalized (obligatory), whereas Polarity Subjunctive has not been grammaticalized; but this does not seem to be a frequent grammaticalization path (as observed by its absence in the case of the genitive), and the parallelism across constructions would be lost. That is, one would have to explain why in Romance only Intensional Subjunctive has become obligatory, but there is no instance in the literature (at least that I could find), where only Intensional Genitive has become obligatory.

In conclusion, although there is indeed a strong correlation between the genitive and the subjunctive, there is not a perfect one-to-one correlation, which complicates a unified analysis. The next chapters, however, will present an analysis that interprets these differences as the result of two competing mechanisms.

3.2 Semantic theories

Even though syntactic accounts were predominant until the 20th century most recent proposals acknowledge the crucial role of semantics in mood selection. Determining the precise semantic contribution, however, has proven difficult. A traditional intuition, which goes back at least to Bello (1847), proposes that the subjunctive is the form of the *irrealis*. Other proposals, among many others, include futurity and non-specificity. See Fábregas (2014) for a summary of different proposals.

The main problem for defining the meaning of the subjunctive stems from its plurifunctionality. Already in languages like Latin, it had many interpretations and uses: potential, deliberative, prospective, volitional, hypothetical, etc. (López Fonseca 2009). Some authors have ascribed this myriad of functions to the fact that the subjunctive in Latin conflates the forms and the meanings of the Indo-European subjunctive and optative moods (Magni 2010). The problem of determining the meaning of mood morphology is, thus, complicated as it interacts with the historical evolution of the forms.⁵

⁵ An in-depth discussion of the semantics of the mood and modal systems in Latin is beyond the scope of this work, but the interested reader is referred to Núñez (1991).

Table 3.3: Summary of the distribution of Mood and Genitive. Cells in gray mean that at least they take that case, but this does not exclude that they could appear with other cases too. The information for those cells is based on linguistic examples and not on grammatical rules. Bolded text indicates that there is a mismatch between mood and case (i.e., there cannot be a one-to-one mapping between ind-acc and case (i.e., there cannot be a one-to-one mapping between ind-acc and sbjv-gen). MS = Mood in Spanish, GR = Genitive in Russian, GL = Grammaticalized Genitive in Lithuanian. LKŽ = Lietuvių Kalbos Žodynas (Dictionary of Lithuanian).

Predicate	Verb	MS	Russian	GR	Reference	Lithuanian	GL	Reference
Doxastic	believe	ind	verit'	acc	Native Speaker	tikėti	acc	Ramoniėnė & Press (2015: 317)
Communication Epistemic	say	ind	govorit'	acc	Native Speaker	sakyti	acc	LKŽ
	know	ind	znat'	acc	Becerra (2006)	žinoti	acc	Bjarnadóttir (2014: 40)
Fiction	dream	ind	mečtat'	(pp case)	Native Speaker	sapnuoti	acc	Ambrasas et al. (1997: 606)
Commissives	promise/vow	ind	obeščat'	acc	Kagan (2007: 159)	prisiekti	acc	LKŽ
Volition	want	sbjv	xotet'	gen/acc	Kagan (2012: 7)	norėti	gen	Bjarnadóttir (2014: 42)
Causative Directives	achieve	sbjv	dobivat' sja	gen/acc	Dahl (1971: 135)	pasiekti	acc	LKŽ
	order	sbjv	trebovat'	gen/acc	Kagan (2012: 7)	reikalauti	gen	Ramoniėnė & Press (2015: 730)
Emotive	fear	sbjv	bojat' sja	gen/acc	Polinsky (1996: 36)	bijoti	gen	Bjarnadóttir (2014: 43)
Emotive-factive	regret	sbjv	sožalet'	(pp case)	Native Speaker	gailėtis	gen	Sigurðsson & Sereikaitė (2023: 172)
Non-assertive	deny	sbjv/(ind)	otricat'	acc	Native Speaker	neigti	acc	LKŽ
Negated predicate	not V	sbjv/ind	ne vižu	gen/acc	Neidle (1988: 34)	nežino	gen	Ambrasas et al. (1997: 503)

From a formal perspective within the framework of possible-world semantics, which will be the focus of this book (see coming subsection for more information), two major approaches could be delineated: those that postulate that the subjunctive mood carries a specific meaning such as evaluation (Baunaz & Puskás 2022), a non-null ordering source (Giorgi & Pianesi 1997), non-veridicality (Gianakidou 2009) or model shift (Quer 1998), among others; and those that propose that it is a default device and the apparent meaning emerges as a result of competition with other forms (Schlenker 2005, Siegel 2009, Faulkner 2022).

The following section describes one representative work of each of these two camps. The accounts chosen are Giorgi & Pianesi (1997), where it is proposed that the subjunctive conveys a comparative meaning, and Schlenker (2005), where the subjunctive mood is analyzed as a default without any meaning. Although the main focus will be on these two texts, references to other influential ideas will be mentioned when necessary. Before tackling the different proposals, however, a brief overview of the framework of possible world semantics will be introduced.⁶

3.2.1 The possible-worlds framework

There are three key ideas from the possible-worlds framework which constitute the cornerstone of most current semantic accounts of mood, and hence are essential in order to fully engage with the debates on the analyses of mood: (i) the notion of possible world, (ii) the analysis of modals as quantifiers over possible worlds, (iii) the extension of this analysis to attitude predicates.

A possible world is a representation of an actual or non-actual state of affairs/sets of circumstances (Chierchia & McConnell-Ginet 1990, Perry 1998, von Stechow & Heim 2011). This notion, which in the philosophical tradition dates back to the work of Leibniz, has played a key role in first-order modal logic and semantics (Menzel 2016). In particular, this idea has been very fruitful in the study of phenomena that involve modal displacement in natural language. For example, the conditional proposition in (16) does not make a claim about the actual world, but about a world in which the car has not broken.

- (16) If the car hadn't broken last week, they would be traveling to Greece.

In order to account for this, the interpretation function (a function that maps each propositional letter of that language to a truth value) from Heim & Kratzer (1998) is relativized to a possible world: $\llbracket \cdot \rrbracket^{w,g}$, see (17). In the absence of shifting

⁶See Portner (2009) and von Stechow & Heim (2011) for a book-length introduction to the key notions discussed in this chapter.

3 Insights from Intensional Mood

devices the sentence is evaluated in the actual world (w_0), see (18). However, an expression such as *in the world of Chants of Sennaar* shifts the world parameter to a different possible world w' , see (19).

- (17) $\llbracket \text{a child speaks Anchorite} \rrbracket^{w,g} = 1$ iff $\exists x [\text{CHILD}(x,w) \wedge \text{SPEAKS}(x,a,w)]$
- (18) $\llbracket \text{a child speaks Anchorite} \rrbracket^{w_0,g} = 1$ iff $\exists x [\text{CHILD}(x,w_0) \wedge \text{SPEAKS}(x,a,w_0)]$
- (19) $\llbracket \text{In the world of Chants of Sennaar, a child speaks Anchorite} \rrbracket^{w_0,g} = 1$
iff the world w' as described in Chants of Sennaar is such that:
 $\llbracket \text{a child speaks Anchorite} \rrbracket^{w',g}$
iff the world w' as described in Chaants of Sennaar is such that:
 $\exists x [\text{CHILD}(x,w') \wedge \text{SPEAKS}(x,a,w')]$

This is known as the extensional view: a sentence returns at a given world w a truth value. However, in addition to this, we'll need the intensional view, whereby a sentence expresses a proposition or thought which can be modeled as a set of possible worlds (or equivalently its characteristic function⁷, see (20)). That is, the proposition a sentence expresses can be associated with the set of possible worlds in which it is true or a function which assigns to each relevant world a truth value (von Fintel & Heim 2011, Chierchia & McConnell-Ginet 1990).

- (20) a. $\llbracket \text{Slinky is running} \rrbracket_{\Phi}^g$
 $\{w: \text{Slinky is running in } w\}$
- b. $\llbracket \text{Slinky is running} \rrbracket_{\Phi}^g$
 $= \lambda w'. \text{RUN}(s,w')$

Under this framework, modals are then interpreted as quantifiers over possible worlds. Following Kratzer (1981), modals bring in quantificational force (existential or universal) and the worlds it quantifies over which are determined by context, see (21).

- (21) Slinky must be running
 $\llbracket [\text{must pro}_{1,<s,<s,t>>}] [\text{Slinky is running}] \rrbracket^{w,g} = 1$
 iff $\forall w' [w' \text{ is compatible with the evidence in } w \rightarrow \text{RUN}(s,w')]$

⁷For any set A , its characteristic function f is the function that, for any $x \in A$, $f(x) = 1$ and for any $x \notin A$, $f(x) = 0$ (Heim & Kratzer 1998: 24). In the following discussions I will be switching from set view to the function view freely.

Key for our purposes is that this modal analysis is usually extended to account for propositional attitude verbs (Hintikka 1961). Expressions like *believe*, *know*, *doubt*, etc., are said to express relations between individuals and propositions. Each of these verbs introduces a modal force (existential/universal) and a particular set of worlds relative to the attitude holder and the evaluation world w .

For example, the verb *believe* brings in universal quantificational force ($\forall$) and the set of worlds it quantifies over ($\cap \text{Dox}_y(w)$), see (22). This set of worlds is obtained by taking the set of propositions a given attitude holder believes in ($\text{Dox}_y(w)$) and intersecting those propositions. Given that a proposition is defined as a set of possible worlds, by taking the big intersection⁸ of all these propositions we obtain a set of worlds, see (23-24).

$$(22) \quad \llbracket \text{believe} \rrbracket^{w,g} = [\lambda p_{\langle s,t \rangle}. [\lambda y_e. \forall w'' [w'' \in \cap \text{Dox}_y(w) \rightarrow p(w'')=1]]]$$

$$(23) \quad \text{Dox}_y(w) = \{p_1, p_2, p_3, \dots\} = \{\{w_1, w_2, w_3\}, \{w_1, w_3\}, \{w_1, w_3, w_8\} \dots\}$$

$$(24) \quad \cap \text{Dox}_y(w) = \{w_1, w_3\}$$

These sets of propositions are commonly referred to as Conversational Backgrounds and have different flavors depending on the attitude predicate. The following is a list of the ones that will be used in this book:

- Epistemic (epi): sets of facts known in w .
- Doxastic (dox): set of beliefs in w .
- Deontic (deo): set of rules in force in w .
- Bouletic (bou): set of desires in w .
- Circumstantial (circ): set of circumstances in w .
- Stereotypical (stereo): set of expectations concerning what w is like.
- Reported Common Ground (rpg): set of what has been said in w .

⁸For any sets A and B , $A \cap B$ represents the intersection of the two sets (i.e., the set containing all the elements shared by A and B). The big intersection of a set of sets A , $\cap A$, is the set consisting of all the elements shared by all the sets in A (O'Donnell & Hall 2006: 194).

$$\bigcap_{i=1}^n A_i = A_1 \cap A_2 \dots \cap A_n$$

- Teleological (teleo): set of means that are possible or necessary for achieving a particular goal in *w*.

Most current accounts derive the distribution of mood based on how the semantic features of the indicative and the subjunctive interact with these conversational backgrounds. In the following subsections, and building on the theory presented in this section, some of the most important theories of the subjunctive mood are discussed.

3.2.2 Subjunctive as meaningful: Non-null ordering source

Giorgi & Pianesi (1997) proposed an analysis of mood morphology couched in Kratzer (1981)'s theory of modality. In this approach, the different interpretations of modal expressions emerge out of the interplay of three components: the Modal Force (MF), the Modal Base (MB) and an Ordering Source (OS). The force indicates whether the modal is existential or universal, the modal base represents the set of worlds where an expression is evaluated and the ordering source orders the worlds according to the degree they adhere to an ideal (likelihood, preference, etc.). Allowing the interpretation to depend on two conversational backgrounds rather than one, this system can capture a wide range of interpretations that the previous accounts of modality could not handle, including graded modal notions (*slight possibility*, *weak necessity*, *better possibility*, etc.), inconsistencies in conversational backgrounds and interactions between modals and conditionals. Take for example the case of the modal verb *must* which can have several different modal interpretations – *epistemic*, *deontic*, *bouletic*, etc. –, see (25). Under this approach to modality, the differences can be accounted for in terms of differences in the conversational backgrounds. For example, sentences (25b)–(25d) all take a *circumstantial* modal base, but importantly they all have different ordering sources, thus deriving different interpretations (Portner 2009: 73).

- (25) a. The book must have been stolen. (MB: Epistemic, OS: Doxastic)
 b. You must pay a fine. (MB: Circumstantial, OS: Teleological)
 c. I must win the race. (MB: Circumstantial, OS: Bouletic)
 d. We all must die. (MB: Circumstantial, OS: empty)

This analysis, which was originally developed to account for the behavior of modal auxiliaries, has been extended to account for attitude predicates as well. For example, a non-factive doxastic-predicate like *believe*, see (26), would have a *doxastic* modal base – the set of propositions that an agent (in this case Gonzalo)

believes about the world of evaluation (i.e., the real world or w_0) – and a null ordering source, as there is no ordering of the relevant worlds according to any specified parameter.

(26) Gonzalo believes ants are fascinating.

(27) $\lambda w_0. \forall w [w \in \cap \mathbf{Dox}_{\text{gonzalo}}(w_0) \rightarrow \text{ants are fascinating in } w]$

In contrast, a predicate like *want*, see (28), would consist of a doxastic (*dox*) modal base **and** a *bouletic* (*boul*) ordering source which ranks the worlds according to the desires of the matrix subject (Heim 1992, von Stechow & Heim 2011).⁹ In this case, a world in which Ana sees the northern lights is preferable to a world in which she doesn't.

(28) Ana wants to see the northern lights.

(29) $\lambda w_0. \forall w \in \cap \mathbf{Dox}_{\text{Ana}}(w_0) [\text{Sim}_w (\lambda w'. \text{Ana sees the northern lights in } w') \\ >_{\mathbf{Boul}_{\text{Ana}}(w_0)} \text{Sim}_w (\lambda w'. \neg \text{Ana sees the northern lights in } w')]]$

In broad lines, Giorgi & Pianesi (1997)'s proposal is that if a predicate contains a non-null ordering source (e.g., *want*) the predicate will select the subjunctive mood, else the indicative will be used. Applying the analysis to Spanish,¹⁰ the account predicts correctly the behavior of a wide array of verbs. Doxastic (30a), epistemic (30b), communication (30c), and fiction verbs (30d) all have been argued to have a null ordering source and they all select the indicative in Spanish.

(30) Ordering source: null $\rightarrow$ indicative

- a. $\llbracket x \text{ thinks/believes that } p \rrbracket^{w_0}$
 $\forall w [w \in \cap \mathbf{Dox}_x(w_0) \rightarrow p(w)]$
- b. $\llbracket x \text{ knows that } p \rrbracket^{w_0}$
 $\forall w [w \in \cap \mathbf{Epi}_x(w_0) \rightarrow p(w)]$
- c. $\llbracket x \text{ says that } p \rrbracket^{w_0}$
 $\forall w [w \in \cap \mathbf{Rpg}_x(w_0) \rightarrow p(w)]$
- d. $\llbracket x \text{ dreams that } p \rrbracket^{w_0}$
 $\forall w [w \in \cap \mathbf{Drm}_x(w_0) \rightarrow p(w)]$

⁹In this approach the modal base is doxastic because the relevant alternatives are those the agent believes will be realizable.

¹⁰I restrict the discussion to Spanish. Portner (1997) shows that the account has other problems in languages like French with predicates such as *hope*, *be possible* and *be probable*.

Volition (31a), directives (31b), emotive-factives (31c) and modal verbs (31d), on the other hand, have been argued to contain ordering sources in their semantics (Portner & Rubinstein 2020), and select the subjunctive mood in Spanish.

(31) Ordering source: non-null $\rightarrow$ subjunctive

- a. $\llbracket x \text{ wants that } p \rrbracket^{w_0}$
 $\forall w \in \cap Dox_x(w_0) [\text{Sim}_w(p) >_{Bou_x(w_0)} \text{Sim}_w(\neg p)]$
- b. $\llbracket x \text{ orders that } p \rrbracket^{w_0}$
 $\forall w \in \cap Rpg_x(w_0) [\text{Sim}_w(p) >_{Deo_x(w_0)} \text{Sim}_w(\neg p)]$
- c. $\llbracket x \text{ regrets that } p \rrbracket^{w_0}$
 $\forall w \in \cap Dox_x(w_0) [\text{Sim}_w(\neg p) >_{Bou_x(w_0)} \text{Sim}_w(p)]$
- d. $\llbracket \text{is probable that } p \rrbracket^{w_0}$
 $\forall w \in \cap Circ(w_0) [\text{Sim}_w(p) >_{Stereo(w_0)} \text{Sim}_w(\neg p)]$

However, there are a few predicates that this analysis does not capture. Among them are: commitment/commissive verbs such as *promise*, which do have an ordering source (Silk 2018: 130) and notwithstanding take the indicative (32), and non-assertive verbs like *deny* which have a null ordering source according to most accounts, but usually select the subjunctive mood in Spanish (33).¹¹

(32) Problematic predicates: non-null $\rightarrow$ indicative

- $\llbracket x \text{ promises that } p \rrbracket^{w_0}$
 $\forall w \in \cap Rpg_x(w_0) [\text{Sim}_w(p) >_{Deo_x(w_0)} \text{Sim}_w(\neg p)]$

(33) Problematic predicates: null $\rightarrow$ subjunctive

- $\llbracket x \text{ denies that } p \rrbracket^{w_0}$
 $\forall w [w \in \cap Rpg_x(w_0) \rightarrow \neg p(w)]$

Several current accounts have tried to modify the ordering source idea to accommodate a wider range of verbs. For example, Baunaz & Puskás (2022) have proposed that the key notion is evaluation rather than comparison, and Portner & Rubinstein (2020) have proposed a unification mechanism whereby two backgrounds may function as one under certain well-defined circumstances. Despite these advancements, the theory still runs into problems accounting for the use of the subjunctive mood after the predicates in (33).¹²

¹¹For simplicity and following Schlenker (2005), *deny* has been analyzed as *claim that not p*.

¹²Moreover, extending the analysis to explain Polarity Subjunctive is not trivial, as the authors acknowledge.

3.2.3 Subjunctive as a default: Referentiality

Schlenker (2005)'s analysis of mood follows the Referential Approach in treating mood morphology as a variable-based logical form.¹³ That is, mood morphology introduces at Logical Form a variable and some features which impose presuppositions. The difference between the indicative and the subjunctive reduces to the indicative carrying the presupposition that the world denoted by w lies in the Context Set (CS)¹⁴ of an attitude holder x' at time t' in world w' ; whereas the subjunctive carries no presupposition. The lexical entries for the indicative and the subjunctive appear in (34a) and (34b), respectively.

- (34) a. $\llbracket Ind_2^{pro1} \rrbracket^g$ is defined only if $g(2) \in g(1)$, that is, only if $g(2) \in CS^{15}$;
 if defined, $\llbracket Ind_2^{pro1} \rrbracket^g = g(2)$
 b. $\llbracket sbjv^{pro1} \rrbracket^g$ always defined;
 if defined $\llbracket sbjv^{pro1} \rrbracket^g = g(2)$

As such, indicative clauses are partial functions whose domain is restricted to the worlds in a Context Set (35), whereas subjunctive clauses are total functions and thus are always defined (36).

- (35) $\llbracket \text{that Mary walks at } Ind_2^{pro1} \rrbracket =$
 $\lambda w' : w' \in Dox_x(w_0). \text{ Mary runs in } w'$

- (36) $\llbracket \text{that Mary walks at } sbjv^{pro1} \rrbracket =$
 $\lambda w' : w' \in Dox_x(w_0). \text{ Mary runs in } w'$

In the case of a plain affirmative sentence such as (37), the anchor would be the CS of the speaker (CS^*). If one attempts to follow the sentence with the continuation 'I don't believe so', the sentence becomes semantically odd.¹⁶ The reason being that the indicative indicates that the world of evaluation lies in the $Dox_{speaker}(w_0)$ but the continuation states that w does not belong to the $Dox_{speaker}(w_0)$. Thus, the analysis predicts the correct behavior for asserted clauses.

¹³In particular, it follows the work by Stone (1997) which proposes that the same mechanism used to account for the anaphoric behavior of pronouns and tenses should be applicable to verbal morphology.

¹⁴The set of worlds compatible with what the speaker presupposes.

¹⁵CS here can refer to the Context Set of the speaker (CS^*) the Context Set of the matrix subject ($Dox_{subject}(w_0)$) or the Context Set of another attitude holder salient in the context ($Dox_x(w_0)$) (Schlenker 2005).

¹⁶The literature refers to this effect as Moore's Paradox (Giannakidou & Mari 2017).

- (37) María camina. (Spanish)
 Mary walk.PRS.IND
 ‘Mary walks.’
- (38) $\lambda w_0: w_0 \in CS^* . WALKS_{w_0}(m)$
- (39) # María camina, pero yo no lo creo.
 Mary walks but I not it believe.
 ‘# Mary walks, but I don’t believe so.’

When looking at embedded clauses the analysis also predicts (mostly) the correct behavior of epistemic predicates like *believe*. In this case, again the embedded clause in the indicative will carry the presupposition that the world of evaluation lies in an accessible CS. There are two possible options: the CS is that of the matrix subject, or the CS is that of the speaker. If the first option is taken, then we derive the correct result (40a). However, if the Context Set of the speaker (CS^*) is taken, we do not obtain the correct result, see (40b). The active domain would be the CS of the speaker but the matrix predicate requires to search for worlds in $Dox_{Gonzalo}(w_0)$. We would expect to find utterances in which uttering (40) would convey the meaning that the speaker is committed to it, but this is not the case:

- (40) Gonzalo believes that ants are fascinating.
- a. $\lambda w_0. \forall w [w \in \cap Dox_g(w_0) \rightarrow \lambda w' : w' \in Dox_g(w'). \text{ants are fascinating in } w']$
 - b. $\lambda w_0. \forall w [w \in \cap Dox_g(w_0) \rightarrow \lambda w' : w' \in CS^*. \text{ants are fascinating in } w']$

This, however, need not be fatal for the analysis if one assumes that there is a mechanism that restricts which is the available CS. One option would be to add a locality constraint, so that given the presence of the matrix subject the embedded clauses never select for the CS^* .

The account also predicts the correct selectional pattern for epistemic verbs (e.g., *know*), communication verbs (e.g., *say*), and commissive verbs (e.g., *promise*). Given that the Epistemic Set and the Reported Common Ground Set are subsets of the Doxastic set of a given individual (assuming, of course, that in the general case people don’t lie), the worlds that need to be accessed are all active, and hence the indicative should be used.

Moving on to embedded subjunctive clauses, Schlenker (2005) proposes that those predicates that select the subjunctive have a semantics that needs to access

worlds outside the active domain. For example, an emotive-factive predicate as in (41) conveys two pieces of meaning: i) Martín believes today is Friday; and ii) if it were not Friday, Martín would be less happy (counterfactual thinking). Because of this second piece of meaning, the worlds that need to be accessed are outside of the active domain (outside Martín's Context Set), and hence the subjunctive must be selected.

- (41) Martín is happy that finally it is Friday.

Counterfactual: if it were not Friday, Martín would be less happy.

This analysis can be extended to explain causative verbs which select the subjunctive in Spanish. Following Lewis (1973)'s Counterfactual Theory of causation, sentence (42) conveys the counterfactual meaning that *if it hadn't rained, the river wouldn't have overflowed* (Menzies & Beebe 2001). Thus, like in the previous case, the worlds that need to be accessed are outside of the active domain (in this case it would be the speaker's Context Set, as there is no intervening matrix subject), and the subjunctive must be used.

- (42) The heavy rains caused the river to overflow.

Counterfactual: If it hadn't rained, the river wouldn't have overflowed.

Additionally, the account might be capable of explaining why directives can select the subjunctive. The worlds that need to be accessed can be outside the $\text{Dox}_{sp}(w_0)$. For example, in a situation in which a speaker gives an order even if they believe it to be impossible (43). In this regard *promise* and *order* are different in the sense that one could order impossible things, but one cannot promise impossible things. See also Portner & Rubinstein (2012) for other relevant differences between these two predicates.

- (43) As punishment, he ordered them to reach the top of the infinite mountain.

- (44) ?? I promised that this weekend will last forever (non-metaphorical)

Despite being able to derive a wide range of predicates, there are some potential problems with this account. First, regarding volitional predicates, under an analysis which postulates a doxastic modal base, it is not entirely clear why these verbs select the subjunctive. It must be said, though, that several authors (e.g., Villalta 2008) have argued that these predicates do not have such a modal base, in which case these predicates would not be a problem for the theory.

Second, looking at fiction verbs like *dream*, the account only holds if one takes a broad definition of CS. That is, if instead of defining the Context Set as the set of worlds compatible with what an individual believes about the real world, one loosens the definition and allows for a fictional CS.

Thirdly, the mechanism as to when each of the different CS are available needs to be refined. For example, in order to account for the behavior of non-assertive predicates like *deny*, Schlenker (2005) proposes that three CSs are available. If the CS chosen is the matrix subject's one, the use of the indicative would lead to a semantic contradiction (assuming the speaker is not lying). That is, using the indicative would convey the meaning that the worlds compatible with what John claims not to be the case are all compatible with what John believes to be the case. Thus, the subjunctive is the preferred option in these cases. However, there is the possibility that instead of anchoring the world of evaluation to the matrix subject, it is anchored to the speaker (47), in which case we obtain a factive reading of *deny*, or to some other salient individual (48), in which case the meaning conveyed will be that John is trying to challenge what is thought or said by that other salient individual in the discourse.

(45) John said that Mary doesn't run.

(46) $\lambda w_0. \text{SAY}_{w_0}(j, \lambda w: w \in \text{Dox}_j w_0. \neg \text{RUN}(m))$

(47) $\lambda w_0. \text{SAY}_{w_0}(j, \lambda w: w \in \text{CS}^*. \neg \text{RUN}(m))$

(48) $\lambda w_0. \text{SAY}_{w_0}(j, \lambda w: w \in \text{CS}'. \neg \text{RUN}(m))$

One problem would be that for epistemic predicates like *think* these other options do not seem to be available, but they would be for *deny*.

Lastly, and to conclude, although the analysis has been extended by Montero & Romero (2023) to account for cases of mood alternations under negated predicates, the proposal is only capable of explaining the behavior of semi-factive verbs and does not explain, at least in its current form, the alternation with non-factive predicates.

Montero & Romero (2023)'s analysis combines the insights from Schlenker (2005) with Romoli (2015)'s scalar approach to the presuppositions of soft triggers. Romoli (2015) proposes to derive "soft presuppositions" (those triggered by, for example, semi-factive predicates) as scalar conversational implicatures. A negative clause (49) has a set of lexical alternatives of which it is the weakest element (49a). To these alternatives the operation called Exhaustification – which takes a proposition p and a set of alternatives $\text{Alt}(p)$ as arguments and outputs

the negation of all innocently excludable alternatives – applies, leading to the result in (49b). Thus, these expressions convey two pieces of information, the asserted part ($\neg \text{KNEW}(j, \text{RAN}(m))$) and the implicated part ($\text{RAN}(m)$).

- (49) John no sabía que María corría/ corriese.
 John not knew that Mary ran.IND ran.SBJV
 ‘John didn’t know that Mary ran.’
- a. Alt: $\neg \text{KNEW}(j, \text{RAN}(m))_{\text{weak}}, \neg \text{RAN}(m)_{\text{strong}}$
 - b. Exh: $\neg \text{KNEW}(j, \text{RAN}(m)) \wedge \text{RAN}(m)$

When the mood morphology is taken into account we get the following meanings. If the indicative is used the asserted part leads only to near trivial presuppositions – that $\text{Epi}_{w_0}(j) \subseteq \text{Dox}_{w_0}(j)$ – the implicated part, however, triggers the emergence of speaker commitment (50b). If the subjunctive is used no such presupposition emerges (51b):

- (50) John no sabía que María corría.
 John not knew that Mary ran.IND
- a. $\lambda w_0. \neg \text{KNEW}_{w_0}(j, \lambda w : w \in \text{Dox}_{w_0}(j). \text{RAN}_w(m))$
 - b. $\lambda w : w \in \text{CS}^* . \text{RAN}_w(m) \rightarrow \text{speaker's commitment to } p$
- (51) John no sabía que María corriese.
 John not knew that Mary ran.SBJV
- a. $\lambda w_0. \neg \text{KNEW}_{w_0}(j, \lambda w : \cancel{w} \in \text{Dox}_{\cancel{w}_0}(j). \text{RAN}_{\cancel{w}}(m))$
 - b. $\lambda w : \cancel{w} \in \text{CS}^* . \text{RAN}_{\cancel{w}}(m)$

Non-factive predicates, on the other hand, do not give rise to an ordered set of alternatives (52). Thus, the only piece of meaning they convey is the assertive part, and no speaker commitment emerges in either case (53-54). According to Maximize Presuppositions (MP!) if two alternatives are truth-conditionally equivalent the one with the strongest presuppositions should be used. In this case, given that only the indicative carries any presupposition (i.e., it’s stronger), the theory predicts that only the indicative would be used in these cases.

- (52) $\neg \text{SAID}(j, \text{RAN}(m))$
- (53) John no dijo que María corría.
 John didn’t say that Mary ran.IND
- a. $\lambda w_0. \neg \text{SAID}_{w_0}(j, \lambda w : w \in \text{Dox}_{w_0}(j). \text{RAN}_w(m))$

- (54) John no dijo que María corriese.
 John didn't say that Mary ran.SBJV
 a. $\lambda w_0. \neg \text{SAID}_{w_0}(j, \lambda w : w \in \text{Dox}_{w_0}(j). \text{RAN}_w(m))$

The account thus predicts that in cases of negation, there would be alternation between the indicative and the subjunctive, and that this difference would reflect a difference in speaker commitment. The account, however, cannot explain why there is also alternation with other negated non-factive predicates.

3.3 Extensions to polarity mood

The distribution of the subjunctive has been argued to be similar to two other syntactic phenomena: V2 in Germanic, and Intensional Genitive/Genitive of Negation in the Slavic/Baltic languages. Although the phenomena in question do have a strikingly similar distribution, the parallelisms are not perfect. Table 3.4 summarizes the similarities and differences between Mood in Spanish (MS), Verb Second in German (V2G), Genitive in Russian (GR), and grammaticalized Genitive in Lithuanian (GL). Semantic approaches also have problems deriving the correct selection pattern for Spanish mood. Importantly, both intensional accounts explored in this chapter have problems accounting for Polarity Subjunctive and non-assertive predicates like *deny* (and the same applies to other intensional accounts). In addition, fiction verbs pose a problem for the theories based on the notion of Context Set, like Schlenker (2005)'s Referential theory.

Despite the problems, something interesting is that by combining the predictions of the referential approach with the predictions of having a grammaticalized genitive (assuming the genitive is the equivalent of the subjunctive and the accusative of the indicative) one obtains a result very similar to the one desired, see Table 3.5.¹⁷ Importantly, the correct pattern would be obtained for non-assertive and negated predicates, which are the two predicates that most semantics accounts have problems explaining.

The key idea that will be explored in more depth in the next chapter is that the present-day Spanish mood pattern may be the result of multiple competing

¹⁷The only major problem might be causative predicates for which there is variation predicted, but mood in Romance does not show such variation. It is interesting, however, that causative would take accusative case in Lithuanian given that in Russian this predicate does show the alternation between genitive and accusative. That is, it seems that the rest of the paradigm of verbs that allow the alternation genitive was grammaticalized, but causatives grammaticalized in the opposite direction. Historical exploration of this development could elucidate the reasons for this difference.

Table 3.4: Summary of the distribution of Mood, V2 and Genitive and the predictions of the semantic proposal. MS = Mood in Spanish, V2G = V2 in German, GR = Genitive in Russian, GL = Grammaticalized Genitive in Lithuanian.

Predicate	MS	Syntactic approaches			Semantic approaches	
		V2G	GR	GL	Ordering	Referential
Epistemic	ind	(no)V2	acc ✓	acc ✓	ind ✓	ind ✓
Communication	ind	(no)V2	acc ✓	acc ✓	ind ✓	ind ✓
Doxastic	ind	(no)V2	acc ✓	acc ✓	ind ✓	ind ✓
Fiction	ind	(no)V2	(pp case)	acc ✓	ind ✓	ind ?
Commissives	ind	?	acc ✓	acc ✓	sbjv ✗	ind ✓
Volition	sbjv	noV2 ✓	gen/acc	gen ✓	sbjv ✓	sbjv ✓
Causative	sbjv	noV2 ✓	gen/acc	acc	sbjv ✓	sbjv ✓
Directive	sbjv	noV2 ✓	gen/acc	gen ✓	sbjv ✓	sbjv ✓
Emotive	sbjv	noV2 ✓	gen/acc	gen ✓	sbjv ✓	sbjv ✓
Emotive-factive	sbjv	noV2 ✓	(pp case)	gen ✓	sbjv ✓	sbjv ✓
Non-assertive	sbjv/(ind)	noV2 ✓	acc	acc	ind ✗	sbjv ✗
neg. semi-factives	sbjv/ind	noV2	gen/acc ✓	gen	ind ✗	ind/sbjv ✓
neg. non-factives	sbjv/ind	noV2	gen/acc ✓	gen	ind ✗	ind ✗

Table 3.5: Predictions of mood selection under a combined system.

Predicate	GL	Referential	Both (GL+Referential)
believe	acc(ind)	ind	ind
say	acc(ind)	ind	ind
know	acc(ind)	ind	ind
dream	acc(ind)	ind	ind
promise	acc(ind)	ind	ind
want	gen(sbjv)	sbjv	sbjv
achieve	acc(ind)	sbjv	ind/sbjv
order	gen(sbjv)	sbjv	sbjv
fear	gen(sbjv)	sbjv	sbjv
regret	gen(sbjv)	sbjv	sbjv
deny	acc(ind)	sbjv	ind/sbjv
not know	gen(sbjv)	ind/sbjv	sbjv/ind
not believe	gen(sbjv)	ind	sbjv/ind

grammars. In particular, one that is semantic and one that is syntactic. This is not a radical solution, as there is ample evidence that hybrid features (semantic and syntactic) are present in the nominal system as well.

For instance, several studies have shown that some nouns, known as hybrid nouns, can trigger either syntactic or semantic agreement in the exponents. The interesting aspect about these nouns is that they can trigger both syntactic and semantic agreement within the same DP/clause (Landau 2016, Wechsler & Zlatić 2000). Sentence (55) illustrates this phenomenon in Russian, where the noun *doctor* triggers masculine syntactic agreement on the adjective *new*, but feminine semantic agreement on the verb (given that the referent is female). It is important to highlight that this type of hybrid agreement is also possible within the nominal phrase. For example, in Russian when there are several adjectives in the nominal phrase, see (56), the one that is furthest away from the noun can, to a certain extent, show semantic agreement, while the one that is closest to the noun shows syntactic agreement (Smith 2017a). Similar patterns are also observed in other languages like Chichewa and Hebrew (Smith 2017a).

- (55) Nov-yj vrač-ъ prišl-a. (Russian)
 new-M doctor arrived-F
 ‘A new (female) doctor arrived.’ (Messick 2020: 2)
- (56) ? U menja očen’ interesn-aja nov-yj vrač. (Russian)
 by me very interesting-F new-M doctor
 ‘I have a very interesting new (female) doctor.’ (Smith 2017a: 4)

In order to account for this phenomenon, some authors have argued that nouns can have sets of different features (e.g., concord vs. index features) (Wechsler & Zlatić 2000), that feature sets have different loci within the DP (Landau 2016) or that nouns can have both semantic and syntactic features (Smith 2017b). Regardless of the theoretical implementation, the general consensus is that nouns can have more than one feature set. In the normal case, both semantic and syntactic features overlap, but in some circumstances this is not the case, triggering hybrid agreement.

The phenomenon has been shown to occur not only with gender features but also with number and person features. Regarding number features, in British English, for instance, sentence (57) shows that demonstratives can get singular syntactic agreement, but the verb can trigger plural semantic agreement (Smith 2017b). Regarding person features, a number of studies have shown that verbal/pronominal agreement with what have been termed as *imposters* can track

the formal (syntactic) features or the notional (semantic) features of the subject (Akkuş 2017, Wood & Sigurðsson 2014, Dudley 2014, Furuya 2019). For example, in Icelandic, Wood & Sigurðsson (2014) have argued that there are cases in which the verb can agree with the 3rd person formal features of the subject or with the notional first person subject, see (58). Similarly, in Spanish the imposter *abajo firmantes* ‘undersigned’ can trigger first person agreement on the verb (Dudley 2014).¹⁸

- (57) This committee have decided on the issue. (British English)
- (58) Undirritaðir [hafa/ ?höfum] haldið þessu fram. (Icelandic)
undersigned.M.PL have.3PL have.1PL held this forth
‘The undersigned(pl) have claimed this.’
(Wood & Sigurðsson 2014: 92)
- (59) Los abajo firmantes pensábamos en vender la casa. (Spanish)
the under signed thought.1PL in sell the house
‘The undersigned were thinking about selling the house.’
(Dudley 2014: 49)

One could even argue that the distinction between semantic and syntactic features has an equivalent in the case system as well. There are clear constructions in which case features are semantically interpretable on the DP. Not only in cases of inherent case,¹⁹ but also in contexts of case alternations such as partitive/accusative in Finnish, absolutive/dative in Warlpiri, etc., see Kiparsky (1998) and Butt (2006).²⁰

In conclusion, there seems to be a tendency for most features in the nominal domain to have both semantic and syntactic instantiations (across languages and within the same language), see (60-63).

¹⁸It has been argued that there are languages in which there is alternation between the two (e.g., Icelandic, Spanish), languages which only track formal features (e.g., Bengali), and languages which only track notional features of the imposter (e.g., Mandarin) (Akkuş 2017).

¹⁹In the generative tradition, inherent case is used to refer to case that is dependent on semantic features and structural to the case that is dependent solely on structural configuration.

²⁰Svenonius (2001) and Pesetsky & Torrego (2004) have argued that nominative and accusative could be understood as uninterpretable Tense and Inner Aspect, respectively. Thus, although case is not interpretable on the DP itself, it would be interpretable somewhere else in the structure. For the present purposes, it is sufficient that case on the DP is sometimes semantically interpretable and sometimes semantically inert.

3 Insights from Intensional Mood

(60) Gender:

- | | |
|---|--|
| a. Syntactic agreement: [<i>uG</i> : <i>fem</i>]
Naša braća.
our.F brother
(Despić 2017: 254)
(Serbian) | b. Semantic agreement: [<i>iG</i> : <i>fem</i>]
Novyj vrač-ъ prišl-a.
new doctor arrived-F
(Messick 2020: 2)
(Russian) |
|---|--|

(61) Number:

- | | |
|--|--|
| a. Syntactic agreement: [<i>uN</i> : <i>pl</i>]
The glasses are.PL broken.
(Hahm 2010: 2)
(English) | b. Semantic agreement: [<i>iN</i> : <i>pl</i>]
The committee decide.PL who is hired.
(Smith 2017b: 824)
(English) |
|--|--|

(62) Case:

- | | |
|---|---|
| a. Syntactic case: [<i>uCase</i> : <i>gen</i>]
Jonas ne-perskaitė laišk-o.
Jonas not-read letter.GEN
(Sigurðsson & Šereikaitė 2023: 2)
(Lithuanian) | b. Semantic case: [<i>iCase</i> : <i>gen</i>]
Ja ne vižu dna.
I not see day.GEN
(Bailyn 1997: 14)
(Russian) |
|---|---|

(63) Person:

- | | |
|---|---|
| a. Syntactic agreement: [<i>uPerson</i> : 3]
Un servidor compró...
a servant(I) bought.3sg
(Dudley 2014: 49)
(Spanish) | b. Semantic agreement: [<i>iPerson</i> : 1]
Los abajo firmantes pensábamos.
the under signed(we) thought.1pl
(Dudley 2014: 49)
(Spanish) |
|---|---|

The present work explores the possibility that something similar might exist at the clausal level, in particular regarding mood features. That is, mood features may have both syntactic and semantic instantiations, as in (64).

- (64) a. Syntactic mood: [*uMood*: *sbjv*] b. Semantic mood: [*iMood*: *sbjv*]

The following chapter will discuss this proposal in more detail and will look at how the competition between grammars could explain both the contemporary mood pattern of Spanish and the diachronic changes that the system has undergone (see Chapter 1).

4 Towards a new theory of mood: The competing grammars approach

4.1 The competing grammars framework

The existence of multiple competing variants/forms is a pervasive feature in natural language. Examples (1)-(3) illustrate this with variable negative concord in Montréal French (Burnett 2016), optional prepositions before bare axial terms in Central Catalan (Real-Puigdollers 2021), and optional object-participle agreement in Galician (Sitaridou 2017).

- (1) J' ai (pas) vu **personne**. (Montréal French)
I have not seen no.one
'I saw no one.' (Burnett 2016: 122)

- (2) (a) fora de la cuina (Central Catalan)
at outside of the kitchen
'outside the kitchen' (Real-Puigdollers 2021: 103)

- (3) Teño **lid-o/os** muitos libros. (Galician)
Have read-M.SG/M.PL many-M.PL book-M.PL
'I have read many books.' (Sitaridou 2017: 114)

While some of this variation is language-internally or sociolinguistically conditioned (Labov 1972), other variants have been argued to be probabilistically determined (Yang 2000, Adger 2006). That is, in some cases speakers may alternate between the competing forms in a non-deterministic manner, with the linguistic and social context playing no or a very small role in the distribution of the variants. Evidence of this comes from oral interviews in which the same speaker alternates between forms probabilistically (Adger 2006, Yang & Montrul 2017) and from historical corpus data where the same writer uses the variants interchangeably across similar contexts (Schiegg & Huber 2023, Blas Arroyo 2023, Wallenberg et al. 2026).

The key idea of the competing grammars framework (Kroch 1989, 1994) is that this non-deterministic variation in natural language (i.e., non-linguistically determined alternatives) can be formalized as the competition between multiple underlying grammars, where the exact definition of grammar is theory-dependent.¹ Kroch (1989)’s foundational work was couched within the Principles & Parameters framework, and grammars were defined as the settings of a given parameter. However, the competing grammars paradigm is in principle theory independent and a grammar could be defined as a rule, construction, etc. (see Wallenberg et al. 2026 for an in-depth description of how the paradigm has evolved in the literature). In this book, the term “grammar” will be used for historical reasons, but nothing in the analysis hinges on this decision.

The grammar competition paradigm has been very influential in the field of diachronic linguistics, as it provides an explanation as to why language change can be gradual. For example, the loss of V2 in French (4) took about 400 years to reach completion (Kroch 1989, Yang 2002). This slow change can be understood as a result of the interaction of the competition of the two grammars (V2 vs. non-V2) at the individual and population level, which provides a possible explanation of why the change in two categorical grammars was not drastic.

- (4) a. Lors **oïrent** ils venir un escroiz de tonnoire. (Old French)
 then heard they come a clap of thunder.
 b. * Puis **entendirent**-ils un coup de tonnerre. (Modern French)
 then heard-they a clap of thunder.
 (Yang 2002: 377)

The framework has been used to explain many cross-linguistic changes. Within the field of Romance linguistics, the competing grammars framework has been deployed to analyze the change in differential object marking in Spanish (von Heusinger 2008), negation in Old French (Ingham 2014), the loss of OV in Latin (Danckaert 2017), the evolution of the definite article in possessive noun phrases in French, Spanish and Portuguese (Simonenko & Carlier 2022), the position of clitics in Medieval French (Olivier 2022a,b), the competition between bare nouns and the *di*+article construction in Italian (Morosi 2025), and fronting in Old Catalan (Pujol i Campeny 2025), among others.

It is important to highlight that the term competing grammar in the literature is used in a broad sense to encompass any instance in which two variants are in

¹Widespread multilingualism and diglossia are often taken as evidence of the psychological reality of this proposal.

competition. This includes syntactic, morphological and also lexical competition. For example, Wagner (1996) studies the transition from the auxiliary *haver* (5a) to *ter* (5b) in the history of Portuguese, using the competing grammars framework.

- (5) a. ... que lhe **aviam** tornado os seus servos. (Portuguese, 14th c.)
 that him had took the their slaves
 ‘... that they had taken their slaves to him.’
 (*Biblia Medieval*, 1958 ed., p.42, apud Wagner (1996: 133))
- b. ... e dizer que não me **tinha** ocorrido! (Portuguese, 20th c.)
 and to-tell that not me has occurred
 ‘... and to say that it didn’t occur to me!’
 (Jorge Reis, *Matai-vos uns aos outros!*, 1977 ed., apud Wagner (1996: 133))

The idea put forth in this chapter is that the mood system in Spanish as well as its diachronic change (see Chapter 1) could be better understood as the result of multiple competing grammars. Of course one must be careful of how exactly that competition is formalized, as one can easily run the risk of exponentially adding grammars into the explanation of linguistic variation. In order to prevent this from happening, the usual procedure is to prefer an analysis of a single grammatical system wherever possible (Santorini 1992: 620). In Section 4.2 the proposed grammar competition will be described and Chapters 5-7 will present further quantitative evidence of why the proposed three-grammar competition is needed to account for the data. Before proceeding with this analysis, however, Section 4.1.1 and Section 4.1.2 will present two additional components – the Variational Learner and Variational Specilization – that are necessary for understanding the proposal in Chapters 5-7.

4.1.1 Variational Learning

One of the most influential formalizations of the competing grammars approach is the Variational Learner model proposed by Yang (2000). In this model, speakers can internalize more than one grammar and assign a probability to each of them. In the course of learning a language these probabilities are adjusted based on the linguistic input until they converge to a given set of values.

In order to illustrate how exactly the probabilities are adjusted, a simple case with two competing grammars will be used. One grammar will be called Grammar 1 (G_1) and the other Grammar 2 (G_2). The set of outputs produced by these

grammars will be called Languages and will be represented as L_1 and L_2 , respectively. Following the notation in Kauhanen (2023), the probabilities/weights a given learner assigns to each grammar will be represented with the notation $P(W_n)$. For simplicity, it will be assumed that in the initial state each grammar is assigned a weight of 0.5 (i.e., the learner is in a neutral state regarding which grammar is better), see (6)-(7).

(6) Grammars: G_1 and G_2

(7) Learner: Learner ($P(W_1)=0.5$, $P(W_2)=0.5$)

In such situations, there will be sentences that will be parsed only by G_1 ($L_1 \setminus L_2$), sentences that both grammars will be able to parse ($L_1 \cap L_2$), and sentences that only grammar G_2 will be able to parse ($L_2 \setminus L_1$).² For example, in the case where an SOV grammar (G_1) is competing with a SVO grammar (G_2), those sentences with a SVO order, see (8a), will logically be only parsed by grammar G_2 , and those sentences with a SOV order, see (8b), by grammar G_1 . However, unaccusative sentences, see (8c), will be parsed by both grammars.

- | | | |
|-----|--|-----------------------|
| (8) | a. [The girl] _S [read] _V [a book] _O | $(L_2 \setminus L_1)$ |
| | b. [The girl] _S [a book] _O [read] _V | $(L_1 \setminus L_2)$ |
| | c. [The girl] _S [slept] _V | $(L_1 \cap L_2)$ |

The fitness parameter α denotes the probability of a speaker of G_1 uttering a sentence not compatible with G_2 ($L_1 \setminus L_2$). The fitness parameter β , in turn, denotes the probability of a speaker of G_2 uttering a sentence not compatible with G_1 ($L_2 \setminus L_1$).

(9) Fitness parameters: $\alpha = P(L_1 \setminus L_2)$, $\beta = P(L_2 \setminus L_1)$

The learner will then select a grammar with which to parse her next interaction and will penalize that grammar (reduce its probability) whenever they encounter a speaker of the opposite grammar who utters a sentence not compatible with theirs. The probability of this happening is calculated as the multiplication of both situations happening. That is, the learner encounters a speaker of the opposite grammar, and that speaker utters a sentence which the learner's current grammar cannot parse.³ In the case that the speaker had chosen grammar G_1 ,

² A Grammar, here abbreviated as G, is understood as the set of rules that generate a language. A language, abbreviated as L, is the set of strings which a grammar can produce/parse.

³ The probability of obtaining outcome A and outcome B in two independent random experiments is product of the two probabilities (Bonamente 2017): $P(A \text{ and } B) = P(A) \cdot P(B)$

the probability of penalizing that grammar would be equal to the probability of encountering a G_2 speaker ($1-p_t$) multiplied by the probability of that speaker uttering a sentence grammar G_1 cannot parse (β). This penalty is referred to as the penalty probability of G_1 (here denoted as $c1$), see (10). Conversely, if the learner had chosen grammar G_2 to try to parse the next interaction, the probability of penalizing that grammar would be equal to the probability of encountering a G_1 speaker (p_t) and the probability of that speaker uttering a sentence G_2 cannot parse (α), see (11). This penalty probability for grammar G_2 will be denoted as $c2$.

$$(10) \quad c1 = \beta \cdot (1 - p_t)$$

$$(11) \quad c2 = \alpha \cdot p_t$$

If, however, the learner chooses a grammar and the speaker they encounter utters a sentence that grammar can parse, then that grammar will be rewarded (its probability will be increased). This is known as the linear reward-penalty scheme (Bush & Mosteller 1955), and has been very influential in modeling language acquisition (Ingason et al. 2013, Danckaert 2017, Kauhanen 2021).

The velocity at which the proportion of G_1 and G_2 speakers changes over time is directly controlled by the relation between α and β , see (12) (Kauhanen 2023). A bigger difference between α and β will lead to faster change, whereas a small difference between the two would lead to slower change. The extreme case is when both fitness parameters are equal, a situation in which there will be no change.⁴

$$(12) \quad s = \log\left(\frac{\alpha}{\beta}\right)$$

Moreover, based on the values of α and β , it can be predicted which grammar will win the competition in the long run. According to Yang (2000)'s Fundamental Theorem of Language Change, the grammar that is predicted to win out in diachrony is the one with the greatest fitness (because this one will be penalized less often by the algorithm), see (13).

- (13) **The Fundamental Theorem of Language Change:** (Yang 2000: 239)
 G_2 overtakes G_1 if $\beta > \alpha$: the advantage of G_2 is greater than that of G_1 .

⁴The logarithm of a number x to the base b is the exponent to which b must be raised to produce x . The reason for taking the natural logarithm (base 10) of $\frac{\alpha}{\beta}$ is that the information is converted into a more intuitive measure which has the following properties: (i) If positive, G_1 is replacing G_2 , (ii) if it equals zero, there is no change and (iii) if negative, G_2 is replacing G_1 (Kauhanen 2023).

This model has been very influential in explaining many diachronic changes (see Wallenberg et al. 2026 for an overview), but there are several historical developments that it is not capable of straightforwardly explaining, e.g., the development of *meltd* and *molten* in the history of English. Out of the need to explain such cases, the theory of Variational Specialization (Wallenberg 2019) was developed. Section 4.1.2 will explain this theory in more detail.

4.1.2 Variational Specialization

The theory of Variational Specialization (Wallenberg 2019) emerged out of the need to account for cases in which the competition between different grammars/variants leads to different outcomes across constructions (i.e., no grammar wins in the grammar as a whole, but each grammar wins in a different environment). For example, the doublets *meltd* and *molten* had the same meaning and function during the Old and Middle English periods, but they specialized. *Meltd* was restricted to perfect/passive participle contexts and *molten* to adjectival contexts. Similarly, the alternation between *if* and *whether* in embedded polar questions (see also Wallenberg & Fruehwald 2013) specialized in the history of English: *Whether* became used mostly in cases in which there was disjunction in the embedded clause; while *if* started to be used in contexts without disjunction.

This specialization is also encountered outside of the morphological/lexical domain. Although not originally mentioned in Wallenberg (2019), in the loss of V2 in the history of English a similar situation could be found: While V2 was present in both interrogative and declarative clauses in Old English, in Modern English interrogative clauses are V2 and declarative clauses are not.⁵ That is, the V2 grammar won in interrogative clauses and the non-V2 grammar in declarative ones.

If the frequencies are tracked on the language as a whole (as proposed in Yang 2000's original implementation of the Variational Learner), how can we explain situations of variation where neither grammar wins overall and the result is a mixed system, as in the cases above? Wallenberg (2019)'s key idea is to extend the Variational Learner model by allowing the grammars to be decomposed, in certain cases, into subsets with the probabilities of each subset being tracked separately. As in the Variational Learner model, the prediction is that the grammar with the greatest fitness will be the one that wins in diachrony, but because the probabilities are tracked separately across contexts, the fitness parameters are also calculated separately across contexts. This idea is summarized in Table 4.1.

⁵This is the reason that has led some authors, for example Westergaard (2007), to argue that English is actually a mixed V2 language.

Table 4.1: Comparison of how probabilities are tracked in Yang (2000)’s Variational model and how they are tracked in the extension in Walenberg (2019)’s theory.

Variational theory	P_{G_1}	P_{G_2}
Language	p_t	$1 - p_t$
Variational Specialization theory	P_{G_1}	P_{G_2}
Context 1	p_t	$1 - p_t$
Context 2	p_t	$1 - p_t$

4.2 The three competing grammars approach to mood

Given the problems current theories have to explain polarity mood (see Chapter 4), the key idea that this book explores is whether the Spanish mood system as a whole could be better understood as a result of the competition between multiple grammars. From the introduction in Chapter 1 (see Sections 1.3-1.4), we know that the Spanish mood system must have undergone several changes, as the selectional patterns and constraints that affect PM and IM in declarative clauses seem to differ when compared to the present-day language. Regarding IM, doxastic (e.g., *believe*), commissive (e.g., *promise*), perception (e.g., *see*) and volitional (e.g., *hope*) verbs used to select both the indicative and the subjunctive to some extent, but in the present-day language they select the indicative. Emotive-factive predicates such as *regret* used to take mostly the indicative, but in the present day language there is a tendency for the subjunctive. Regarding PM environments (which are licensed when indicative-selecting verbs are negated), Modern Spanish mostly takes the subjunctive when the matrix verb is first person and present tense, but this tendency is not as clear in Old Spanish. In all other persons and tenses (i.e., present third person, past first person, etc.) both Old Spanish and Modern Spanish allow the indicative and the subjunctive. Table 4.2 summarizes these changes.

To explain these changes, at the very least, two grammars are needed: one explaining the Old Spanish system, and another explaining the Modern Spanish one. However, as was shown in Chapter 2 and Chapter 3, no current account explains the Modern Spanish data fully. This chapter therefore explores whether the changes could be explained as a result of three competing grammars.⁶

⁶See also Chapter 5 for more evidence that a two competing grammar system might not be enough.

4 Towards a new theory of mood: The competing grammars approach

Table 4.2: Intensional and polarity mood selectional patterns in Modern and Old Spanish declarative clauses. Predicates in bold indicate that they have undergone a change from Old Spanish to Modern Spanish. Parentheses indicate that the option is possible but not frequent.

	Verbs	Old Spanish	Modern Spanish
Intensional Mood	Epistemic	ind	ind
	Doxastic	ind/sbjv	ind
	Certainty	ind	ind
	Perception	ind/(sbjv)	ind
	Communication	ind	ind
	Commissive	ind/sbjv	ind
	Fiction	ind(?sbjv)	ind/(sbjv)
	Volition I (<i>want</i>)	sbjv	sbjv
	Volition II (<i>hope</i>)	ind/(sbjv)	sbjv
	Directive/Request	sbjv	sbjv
	Causative	sbjv	sbjv
	Dubitative	sbjv	sbjv
	Emotive-factive	ind	sbjv/(ind)
Polarity Mood	Weak Epistemic	sbjv	sbjv
	V.1ST. PRES	ind/sbjv	sbjv
	V.OTHER	ind/sbjv	ind/sbjv

4.2.1 G₀: Mood as a hedging device

Firstly, a mood system able to generate the Old Spanish data would be needed. Given that inferring semantic meaning for early stages of a given language is a highly complicated task, this section attempts to see if any present-day system has a similar pattern of mood selection as the one observed in Old Spanish. The intention is to see if by looking at a similar system for which there are native speaker intuitions, a clearer understanding of the Old Spanish data could be obtained.

Out of the other Romance languages (see Section 4.3 in Chapter 1 and also Table 4.3 in this chapter) none has the exact same mood selection pattern as Old Spanish. Romanian presents an interesting case as it allows the indicative and the subjunctive with commissive (e.g. *promise*) and volitional II (e.g. *hope*) verbs (Farkas 1992, Portner & Rubinstein 2020), which are key environments in which Old Spanish differs from Modern Spanish. However, there are some differ-

ences with epistemic, doxastic and dubitative predicates. In Romanian, epistemic predicates such as *know* can take both the indicative and the subjunctive (Dindelegan & Maiden 2013: 44), but in Old Spanish they only appear with the indicative (Jensen & Lathrop 1973). Moreover, in Romanian, doxastic predicates such as *believe* only take indicative complements in declarative clauses (Farkas 1992), but in Old Spanish *believe* could take both the indicative and the subjunctive complements in those same environments (Jensen & Lathrop 1973). Lastly, in Romanian, dubitative predicates take indicative complement clauses (Dindelegan & Maiden 2013: 45), but in Old Spanish these predicates mostly took the subjunctive (Jensen & Lathrop 1973). In the other Romance languages investigated in this book, there are also significant differences in terms of mood selection. Table 4.3 presents an overview. The highlighted grids indicate that the selection is the same as in Old Spanish. The verb *dream* has been grayed out, because the data from Old Spanish data was not fully conclusive. It is, therefore, hard to tell whether the underlying system would match any of the ones found in the other Romance languages.

Given that no Romance language seems to have an exact match to the Old Spanish data, an alternative route is to expand the scope of investigation, and explore to what extent systems outside of the Romance family have a similar pattern of mood selection. Subsection 4.2.1.1 presents the mood system in Bulgarian. The intention is not to claim that the two systems work in the exact same way, as there are clear differences in terms of how mood is encoded. For example, in Bulgarian, contrary to Spanish, the infinitive is not an available option. That is, what is covered in Bulgarian by the subjunctive construction covers more situations than the subjunctive in Spanish. The intention is therefore not to propose a unified analysis of the Bulgarian and Old Spanish mood systems, but rather to obtain insights and ideas as to what the meaning of mood in Old Spanish might be.

4.2.1.1 Similarities between Old Spanish and Bulgarian mood

Most Romance languages express mood distinctions as suffixes that attach to the verb, but mood exponents can also appear in other locations. For example, in Greek, as well as in several Balkan languages, mood has been argued to be expressed by means of a complementizer distinction (Giannakidou & Mari 2021, Baunaz & Puskás 2022). Looking at Bulgarian in particular, Smirnova (2011) has argued that the complementizer *če* and *da* could be analyzed as mood markers, *če* being the equivalent of the indicative and *da* of the subjunctive in the Romance languages (see Smirnova 2011's Chapter 8 for in-depth argumentation as to why they should be analyzed as mood markers). For example, a verb such as *iskam*

Table 4.3: Mood selection in affirmative declarative clauses across several Romance languages. Parentheses indicate that although possible, the choice is not as frequent.

Verb	Catalan ¹	Galician	E. Portuguese	Italian	Romanian	French	Old Sp.
Epistemic (<i>know</i>)	ind	ind ²	ind ⁵	ind ⁹	ind/(sbjv) ¹³	ind ¹⁶	ind
Doxastic (<i>believe</i>)	ind	ind(sbjv) ⁴	ind/sbjv ⁵	sbjv(ind) ⁹	ind ¹⁴	ind ¹⁶	ind/sbjv
Certainty (<i>be certain/sure</i>)	ind	ind ²	ind ⁶	ind(sbjv) ⁹	ind ¹⁴	ind ¹⁴	ind
Perception (<i>see</i>)	ind	ind ²	ind ⁷	ind ¹²	ind ¹⁵	ind ¹⁸	ind(sbjv)
Communication (<i>tell</i>)	ind	ind ^{13,12}	ind ⁵	ind(sbjv) ⁹	ind ¹³	ind ¹⁶	ind
Commissive (<i>promise</i>)	ind	ind ⁴	ind ¹²	ind ¹²	ind/sbjv ¹⁴	ind ¹⁶	ind/sbjv
Fiction (<i>dream</i>)	ind	ind/(sbjv) ¹²	ind ⁵	ind(sbjv) ⁹	ind ¹⁴	ind/(sbjv) ¹⁹	ind/(?sbjv)
Volition I (<i>want</i>)	sbjv	sbjv ²	sbjv ⁵	sbjv ⁹	sbjv ¹³	sbjv ¹⁶	sbjv
Volition II (<i>hope</i>)	sbjv/(ind)	sbjv ¹²	ind/sbjv ⁸	ind/sbjv ⁸	ind/ sbjv ⁸	ind ⁸	ind/(sbjv)
Directive (<i>order</i>)	sbjv	sbjv ²	sbjv ⁵	sbjv ⁹	sbjv ¹⁴	sbjv ¹⁶	sbjv
Causative (<i>make</i>)	sbjv	–	sbjv ⁵	–	sbjv ¹³	–	sbjv
Dubitative (<i>doubt</i>)	sbjv	sbjv ¹²	sbjv ⁵	sbjv ¹⁰	ind ¹³	sbjv ¹⁴	sbjv
Emotive-factive (<i>regret</i>)	sbjv	sbjv ¹²	sbjv ⁵	sbjv ¹¹	ind ¹⁴	sbjv/(ind) ¹⁷	ind
Weak epistemic (<i>be possible</i>)	sbjv	sbjv ¹²	sbjv ⁶	sbjv ⁹	sbjv ¹⁴	sbjv ¹⁴	sbjv

References: 1. Institut d'Estudis Catalans (2016: 933–939; 1006–1008) 2. Álvarez & Xove (2002: 272–274) 3. Frias Conde (2015) 4. Jardon & Björnsdóttir (2026) 5. Marques (2010) 6. Becker (2010a) 7. Martins (2008) 8. Portner & Rubinstein (2020) 9. Giannakidou & Mari (2021) 10. Manzini (2008) 11. Giorgi (2009) 12. Native speaker 13. Dindelegan & Maiden (2013) 14. Farkas (1992) 15. Becker (2010b) 16. Silk (2018) 17. Schlenker (2005) 18. Achard (2002) 19. Baunaz & Puskás (2022: 23)

‘want’ selects for the *da* subjunctive-like complementizer, whereas a verb such as *znam* ‘know’ selects *če*, see (14)-(15).

- (14) Maria *iska* [**da** živee v Sofia].
 Maria want SBJV live in Sofia
 ‘Maria wants to live in Sofia.’ (Smirnova 2011: 185)
- (15) Maria *znae*, [**če** živee v Sofia].
 Maria knows that(IND) live in Sofia
 ‘Maria knows that she lives in Sofia.’ (Smirnova 2011: 185)

Interestingly the pattern of mood selection in Bulgarian (Smirnova 2011, Wiemer 2021) is very similar to the one in Old Spanish, both regarding IM and PM. The parallelisms between the two systems and their differences with respect to present-day peninsular Spanish are summarized in Table 4.4. As can be seen, Old Spanish and present-day Bulgarian have a very similar selectional pattern. The only slightly more complicated cases are the doxastic predicates like *believe* and particularly fiction verbs like *dream*. For completeness, however, before discussing these verbs, in what follows, the empirical data for the predicates in Table 4.4 will be provided in the examples (16)-(27).

- (16) **Epistemic** (Bulgarian):
 Maria *znae*, [**če** živee v Sofia].
 Maria knows that(IND) live in Sofia
 ‘Maria knows that she lives in Sofia.’ (Smirnova 2011: 185)
- (17) **Certainty** (Bulgarian):
 Sigurna *săm* [**če** ima teč v rezervoara].
 certain be that(IND) have leak in oil.tank
 ‘I’m certain that there is a leak in the oil tank.’ (Smirnova 2011: 272)
- (18) **Perception** (Bulgarian):
- a. *Viždam* [**če** idvat].
 see that(IND) come
 ‘I see that they are coming.’ (Smirnova 2011: 249)
 - b. *Viždam* gi [**da** idvat].
 see them SBJV come
 ‘I see them coming.’ (Smirnova 2011: 249)

Table 4.4: Intensional and polarity mood selectional patterns in Modern and Old Spanish and Bulgarian declarative clauses. Predicates in bold indicate that they have undergone a change from Old Spanish to Modern Spanish. Parentheses indicate that the option is possible but not frequent. Cells in Bulgarian that match Old Spanish have been colored.

	Verb Class	Bulgarian	Old Spanish ^{5,6}	Modern E. Spanish
IM	Epistemic	ind ¹	ind	ind
	Doxastic^a	ind/sbjv ²	ind/sbjv	ind
	Certainty	ind ¹	ind	ind
	Perception	ind/sbjv ¹	ind/(sbjv)	ind
	Communication	ind ²	ind	ind
	Commissive	ind/sbjv ²	ind/sbjv	ind
	Fiction	sbjv ³	ind(?sbjv)	ind/(sbjv)
	Volition I (<i>want</i>)	sbjv ¹	sbjv	sbjv
	Volition II (<i>hope</i>)	ind/sbjv ²	ind/(sbjv)	sbjv
	Directive	sbjv	sbjv	sbjv
	Causative	sbjv ⁴	sbjv	sbjv
	Dubitative	sbjv ¹	sbjv	sbjv
	Emotive-factive	ind ¹	ind	sbjv/(ind)
	Weak epistemic	sbjv ¹	sbjv	sbjv
PM	V.1ST. PRES	ind/sbjv ^{1,3}	ind/sbjv	sbjv
	V.OTHER	ind/sbjv ¹	ind/sbjv	ind/sbjv

References: 1. Smirnova (2011) 2. Wiemer (2021) 3. P.c. Bozhil Hristov 4. Bezinska et al. (2018) 5. Jensen & Lathrop (1973) 6. Real Academia Española (2013)

^aAlthough Smirnova (2011) states that *believe* in Bulgarian only takes the indicative in affirmative clauses, Wiemer (2021), among others, shows there are a few contexts in which it is possible to use the subjunctive with this verb even in affirmative clauses.

(19) **Communication** (Bulgarian):

Kazvat *če* decata pejat.

say that(IND) children sing

‘They say that the children sing.’ (Wiemer 2021: 70)

(20) **Commissive** (Bulgarian):

a. *Obeštavam* *če* *šte* *băda* naj-poslušното дете v Bălgarija.

promise that(IND) FUT be.FUT most-obedient child in Bulgaria

‘I promise that I will be the most obedient child in Bulgaria.’ (Wiemer 2021)

b. Azi godina ne bjax poslušen no *obeštavam* *da* *băda* prez
this year not was obedient but promise SBJV be.FUT during
sledvaštata.

next

‘This year I haven’t been obedient, but I promise that I will be next year.’ (Wiemer 2021)

(21) **Volition I** (Bulgarian):

Maria *iska* [*da* *živee* v Sofia].

Maria want SBJV live in Sofia

‘Maria wants to live in Sofia.’ (Smirnova 2011: 185)

(22) **Volition II** (Bulgarian):

a. Ivan se *nadjava*, *če* *šte* *spečeliš*.

Ivan REFL hope that(IND) FUT win

b. Ivan se *nadjava* *da* *spečeliš*.

Ivan REFL hope SBJV win

‘Ivan hopes that you will win.’ (Wiemer 2021: 83)

(23) **Directive** (Bulgarian):

Ivan kara Maria [*da* *tantsuva*].

Ivan urges Maria SBJV dance

‘Ivan urges Maria to dance.’ (Smirnova 2011: 213)

(24) **Causative** (Bulgarian):

Ivan ja kara *da* go piše.

Ivan her make SBJV it write

‘Ivan is making her write it.’ (Bezinska et al. 2018)

- (25) **Dubitative** (Bulgarian):

Sămnyavam se [da ima teč v rezervoara].

doubt REFL SBJV be leak in oil.tank

‘I doubt that there is a leak in the oil tank.’ (Smirnova 2011: 280)

- (26) **Emotive-factive**(Bulgarian):

Maria sǎžaljava, če živee v Sofia.

Maria regrets that(IND) live in Sofia

‘Maria regrets that she lives in Sofia.’ (Smirnova 2011: 196)

- (27) **Weak Epistemic** (Bulgarian):

Vǎzmožno e [da ima teč v rezervoara].

possible be SBJV be leak in oil.tank

‘It is possible that there is a leak in the oil tank.’ (Smirnova 2011: 280)

Regarding doxastic predicates, Smirnova (2011) states that these verbs do not allow for mood variation in Bulgarian, see (28). However, Mitkovska & Bužarovska (2015) and also Wiemer (2021) argue that there is some variation with doxastic predicates, see examples in (29) and (30). They argue that when the subjunctive is used after these verbs, they additionally convey a volitive/deontic meaning. From this, it can be concluded that there might be some contexts in which these predicates do allow for mood variation (although there might be subtle speaker differences, and the subjunctive option might be a bit more marked than the indicative counterpart).

- (28) Bulgarian: (Smirnova 2011: 230)

#Vjarvam [da ima teč v rezervoara].

believe SBJV have leak in oil.tank

‘I believe there is a leak in the oil tank.’

- (29) Bulgarian: (Mitkovska & Bužarovska 2015: 191)

- a. Iskreno vjarvam, če v tozi mač šte ima sportmenstvo.

‘I sincerely believe that that game will have sportsmanship.’

- b. Mnogo mi se iska da e taka i iskreno vjarvam da se sluči v maksimalno kratki srokov!

‘I really want this to be true and I sincerely believe it to happen in the shortest possible time.’

(30) Bulgarian: (Wiemer 2021: 40)

- a. *Mislja, če* rešix vsički zadači na testa.
think that(IND) solve all task on test
'I think I (have) solved all tasks from the test.'
- b. *Mislja da* izlezna malko na vāzdux.
think SBJV go.out a.bit on air
'I think I'll go a little bit out into fresh air.'

Regarding polarity mood, in Bulgarian negated epistemic verbs, e.g., *not know*, and negated doxastic verbs, e.g., *not believe*, accept both the indicative and the subjunctive even with a first person construction and a present tense verb, see (31a) and (31b) (Smirnova 2011). This mood variation was also possible in Old Spanish, see (32), but in present-day peninsular Spanish, the indicative is used in these constructions.⁷

(31) Bulgarian:

- a. *Ne vjarvam [če/da ima teč v rezervoara].*
not believe that(IND)/SBJV be leak in oil.tank
'I don't believe that there is a leak in the oil tank.' (Smirnova 2011: 233)
- b. *Ne znam [če/da ima teč v rezervoara].*
not know that(IND)/SBJV be leak in oil.tank
'I don't know if there is a leak in the oil tank.' (p.c. Bozhil Hristov)

(32) Old Spanish: *Diccionario histórico de la lengua española* (CDH)

- a. *No creo que codicias la vida.*
not believe that lust.IND the life
'I don't believe you lust for life.'
(1496, *Tratado que hizo Nicolás Núñez sobre el que Sant Pedro compuso de Leriano y Laureola llamado Cárcel de amor*)
- b. *No creo que lo huuiesse consentido.*
not believe that it had.SBJV consented
'I don't believe they would have consented to it.'
(1494, *Fernando al papa*)
- c. *No se que le llaman verdadero señor de Roma.*
not know that him call.IND true king of Rome
'I don't know if they call him true king of Rome'

⁷The verb *know* when in present tense and in first person no longer accepts *that*-complement clauses. It only accepts *if*-complement clauses and *what*-complement clauses.

4 Towards a new theory of mood: The competing grammars approach

(1499, *Traducción de la Crónica de Aragón de fray Gualberto Fabricio de Vagad*)

- d. Y una palabra ociosa no sé que se **perdone**.
and a word idle not know that CLT forgive.SBJV
'and I don't know if an idle word can be forgiven/is usually forgiven'
(1527, *Tercera parte del libro llamado Abecedario Espiritual*)

- (33) Present-day peninsular Spanish:
No *creo* que ***codicias/codicies** la vida.
not believe that lust.IND/SBJV the life
'I don't believe you lust for life.'

Therefore, PM in Bulgarian, similar to Old Spanish and unlike Modern Spanish, does not seem to be affected by the first person effect described in Chapter (2).

Lastly, fiction verbs do not show an exact match to the Old Spanish data. Smirnova (2011) provides an example of the verb *dream* taking the subjunctive, but does not mention whether the verb can take the indicative as well, nor does the work include this type of verb in the theoretical analysis. A native speaker (p.c. Bozhil Hristov) was consulted, and reported that the use of the indicative after this verb would be rather marked. This is a very interesting result given that the Romance languages all select the indicative with this verb.

- (34) **Fiction** (Bulgarian):
Maria mečtae [**da** živee v Sofia].
Maria dreams SBJV live in Sofia
'Maria dreams that she lives in Sofia.' (Smirnova 2011: 196)

In conclusion, there seems to be a very strong parallelism between the mood system in Old Spanish and present-day Bulgarian, as shown in Table 4.4 above. There are some clear differences (e.g., fiction verbs taking the subjunctive, lack of infinitive in the language, etc.), so a unified account is not possible. However, out of the languages investigated in this book, Bulgarian is the closest to Old Spanish (at least in terms of the indicative and the subjunctive selection in complement clauses), and Smirnova (2011) does not take fiction verbs into account in her analysis. For these reasons, it will be explored to what extent the analysis proposed in Smirnova (2011) could help elucidate the type of meaning that mood morphology may have had in Old Spanish, bearing in mind that the intention is not to propose a unified analysis, but solely draw ideas from languages with similar mood selection patterns.

Section 4.2.1.2 briefly explains the proposal by Smirnova (2011) to explain the Bulgarian pattern and Section 4.2.1.3 attempts to expand this analysis to Old Spanish.

4.2.1.2 Bulgarian mood: Weak vs. strong commitment

In order to account for the mood patterns in Bulgarian, Smirnova (2011) proposes that in this language the indicative is used to express strong commitment on the part of the attitude holder, whereas the subjunctive is used to express weak commitment. This would explain why a non-factive negated verb like *believe* when taking an indicative embedded clause cannot be followed by a hedging clause, but it can when taking a subjunctive complement clause. For example, in example (35) the continuation *but there might be* is odd, because the sentence with the indicative conveys strong commitment to $\neg p$, whereas the continuation expresses weak commitment towards p . In contrast, example (36) with a subjunctive complement clause can be followed by *but there might be*, because the subjunctive only conveys weak commitment towards $\neg p$, and thus does not clash with the meaning of the second part of the sentence.⁸

(35) Ne vjarvam [če ima teč v rezervoara], #no može i da ima.
not believe that(IND) be leak in oil.tank but can and SBJV have
'Intended: I don't believe that there is a leak in the oil tank, but there might be.' (Smirnova 2011: 238)

(36) Ne vjarvam [da ima teč v rezervoara], no može i da ima.
not believe SBJV be leak in oil.tank but can and SBJV have
'Intended: I don't believe that there is a leak in the oil tank, but there might be.' (Smirnova 2011: 238)

Similarly, a perception verb like *see* can take both the indicative and the subjunctive, but when the indicative is used it conveys the idea that there is less doubt on the part of the attitude holder about what they saw. For instance, examples (37) and (38) taken from Smirnova (2011: 250) illustrate this point. Because in (37) the attitude holder cannot clearly see, the subjunctive is used. In contrast, in (38), because the attitude holder clearly sees the people who are coming, the use of the indicative is preferred.

⁸Smirnova (2011) proposes that the same distinction emerges with negated semi-factive predicates, but the examples provided in the work are all in past tense. Notwithstanding, a similar contrast seems to be present even with a first person and present tense verb, as in (31b). In these cases, *če* conveys a more factual meaning of the complement clause, while *da* is more modalized and casts doubt on the content of the embedded proposition (p.c. Bozhil Hristov).

(37) Context: You have a meeting with your friends and are waiting for them outside. **It is foggy and you cannot see well.** You see a group of people approaching, who might be your friends. You tell your sister:

- a. Viždam gi [da idvat].
see them SBJV come
'I see them coming.'
- b. #Viždam [če idvat].
see THAT(IND) come
'I see that they are coming.'

(38) Context: You are waiting for your friends. **You have good eyesight and can clearly see that the people approaching are your friends.** You tell your sister:

- a. #Viždam gi [da idvat].
see them SBJV come
'I see them coming.'
- b. Viždam [če idvat].
see THAT(IND) come
'I see that they are coming.'

According to this author, the key difference between Bulgarian and the Romance languages is that whereas in Romance languages the subjunctive indicates a lack of commitment, in Bulgarian it indicates that there is commitment but that it is weak. The subjunctive is, thus, similar to a hedging device,⁹ whereby the speaker assigns non-maximal certainty to the prejacent.

Smirnova (2011) does not mention whether the weak vs. strong analysis also applies to commissive predicates such as *promise*, desire predicates such as *hope* or doxastic predicates such as *believe*. Wiemer (2021) acknowledges that “especially for these verbs, it is difficult to pin down factors that influence the choice of *če* vs. *da*” (Wiemer 2021: 40), but suggests that the variation with these predicates may also be used to highlight the epistemic or the volitive component of the predicates, as in (39) and (40).

(39) Bulgarian: (Wiemer 2021: 82)

- a. Ivan se *nadjava*, *če* *šte* *spečeliš*.
Ivan REFL hope that(IND) FUT win

⁹The term “hedging” in linguistics is attributed to Lakoff (1973) where it is described as an expression “whose job is to make things fuzzy”.

- b. Ivan se *nadjava* **da** spečeliš.
 Ivan REFL hope SBJV win
 ‘Ivan hopes that you will win.’

(40) Bulgarian: (Wiemer 2021: 40)

- a. *Mislja*, **če** rešix vsički zadači na testa.
 think that(IND) solve all task on test
 ‘I think I (have) solved all tasks from the test.’
 b. *Mislja* **da** izlezna malko na vāzdux.
 think SBJV go.out a.bit on air
 ‘I think I’ll go a little bit out into fresh air.’

In any case, for the purposes of this work, it is sufficient to know that in the case of negated semi-factive and non-factive predicates, the key distinction is that of weak vs. strong commitment.

4.2.1.3 Extending the analysis to Old Spanish

Similar hypotheses regarding the meaning of the subjunctive in Old Spanish have been phrased by Jensen & Lathrop (1973: 59). These authors state that “[c]reer, when followed by the subjunctive in Old Spanish, expresses an attenuated belief, a subjective opinion”. As evidence they provide examples as in (41), where the speaker is expressing doubt/uncertainty about their belief:

- (41) Nazaret creo que sea.
 Nazaret believe that is.SBJV
 ‘I believe that it may be Nazaret.’

That is, it seems that in Old Spanish the subjunctive could be used as a hedging device to indicate that the subject was only expressing weak commitment.

Moreover, similar to Bulgarian, in the case of commissive verbs like *promise*, the use of the subjunctive has been argued to convey a volitive meaning which is not present when the indicative is used:

“Verbs that indicate guarantees and promises, such as *dar fidanza*, *poner fiador*, *asegurar*, *poner*, *prometer*, etc., very often contain a strong volitive element and therefore take a subjunctive. [...] The verb *prometer*, if void of any element of volition, takes the indicative” (Jensen & Lathrop 1973: 51).

In these two languages, Old Spanish and Bulgarian, mood seems to convey a distinction between weak vs. strong commitment, and additionally the subjunctive may be used to express volition.

4.2.2 G₁: Mood as a referential pronoun

So far, this chapter has argued that the Old Spanish mood system could be explained by proposing that the subjunctive is used in the language as a hedging device. There remains the question of how to explain the present-day mood pattern. As was explained in Chapter 2 and Chapter 3, however, this question is not as easily answered, given that no current theory is able to explain both IM and PM in Modern Spanish. Instead, it was proposed that in the present-day language there might be two competing grammars: one in which mood is semantic, and another in which mood is a syntactic marker. In this section, the semantic mechanism described in Chapter 3.2.3 is briefly reviewed.

Following Schlenker (2005) and Montero & Romero (2023), the key idea is that mood morphology can be formalized as a variable-based logical form. The indicative carries a presupposition that the world denoted by w lies in the Context Set of an attitude holder x' at time t' in world w' , see (42a), whereas the subjunctive is a default form without any presuppositions (42b).

- (42) a. $\llbracket Ind_2^{pro1} \rrbracket^g$ is defined only if $g(2) \in g(1)$, that is, only if $g(2) \in CS^{10}$;
 if defined, $\llbracket Ind_2^{pro1} \rrbracket^g = g(2)$
 b. $\llbracket Sbjv_2^{pro1} \rrbracket^g$ always defined;
 if defined $\llbracket Sbjv_2^{pro1} \rrbracket^g = g(2)$

Regarding complement clauses, this proposal predicts that the subjunctive would be used whenever the matrix predicate has a semantics that requires accessing worlds outside the active domain. In this sense, the theory predicts the correct mood pattern for a wide range of verbs such as emotive-factives, causatives, directives and volitional predicates, all of which select the subjunctive as expected. In all other cases, both the indicative and the subjunctive could be used. However, according to Maximize Presupposition (MP!) (Heim 1991), if two alternatives are truth-conditionally equivalent the one with the strongest presuppositions should be used. In this case, then, the indicative is expected to occur with other predicates. This is indeed what we see with doxastic predicates, epistemic predicates and communication verbs, among others (see Section 3.2.3 for an explanation of how the analysis derives each verb).

The proposed analysis, however, encounters problems when trying to explain why non-factive predicates (e.g. doxastic verbs) allow both the indicative and the subjunctive mood when negated. Under the proposed account, given that (43)

¹⁰CS here can refer to the Context Set of the speaker (CS*) the Context Set of the matrix subject ($Dox_{subject}(w_0)$) or the Context Set of another attitude holder salient in the context ($Dox_x(w_0)$).

and (44) are truth-conditionally equivalent, by MP! the indicative should be the preferred option, contrary to fact, as in the present-day language both sentences can be used.

- (43) John no creía que María corría.
 John didn't believe that Mary ran.IND
 a. $\lambda w_0. \neg \text{BELIEVED}_{w_0}(j, \lambda w : w \in \text{Dox}_{w_0}(j). \text{RAN}_w(m))$
- (44) John no creía que María corriese.
 John didn't believe that Mary ran.SBJV
 a. $\lambda w_0. \neg \text{BELIEVED}_{w_0}(j, \lambda w : \cancel{w} \in \text{Dox}_{w_0}(j). \text{RAN}_w(m))$

Because of this problem, Chapter 3 explored the possibility that this semantic grammar, which selects the indicative with negated doxastic, might be in competition with a syntactic one that selects the subjunctive mood in these environments. Section 4.2.3 will introduce this other grammar.

4.2.3 G₂: Mood as a grammaticalized case

Chapter 3 put forth the hypothesis that the subjunctive morphology in Spanish might in certain environments be becoming grammaticalized, such that the subjunctive occurs whenever negation appears in the matrix clause. The idea was inspired by how Genitive of Negation is becoming obligatory in some Baltic/Slavic languages, and it was an attempt to extend the parallelism between the nominal and verbal domains.

In the case of Genitive of Negation, it has been argued that in Lithuanian and Polish genitive case in negated clauses is a purely syntactic phenomenon with no semantic contribution (Sigurðsson & Šereikaitė 2023, Arkadiev 2016, Witkoś 2008, Błaszczak 2001). Although there are many different implementations of this idea, the core intuition is that genitive case in negated contexts works similarly to other structural cases (e.g., accusative case). Agreement-based theories of case (Chomsky 2001, Hornstein et al. 2005, Deal 2025) propose that structural cases are a reflex of the Agreement operation between a Probe (a head with uninterpretable features) and an active Goal (an element with matching interpretable features and which itself has some other uninterpretable feature unchecked). For example, in (45), the ν head is the Probe which carries uninterpretable and unvalued ϕ features. The DP can function as its goal as it carries interpretable and valued ϕ features and itself has a pair of uninterpretable features unchecked (the case features). The two elements can enter into an Agree relation whereby the

ϕ features of the v head are valued and deleted and the uninterpretable case features of the DP are valued and deleted.¹¹

(45) ... loves him

- a. $[_{vP} \ v \quad \quad \quad \text{DP} \quad \quad \quad]$
 $\quad \quad \quad \text{u}\phi[\], \text{iCase}_{[acc]} \quad \text{i}\phi_{[3SG.M]}, \text{uCase}[\]$
- b. $[_{vP} \ v \quad \quad \quad \text{DP} \quad \quad \quad]$
 $\quad \quad \quad \text{u}\phi_{[3SG.M]}, \text{iCase}_{[acc]} \quad \text{i}\phi_{[3SG.M]}, \text{uCase}_{[acc]}$

Similar to other analyses of accusative case in the Minimalist Program, then, Genitive of Negation would be analyzed in a similar manner. The uninterpretable and unvalued case features of the DP are checked and deleted as a result of ϕ agreement between the DP and the Probe. In contrast to the classical Agree operation in the Minimalist Program, which assumes the existence of only one Probe (Chomsky 2001), some theorists working on Genitive of Negation (e.g., Witkoś 2008) have proposed an alternative theory in which a given Probe can be decomposed into multiple heads. More specifically, Witkoś (2008) proposes that there exists two types of Probes: complete ones, which have both ϕ and case features (as in the Standard Theory of Agree), and incomplete ones, which have the feature sets separated across several heads. Witkoś (2008) proposes that in the case of Genitive of Negation the Probe is double and composed of a v head and the Polarity head. For illustration purposes, a simplified analysis of this idea is represented in (46-47) (see Witkoś 2008 and Sigurðsson & Šereikaitė 2023 for more elaborated analyses). According to this theory, the value given will be accusative if polarity is positive (46b), but genitive if negative (47b). These agreement mechanisms would be mandatory, thus deriving a system as in Lithuanian where genitive case has been argued to be compulsory after negation.

(46) **Affirmative clause:**

- a. $[_{PolP} \ \text{Pol}_{[aff]} \quad \dots \ [_{vP} \ v \quad \text{DP} \quad \quad \quad] \]$
 $\quad \quad \quad \text{iCase}_{[acc]} \quad \quad \quad \text{u}\phi[\] \quad \text{i}\phi_{[val]}, \text{uCase}[\]$
- b. $[_{PolP} \ \text{Pol}_{[aff]} \quad \dots \ [_{vP} \ v \quad \text{DP} \quad \quad \quad] \]$
 $\quad \quad \quad \text{iCase}_{[acc]} \quad \quad \quad \text{u}\phi_{[val]} \quad \text{i}\phi_{[val]}, \text{uCase}_{[acc]}$

(47) **Negated clause:**

- a. $[_{PolP} \ \text{Pol}_{[neg]} \quad \dots \ [_{vP} \ v \quad \text{DP} \quad \quad \quad] \]$
 $\quad \quad \quad \text{iCase}_{[gen]} \quad \quad \quad \text{u}\phi[\] \quad \text{i}\phi_{[val]}, \text{uCase}[\]$

¹¹Some accounts do not postulate an iCase feature on the v head, but rather assume the uCase feature of the DP is deleted as a by-product of the Agree operation itself (Hornstein et al. 2005).

- b. $[_{PolP} Pol_{[neg]} \dots [_{vP} v \quad DP \quad]]$
 $iCase_{[gen]} \quad u\phi_{[val]} \quad i\phi_{[val]}, uCase_{[gen]}$

It is important to mention that the above formalization is used for illustrative purposes only. How Agree works and its exact implementation is subject to a lot of ongoing debate (Carstens & Diercks 2013). For example, the above implementation follows Chomsky (2001)'s proposal that there are two types of feature: (i) uninterpretable and unvalued ($uF, []$) and (ii) interpretable and valued ($iF, [val]$). However, it has been proposed in the literature that there might be other possible feature combinations available in grammar, namely: (iii) uninterpretable and valued ($uF, [val]$) and (iv) interpretable and unvalued ($iF, []$) (Pesetsky & Torrego 2008, Carstens 2010). Following these authors one could, then, propose (48) as an alternative formalization of how Agree works in the case of Genitive of Negation. It has the advantage that now the case feature on *Pol* is not interpretable, a welcome result, but raises questions regarding how uninterpretable features are deleted before they are sent to the interfaces (for a possible explanation the interested reader is referred to Carstens 2010).

- (48) $[_{PolP} Pol_{[aff]} \dots [_{vP} v \quad DP \quad]]$
 $uCase_{[acc]} \quad u\phi[] \quad i\phi_{[val]}, uCase[]$

Another way of formalizing Genitive of Negation could be via Marantz (1991)'s Dependent Case Theory. For the purposes of this chapter, the Agreement-based theories will suffice, but in Chapter 7, the benefits of using Dependent Case Theory will be further discussed.

Leaving aside the debate on the exact formalization of Genitive of Negation, the idea put forth in this chapter is that a similar mechanism could explain mood morphology in Spanish.

Given that CPs trigger verbal agreement (McCloskey 1991), one could argue that they have interpretable ϕ features, like DPs. For example, in sentence (49) the coordinate subject clauses trigger plural morphology on the matrix verb (McCloskey 1991).

- (49) That the march should go ahead and that it should be canceled **have.PL**
 been argued by the same people at different times (McCloskey 1991: 564)

If so, then one could postulate an agreement between the double Probe (formed by *v* and *Pol* head) and the CP, similar to the one proposed for accusative/genitive case: The uninterpretable and unvalued Mood features of the CP are checked and deleted as a result of ϕ agreement between the CP and *v*; if the polarity

of the clause is affirmative then the indicative is assigned, and if negative, the subjunctive mood.

(50) Syntactic Mood:

- a. $[_{Pol} Pol_{neg} \dots [_{vP} v \dots CP \dots]]$
 $iMood_{[sbjv]} \quad u\phi[] \quad i\phi_{[val]}, uMood[]$
- b. $[_{Pol} Pol_{neg} \dots [_{vP} v \dots CP \dots]]$
 $iMood_{[sbjv]} \quad u\phi_{[val]} \quad i\phi_{[val]}, uMood_{[sbjv]}$

4.3 Discussion

This chapter has proposed that the mood patterns observed for present-day Spanish and Old Spanish can be better understood as the competition between three grammars: G_0 , in which mood is used as a hedging device and indicates degree of commitment (following Smirnova 2011's analysis of mood in Bulgarian); G_1 , in which the indicative is used to express that the world of evaluation lies in some accessible Context Set (following Schlenker 2005's analysis of mood in French); and G_2 , in which mood is analyzed as a purely syntactic marker (drawing inspiration of similar grammaticalization occurring with Genitive of Negation, as argued by Witkoś 2008). The proposed grammar competition is intended to account for both IM and PM synchronically and diachronically. G_0 is intended to explain the mood pattern observed in Old Spanish across both Intensional and Polarity environments laid out in Table 4.4. For the present-day language, the proposal is that G_0 lost the competition against G_1 and G_2 , and that in the present-day system G_1 and G_2 are still competing, but winning across different contexts, applying Variational Specialization (Wallenberg 2019). Given that in IM contexts these last two grammars produce similar outcomes, shown in Table 3.4 in Chapter 3, the competition is not as noticeable in these environments. In the case of PM in negated non-factives such as *believe*, the idea is that G_2 is winning the competition (although it might still be ongoing), whereas in the case of semi-factive verbs G_1 won the competition, as shown in Table 4.5.

Given that in PM environments the proposed theory makes testable predictions, the rest of this book focuses on the behavior of this type of mood to determine if the proposed theoretical account is feasible on empirical grounds. In particular, the rest of this book will look at the five effects that have been argued to affect PM in the present-day languages (see Chapter 2), and how these constraints have evolved diachronically. Chapter 5 deals with the effect of verb

Table 4.5: Predictions of mood selection in negated declarative clauses (PM). Filled cells indicate that the grammar would produce the mood morphology found in Modern Spanish.

Verbs	Construction	G_1	G_2	Modern Spanish Mood
non-factive	First Person Present (FP)	ind	subj	subj
	Other (O)	ind	subj	subj/(ind)
semi-factives	First Person Present (FP)	subj	subj	subj
	Other (O)	ind/subj	subj	ind/subj

class in more detail, Chapter 6 explores the first person effect, and lastly Chapter 7 deals more generally with the other effects: contrastiveness, clause type and NPIs.

5 Explaining Polarity Mood: The verb class effect

Chapter 2 has shown that there is no agreement in the literature regarding the semantic effect of mood after negated predicates. Some have argued that it indicates the amount of speaker commitment (Quer 1998), some that it might be a marker of the scope of negation (Reider 1990, Borgonovo 2003) and some that it is a marker of information status (Faulkner 2021). Recent experimental data, however, has shown that the conveyed meaning might depend on the verb class of the matrix predicate (Montero & Romero 2023). The experiment found that for:

1. semi-factive verbs: mood had a significant effect on speaker commitment;
2. non-factive verbs: mood had no significant effect on speaker commitment.

This chapter aims to provide a better understanding of this asymmetric behavior, by looking at the phenomenon in diachrony. In particular, it aims to determine when this split across predicates emerged, and whether mood has any other semantic effect in the case of non-factive verbs.

The chapter is divided into two main sections. The first part provides quantitative data showing that there is also a difference diachronically across verb classes: with non-factives the use of the indicative is decreasing in all constructions, whereas with semi-factives, the use of the indicative only decreased after first person constructions. The second part shows how the competing grammars approach (Yang 2000, Kroch 1989) combined with the theory of Variational Specialization (Wallenberg 2019) could explain this divergent behavior across predicates and the apparent lack of meaning following non-factive verbs.

5.1 The effect of verb class: A quantitative diachronic study

The only study, to my knowledge, looking at the development of Polarity Subjunctive (PS) is B-Violette (2019). This work carried out a qualitative diachronic

investigation from Latin to Modern French, showing how the phenomenon of PS emerged first with cognitive-factive verbs and spread during the Middle French period to other predicate types. However, the work did not compare the behavior of different constructions (e.g., first person constructions vs. non-first person), nor did it look in detail at the meanings and usages of PS.

This chapter complements this existing literature by providing a quantitative analysis of the phenomenon in the history of Spanish, comparing different structures and usages. Its goal is to obtain a better understanding of the behavior and meaning of PS across different verb classes. Given the subtle semantic differences that PS has been argued to convey –i.e., speaker commitment (Quer 1998), information status (Faulkner 2021) and scope of negation (Borgonovo 2003, Reider 1990) – and the disagreement as to which is the correct meaning even in the present-day language where we have access to native speaker’s intuitions, the task of determining its interpretation in earlier stages of the language is tricky to tackle. There are, however, a few strategies that can be used to accomplish this task.

One strategy is to look at constructions whose ungrammaticality in the present-day language is semantically driven, and observe their behavior in diachrony.¹ If these constructions were grammatical in previous stages of the language, we then can argue that whichever reason is causing the ungrammaticality in the present-day language was absent at previous stages. That is, encountering a given expression relatively frequently can be taken as an indication that that construction was grammatical and it is not the result of a copying error or similar.

For example, the literature has argued that the use of the indicative is ruled out when the clause has a first person matrix subject and a verb in present tense, see (1), and that this ungrammaticality is semantically driven: the speaker commitment conveyed by the indicative mood clashes with the semantic contribution of the matrix predicate (see Ridruejo Alonso 1999: 3223; Quer 1998: 61; among others).

- (1) a. No creo que *está/esté lloviendo.
 not believe that be.PRS.IND/PRS.SBJV raining
 ‘I don’t think that it is raining.’
 b. No digo que *está/esté lloviendo.
 not say that be.PRS.IND/PRS.SBJV raining
 ‘I am not saying that it is raining.’

¹See Abrusán (2019) for a summary of the different phenomena that lead to semantic ungrammaticality.

- c. No veo que *está/esté lloviendo.
 not see that be.PRS.IND/PRS.SBJV raining
 ‘I don’t see it raining.’

Sentence (1a) with the verb *believe* would result in the speaker having contradictory thoughts: the speaker will be committed to *p* and at the same time not believe/be agnostic about *p*, see (2a). In the case of the verb *say*, see example (1b), Ridruejo Alonso (1999) has argued that the reason for the ungrammaticality is that the speaker asserts *p* while at the same time denying that the speaker asserts *p*. The case of *see*, see sentence (1c), is a bit more complicated and the explanation will depend on where negation is interpreted in the clause. If negation is interpreted low, the sentence could be paraphrased as in (2c), which one could argue conveys at the very least a very odd scenario in which the speaker is committed to *p* despite having evidence against *p*. If negation is interpreted high as in (2d), it would convey that *the speaker has evidence for p* (hence the reason why they are committed), and that *the speaker lacks evidence for p*. Note that the evidence could be visual but need not: for example, in sentence (3), the evidence is cognitive, despite the use of the verb *see*. Thus, the verb seems to convey that there is (lack of) evidence in a general sense rather than it just referring to having visual evidence (see Albelda Marco & Jansegers 2019 for a study of the different meanings that the verb *see* can convey in Spanish). The ungrammaticality of all the sentences in (1), thus, could be argued to be due to a semantic incompatibility between the indicative and the matrix negated predicate.

- (2) a. ?? I believe it is raining and I don’t believe it is raining.
 b. ?? (I say) it is raining and I don’t say it is raining.
 c. ?? (I believe) it is raining and I see that it is not raining.
 d. ?? (I believe) It is raining and I don’t see that it is raining.
- (3) Veo que estamos cerca de alcanzar una solución.
 See that be.PRS.IND close of reach a solution
 ‘I see we are coming closer to a solution.’

Interestingly, this state of affairs was not the same in earlier stages of the language, see the examples in (4) extracted from the *Corpus del Diccionario Histórico de la lengua española* (CDH). This suggests that the indicative mood may not

5 Explaining Polarity Mood: The verb class effect

have conveyed speaker commitment at that time,² as otherwise the sentences in (4) would be ungrammatical.³

- (4) a. y no creo que haremos negocio
and not believe.1SG.PRS that do.IND business
'I don't think we will do business.'
(1883–1889, José Zorrilla, *Cartas íntimas e inéditas*, CDH)
- b. Yo no digo que no le hay.
I not say.1SG.PRS that not CLT have.IND
'I don't say that there aren't.'
(1559, Juan de Timoneda, *La Comedia de los Menemnos*, CDH)
- c. Non veo que los de Mérida valen tanto.
not see.1SG.PRS that the of Merida are.worth.IND so
'I don't think the ones from Merida are worth so much.'
(1430, Pedro de Corral, *Crónica del rey don Rodrigo, postrimero rey de los godos*, CDH)
- d. No sé que nos llevamos ventaja en esto.
not know.1SG.PRS that us carry.IND advantage in this
'I don't know if we have an advantage in this.'
(1598, Baltasar Álamos de Barriento, *Discurso político al rey Felipe III*, CDH)

²In contrast to the other examples in (4), in example (4a) the verb is in future tense. While it could be the case that in this example the future is used to encode a modal interpretation, the example suffices for the purposes of this section in the sense that in Modern Spanish the indicative is disallowed in these environments (even in future tense).

³Here we have also included the verb *know* to illustrate that this verb could take non-interrogative clauses in the past even with a first person construction and present tense, a tendency that is observed as late as 1926 in the corpus, see (i.a). In present-day Spanish, these sentences are ungrammatical (at least in my idiolect) and they only allow an interrogative embedded clause with the *if* complementizer:

- (i) a. No sé que se pueda valorar el mundo sin creer en el
not know that CLT can.SBJV appreciate the world without believing in the
ultramundo.
underworld
'I don't know if one can appreciate the world without believing in the underworld.'
- b. No sé si/ *que ha venido.
not know if that has come
'I don't know if they have come.'

5.1 The effect of verb class: A quantitative diachronic study

By tracking the frequencies of mood following these constructions (one isolated example cannot be taken as full evidence that the construction was grammatical as it might have been the result of a copying error, a translation effect, etc.), this strategy then allows us to discern when in time the speaker commitment meaning emerged and whether there were differences across verb classes.

Another strategy, which will be particularly useful for our given purposes, is to look at the stability of the alternation in declarative clauses more generally. The language acquisition literature has pointed out that semantically vacuous alternations are hard to acquire and thus that they are rather rare in language (e.g., Clark 1987's Principle of Contrast). However, when they do occur they tend to specialize (semantically, sociolinguistically, etc.) or become unstable leading to the extinction of one of the forms (e.g., Geeraerts 1999's Isomorphic Principle and Wallenberg 2019's theory of Variational Specialization). For instance, the morphological doublet *shined-shone* was in competition in the history of English until it specialized, *shined* becoming transitive and *shone* intransitive, see (5). In contrast, the doublet *welk-walked* also competed in diachrony but no specialization in meaning occurred, which led to the eventual disappearance of one of the forms, see (6) (Taylor 1994, Kroch 1994).

- (5) Early English: *shined-shone*
Present-Day English: *shined* (transitive) - *shone* (intransitive)
- (6) Early English: *welk-walked*
Present-Day English: *walked*

By tracking the frequencies of usage of the indicative and the subjunctive forms, this second strategy allows us to tackle the question of whether a given alternation is meaningful or not. If it is, we expect to see specialization and subsequent stability diachronically. If it is not, we expect to see the loss of one of the variants. For illustration purposes, the expected patterns of behavior are shown in Figure 5.1. The *x*-axis indicates time and the *y*-axis the proportions of mood. 0 indicates that all instances were in the subjunctive and 1 that they were all in the indicative. A value of 0.25 in the graph then indicates that the indicative was used 25% of the time and the subjunctive 75%. In the image on the left the expected behavior if the alternation specializes is plotted: initially, the two forms would have competed in the two environments, but as time progresses one form would start to be used in a given environment and the other form in the other environment. In the image on the right the expected patterns if the competing forms had no meaning difference and no specialization occurred are plotted: one form (in this case the indicative) is lost across all environments.

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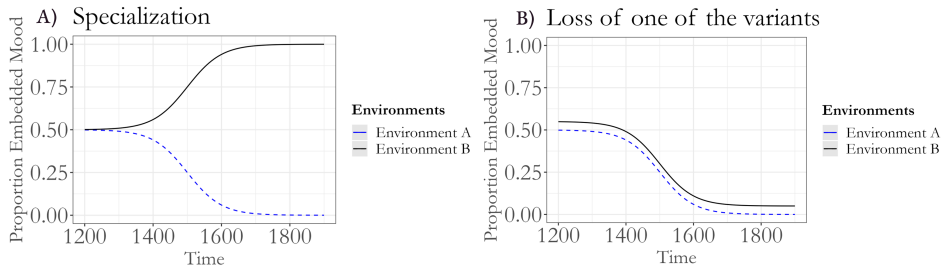


Figure 5.1: Possible diachronic developments: specialization or loss.

In sum, these two strategies (looking at semantically driven ungrammatical contexts, and looking at the stability of the phenomenon) can be used to determine when speaker commitment emerged, and in the cases in which there might be no speaker commitment whether the mood alternation carries any other meaning (in which case we predict some form of specialization and stabilization) or not (in which case we expect the alternation to be unstable).

5.1.1 Study design

In order to investigate the development of PS across different verb classes, the *Corpus del diccionario histórico de la lengua española* (CDH) was used (Real Academia Española 2013). This corpus was chosen because (i) it contains a wide range of texts published in Spanish from the 12th century until the year 2000 (32 million different fragments in total), all of which have been automatically lemmatized (ii) the corpus has texts from different genres including fiction (prose and poetry) and non-fiction works (scientific, religious, etc.), allowing the investigation of a wide range of contexts and usages, and (iii) the corpus interface allows the user to restrict the search to texts of a given country, region or dialect, which facilitates the study of dialectal differences. This last aspect was particularly important given that it has been shown that there is dialectal variability regarding mood morphology (Gallego & Alonso-Marks 2014, Busch 2017: among others), and I wanted to restrict the investigation to Peninsular Spanish. All other major Spanish diachronic corpora (see Enrique-Arias 2013 for a summary) do not meet all of these requirements: either the interface does not allow the user to restrict the searches based on country or region – e.g., Mark Davies’ *Corpus del Español: Historical* Davies (2009) –, the size is not big enough – e.g., the *Corpus de documentos españoles anteriores a 1900* CODEA+2022 (GITHE 2022) is smaller than the CDH, consisting of 4,023 fragments –, or are genre specific – e.g., the corpus

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Biblias Medievales (Enrique-Arias & Mena 2008–2026) only includes religious texts –.

For the present study, four different constructions were searched for, see (7). They all consisted of negation (*no/non*),⁴ which was immediately followed by a semi-factive (*know* or *see*) or non-factive (*believe* or *say*) verb, which in turn immediately selected for a *that*-complement clause. Additionally, the search was restricted to include only texts published in Spain. This was done to avoid dialectal effects.

- (7) a. negation + believe + *that*-complement clause: *no/non [creer] que...*
 b. negation + say + *that*-complement clause: *no/non [decir] que...*
 c. negation + know + *that*-complement clause: *no/non [saber] que...*
 d. negation + see + *that*-complement clause: *no/non [ver] que...*

The choice of predicates was based on two criteria. First, their frequency in the corpus: they all had the highest token frequency in the corpus within their verb class.⁵ Second, they gave a balanced amount of tokens per verb class such that in total there was a relatively balanced amount of tokens for semi-factive and non-factive verbs, see Table 5.1:

Table 5.1: Number of sentences for each predicate in the CDH corpus.

semi-factive	Total Sentences	non-factive	Total Sentences
<i>saber</i> ‘know’	3,975	<i>creer</i> ‘believe’	3,844
<i>ver</i> ‘see’	1,793	<i>decir</i> ‘tell’	2,521
Total semi-factive	5,768	Total non-factive	6,365

5.1.2 Corpus annotation

After running the searches, the sentences could be downloaded and annotated. The total number of sentences was 12,133.

Each sentence was annotated with (i) metalinguistic information, (ii) information on the matrix clause and (iii) information on the embedded clause. The

⁴See Moreno Bernal & Horcajada (1997) for the development of the two forms in the history of Spanish.

⁵The other verbs used in Montero & Romero (2023) had a lower frequency and thus were not annotated: *recordar* (70), *notar* (18), *descubrir* (11), *pensar* (2,050), *sospechar* (94) and *contar* (12).

following are all the different categories and labels that were used (see also Section 5.1.3 for a summary of the annotated features and which sentences were used for analysis):

1. **Metalinguistic information:**

- Period: 1200–1900, divided into 8 periods of 100 years each.
- Year: 1200–1999.

2. **Matrix clause information:**

- Clause type: *non-interrogative*, *interrogative* and *NA*.
- Verb class: *Semi-factive*, *Non-factive* and *Other*.
- Verb finiteness: *finite* and *non-finite*.
- Verb additional information:⁶ *indicative*, *subjunctive*, *imperative*, *gerund*, *infinitive* and *participle*.
- Construction: *first present* and *other*.

3. **Embedded clause information:**

- Embedded clause: *ambiguous* and *unambiguous*.
- Embedded verb: *indicative*, *subjunctive*, *infinitive* and *NA*.

Starting with the (i) **metalinguistic information**, the only categories that were annotated were the year of composition (ranging from 1200 to 1999) and period (the data was divided into 8 periods of 100 years each). In the cases when the year given ranged between two periods, the mean year of composition was calculated and the resulting date was used to determine the period. For example, the work *El halcón de Federico* was composed between 1599 and 1605. In this case the text was assigned to the period 1600:

- (8) *El halcón de Federico* (1599 – 1605)
Mean Year: 1602
Period: 1600

⁶Technically speaking one should have created two columns in the data set: one for finite verbs with the coding of *indicative*, *subjunctive* and *imperative* and *NA* (for all non-finite verbs); and another column with *gerund*, *infinitive* and *participle* for non-finite verbs and *NA* for the rest verbs. For simplicity, however, the two columns were combined into one, thus skipping over the *NA* label which was not going to be used in the data analysis.

With regard to (ii) **matrix clauses**, they were annotated according to 5 criteria: clause type, verb class, verb finiteness, verb type, and construction.

First, the data was divided into *interrogative* and *non-interrogative* clauses. This was done because there seems to be an effect of clause type on whether the embedded clause allows mood alternations or not. For example, in sentence (9), despite the presence of negation, the embedded verb can only appear in the indicative and does not alternate with the subjunctive. Thus, in order to get a better understanding of the phenomenon of Polarity Subjunctive in plain declarative clauses, without other factors affecting the behavior, interrogative clauses were separated.

- (9) ¿No creías que ya no *estuvieras/estabas enamorada de Dani?
Not believe that no.longer not be.PST.SBJV/PST.IND in.love of Dani
'Didn't you think you were no longer in love with Dani?'

An additional subcategory *NA* was created because the corpus contains sentences extracted from Spanish grammar books indicating that a given sentence is (un)grammatical, see (10). Only 4 sentences were excluded following this criterion.

- (10) *No creo que muchos de los niños prueban gota de vino.
Not believe that many of the kids try drop of wine
(1970, *Una restricción que impone la estructura superficial del español a las oraciones doblemente negativas*, María Luisa Rivero, CDH)

After this initial separation, matrix predicates were separated into three categories: *Semi-factive*, *Non-factive* and *Other*. The verbs *know* and *see*⁷ were assigned to the first category, and the verbs *believe* and *say* to the second one. The last category *other* was created to include possible parsing errors in the corpus. For example, sentence (11) was included in the output file because there was an error in the original corpus, and the verb *dictate* was lemmatized as the verb *say*. This example was the only error of this type that we could find in the data.

- (11) La política no dicta que sea lenta.
the policy not dictate that be slow
'The policy does not dictate that it should be slow.'
(1862, *Nociones del arte militar*, Francisco Villamartin, CDH)

⁷The verb *see* is polysemous and can take many interpretations. For example, in the *Diccionario de la lengua española* (Real Academia Española 2024), 14 different uses are reported. The boundaries between the meanings, while clear in prototypical examples, are harder to distinguish in real-world data. For this reason, no further information regarding its interpretation in the sentence was added.

5 Explaining Polarity Mood: The verb class effect

The choice of these two verb classes was based on two criteria. First, in order to explore the effect of negation on mood, the verb class had to be one in which the verbs select the indicative in affirmative clauses. That is, verbs that select the subjunctive in affirmative clauses like *want*, also select the subjunctive under negation. Hence the reason for excluding them in the analysis. These reduces the options to epistemic, doxastic, certainty, perception, communication, commissive and fiction verbs. Epistemic, certainty and perception verbs are considered semi-factive predicates; while the rest are non-factive predicates.

- (12) Ana (no) quería que María [viniera / *venía].
Ana not wanted that Maria come.PST.SBJV come.PST.IND
'Ana (didn't) want(ed) Mary to come.'

Second, within these groups, the chosen verbs needed to be frequent in corpora (see previous section), and ideally would be ones for which there is present-day experimental data. In this way the results could be compared more straightforwardly to the present-day usage and meaning.

In addition to this grouping, matrix predicates were labeled as: *indicative*, *subjunctive*, *imperative*, *gerund*, *infinitive* or *participle*. This was done because the type of mood and non-finiteness of matrix predicate may also have an effect on the mood the embedded clause can have. For instance, despite the presence of negation in sentence (13), the only grammatical option is the use of the indicative. Borgonovo (2003) has argued that this is due to the matrix verb having the subjunctive morphology.

- (13) Lo dije para que no **pensaras** que Pedro era/*fuera
it said so that not think.PST.SBJV that Peter be.PST.IND/PST.SBJV
culpable.
guilty
'I said it so you wouldn't think Peter was guilty.' (Borgonovo 2003: 18)

Moreover, a verb such as *pensar* 'think' in affirmative clauses only selects indicative complements, but if the verb is a gerund it can allow the use of the subjunctive as well, as in (14).

- (14) **Progressive:** (Real Academia Española 2010: 479)
Inicialmente estaba **pensando** que participaran los dos
Initially was thinking.GERUND that participate.SBJV the two
equipos.
teams
'Initially I was thinking that the two teams should participate.'
(Razón [Esp.] 2/9/2002, CDH).

Finally, matrix clauses were annotated depending on whether they were first person and present tense constructions – Construction FP – or not –Construction O –. This was done because, as discussed in the previous chapter, the literature has observed that when the matrix subject is first person and the verb is present tense, mood alternations are more constrained. Table 5.2 illustrates how the annotation was done.

Table 5.2: Annotation of matrix clauses according to Construction: 1st present vs. other.

Tense/Person	1sg	2sg	3sg	1pl	2pl	3pl
Present	1st present	other	other	1st present	other	other
Past	other	other	other	other	other	other
Future	other	other	other	other	other	other
Conditional	other	other	other	other	other	other

The choice of annotating for construction type and not tense and person combinations was based on three criteria. First, some of these combinations are rather infrequent (e.g., 2nd person future) in comparison to others (e.g., 1st person present), hence from a statistical point of view we might not have enough data for some combinations to see if there is a statistically significant effect. Second, under this annotation, construction type (the independent variable in the statistical analysis) has only 2 levels (FP vs. O); if one annotates for each of the tenses and person combinations, the independent variable in the statistical analysis would have had 20 levels. Given the small data set, the statistical models would not have been able to converge. Third, there is no work, theoretical or descriptive, that I am aware of, that states that any other combination of person and tense (other than first person and present) behaves differently than the other combinations in Spanish.

Regarding the annotation of (iii) **embedded clauses**, they were annotated for two categories: the type of embedded clause and the type of embedded verb. Although the search specified that the complement clause should be declarative, in Spanish the only difference between a declarative *that*-complement clause and an interrogative *what*-complement clause is the acute accent *que* vs. *qué* and there were instances in which the corpus outputted the sentence as a *that*-complement clause, when it was a *what*-complement clause. This is illustrated in (15) and (16), where, as the translation shows, the only possible interpretation of the sentence is one in which the embedded clause is interrogative but nonetheless what should be the interrogative pronoun appears without the acute accent.

- (15) Non se que me tengo que creer dellas...
 not know that myself have to believe of.them
 ‘I don’t know what (*that) I should believe from them.’
 (1487, *Fernando al conde de Tendilla sobre la provisión de obispados y abadías, concesión de beneficios*, Anónimo, CDH)
- (16) yo non se que es lo que en esta misera vida tiene de prouecho
 I not know that is it that in this miserable life has of profit
 ‘I don’t know what (*that) is it that this miserable life has of profit.’
 (1424–1520, *Cancionero de Juan Fernández de Íxar*, Anónimo, CDH)

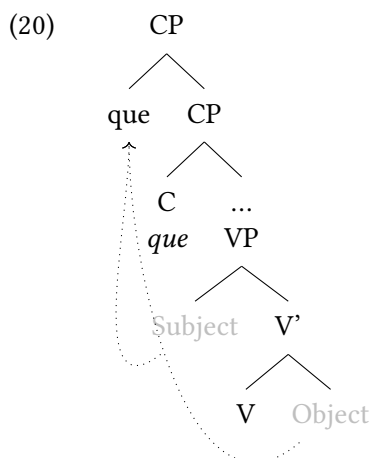
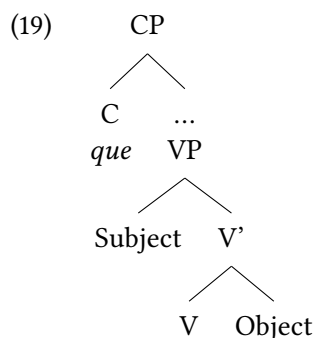
This was problematic because in present-day Spanish embedded interrogative clauses force the embedded clause to appear in the indicative mood irrespective of negation, see (17).⁸ Moreover, most of the verbs chosen for analysis take interrogative clauses when negated – *know*, *say* and *see* are all responsive predicates –, the only exception being the verb *believe* which is anti-rogative and does not take interrogative embedded clauses, see (18).

- (17) a. Sabe qué se ha/*haya roto.
 know what CLT have.PRS.IND/PRS.SBJV broken
 ‘They(sg) know what has been broken.’
 b. No sabe qué se ha/*haya roto.
 not know what CLT have.PRS.IND/PRS.SBJV broken
 ‘They(sg) don’t know what has been broken.’
- (18) a. (No) dice qué se ha roto.
 not say what CLT has.IND broken
 ‘They(sg) (don’t) say what has been broken.’
 b. (No) ve qué se ha roto.
 not say what CLT have.PRS.IND/PRS.SBJV broken
 ‘They(sg) (don’t) see what has been broken.’
 c. *(No) cree qué se ha roto.
 not believe what CLT have.PRS.IND/PRS.SBJV broken
 ‘They(sg) (don’t) believe what has been broken.’

In order to circumvent this problem, embedded clauses were divided into ambiguous and non-ambiguous. To make the annotation as reliable as possible, a

⁸Previous stages of the language as well as some modern varieties do allow the use of both the indicative and the subjunctive in these environments (Nieuwenhuijsen 2001, 2016).

syntactic criterion was followed: if both the subject and object of the sentence were overtly expressed, as in (19), the sentence was marked as unambiguous given that in this case *que* cannot be interpreted as an interrogative pronoun (as *qué* can only pronominalise a subject or an object). If, on the other hand, either the (inanimate) subject or the object had not been overtly expressed in the sentence, the sentence was marked as ambiguous, as in these cases the sentence could be a declarative clause or an interrogative one, see (20).



In the following sections only unambiguous sentences are taken into account, as they represent a more realistic (less noisy) sample of the phenomenon. This means that possibly some sentences that were declarative clauses have been excluded, but the opposite choice – including ambiguous sentences – would have entailed that some sentences that were interrogative would have been included in the results. That is, we have opted to have a lower number of tokens which we are sure represent the phenomenon we are interested in (this sample was large enough to be able to carry out statistical analyses), as opposed to increasing the data set and introducing some noise from interrogative clauses.

It must be highlighted at this point, though, that the amount of unambiguous sentences was about four/five times the amount of ambiguous sentences (1,580 ambiguous sentences out of 7,042; that is, 22% of the data), and most importantly that this methodological choice did not affect the overall diachronic development. The diachronic patterns are mostly preserved irrespective of whether ambiguous sentences are included or not in the analysis. See Appendix A for a comparison of the results with and without these sentences.

Finally, the embedded verbs (if any) were divided into four categories: *indicative*, *subjunctive*, *infinitive*⁹ or NA (if no verb appeared in the embedded clause as in (21)).

(21) No digo que no.

not say that no

‘I don’t say no.’

(1833, *Julia Comedia en dos actos del célebre Scribe arreglada a nuestra escena*, Mariano José de Larra, CDH)

For the annotation of both embedded and matrix verbs into *indicative*, *subjunctive* and *infinitive*, the following resources were employed. For the earlier stages of the language the following grammars/textbooks were used: Lapesa (1968), Hanssen (1913), Andrés-Suárez (1994) Menéndez Pidal (2005), Penny (2006), Company Company (2006) and Pons Rodríguez (2010). For the present-day language the Real Academia Española (2010) and the RAE dictionary (<https://dle.rae.es/>) were used.

In all these grammars/resources conditionals are included within the indicative group, and we also followed this categorization. The main reason for why these grammars consider the conditional to be a tense within the indicative mood is that it occurs in the same environments as other indicative tenses and is rejected in environments that require the subjunctive: “*el condicional aparece en entornos sintácticos en los que se selecciona el modo indicativo (Prometió que iría) y se rechaza en los que seleccionan el subjuntivo: Deseamos que *irían fueran ustedes.*” (Real Academia Española 2010). Another reason that has put forth in the literature for grouping the conditional with the indicative is that the future and the imperfect, like the conditional, also have non-temporal uses (De Mulder 2010). It is also important to note that diachronically there was a change in the mood system concerning the use of imperfective forms. In present-day Spanish there are two imperfective subjunctive forms: (*estuvie*)-*ra* vs. (*estuvie*)-*se*. However, the first form with *-ra* came from the Latin pluperfect indicative (*amaveram*) and only became a past tense subjunctive marker at a later point, see (1)–(2) (Jensen & Lathrop 1973). Thus, in Old Spanish, the *-ra* form could be used with the value of the pluperfect indicative or the subjunctive. The indicative use decreased during the 14th and 15th centuries, and by the 17th century there were only a few isolated examples of the indicative (Jensen & Lathrop 1973, Fukushima 2015).

⁹Most sentences in which the embedded verb was infinitive were actual cases of embedded interrogative clauses. Given that there was not enough data that was unambiguous with a non-finitive verb, the data analysis section only considers the indicative and the subjunctive embedded clauses.

1. amaveram (pluperfect indicative) > amara (imperfect subjunctive)
2. amavissem (pluperfect subjunctive) > amase (imperfect subjunctive)

The competition of the two forms (*ra* vs. *-se*) nonetheless is not equally distributed across the different dialects in the Peninsula, and in certain regions (e.g., Galicia), the *-se* form is much more widely used than the *-ra* form (Lara Bermejo 2019).

Because of this, two different annotations were carried out: in the first one the *-ra* form was included as a subjunctive form (as in several historical periods this form has been/is unambiguously subjunctive), in the second annotation the *-ra* form was labeled differently than the other subjunctive forms indicating that it could be ambiguous between the indicative and the subjunctive form (which would be the most appropriate characterization in periods in which there was still alternation, namely before the 17th century).

For the analysis, a decision had to be made regarding whether to include *-ra* forms in the results or not. We decided to include them because of the following reasons: First, after the 17th century the *-ra* form, for the most part, is no longer ambiguous, and so by excluding these tokens, we would be excluding tokens that were unambiguous in that period. Second, given that there is dialectal variation in usage of the forms, by excluding one of the forms completely we would be biasing the results for a given dialect. Third, the effect of including the *-ra* form in the data has the biggest effect after 1700, a moment when the form is no longer ambiguous, see Appendix B for more information on this.

It must be emphasized here that this methodological choice did not have a major effect on the overall results, as the imperfective *-ra* forms constitute a small fraction of all the subjunctive forms: 8% (62/837) from 1200–1600 when they could be ambiguous, 14% (273/2023) from 1700–2000 when they have been argued to be mostly unambiguously subjunctive forms. A comparison of including and excluding the *-ra* forms can be found in Appendix B, where it is shown that the diachronic patterns are unaffected by this methodological decision as well.

5.1.3 Data extraction

In summary, each sentence was annotated with information regarding the clause type and verb type of both its matrix clause and embedded clause. The annotated categories appear illustrated in Figure 5.2. Given the complexity of the phenomenon, as a first step we wanted to understand the behavior of PS in plain matrix declarative clauses with an indicative matrix verb (the focus of Chapter 5 and

Chapter 6), and only later add the effect of the matrix predicate's mood (Chapter 7).

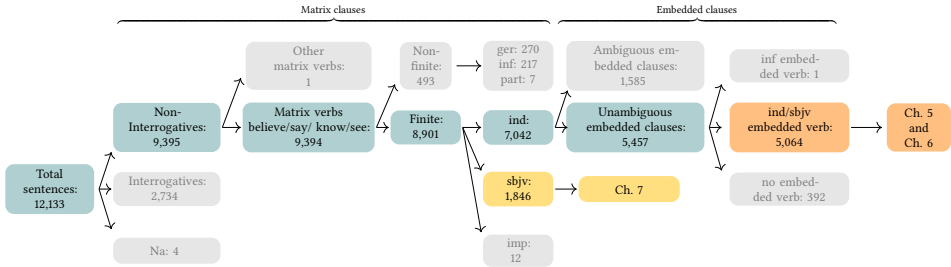


Figure 5.2: Data extraction process. Numbers indicate the number of tokens per category. Data in gray boxes has been excluded from analysis in the present book. The data in the orange box will be used in Chapter 5 and 6 and (some of) the data in the yellow box in Chapter 7.

The next section discusses the results obtained for plain matrix declarative clauses with the matrix verb in the indicative.

5.1.4 Results

5.1.4.1 Overall results

Figure 5.3 shows the results of the development of PS in plain matrix declarative clauses with a matrix indicative verb. The x -axis represents the period and the y -axis the proportion of the indicative mood in each period on a scale from 0 (meaning all instances were in the subjunctive) to 1 (all instances were in the indicative). On the panel on the left the proportions for semi-factive verbs are plotted, and on the panel on the right the proportions for non-factive verbs. The size of the dot indicates how much data there was for that given period (the exact amount of tokens and the means can be found in the table annexed to the figure). The color indicates the construction: black for constructions with a first person matrix subject and a present tense verb (Construction FP), and blue for all other constructions (Construction O). See Section 5.1.2 for more information on how the different constructions were annotated.

Starting with semi-factive verbs, one observes that although the proportion of mood during the period 1200–1400 was relatively similar across Construction FP and Construction O, starting in 1500 the two constructions took divergent paths. In Construction FP (i.e., sentences with a first person matrix subject and the matrix verb in present tense) the use of the indicative declined and by 1700

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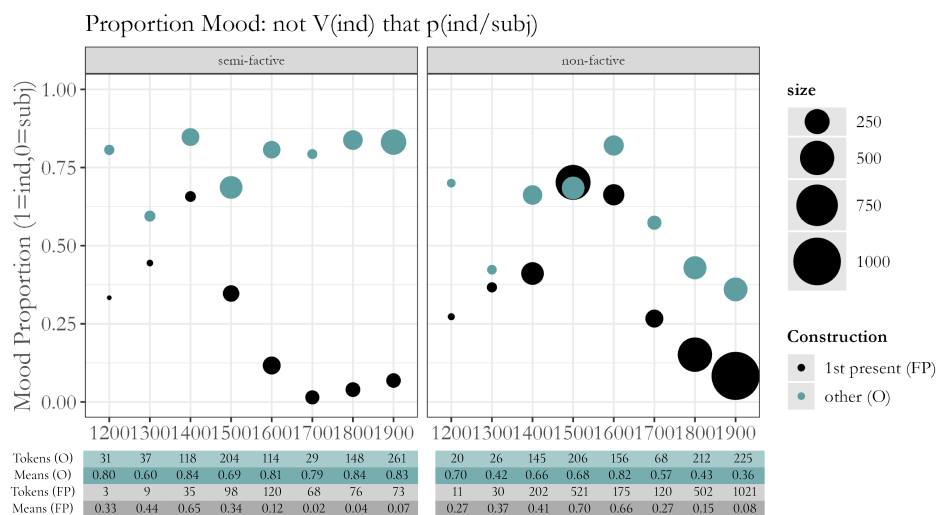


Figure 5.3: Proportion of mood in declarative clauses per verb class.

the use of the subjunctive was predominant (with only a 5% rate of the indicative). In contrast, in Construction O (non-first person and present tense constructions) the use of the indicative did not decline and remained relatively stable at around 75–80%.

Looking at non-factive verbs, one observes a different pattern. There is an up-and-then-down tendency for both Construction FP and Construction O, with a peak in indicative usage during 1500 and 1600. That is, both constructions followed a similar diachronic trajectory (it must be pointed out, though, that Construction FP seems to have for the most part a lower proportion of indicative than Construction O).

Overall, then, we observe that the use of the indicative in Construction FP has declined across both verb classes, indicating that at earlier stages of the language the indicative might have not conveyed speaker commitment, but that this decline happened at markedly different times across verb classes (200 years difference). This raises questions about whether the decline should be given a unified account, a point we will return to in the following sections.

Additionally, we observe that the only environment in which the proportion of mood has been stable is Construction O with semi-factive verbs. All other environments have not been as stable and all show a decline in the use of the indicative during the later periods: with semi-factive verbs in Construction FP the use of the indicative started to decline during the 1500s, and for non-factive

verbs the use of the indicative declined across constructions during the 1700s. This raises the question of why the alternation was only stable with Construction O with semi-factive verbs but not non-factive verbs and indicates that, in addition to an effect of Construction and Time, there might also be an effect of Verb Class interacting with these other two variables. These results, if statistically significant, are interesting insofar as none of the existing theoretical approaches to PS reviewed in Chapter 2 seem to predict such an (interacting) effect of Verb Class.

In order to test whether the effect of Verb Class is statistically significant, a generalized additive model (GAM) was conducted (Wood 2017a, Wieling 2018).

This type of model has been used within linguistics in a wide range of contexts: sociolinguistics (Wieling et al. 2011), phonetics (Zahner-Ritter et al. 2022), and historical linguistics (Baayen et al. 2017); and has been gaining popularity to analyze time dependent language change more generally. The benefit of using these models is that they can capture nonlinear relationships by means of fitting a smooth function, which, given the nonlinearity of the data we are analyzing (see Figure 5.3), makes this model well suited to investigate the phenomenon at hand.

Using the R package *mgcv* (Wood 2017b) three GAM models were fitted, specifying in each Embedded Mood as the binary dependent variable (the subjunctive was coded as a 0 and the indicative as a 1). In the first model no effect of verb class was included: Construction (with two levels: FP or O) was a categorical predictor and Year a smooth term which varied across the levels of the Construction variable. In the second model a main effect of verb class as a categorical predictor was introduced. Finally, the last model incorporated a three way interaction between all three effects. In order to do so, a new categorical variable called Environment was constructed. It consisted of four levels: Semifactive-ConstructionFP, Semifactive-ConstructionO, Non-factive-ConstructionFP and Non-factive-ConstructionO. Year was then included as a smooth term which varied across the levels of the Environment variable.

To test the hypothesis that the interaction with Verb Class is important, we performed a model comparison using an AIC comparison.¹⁰ As can be seen in Table 5.3, the interaction term improved the model's performance significantly (300 difference in the AIC score). Thus, corroborating the hypothesis that verb class should be taken into account as an important factor determining the behavior of PS in diachrony.

¹⁰The Akaike information criterion (AIC) is an estimator of prediction error, which is commonly used for model selection (Emiliano et al. 2014).

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Table 5.3: AIC model comparisons

Model	Df	AIC
No Effect of VerbClass	14.36729	5370.142
Main Effect VerbClass	15.41804	5363.737
Three way Interaction	20.15714	5045.599

This last model with the interaction with Verb Class accounted for 34% of the variance (adjusted R-square 0.34), which, given the nature of the data we are handling (i.e., the data comes from old manuscripts, from different authors, different regions and ages, etc.), is a relatively high level of explanation.

Visual inspection of the model’s predictions confirms that the model fits the data well. The predictions were plotted using `ggplot2` (Wickham & Carson 2016) and `tidymv` (Coretta 2023), and can be found in Figure 5.4. Similar to the previous graph, the *x*-axis indicates time and the *y*-axis the predicted proportion of the indicative mood. The panels plot the proportions for semi-factive verbs (left) and non-factive (right). The color/line type of the curve indicates the type of construction: Construction O is blue/dotted and Construction FP is black/continuous.

As can be seen, the diachronic patterns described in the previous section are captured by the model: with semi-factive verbs, the proportion of the indicative in Construction O is predicted to stay stable and in Construction FP it is predicted to decline starting in 1500; with non-factive verbs the prediction is an up-and-then-down trajectory with a peak in 1500–1600 and subsequent fall in 1700.

In order to better understand the effect of this interaction, a difference curve was calculated using the functions `get_difference` and `plot_difference` from the package `tidymv` (Coretta 2023), see Figure 5.5. These curves are obtained by subtracting one trajectory from another and provide useful information on when exactly the two constructions differ and in what ways their differences have been changing over time. In the left panels in Figure 5.5, the plotted curve is the one obtained by subtracting the log-odds of Construction O minus Construction FP in semi-factive verbs, and on the right panels the plotted curves are the ones obtained by similarly subtracting Construction FP from B but this time with non-factive verbs. On the top row (1), the curves are plotted on a log-scale and in the bottom row (2) the *y*-scale represents the differences in proportions.¹¹ The gray

¹¹The function used to calculate the GAM model transforms the data into log-odds by default and hence the data had to be back-transformed using the inverse log. See supplementary materials for more information on how this was done.

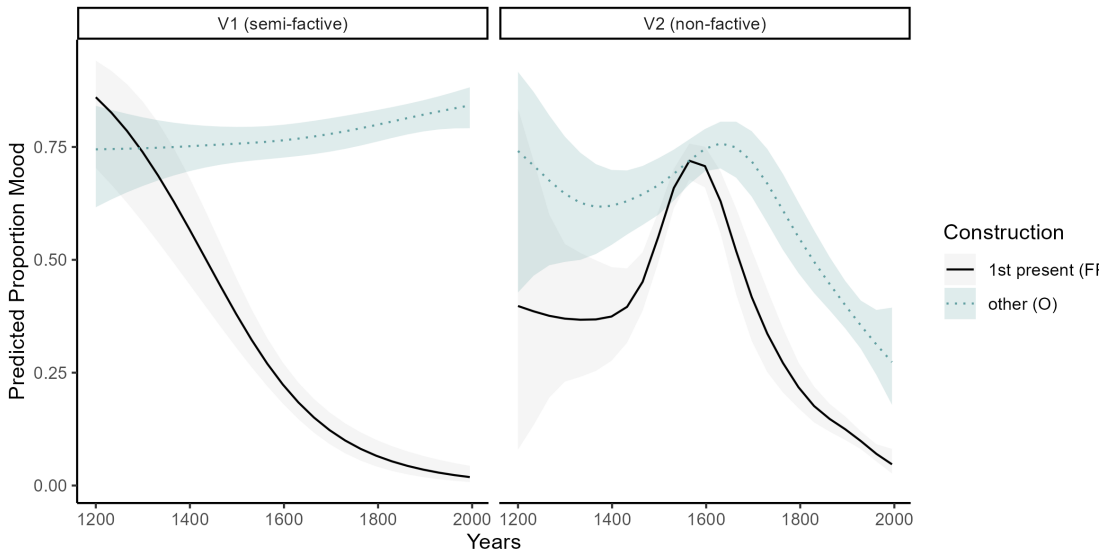


Figure 5.4: Predicted proportion of mood in declarative clauses per verb class (GAM).

band around each curve indicates the 95% confidence interval (CI), and the red area indicates the years in which the difference between the two constructions is significantly different from 0.

As can be seen for both semi-factive and non-factive verbs, Construction FP and O have been mostly different diachronically, the only periods in which they have not been different are 1200–1300 (for both semi-factive and non-factive verbs) and late 1500 to early 1600 (only for non-factive verbs). This already shows that the effect of Construction is rather robust in all the periods of the language. Moreover, looking at the graph one can better understand the nature of the interaction between Construction and Verb Class. In the case of semi-factive verbs the difference in Construction FP and O increases logarithmically as time progresses. In contrast, with non-factive verbs the difference across construction FP and O does not increase with time, but rather stays stable at around 25–30% (1.5 in the log scale), only having a dip in 1600, when the constructions behaved similarly. That is, with non-factive verbs the proportion of the indicative is higher in Construction O than in Construction FP, but in contrast to semi-factive verbs, this difference is rather constant.

In summary, these results show that the proportion of mood has not been stable across most constructions (the only exception being Construction O with

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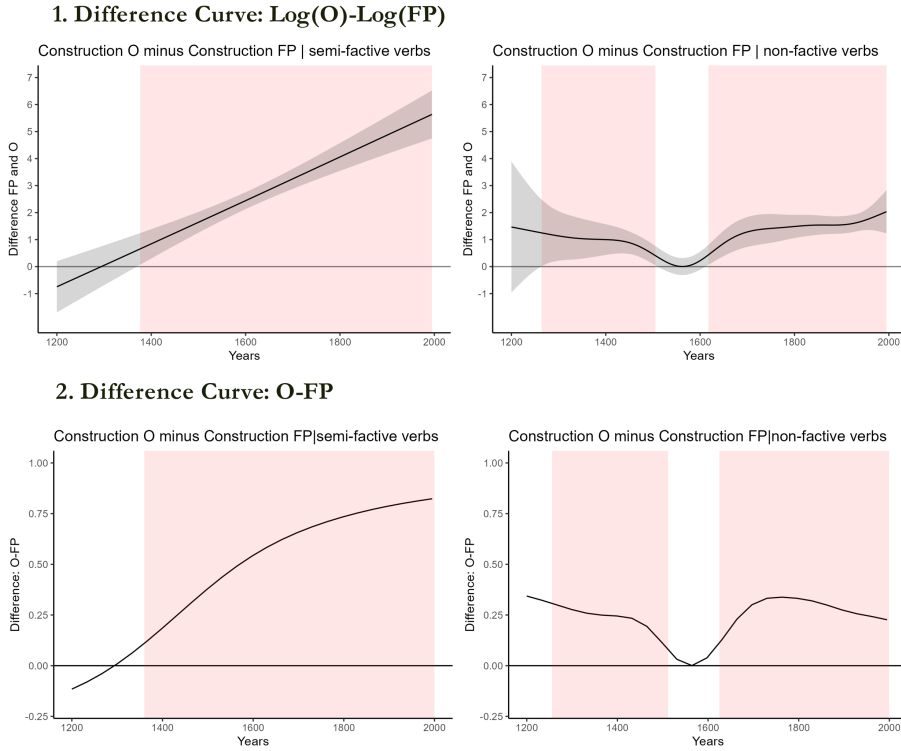


Figure 5.5: Difference curves.

semi-factive verbs), and displays a highly nonlinear behavior which can partially be accounted for by means of a three way interaction of [Year $\times$ Construction $\times$ Verb Class]. As Year increases the use of the indicative declines, but it does so differently depending on the verb class and the construction. Moreover, the effect of Construction has a different behavior depending on the verb class of the matrix predicate and the year: the difference across constructions increases with semi-factive verbs as time passes by, but in the case of non-factive verbs the difference across constructions stays diachronically rather constant (with only a dip in 1600). So far, however, we have not looked at the internal variability within the verb classes. The next subsection is devoted to exploring this.

5.1.4.2 Variability within predicates

Although the previous section shows how there are differences across verb classes, it does not explore variability within the predicates themselves, an as-

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pect which is important if we want to put forth the idea that the difference is caused by the verb class of the predicates. In this section, we aim to determine how much variability there is within groups. Figure 5.6 shows the results for each individual verb. As in the previous graphs, the size of the dot indicates the number of tokens and the color the construction. On the left column the semi-factives verbs *know* and *see* appear, and on the right column the non-factive predicates *believe* and *say*.

Starting with the similarities, with semi-factive verbs (left column) the use of the indicative remained stable in Construction O, but declined in Construction FP (both with *know* and *see*). With non-factive verbs (right column) the use of the indicative in both Construction FP and O has not been stable and has been declining since the 16th–17th century (both for *believe* and *say*). Thus, there seems to be a similar pattern of behavior within verb classes.

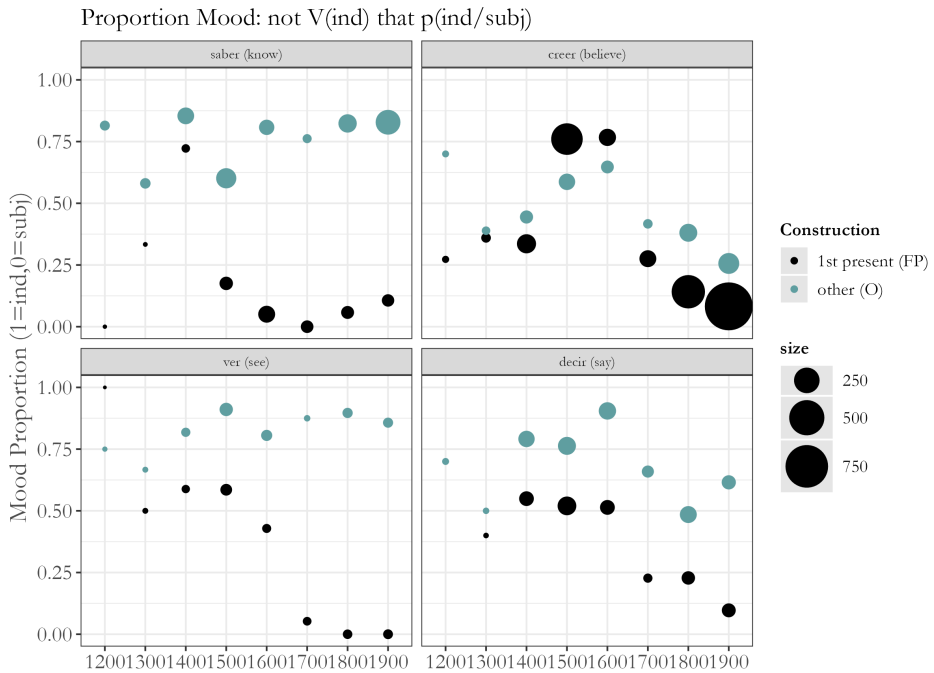


Figure 5.6: Proportion of mood in declarative clauses per predicate.

Despite these similarities, there is also some variability. For example, the decline of the indicative in Construction FP happened earlier with the verb *know* (1500) than with the verb *see* (1600). Also, the verb *say* seems to have been stable

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until 1600 a moment after which the use of the indicative decreased across constructions, but the verb *believe* had an up-and-down trajectory, increasing the indicative use until 1600 and decreasing afterward.¹²

The first difference (the different times of the indicative loss for the verb *know* and *see*) is not unexpected, as there are many instances of linguistic changes that show similar delays across environments. For instance, in the emergence of periphrastic *do*, Kroch (1989) shows that the time of introduction of *do* is different across contexts (declarative, questions, affirmative intransitive questions, etc.), see Figure 5.7. Kauhanen & Walkden (2018) expanded on this work and showed that although there can be differences across constructions, in order to be considered part of the same underlying change the time difference between the tipping points (the point of greatest growth in a logistic) cannot be larger than $1.76(1/|s|)$ (where s indicates the speed of change), see (22). This predicted time range is indicated in Figure 5.7 by the horizontal line, and as can be seen, in the case of the emergence of *do*-support the changes all lie within the expected time range.

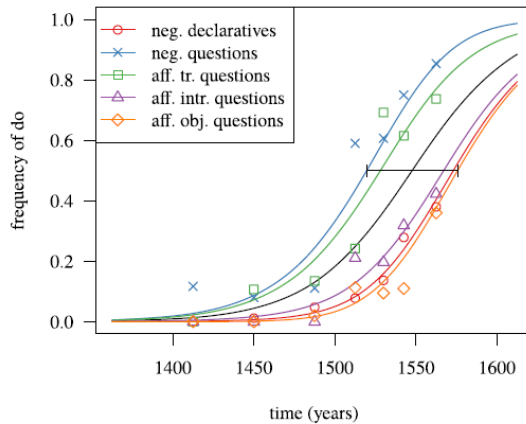


Figure 5.7: The emergence of periphrastic *do* in the history of English. Image taken from Kauhanen & Walkden (2018) [creative common license: <http://creativecommons.org/licenses/by/4.0/>].

Time Separation Theorem (Kauhanen & Walkden 2018: 501): For any two contextual reflexes of an underlying change from G_1 to G_2 approximated by a logistic with slope s , the maximal time separation at tipping points is:

¹²These differences are statistically significant: comparing the AIC of using verb rather than verb class in the model shows that the model would be considerably better by taking into account individual predicates rather than verb classes (4873 vs. 5045).

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$$(22) \quad \Delta s = \frac{2}{|s|} \log\left(\frac{1}{\sqrt{2}-1}\right) \approx 1.76 \frac{1}{|s|}$$

In order to determine if the difference in time between *know* and *see* in the data is expected and could be explained by means of a single underlying change, we can follow these authors and investigate: (i) whether the speed of change of the two constructions are similar, i.e. the Constant Rate Effect (Kroch 1989), and (ii) whether their time difference lies within the expected range, i.e. the Time Separation Theorem (Kauhanen & Walkden 2018).

Firstly, a logistic equation, see (23), was fitted to the data using the nonlinear least squares fitting function (nls) in R. In this equation s controls the speed of change (how steep the curve will be), and k indicates the tipping point (the moment of greatest growth).

$$(23) \quad \frac{1}{1 + e^{-s(t-k)}}$$

Initially, the logistic was fitted to all the data (see Figure 5.8 on the left). However, the fitting was not perfect given that in the years previous to 1400/1500 the data does not follow the expected behavior of a logistic (this is especially salient for the verb *see* for which there is more data and we are more confident that this deviation is not noise from a low number of tokens). Because of this, another logistic was fitted, but this time only with the data in which we start to see a decline in the indicative. For the verb *know* we included data from 1400 onward and for *see* from 1500 onward. As can be seen on the right image in Figure 5.8, this fit was a better one than the one obtained before, especially for the periods we are interested in (when the decline occurs).

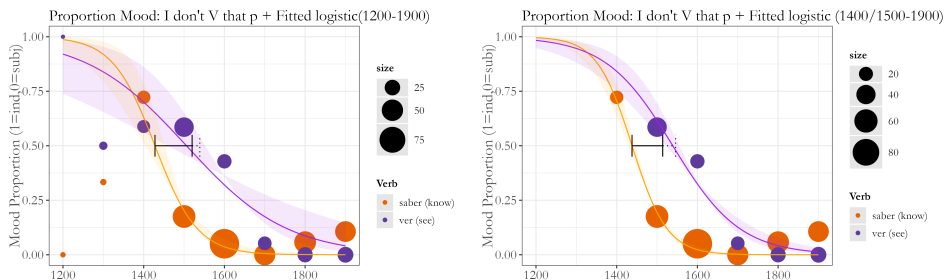


Figure 5.8: Fitted Data and their predicted maximum time separation. The horizontal black line indicates the predicted maximal separation time, and the dotted horizontal line the error in that measurement.

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The obtained values for the s (the speed) and k (the tipping point) parameters for the two models can be found in Table 5.4. Starting with the speed of change, both constructions do have different speeds but the difference can be argued to be non-significant as their confidence intervals overlap: 0.023 ± 0.008 for *know* and 0.012 ± 0.006 for *see*.¹³ Thus, corroborating the idea that these two changes might be caused by the same underlying trigger.

Looking at the maximal difference in the tipping points rather different measures for the two changes were obtained (the verb *see* allows for more variation). Given that the model fit of the verb *know* seems to be better, and also the maximum time is smaller and more conservative with this verb, we took the estimates of the verb *know* to estimate the range of time that can happen between changes. With the first model the upper bound is 93 years and with the second model 76 years. This predicts the upper bound year to be 1521 (1428+93) for the first model and 1514 (1438+76) for the second model. The measures obtained for the tipping point of the verb *see* are not outside of this range if we take into account the error in the measures: 1506 ± 46 and 1539 ± 32 , respectively. This is more clearly illustrated in Figure 5.8, where the upper bound limit is marked with a horizontal black line.¹⁴ As can be seen, the two changes lie within the expected limits.

Table 5.4: Fitted nls model with confidence intervals.

	Data (1200–1900)		Data (1400/1500–1900)	
	<i>know</i>	<i>see</i>	<i>know</i>	<i>see</i>
s	0.019 ± 0.006	0.008 ± 0.004	0.023 ± 0.008	0.012 ± 0.006
k	1428 ± 24	1506 ± 46	1438 ± 18	1539 ± 32
Max sep t	93	220	76	146

In conclusion, the differences in time between *know* and *see* are not unexpected (both constructions show a non-significant different speed and lie within the expected time separation) and current models correctly predict the variation observed.

¹³The confidence interval for the first model does not overlap, but by very little (0.001): $[0.025 - 0.013]$ vs. $[0.012 - 0.004]$. Given that this model seems to fit the downward change worse, this is not too problematic.

¹⁴The error of the time range was calculated using the lower estimates. For model 1, the range is 1521–1539. This last number is the result of computing $1404 + (1.76(1/0.013))$. For the second model, the range is 1511–1537, this last year was calculated by computing $1420 + (1.76(1/0.015))$.

The second difference (that before 1600 the use of the indicative was increasing following the verb *believe* but after the verb *say* the proportions were stable) is harder to account for in a model that only takes into account the verb class of the predicate. One option is that by looking at individual predicates, we have decreased the number of tokens per period and hence introduced more noise. The variation that we see then would be just noise and need not have a theoretical explanation. This would not be unexpected as the differences are not very large. Another option is that this difference is indicative of a difference in the nature of the predicates. For example, the verb *believe* in affirmative clauses takes only the indicative but the verb *say* can take the subjunctive in certain environments even in affirmative clauses. Thus, there seems to also be some differences across predicates within the same verb class, which could perhaps explain the variation. For current purposes and given that there are not many other predicates for which we would have enough diachronic data to compare with, we will assume that the first alternative may explain the data.

In conclusion, the results in this section show that although there is internal variability, it is within a reasonable range and the predicates in each group are not showing radically different behaviors. The small differences are predicted by current linguistic models of language change (e.g., the case of variation within semi-factive verbs), or may be due to noise from a lower number of tokens. These results then provide support for the model presented in the previous section, in which the differences in mood proportion can be explained by means of an interaction of three factors: time, construction and verb class. The next section discusses the theoretical implications of such findings.

5.1.5 Discussion

The overall results show that in the case of semi-factive verbs, the use of the indicative declined in Construction FP in the 15th century and by the 18th century most constructions appeared with the subjunctive morphology. Construction O, in contrast, did not decrease in the use of the indicative, and the proportions of usage remained rather stable diachronically with only a very subtle increase.

These results are in line with the presuppositional approach (see Quer 1998 among others) insofar as, under this approach, the use of the indicative in Construction FP but not Construction O would lead to a semantic contradiction. For instance, the verb *know* with the indicative would lead to a contradiction in (25a) but not in (24a), as indicated by the paraphrases. Similarly, the verb *see* both with and without the neg-raising inference leads to a contradiction with a first person construction: with a neg-raising inference the sentence with the indicative can

be paraphrased as *we are finding a solution and I see that we are not finding a solution*; with a non-neg-raising inference the sentence would convey that the speaker has evidence for *p* (under the hypothesis that strong commitment is justified belief), and that the speaker lacks evidence for *p*, as they cannot see that *p*. In contrast, although Construction O leads to an odd/less likely situation when the verb neg-raises *we are finding a solution and Mary sees that we are not finding a solution*, the second reading is acceptable: *we are finding a solution and Mary doesn't see we are finding a solution*.

(24) **Construction O**

- a. *María no sabe que está lloviendo.*

Mary doesn't know that it is.IND raining $\leadsto$ it is raining

Paraphrase: (I know) It is raining and Mary doesn't know it is raining.

- b. *María no ve que vamos a encontrar una solución.*

Mary doesn't see that we are.IND going to find a solution. $\leadsto$ we are finding a solution

Paraphrase 1: ? (I know) we are finding a solution and Mary sees we not are finding a solution.

Paraphrase 2: (I know) we are finding a solution and Mary doesn't see we are finding a solution.

(25) **Construction FP**

- a. **No sé que está lloviendo.*

I don't know that it is.IND raining $\leadsto$ it is raining

Paraphrase: ?? (I know) It is raining and I don't know that it is raining.

- b. **No veo que vamos a encontrar una solución.*

I don't see that we are.IND going to find a solution. $\leadsto$ we are finding a solution

Paraphrase 1: ?? (I know) we are finding a solution and I see that we are not finding a solution.

Paraphrase 2: ?? (I know) we are finding a solution (and I can defend this view with evidence) and I don't see we are finding a solution (there is no evidence indicating whether we will find a solution).

Under this approach, then, one could argue that the difference in speaker commitment (the indicative conveys speaker commitment, the subjunctive is agnostic towards it) may have emerged during the 15th century, and would explain why

from the 18th century onward the use of the indicative only in Construction FP is for the most part ruled out.

The situation is different, however, when we look at non-factive verbs. With these predicates, the use of the indicative decreased both in Construction FP and Construction O during the 17th century and it seems that this change is still ongoing. The presuppositional account does not predict this behavior, as one would expect that, similar to semi-factive verbs, the alternation would be stable after Construction O, and the decline in Construction FP would have occurred earlier. That is, one would have to explain why speaker commitment emerged with semi-factive verbs during the 15th century and was fully operational by 1700, but in other constructions the change did not occur until 200 years later. A unified explanation of these two changes would have to postulate that the change in semi-factive constructions ran into completion before it even took off with non-factive verbs.

The obtained results, thus, provide further evidence for the three-grammar competition proposal put forth in Chapter 4. As was argued there, at least two grammars are needed to account for the Old Spanish and Modern Spanish mood data. The fact that in the change from Old Spanish to Modern Spanish we observed two different changes happening at markedly different times, provides additional support for introducing a third grammar into the system. One for the change that started in the 15th century and another for the change that started in the 17th century. Chapter 6 will describe in more detail the need for a three-grammar competition.

Moreover, the results directly connect with the experimental data found in Montero & Romero (2023). While there was a speaker commitment effect for semi-factive verbs, the experiment did not find such an effect for non-factive verbs. Thus, one could maintain that speaker commitment originated during the 15th century with semi-factive verbs, but a different explanation for the decline of the indicative with non-factive verbs would be needed.

Other semantic approaches to PS also do not predict the three way interaction between verb class, time and construction found in the data. The information status approach (Faulkner 2021) and the Neg-raising one (Reider 1990) could have potentially explained the time effect as a result of the indicative acquiring a new meaning. However, there is nothing in the theories predicting an effect of Construction. There is no *a priori* reason why new information should be used more frequently with non-first person subjects or that the neg-raising reading inference should be obligatory with a first person subject. Similarly, the effect of verb class (in particular a distinction between semi-factive and non-factive) is not predicted by these accounts. The neg-raising approach if anything would predict a

difference between the neg-raising verbs *see* and *believe* and the non-neg-raising ones *say* and *know*, but this is not the pattern found in the data. The information status account would equally not predict, at least to my understanding, a decline in the use of new information only after non-factive verbs.

In conclusion, the obtained results pose interesting challenges to all current approaches to PS, and raise the following questions:

1. Why did the decline of the indicative in Construction FP (first person and present tense) happen at different times across verb classes (1400 for semi-factives, 1700 for non-factives)?
2. Why did the decline in the indicative in non-factive verbs start in 1700 in both Construction FP and O?
3. Why has the proportion of the indicative in Construction O remained stable with semi-factive verbs but not with non-factive verbs?

The first question already suggests that we might need two different explanations for the decline of the indicative in Construction FP: one trigger for semi-factive verbs and another for non-factive verbs. The second question directly links to the first one, and indicates that the different trigger for the loss of the indicative in Construction FP with non-factive verbs may also be causing the decline in Construction O with non-factive verbs. Lastly, the third question suggests that whatever the cause of the decline in Construction O (and possibly also FP) with non-factive verbs it must be caused by something construction O with a semi-factive verb lacks/has.

The next section explores how the competing grammars approach, first presented in Chapter 4, could provide new insights into these questions and how coupled with the theory of Variational Specialization (Wallenberg 2019) it could predict the three way interaction of time, verb class and construction that was described in this section.

5.2 Towards an explanation of the verb class effect

The general idea that this section explores in order to account for the data presented in the previous section is the following: a semantic mechanism (G_1) which indicated an agent's commitment to p (or more precisely, following Schlenker 2005, that the word w lies in some accessible Context Set), and a syntactic one

(G_2) whereby the subjunctive mood was grammaticalized into a marker of negation (similar to the grammaticalized Genitive of Negation case in Lithuanian).¹⁵ With semi-factives these two grammars stood in a superset-subset relation (see Section 5.2.1), and as a consequence the superset grammar (G_1 , semantic selection) won. With non-factives these two grammars stood in a disjoint relation (see Section 5.2.1), hence taking longer for one grammar to win the other, the competition still ongoing in the present-day language, but leaning towards grammar G_2 (syntactic selection). This led to a situation in which the two grammars were preserved in the language: G_1 winning with semi-factive verbs and G_2 winning with non-factive verbs.

The proposed approach has several benefits in comparison to previous approaches to PS. Firstly, the three way interaction between class, time and construction can be explained as a result of competition between grammars having different outcomes across constructions. Secondly, the different times for the loss of the indicative in Construction FP across verb classes observed in the previous section follow from the proposed account (see question 1 above). Under this account, there are two different triggers leading to the loss of the indicative: One, a semantic clash between first person and speaker commitment which affected Construction FP with semi-factive verbs during the 1500s. Two, grammar G_2 (syntactic selection) started winning over G_1 with non-factive verbs during the 1700–1900s which affected both Construction FP and O with these verbs (indirectly also answering question 2 above). Thirdly, under this account the fact that in Construction O with semi-factive verbs the use of the indicative was stable (question 3 above) is expected: with semi-factive verbs semantic selection won over syntactic selection, and given that the semantic mechanism allows both the indicative and the subjunctive after these predicates (see Chapter 4), the alternation would be semantically triggered and stable. This would also explain the experimental results found in Montero & Romero (2023), where mood after negated predicates seemed to be semantically driven with semi-factive verbs but not with non-factive verbs.

The rest of this section explains in more detail how the idea just presented works. Section 5.2.1 summarizes the G_1 and G_2 grammars presented in more detail in Chapter 4 (the interested reader is referred back to that chapter for a more in-depth description of the account). Section 5.2.2 explains how the original formulation of variational learning does not fully account for the observed

¹⁵For ease of explanation, this section will first focus on the two-grammar competition between G_1 and G_2 . In the following chapter the third grammar will be added into the system to explain the results.

diachronic data. Finally, Section 5.2.3 explains how Wallenberg (2019)’s theory of Variational Specialization (an extension of the original Variational Learner proposed in Yang 2000) could better account for the data.

5.2.1 The competing grammars approach to mood

The approach proposed in Chapter 4 explored the possibility of there being two different mechanisms for mood morphology in the present-day language: one in which mood morphology is semantically interpretable, and another in which mood is uninterpretable and whose feature needs to be eliminated by the syntactic interface before the sentence is sent to LF. The following sections summarize how these two mechanisms work more formally, and the predictions they make regarding how verb class and construction type can influence mood selection.

5.2.1.1 G₁: semantic mechanism

Following Schlenker (2005) and Montero & Romero (2023), see also Section 3.2.3 in Chapter 3, under the competing grammars approach semantic mood is analyzed as a pronoun which carries presuppositions: the indicative presupposes that the world of evaluation is in some accessible Context Set and the subjunctive is a default form carrying no presuppositions. The differences across verb classes and constructions emerge out of the interplay between this presupposition and the contributions that different verbs and parts of the clause make to the final truth-conditional meaning of a sentence.

The key semantic difference between a semi-factive verb and a non-factive verb is that the former but not the latter carries a factive presupposition (26)-(27). A semi-factive verb like *know* is composed of two pieces of meaning: an asserted part *John doesn’t know that Mary runs*, and a presupposed component *Mary runs*; whereas a non-factive verb is only composed of an asserted part *Mary didn’t say that Mary runs*, see Chapter 3 for more details.

- | | |
|--|---|
| <p>(26) semi-factive verbs:
 John didn’t know that Mary ran.
 $\neg \text{KNEW}(j, \text{RAN}(m)) \wedge \text{RAN}(m)$</p> | <p>(27) non-factive verbs:
 John didn’t say that Mary ran.
 $\neg \text{SAID}(j, \text{RAN}(m))$</p> |
|--|---|

Montero & Romero (2023) make use of this key distinction to derive a presuppositional difference across verb classes. With **semi-factive verbs**, speaker commitment is obtained when the the indicative mood presupposition ($w \in \text{CS}$) is added to the second conjunct in (26) (component *b* above), see (28), but no

speaker commitment is obtained with the subjunctive (the speaker can but need not be committed), see (29), as the subjunctive is a default carrying no presuppositions.

- (28) Juan no sabía que María corría.
 John not knew that Mary ran.IND
 ‘John didn’t know that Mary ran.IND.’
 Paraphrase: John didn’t know p and (I know) p .
 a. $\lambda w_0. \neg \text{KNEW}_{w_0}(j, \lambda w : w \in \text{Dox}_{w_0}(j)). \text{RAN}_w(m)$
 b. $\lambda w : w \in \text{CS}^* . \text{RAN}_w(m) \rightarrow \text{speaker's commitment to } p$
- (29) Juan no sabía que María corriese.
 John not knew that Mary ran.SBJV
 ‘John didn’t know that Mary ran.SBJV.’
 a. $\lambda w_0. \neg \text{KNEW}_{w_0}(j, \lambda w : w \in \text{Dox}_{w_0}(j)). \text{RAN}_w(m)$
 b. $\lambda w : w \in \text{CS}^* . \text{RAN}_w(m)$

The speaker commitment presupposition triggered by the indicative, however, would clash with the asserted part when the subject is first person and the verb is in present tense, as the sentence would lead to two contradictory pieces of meaning, see (30). On the one hand, the meaning of *know* is essentially to express commitment to the truth of its complement and its negation leads to the negation of that commitment on the part of the matrix subject. On the other hand, however, the presuppositions of the indicative when included in the second conjunct of (26) (component *b* of meaning), would trigger the meaning of speaker commitment. Thus, the combination of both statements ends up meaning that the speaker is not committed to p and the speaker is committed to p .

- (30) *No sé que María corre.
 No know.1SG.PRS that Mary runs.IND.
 ‘I don’t know if Mary runs.IND.’
 Paraphrase: I don’t know p and (I know) p .
 a. $\lambda w_0. \neg \text{KNOW}_{w_0}(sp, \lambda w : w \in \text{Dox}_{w_0}(sp)). \text{RUN}_w(m)$
 b. $\lambda w : w \in \text{CS}^* . \text{RUN}_w(m) \rightarrow \text{speaker's commitment to } p$

The prediction is, then, that with semi-factive verbs there would be variation between the indicative and the subjunctive in Construction O (non-first person) – and the the alternation would be semantically triggered – but not in Construction FP (first person).

It is importantly to highlight that this mechanism relies on a key difference between soft presuppositional triggers and hard presuppositional triggers (Abusch 2002, Djärv et al. 2018). Semi-factive predicates are considered to be ‘soft’ presupposition triggers. As such, their factive presupposition are derived pragmatically and can be easily suspendable (31a). Emotive-factive predicates are considered to be ‘hard’ presupposition triggers. For these verbs, the factive presupposition is lexically encoded and hence harder to suspend (31b) (Djärv et al. 2018). This key difference accounts for the differences in terms of mood selection between semi-factive and emotive-factive predicates. For a more in-depth description of the analysis the reader is referred to Montero & Romero (2023) and Section 3.2.3 in Chapter 3.

- (31) a. If I **discover** that the proposal offended them, I will apologize
 $\nrightarrow$ the proposal offended them
 b. If I **regret** later that the proposal offended them, I will apologize
 $\sim \rightarrow$ the proposal offended them
 (Djärv et al. 2018: 368)

With **non-factive verbs**, in contrast, no speaker commitment presupposition would emerge as these verbs are composed only of the asserted component (the first element *a*), see (32). In this case, because the indicative can be used and the subjunctive alternative would not convey a meaning difference ((32) and (33) are equivalent), by Maximize Presupposition! (Heim 1991) the indicative must be used. Moreover, no contradiction would emerge with these predicates, even in the case of a first person subject (34), as the indicative version would only state that the matrix subject (and also the speaker) did not assert *p*.

- (32) Juan no dijo que María corría.
 John not said that Mary ran.IND
 ‘John didn’t say that Mary ran.IND.’
 a. $\lambda w_0. \neg \text{SAID}_{w_0}(j, \lambda w : w \in \text{Dox}_{w_0}(j). \text{RAN}_w(m))$
- (33) Juan no dijo que María corriese.
 John not said that Mary ran.SBJV
 ‘John didn’t say that Mary runs.SBJV.’
 a. $\lambda w_0. \neg \text{SAID}_{w_0}(j, \lambda w : w \in \text{Dox}_{w_0}(j). \text{RAN}_w(m))$
- (34) No digo que María corre.
 not say.1SG.PRS that Mary runs.IND
 ‘I don’t say that Mary runs.IND.’
 a. $\lambda w_0. \neg \text{SAY}_{w_0}(sp, \lambda w : w \in \text{Dox}_{w_0}(sp). \text{RUN}_w(m))$

To summarize, the proposed analysis makes the following predictions regarding mood selection: with semi-factive verbs in Construction FP the expected mood is the subjunctive (as this is the only mood that would not lead to a contradiction), and in Construction O the expected mood is both the indicative and the subjunctive, the choice being based on whether the speaker intends to convey speaker commitment or not; with non-factive verbs the prediction is that they will always select the indicative, irrespective of construction, as by Maximize Presupposition! when two alternatives are truth-conditionally equivalent the one with the strongest presuppositions (in this case the indicative option) should be used. These predictions are summarized in Table 5.5.

Table 5.5: Semantic account: mood predictions across verbs and constructions in negative clauses. FP = First Person; O = Other.

	Construction FP	Construction O
semi-factive	sbjv	ind/sbjv
non-factive	ind	ind

The next section compares these predictions with the ones made by the syntactic grammar.

5.2.1.2 G_2 : syntactic mechanism

The second grammar is one in which mood has been grammaticalized and carries no meaning. The key idea that was explored in Chapter 4 is that the subjunctive mood may have become a marker indicating that negation appears higher in the clause: if a sentence is affirmative it gets the indicative mood and if it is negative it gets the subjunctive mood. Drawing inspiration from the analysis of Genitive of Negation as a grammaticalized case in Lithuanian, this mood feature could be represented by an uninterpretable feature $uCase_{\square}$ at narrow syntax which needs to be deleted before the sentence is sent to the interfaces.

Extending Witkoś (2008)'s analysis of genitive case, I explore the possibility that the heads *Pol* and *v* create a double Probe for agreement purposes in Spanish. The uninterpretable and unvalued mood features of the CP are checked and deleted as a result of ϕ agreement between the CP and the double Probe: the value assigned will be the indicative if the sentence is affirmative and the subjunctive if negative.

(35) **Affirmative clause:**

- a. $[_{PolP} Pol_{[aff]} \dots [_{vp} v \quad CP \quad]]$
 $iMood_{[ind]} \quad u\phi_{[]} \quad i\phi_{[val]}, uMood_{[]}$
- b. $[_{PolP} Pol_{[aff]} \dots [_{vp} v \quad CP \quad]]$
 $iMood_{[ind]} \quad u\phi_{[val]} \quad i\phi_{[val]}, uMood_{[ind]}$

 (36) **Negated clause:**

- a. $[_{PolP} Pol_{[neg]} \dots [_{vp} v \quad CP \quad]]$
 $iMood_{[sbjv]} \quad u\phi_{[]} \quad i\phi_{[val]}, uMood_{[]}$
- b. $[_{PolP} Pol_{[neg]} \dots [_{vp} v \quad CP \quad]]$
 $iMood_{[sbjv]} \quad u\phi_{[val]} \quad i\phi_{[val]}, uMood_{[sbjv]}$

Once the feature is deleted and sent to the interfaces, there will be no mood feature at LF to be interpreted and hence the semantic derivation would proceed as if there were no mood features involved.¹⁶

This grammar is not predicted to interact with verb class or construction type (FP vs. O). Table 5.6 shows a comparison of the different predictions these two grammars would make in negated constructions (Constr.).

Table 5.6: Comparison of the predicted selectional patterns of the semantic and syntactic grammars in negative sentences.

	G ₁ : semantic		G ₂ : syntactic	
	Constr. FP	Constr. O	Constr. FP	Constr. O
semi-factive	sbjv	ind/sbjv	sbjv	sbjv
non-factive	ind	ind	sbjv	sbjv

The next section attempts to use Yang (2000)'s variational learning framework to discern how these two grammars will compete in diachrony.

5.2.2 Variational mood learning

Yang (2000)'s variational learning proposes that speakers can internalize more than one grammar and attach a probability to each of these grammars (see Chapter 2 section 2.2.3 for a description of this model and also see Yang 2010). Language change occurs when the probability distributions of one generation of

¹⁶In Chapter 4 other alternative formalizations were proposed. The interested reader is referred to that chapter, and also to Chapter 7.

speakers are different from the probability distributions of the previous/next generation. The directionality and trajectory of this change will be determined by the fitness parameters of each of the grammars.

In the case of a two-grammar competition scenario, like the one we are dealing with in this chapter,¹⁷ the fitness of a grammar can be defined as the probability of a speaker producing a sentence the other grammar cannot parse. Let L_1 stand for the set of sentences that G_1 can parse, and L_2 the set of sentences that G_2 can parse. Then, the fitness parameter of G_1 (α) will be equal to the probability of hearing a sentence which only belongs to L_1 ($L_1 \setminus L_2$). The fitness parameter of G_2 (β) will be equal to the probability of hearing a sentence which only belongs to L_2 ($L_2 \setminus L_1$).

$$(37) \quad \alpha \text{ (fitness } G_1) = P(L_1 \setminus L_2)$$

$$(38) \quad \beta \text{ (fitness } G_2) = P(L_2 \setminus L_1)$$

According to Yang (2000)'s Fundamental Theory of Language Change, the grammar that is predicted to win out in diachrony is the one with the greatest fitness (because this one will be penalized less often by the algorithm), see (39).

(39) **The Fundamental Theorem of Language Change:** (Yang 2000: 239)

G_2 overtakes G_1 if $\beta > \alpha$: the advantage of G_2 is greater than that of G_1 .

In order to determine the predicted outcome for our given case study, one could attempt to calculate the fitness of each of the grammars. This could be done by first looking at the types of sentences each language can parse and then using data from purely G_1 or G_2 languages to calculate the probabilities of those sentences being uttered. Table 5.7 shows the different types of sentences we are interested in, and to which set they belong: sentences in the set $L_1 \setminus L_2$ are only parsed by G_1 , and if one could calculate its probability it would give as the fitness value of G_1 (α); sentences in the set $L_1 \cap L_2$ are sentences which are ambiguous and which either grammar could parse; sentences in $L_2 \setminus L_1$ are sentences that only grammar G_2 can parse, and calculating the probability of of this set will give us the value for the fitness of G_2 (β).

As a next step, then, one would need to look at frequencies of these sentence types in purely G_1 and G_2 languages.

However, there is a problem with this approach. Regardless of the results obtained for α and β , they are not going to be able to account for the data of PS in

¹⁷ In the next chapter we will see that we may need to introduce a third grammar to fully account for the data.

Table 5.7: Types of sentence each grammar is able to parse in negative sentences.

Sentence type	$L_1 \setminus L_2$	$L_1 \cap L_2$	$L_2 \setminus L_1$
semi-factive+Construction O + ind	✓	–	X
semi-factive+Construction O + sbjv	–	✓	–
semi-factive+Construction FP + sbjv	–	✓	–
non-factive+Construction FP + ind	✓	–	X
non-factive+Construction FP + sbjv	X	–	✓
non-factive+Construction O + ind	✓	–	X
non-factive+Construction O + sbjv	X	–	✓

Spanish, as based on the results from the previous section neither grammar seems to be winning completely. Rather G_1 seems to be winning with semi-factive verbs and G_2 with non-factive verbs. Because in Yang (2000)’s implementation of the Variational Learner the probabilities of each grammar are calculated over the language as a whole, once one grammar is on the rise it is predicted to win out across all environments. This has the consequence that long-term stable competition between two grammars is not predicted to occur, but this is precisely the situation we seem to be dealing with.

In the next section it is shown that this problem need not be fatal for the competing grammars approach, if one implements Wallenberg (2019)’s theory of Variational Specialization, whereby the probabilities of each grammar can be tracked separately for individual constructions.

5.2.3 Variational mood specialization

Applying the Theory of Variational Specialization (Wallenberg 2019), see Section 4.1.2 for an overview, to the problem of Polarity Mood, we could hypothesize that the grammar’s probabilities are tracked separately in construction with a semi-factive verb and a non-factive verb and that in the case of constructions with semi-factive verbs grammar G_1 (semantic mood) would win, but with non-factive verbs grammar G_2 (syntactic mood) would win, see Table 5.8.

This hypothesis is interesting in the sense that it has clear predictions that can be tested against empirical data. In order for grammar G_1 to win in constructions with a semi-factive verb the α value has to be greater than the β parameter. Looking again at the types of sentences each grammar can parse, we can observe that

Table 5.8: Variational Specialization (Wallenberg 2019) applied to Polarity Mood in negative sentences.

Variational specialization theory	P_{G_1}	P_{G_2}	Winning Grammar
semi-factive verbs	p_t	$1 - p_t$	G_1
non-factive verbs	p_t	$1 - p_t$	G_2

this is the case: with semi-factive verbs L_2 is a subset of L_1 , and as such the probability of the grammar G_2 producing a sentence that the other grammar cannot parse is 0 ($\beta = 0$). The probability of Grammar G_1 producing a sentence that the other grammar cannot parse would be given by the probability of producing a sentence with a semi-factive verb a non-first person construction and the indicative mood. That is, a sentence that belongs to the set $L_1 \setminus L_2$ in Table 5.7 above. Because this probability is not equal to zero, α is always going to be greater than β (which is 0), and hence grammar G_1 is predicted to win out in these constructions.

In the case of non-factive verbs, the situation is a bit more complicated. In order for grammar G_2 to win the β value would have to be bigger than α . In this context, grammar G_2 and grammar G_1 are in a disjoint relationship and, contrary to the prediction, no grammar is predicated to win out in diachrony (they should remain in stable competition). This need not be fatal for the analysis, if one allows for extra-linguistic factors to play a role in the system. One hypothesis is that changes at the population level (drastic changes in population numbers, contact with speakers of other languages, etc.), may have created an asymmetry in the system.

The idea is as follows: the probabilities α and β are, in purely linguistic terms, equal. However, certain types of speakers might have cognitive biases/preferences for one or the other grammar. In particular, this section explores the possibility that children may learn the syntactic system easier than the semantic one, but that adults would learn the semantic mechanism better than the syntactic one. If this were the case, then a change in the amount of children vs. adult learners at the population level could have as a consequence a preference for one of the grammars: a population with more adult learners would tend towards grammar G_1 (semantic selection), whereas a population with more children would lean towards grammar G_2 semantic selection. In our given case study, this would mean that between 1200–1600 the population might have been characterized by a high number of adult speakers, whereas between

1600–1900 the number would have been reduced. Before exploring whether this prediction is on the right track, a more in-depth explanation as to why these two types of learners may have such cognitive preferences will be discussed.

Starting first with children, it has been reported that Spanish speaking children start using the subjunctive by the age of 2 (López Ornat et al. 1994). However, the acquisition literature also reports that children start realizing that other's beliefs may differ from their own at the age of 4–6 (Perner et al. 2015). That is, the necessary underlying concepts for acquiring grammar G_1 (that the world of evaluation lies in some accessible CS and that that CS can be different from one's own) are acquired at a later stage than the first uses of the subjunctive. This may give a preference to the syntactic system (at least at the earliest years of acquisition), as the key semantic concepts required to acquire G_1 are not acquired yet.¹⁸

On the other hand, adult learners would already understand the required semantic concepts to learn G_1 (semantic selection), but may struggle with a purely syntactic system. The Interpretability Hypothesis (Tsimpli & Hulk 2013) states that uninterpretable features tend to be harder to acquire for adult L2 learners (see Walkden & Breitbarth 2019 for how this might affect language change). If indeed grammar G_2 involves uninterpretable features, as was suggested in the previous sections, then the expectation is that adult L2 learners would struggle more with this grammar.

From an acquisitional perspective, then, it does not seem far-fetched to propose that children might have a preference for the syntactic system (at least in the earliest stages of acquisition), whereas in the case of adults the opposite might be the case.¹⁹ The next question to answer then would be: was there a higher proportion of L2 speakers in pre-17th century than in the following centuries? If so, then one would have an explanation for the non-monotonic behavior of non-factive predicates. Future work from a more sociolinguistic perspective would be able to test whether the analysis is correct. For the time being it is sufficient to say that, at the very least, there was contact with Andalusí Arabic (711–1610) and Portuguese (1580–1640) L2 speakers (Ruhstaller & Gordón Peral 2017, Vicente 2020, Döhla 2014).

¹⁸Interestingly a similar development has been reported in the case of gender acquisition in Russian and several other languages: children first acquire a formal grammatical analysis which is gradually replaced by a semantic (sex-based) distinction (Rodina 2008).

¹⁹It must be highlighted that the acquisition literature on this topic is very extensive, and there is still much ongoing debate regarding what areas are the most problematic for adult L2 learner (see for example Sorace 2011 for the proposal that L2 learners struggle more with phenomena at the interfaces). This short section cannot do justice to in-depth complexities of this topic, but rather presents possible avenues of research to be further investigated.

5.2.4 Conclusion

This chapter has looked at the effect of Verb Class in the frequencies of mood on the diachrony of Spanish. The first part has provided quantitative evidence showing how the frequencies of mood selection seem to be affected by three main (interacting) factors: Year, Construction and Verb Class. Moreover, the data has shown that (i) the only construction in which the proportion of the indicative has been stable is in Construction O (non-first person matrix subject) and a semi-factive verb, and (ii) the decline in the use of indicative across Construction FP (first person) happened at markedly different times for the different verbs (the decline started in 1400 for semi-factive verbs and at 1600 for non-factive verbs). The second part of the chapter has shown how these observed tendencies could be accounted for by means of applying the theory of Variational Specialization (Wallenberg 2019) to the phenomenon of polarity mood. The general proposal that has been put forth is that in semi-factive constructions the semantic grammar has won, and with non-factive verbs the syntactic grammar is winning out. This would explain the differences across predicates both in terms of their semantic contribution (Montero & Romero 2023) and their diachronic behavior. The next chapter will focus on the effect of first person constructions, and how we can account for this effect across different verb classes.

6 Explaining Polarity Mood: The first person effect

The previous chapter has shown that there is a significant difference in the frequencies of the indicative used in Construction FP (first person and present tense) and Construction O (all other constructions) diachronically, but that this effect varied based on the type of matrix verb. With semi-factive verbs the difference across constructions increased as time passed. This, it was argued, may have been a result of the indicative mood starting to be used to indicate speaker commitment: as the new meaning emerged, it led to a semantic clash with Construction FP but not with Construction O; hence the use of the indicative decreased in the former but not the latter context. In contrast, with non-factive verbs the difference across constructions was stable, Construction O taking consistently 25% more indicative than Construction FP. During the 18th century the indicative decreased in both Construction FP and O, a change that was hypothesized to be the result of the grammaticalization of the mood system: G_2 (syntactic mood) started to win over. This explained why the indicative mood decline happened (i) at a later point with non-factive verbs than with semi-factive verbs, and (ii) in both Construction FP and O when the verb was a non-factive verb. These ideas are summarized below:

(1) **Proposal:**

- Semi-factive verbs Construction FP: decline in the indicative starting in 1400. → Change 1: speaker commitment (grammar G_1 wins)
- Non-factive verbs Construction FP and O: decline in the indicative starting in 1700. → Change 2: grammaticalization (grammar G_2 wins)

The first person effect could be then understood as the result of two historical effects: (i) there was a difference between first and non-first person constructions from the beginning of the language in the sense that the indicative was 25% more frequent with non-first person constructions and this difference has been preserved in the language, and (ii) the indicative acquired a presupposition whereby the world of evaluation lies in some accessible context set. This

increased the already existing difference between constructions, as it led to a semantic contradiction in the case of a first person pronoun with a semi-factive verb.¹

This proposal, however, relies on the assumption that the speed of change of Construction FP and O with non-factive verbs are the same, as they are, under this account, caused by the same underlying process (see Kroch 1989); and raises the question of what triggered the 25% difference across constructions during the earlier stages of the language.

The first part of this chapter tackles the issue of the speed of change. It shows that the speed is similar in the case of the verb *say* but not with the verb *believe*. Notwithstanding, it is later shown that in the case of the verb *believe* there is a further linguistic change (the presence of an overt complementizer) affecting the overall speed without which the speed of change of Construction FP and O prove to be similar. These results, thus, indicate that the proposed analysis is feasible, and that the reason why the indicative is decreasing in Construction FP and O with non-factive verbs can be given a unified account.

The last part of this chapter is devoted to answering the question of what caused the initial difference across constructions. It explores the possibility that the asymmetry existed because in Old Spanish the indicative mood might have worked, similar to present-day Bulgarian, as a marker of strong commitment and the subjunctive as a hedging device to indicate weak commitment (Smirnova 2011). Given that hedging devices are more acceptable with a first person and present tense matrix verb (Koev et al. 2021), the observed higher frequency of the indicative for Construction O could be a result of hedging being slightly less acceptable in these constructions.

6.1 The speed of change

6.1.1 The constant rate effect (Kroch 1989)

There are two competing hypotheses in the historical linguistic literature for predicting the speed of change of a given variant. The first one, which dates at least back to Bailey (1973), postulates that change will happen faster in those

¹Interestingly, Quer (1998: 61) mentions in a footnote that the ungrammaticality of having the indicative with a first person subject and a present tense verb is perceived as stronger in the case of the verb *see* than the verb *believe* in Catalan. This aligns well with the proposal presented in this section, as the grammaticality with the verb *see* is argued to be a result of a semantic clash whereas the perceived ungrammaticality with the verb *believe* is an artifact of a frequency difference in Latin.

contexts which are more frequent. The other one, which was first formulated in Kroch (1989), states that the speed of change will be similar across contexts irrespective of frequencies, see (2).

- (2) **The Constant Rate Hypothesis:** *when one grammatical option replaces another with which it is in competition across a set of linguistic contexts, the rate of replacement, properly measured, is the same in all of them.*
(Kroch 1989: 200)

The Constant Rate Effect (CRE) emerged out of the observation that in the history of English the spread of *do*-support occurred at similar speed across constructions (question, negative declaratives, etc.), and has by now acquired a wide range of empirical support from similar situations in other languages and linguistic phenomena. Santorini (1993) shows that the CRE also holds in the change in IP-headedness in the history of Yiddish; Wagner (1996) shows that the change from *haver* to *ter* in Portuguese also follows the CRE; in Spanish von Heusinger (2008) has shown that the same applies for the emergence of Differential Object Marking (DOM); and in Egyptian Gardiner (2015) shows that the change in possessives also followed the CRE (see Wallenberg et al. 2026 and Zimmermann 2023 for a state-of-the-art of the CRE in linguistics).

There is, however, controversy regarding which is the best methodology to determine whether the difference between two speeds is statistically significant (e.g., using likelihood ratios vs. logistic regressions), whether historical studies testing the CRE had enough power, and why in certain situations (see for example Simonenko et al. 2019) the CRE has been shown not to hold (see Wallenberg et al. 2026 and Zimmermann 2023 for in-depth discussion of these issues). Moreover, most current studies testing the CRE have only investigated successful changes (see Section 6.1.2 for a description of other linguistic changes), which raises the question of whether the CRE should apply more generally to other types of changes as well.

Leaving these questions aside, the hypothesis makes interesting predictions regarding our case study. If the decrease in the indicative in non-factive verbs in Construction FP and O was caused by the same underlying change (G_2 -syntactic mood- winning over), then we expect to see the decrease in the indicative happening at similar speed across constructions (provided the CRE is correct). The following sections test whether this prediction is borne out. The next part discusses how the speed of change can be calculated for different types of linguistic changes, and the following sections apply this to calculate the speed of change of mood across different constructions and verbs.

6.1.2 Types of linguistic changes (Postma 2010)

Postma (2010) proposed three types of linguistic changes: successful changes which can be modeled as S-curves (logistic curve), inherent failed changes, which have been modeled as the first derivative of the logistic equation (Hubbert curve), or the accidental failed change, which is also modeled as a Hubbert curve but where the activation and slope parameters of the two S-curves can be different (see Figure 6.1).²

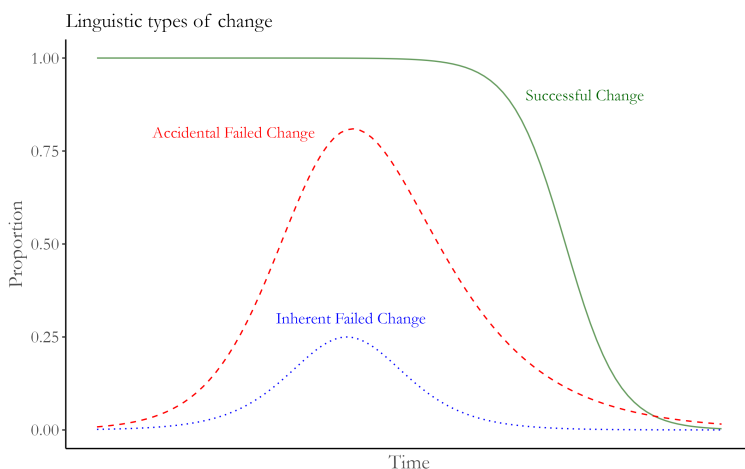


Figure 6.1: Types of linguistic changes according to Postma (2010).

That is, both the inherent and the accidental failed changes are understood as the combination of two successful changes (S-curves), see Figure 6.2, but because in the inherent failed change the activation time for both S-curves is the same, the peak of an inherent failed change will coincide with half the inflection point of the corresponding S-curve.

When looking at the results for the verbs *say* and *believe*, here repeated in Figure 6.3, one observes that while *say* is rather hard to categorize by just eyeballing the graph (it could be a failed change or a successful change) the verb *believe* clearly resembles an accidental failed change – it underwent an increase in the indicative mood until 1600, a moment after which the indicative started to decrease, but the peak of the curve is above half of the inflection point of the S-curve, thus occurring too high for considering it an inherent failed change – see Section 6.1.4 for more details.

²That is, in the accidental failed (AF) change S can be different $AF = S_a(1 - S_b)$; whereas in the inherent failed (IF) change the S parameter needs to be the same $IF = S_a(1 - S_a)$.

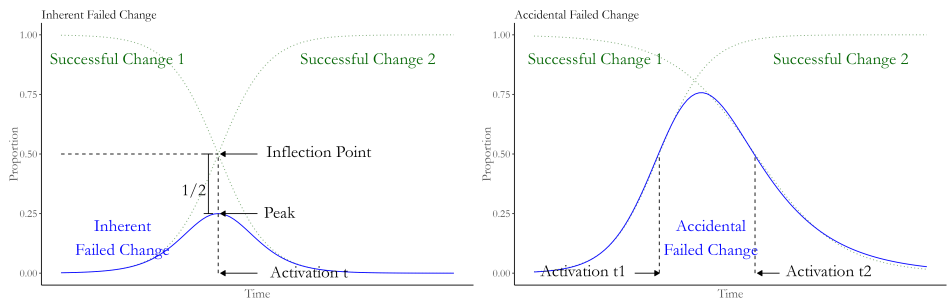


Figure 6.2: Types of failed changes according to Postma (2010).

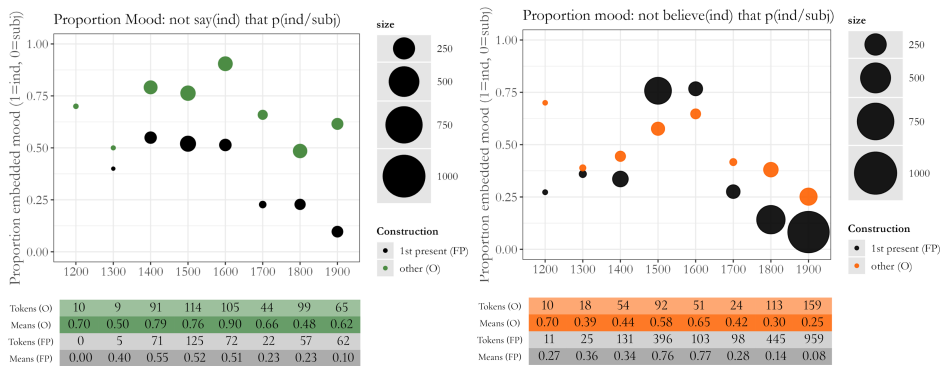


Figure 6.3: Development of Polarity Subjunctive after non-factive verbs. Indicative = 1, Subjunctive = 0, Construction FP = first person and present tense, Construction O = all other constructions, see Table 5.2

Based on this observation, in Section 6.1.3 the speed of these changes is calculated.

6.1.3 The verb *say*

The verb *say* is difficult because by eye-balling the data its behavior could either be classified as a successful change or as a failed change. In order to determine which would be the best way of analyzing the phenomenon, the data was fitted first to a logistic (the common procedure for fitting successful changes) and then to a Laplace curve (see below for a discussion of why this curve was chosen to describe a failed change rather than the Hubbert curve). The model fit of the two models was then compared. As it will be shown, the change could be best understood as an instance of a failed change (a welcome result as it would mean

that both non-factive verbs can be given a unified account) and most importantly the speed across constructions does not differ from each other, irrespective of the model chosen. The following sections describe how these conclusions were reached.

The common procedure to calculate the speed of change of a given successful change is to fit a logistic equation, see (3), to the data. In this equation, $M(t)$ represents the frequency of the indicative at time t , the k parameter indicates the time of greatest growth (known as *tipping point*), and the s parameter controls the speed of change. Note that by substituting s by $1/s'$ we can obtain the change in years, a measure which will be easier to understand. Finally, in this equation the maximum frequency is determined by the m parameter. Given that the frequencies of the indicative never reach 1 with the verb *say*, by modifying the m value in the equation we can determine the maximum frequency it can reach.

$$(3) \quad M(t) = \frac{1}{\frac{1}{m} + e^{-s(t-k)}} = \frac{1}{\frac{1}{m} + e^{-\frac{(t-k)}{s'}}}$$

The behavior of a logistic curve and how the different parameters affect it has been plotted in Figure 6.4. In the left image, the effect of changing the values of the s' parameter is plotted. The bigger the value of s' , the slower the change will be. Note that using s instead of s' will give the same outcome, but the numbers are harder to interpret, hence the choice of using s' through this chapter instead. The figure in the middle shows the effect of changing the k parameter: a k of 1500 indicates that the moment of greatest growth was the year 1500, a k of 1700 that the greatest growth happened in 1700, etc. Finally, the figure on the right shows how the m parameter affects the overall shape. A value of 1 indicates that the maximum frequency is 1, and the curve would tend towards that state. This type of behavior is the one expected in cases in which the two competing grammars are mutually exclusive. That is, grammar G_1 only allows for variant 1 and G_2 for variant 0. However, there are instances, as the one we are dealing with in this chapter, in which this may not be the case, and the initial grammar allows for both variants. The value assigned to m will be the maximum frequency that the system can reach.

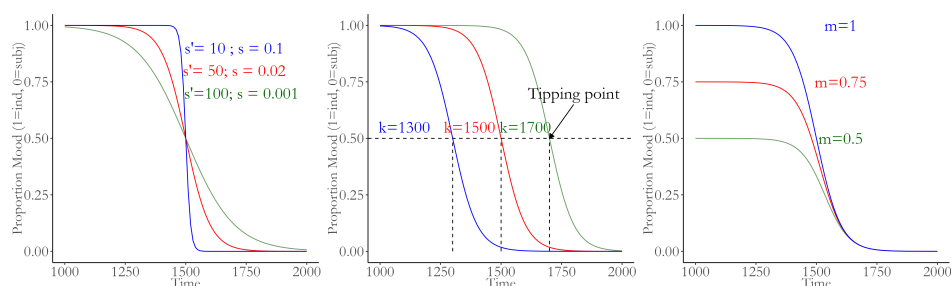


Figure 6.4: Image illustrating how the different parameters affect the shape of the logistic.

In order to calculate the speed of change for the verb *say* in the different constructions, a logistic was fitted to the data using the nonlinear least squares function (`nls`) in R. The parameters obtained appear in Table 6.1. As can be seen, the speed of change across constructions (s') are not statistically different, as their confidence intervals do overlap.

Table 6.1: Values obtained for the best fitting logistic with the confidence intervals.

Parameter	Model 1 (Construction FP)	Model 2 (Construction O)
k	1700 ± 96	1946 ± 102
s'	91 ± 70	164 ± 180
m	0.56 ± 0.12	0.83 ± 0.14

However, visual inspection of the model fit, shows that it might not be the best model for the data. This is because, even if the fit itself is relatively good, the confidence interval for Construction O is too broad (see light green band on the graph), indicating a problem with the model. Thus, a better model should be applied to the data.

As an alternative it was suggested that the change could also be fitted to a non-monotonic curve (such as the Hubbert curve). The following paragraphs show how this alternative hypothesis was tested.

As was mentioned previously, failed changes are usually modeled using the Hubbert curve (the first derivative of the logistic equation). For ease of mathematical computation in the sections to come (see Section 6.1.5), however, a simpler equation was chosen to model the failed change: the Laplace curve. This curve is obtained by combining two exponential equations, e^x and e^{-x} and taking the

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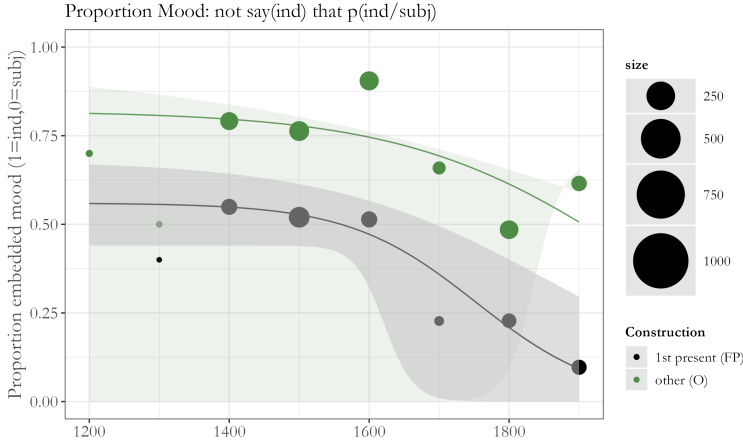


Figure 6.5: Data and the model fit for the verb *say*, using the logistic equation. The gray band represents the CI for Construction FP and the light green band the confidence interval for Construction O.

negative absolute value of the x variable: $e^{-|x|}$. The Laplace function (Figure 6.6 on the left) could be used as a close approximation to the Hubbert function (Figure 6.6 on the right) with the advantage that the equation is simpler, thus facilitating computation (this would be particularly important given that in the following section we will need to add multiple effects, see Section 6.1.5).

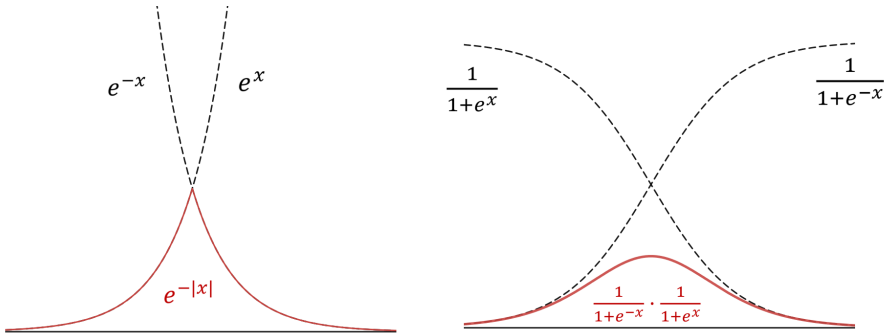


Figure 6.6: Comparison of the Laplace and Hubbert function.

In order to calculate the speed of change, the data was fitted to equation (4):

$$(4) \quad M(t) = m \cdot e^{-\frac{|t-k|}{s'}};$$

where $1/s'$ provides the slope of the curve (rate of change), k determines where the exponential function intersects on the x -axis (where the function reaches its maximum value), and m determines where the exponential functions intersect on the y -axis. A visual representation of how the different parameters affect the shape of the curve can be found in Figure 6.7.

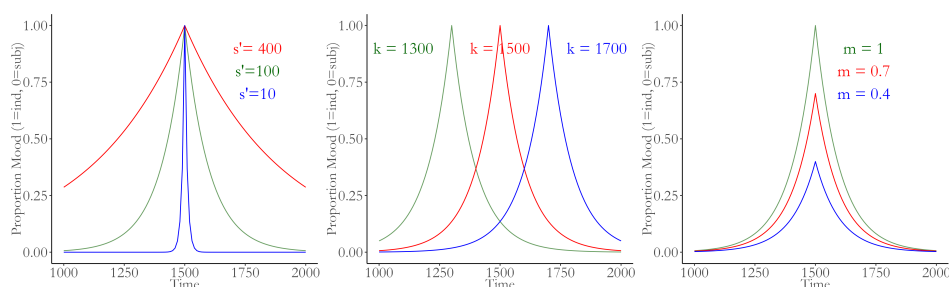


Figure 6.7: Laplace function and how different parameters affect the shape of the function.

The fitting was done with the nonlinear least squares function (`nls`) in R. The parameters obtained appear in Table 6.2. As can be observed, the speed of change across these constructions does not differ statistically from one construction to the other as the confidence intervals for the s' parameter overlap.

Table 6.2: Values obtained for the best fitting Laplace curve with the confidence intervals.

Parameter	Model 1 (Construction FP)	Model 2 (Construction O)
k	1512 ± 48	1533 ± 34
s'	325 ± 166	637 ± 232
m	0.60 ± 0.12	0.90 ± 0.08

Moreover, visual inspection of the model fit shows that this time the model is fitting the data well and the confidence intervals are behaving as expected. This suggests that this model might be a better alternative to explain the data (at least for Construction O). This is also a welcome outcome from a theoretical point of view, given that if the proposal is on the right track both the verb *believe* and the verb *say* should be behaving similarly as they are given a unified theoretical account.

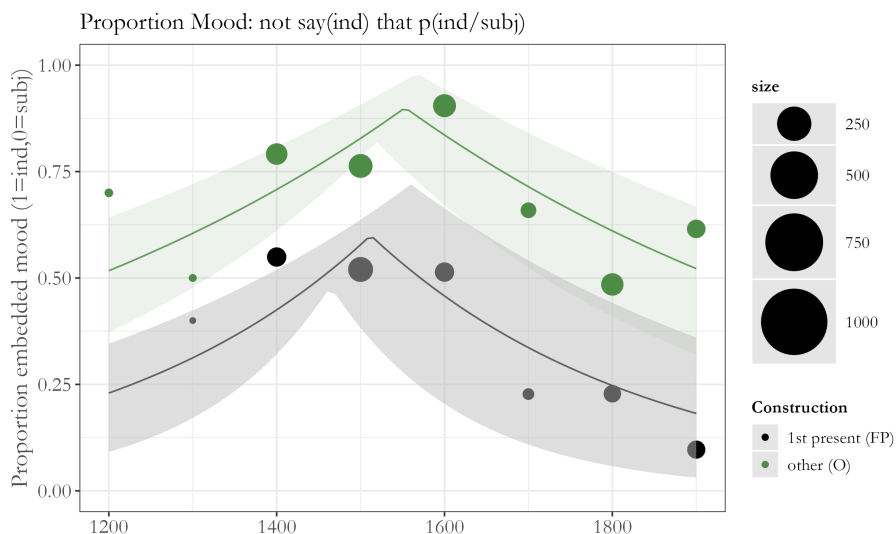


Figure 6.8: Data and the model fit for the verb *say*, using the Laplace function. Confidence intervals are marked with the colored band areas.

Comparing the AIC of the two models, one finds that although the model with the Laplace curve is better for Construction O – 645 (Laplace) vs. 654 (Logistic) – it is worse for Construction FP – 550 (Laplace) vs. 541 (Logistic) –. However, given that the Laplace curve was used as an approximation to the Hubbert curve, this does not entirely discard the possibility of analyzing the change in Construction FP, on par with Change O, as a failed change. Fitting the model again to the data but with the Hubbert curve³ shows that the Hubbert model is better again with Construction O (649 vs. 654), but importantly there is no difference between the Hubbert and the logistic in Construction FP (541 vs. 542). This shows that it is feasible to analyze both changes as an instance of a failed change by means of the same model (a non-monotonic curve). For reasons of comparability with the following sections (and ease of mathematical computation in the following section), the Laplace curve will be preserved, bearing in mind that it is used just as an approximation.

In conclusion, this section has shown that the speed of change across constructions are not different, and that the change in mood after *not say* can be better understood as a failed change, along the lines discussed in Chapter 5. The next

³See Appendix C.

section looks in detail at the behavior of the other non-factive predicate: the verb *believe*.

6.1.4 The verb *believe*

In the case of the verb *believe*, and following again Postma (2010)'s categorization, the change resembles an accidental failed change, as the data is clearly not an S-curve and the inherent failed change forces the peak of the curve to coincide with half of the inflection point of the initial S-curve, occurring too low for our given purposes. In order to approximate the speed of change, the Laplace curve was used. It was done with the nonlinear least squares function (nls) in R. The parameters obtained appear in Table 6.3.

Table 6.3: Parameters of the fitted models with the confidence intervals.

Parameters	Model 1 (Construction FP)	Model 2 (Construction O)
m	1.00 ± 0.08	0.68 ± 0.12
k	1542 ± 8	1551 ± 42
s'	142 ± 14	398 ± 154

These results show that although the time of greatest growth for construction FP and O seems to have been at a similar point in time (≈ 1550), their speed of change (the s' parameter) differ, with the speed of change for construction FP being twice as fast than for construction O. Moreover, their confidence intervals do not overlap, indicating that they are statistically different.

In order to see if the models were fitting the data well, they were plotted against the data, see Figure 6.9. Looking at the results, we observe that the models do appear to be a very good fit for the data, with the possible exceptions of the period comprising 1200 and 1300, but for these two periods the data set was below 15 tokens and so overall not too reliable.

These results are interesting in light of the Constant Rate Effect (Kroch 1989), which states that two changes caused by the trigger should have similar speed of change. There are three possible ways of interpreting these results: (i) that the two changes are caused by different underlying linguistic changes – thus suggesting that a different analysis to the one proposed in the previous chapter is needed –, (ii) that failed changes behave differently than successful changes and their speed can be different, or (iii) the two constructions were affected by the

6 Explaining Polarity Mood: The first person effect

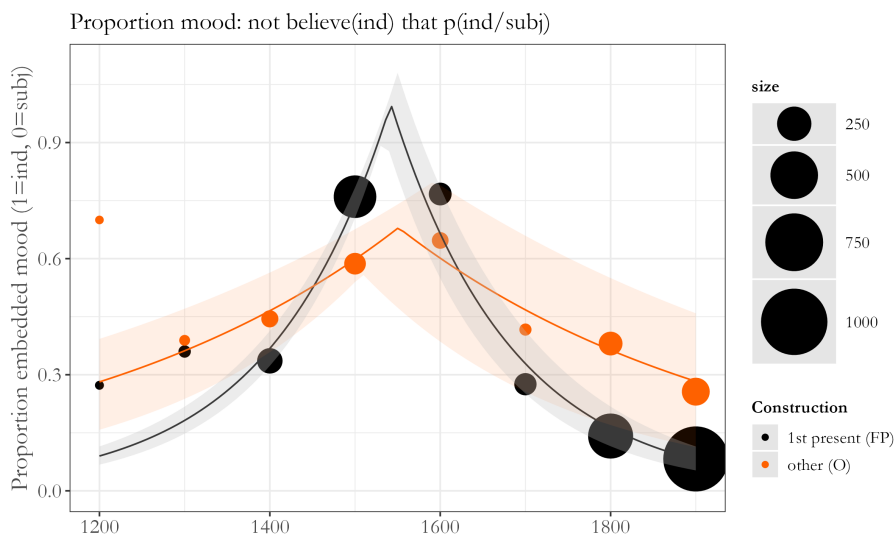


Figure 6.9: Data and model fit for the verb *believe*, using the Laplace function.

same underlying linguistic change, but in addition, there was the effect of another linguistic change which only affected Construction FP, creating the observed asymmetry.

The next section provides evidence for hypothesis (iii): first person constructions were affected by a change in complementizer drop which could have indirectly affected the proportions of mood. This is a welcome result insofar as it allows us to retain the benefits of the current analysis without a multiplication of different underlying linguistic changes affecting mood. Moreover, the Constant Rate Effect need not be constrained to a particular kind of change (e.g., successful changes) but would be a characteristic of language change more generally.

6.1.5 Cumulative effects

6.1.5.1 A parallel change: complementizer drop

The literature has reported that at a similar point in time as the observed change in mood after *not believe* there was a change in the type of complementizer that the language could use (Mazzola et al. 2022). In present-day Spanish, the complementizer *que* ‘that’ is usually obligatory after the predicate *no creer* ‘not believe’

if the embedded verb is in the indicative, see (5). This was not the case during Medieval Spanish, where one finds multiple examples of omitted complementizers as in (6), even when the embedded verb is in the indicative mood:⁴

- (5) Yo no creía *(que) estaba enfermo.
 I not believed that was.IND sick
 ‘I didn’t believe (that) they/s/he was sick.’
- (6) ... y ahora no creo está allí.
 ... and now not believe be.IND there
 ‘and now I don’t believe it is here.’
 (1577, *Epistolario* by Santa Teresa de Jesús)

Most quantitative analyses of the phenomenon have not focused on the verb *believe* and they have not looked in detail at the effect of negation. Thus, in order to understand better this development after *not believe*, a search was conducted using the CDH corpus (Real Academia Española 2013). All instances of the verb *believe* negated and followed by either a verb or a complementizer immediately followed by a verb were found, as in (7):

- (7) not believe (that) V

In order to keep comparability across studies, the methodology used in the previous chapter was followed, see Section 5.1.2 in Chapter 5. The sentences were manually annotated with (i) meta-linguistic information, (ii) information on the matrix clause, and (iii) information on the embedded clause.

Starting first with the metalinguistic information, the only categories that were annotated were the year (between 1200–1999) and period (the data was divided into 8 periods of 100 years each). As in the previous study, if a text ranged between two periods, the mean year was used to determine the period.

With regards to matrix clauses (ii), they were annotated according to 5 criteria: clause type, structure, verb finiteness, verb mood and construction. The data was divided, parallel to the previous study, into interrogative and non-interrogative clauses. Then, the data was grouped according to their structure into three groups: *no creer que V* (‘not believe that Verb’), *no creer V* (‘not believe

⁴There are also cases of omitted complementizers if the embedded verb is in the subjunctive. This complementizer deletion mechanism seems to be a different one from the one observed in the indicative clauses. The reason we argue this is that in present-day Spanish the complementizer can be sometimes omitted with the condition that the embedded verb is in the subjunctive (Etxepare 1996).

V') and *Other*. This last category was created to include data that did not follow the other two structures. For example, in (8) and (9) the words *a* and *nada* appeared tagged in the corpus as the verbs *have* and *swim*, respectively. However, these words do not function as verbs in the examples in (8) and (9): *a* here is a differential object marking, and *nada* is a n(egative)-word meaning *anything*.

- (8) Quien no cree a buena madre...
 who not believe to good mother
 'Whoever does not believe in a good mother..
 (1492, *Cancionero de Pero Guillén*, Anonymous)
- (9) No cree nada de usted.
 not believe nothing of you
 'He doesn't believe anything from you.'
 (1888, *Miau*, Benito Pérez Galdós)

The data was then annotated according to whether the matrix verb was finite or not, and if it was whether it had the indicative or the subjunctive morphology. Finally, as in the previous study the sentences were divided depending on whether they had a first person matrix subject and the verb was in present tense (the equivalent to construction FP in the study of mood in the previous chapter) or not (the equivalent to Construction O), see Table 5.2 here repeated as Table 6.4.

Table 6.4: Annotation of matrix clauses according to Construction: 1st present vs. other.

Tense/Person	1sg	2sg	3sg	1pl	2pl	3pl
Present	1st present	other	other	1st present	other	other
Past	other	other	other	other	other	other
Future	other	other	other	other	other	other
Conditional	other	other	other	other	other	other

With regards to embedded clauses (iii), they were annotated according to two criteria: the mood of the embedded verb which was assigned *indicative*, *subjunctive* or *NA* (if the verb was a non-finite verb), and the type of embedded complementizer (null or non-null). In this study given that the verb *believe* does not take interrogative embedded clauses, there was no need to annotate whether the embedded clause was ambiguous or not as all were unambiguous sentences, see

Section 5.1.2 in Chapter 5 for a discussion of why this was problematic with the other predicates.

After the annotation was completed, the relevant data were extracted. Only clauses which had the desired structure *no creer que V/no creer V* were included. Moreover, only declarative clauses with a matrix indicative verb were included. This was done to maximize comparability across this study and the study on mood which also focused on these structures. Finally, only those embedded clauses which had a finite verb were included. This was done because in clauses with a non-finite embedded verb, see (10), the absence of a complementizer is not indicative of complementizer-drop but rather a sign that the clause is a control structure. Evidence of this comes from the fact that the use of an overt complementizer is ungrammatical, see (10).

- (10) *No creo que estar cansado.
not believe that be.INF tired
'I don't think I'm tired.'

A schematic representation of how the data were extracted can be found in Figure 6.10:

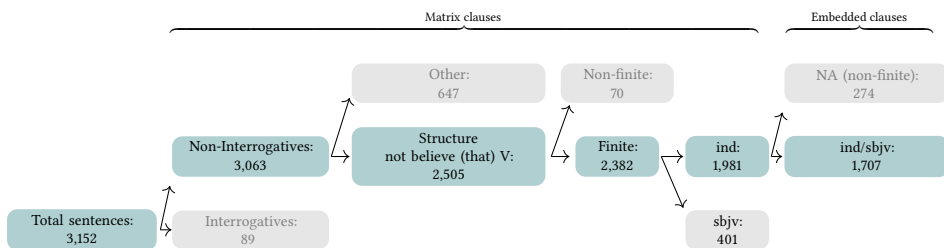


Figure 6.10: Data Extraction Process. Numbers indicate the number of tokens per category.

The final results are displayed in Figure 6.11. In the left panel is the behavior of null complementizers when the embedded verb is the indicative and on the right when the embedded verb is in the subjunctive. The y-axis represents the proportion of null complementizers: 1 indicates that in all instances the complementizer was dropped, 0 that it was never dropped. As before, the x-axis represents the period and the size of the dot the amount of data (the exact numbers can be found in the annexed table).

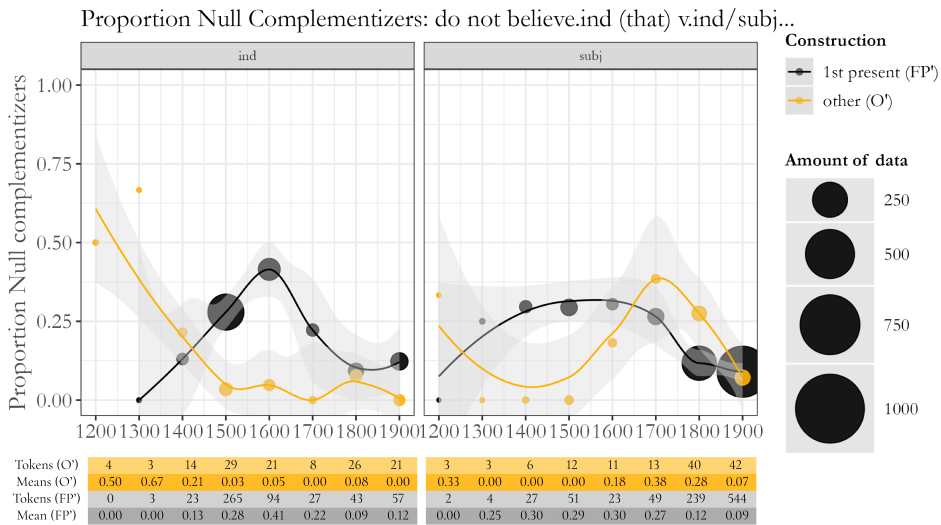


Figure 6.11: Proportion of null (1) vs. overt (0) complementizers comparing first person present with non-first person present. Smoothing was done with LOESS.

As can be seen when the embedded verb is in the indicative in Construction FP' (first person with a present tense) the proportion of null complementizers follows a very similar trajectory as the change in mood observed in the previous section: there is an increase during the early stages with a peak in the 17th century, a period after which there is a decrease. Importantly, in all other constructions this behavior is not encountered. In Construction O' with an indicative embedded the verb, the proportions of null complementizers decreased during 1300 and remained stable ever since with a very low frequency of complementizer drop. Construction O' with an embedded subjunctive verb shows an up-and-then-down tendency, but this change happened later with the peak in 1700. Finally, Construction FP' with an embedded subjunctive verb was stable at 30% and decreased during 1800. That is, the only construction that follows a similar pattern to mood selection is Construction FP' with an embedded indicative verb. Furthermore, although the number of tokens is not very high, the confidence bands of the FP' and O' construction with an indicative embedded verb during the relevant period (i.e., 1500–1600) do not overlap, indicating that the change observed for first-person constructions is statistically different from the one with non-first person constructions, despite the lower numbers for Construction O'.

The different trajectories of complementizer drop depending on the mood of the embedded verb suggests that there might be two independent mechanisms for complementizer drop: one strategy that is used when the embedded verb is in the indicative and another when the verb is in the subjunctive. This point of view has also been defended by Etxepare (1996) on independent grounds.⁵ This work shows that these two types of complementizer drop behave differently in the present-day language as well. For instance, complementizer drop in present-day Spanish is only possible with indicative embedded predicates if there is movement involved, see (11), but is optional if the verb is in the subjunctive, irrespective of whether there is movement involved, see (12).

- (11) **Los paquetes** que Juan cree Correos envió **los paquetes**
 The parcels that John believes postal.service sent the parcels
 ayer.
 yesterday
 ‘The parcels that Juan believes the Postal Service sent yesterday.’
- (12) **Les rogamos disculpen** las molestias.
 CLT beg forgive.PRS.SBJV the inconveniences
 ‘We apologize for the inconveniences.’

Leaving aside an explanation of complementizer-drop in the subjunctive embedded clauses, this section explores the possibility that C-drop in clauses with an indicative embedded verb might indicate that the complement clause is more independent in nature. As proposed in Torrego & Uriagereka (1992) there might be two types of complementation in Spanish, one in which there is an overt null object and the complement clause is an independent clause, see (13), and another in which there is no null pronoun and the complement clause is the only complement, see (14). Additionally verbs may take a full DP with a relative clause as complement (15), but this is a different strategy than the one proposed in (13) by Torrego & Uriagereka (1992).

- (13) Not V [DP-Complement] [Independent Clause]
- (14) Not V [Complement Clause]
- (15) Not V [DP [Relative Clause]]

⁵The null complementizer that we see in indicatives must be kept separate from the null complementizer we see in subjunctives and complements of some factive verbs.” (Etxepare 1996: 472)

Evidence that (13) might have been available, at least in Medieval Spanish, is provided by the existence of clauses such as (16), where the pronoun is actually overtly pronounced.⁶

- (16) No *lo* creas [que lo diré yo].
 not it believe that it say I
 ‘Don’t you believe I will say it’
 (1534, *Segunda Celestina* by Feliciano de Silva, CDH)

The idea is that in cases when strategy (13) is used, the sentence can optionally drop the complementizer, similar to other independent clauses in Spanish which can drop the complementizer, see (17) (Fernández 2000, Gras & Sol Sansiñena 2015, Kocher 2023).⁷ That is, we interpret the increase in null complementizers in sentences with an indicative verb as a sign that the structure (13) was becoming more frequent.

- (17) (Que) Juan se ha comprado una casa.
 that John CLT has bought a house
 ‘John has bought a house.’

The question that arises is the following: did this change in structure have any impact on mood selection? Torrego & Uriagereka (1992) argued that there is an impact: the indicative clauses are cases of strategy (13) and the subjunctive clauses of strategy (14). I’ll depart from these authors in this respect and rather claim that whereas strategy (13) only allows for the indicative mood, the second strategy (14) allows for both the indicative and the subjunctive mood.⁸

⁶Sentence (16) is analyzed as strategy (13) and not as a relative clause, because the pronoun *lo* precedes the verb and it is separated from the rest of the clause. There is no analysis/phenomena, that I am aware of, that allows the N or D heads to move without their relative clause in Spanish.

- (i) a. *Le lo dije que había oído.
 him it said that had heard
 b. Le dije lo que había oído.
 him said it that had heard
 ‘I told him what I had heard.’

⁷There is no consensus in the literature regarding the exact meaning conveyed by the complementizer in these cases. Some have argued it is an expletive, some that it is a marker of background information, others that it is part of a quotative, etc. See Gras & Sol Sansiñena (2015) and Kocher (2023) for a summary of different proposals.

⁸In principle the analysis would also hold if one argues that the competition is between strategy (14) and (15).

(18) Not V [DP- Complement] [Independent Clause] → Indicative

(19) Not V [Complement Clause] → Indicative/Subjunctive

The reason I argue this is that it follows naturally from the competing grammars account proposed. In the case of (18) semantic selection is allowed given that the complement clause is an adjunct, and it is well known that semantic selection can occur in relative clauses (i.e., adjunct clauses), see (20), and in other independent clauses; however, case assignment is subject to locality and given that it has to assign case to the null DP, it would not reach the adjunct/independent clause. Thus, G_1 (semantic mood) can be licensed in both (18) and (19), and in this case would require the indicative mood. G_2 (syntactic mood) is only licensed in (19), and in this case would require the subjunctive mood. The prediction is that an increase in the first strategy would result in an increase of the indicative mood.

(20) Busco a una persona que es/sea simpática.
 Search to a person that be.PRS.IND/PRS.SBJV nice
 ‘I search for a person (any person) that is nice.’

This section thus puts forth the idea that the different speed in mood selection between constructions FP and O reported in Section 6.1.4 might be a result of this extra parallel change in complementizer deletion affecting only first person constructions with an indicative verb. The idea is illustrated in Figure 6.12.

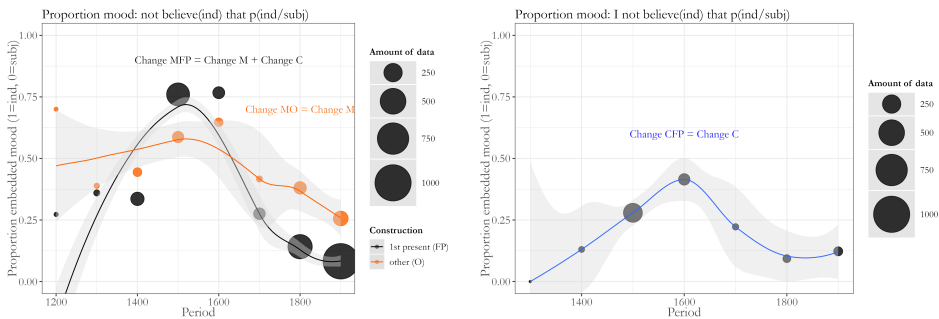


Figure 6.12: Addition of Multiple Changes. Change MFP = change in mood in first person present tense constructions; Change MO = change in mood in all other constructions (non-first person and present tense); CFP = change in complementizer drop in first person constructions. Change M = change in mood selection (semantic vs. syntactic); Change C = Change in complementizer which is interpreted as a change in the merging position of the CP.

The proposal is that superficially there are three changes each with a different speed: a change in mood in first person and present tense constructions (Change MFP), a change in mood in all other constructions that are not present and first person (Change MO) and a similar change at a similar time in complementizer drop in first person and present tense constructions with an indicative verb (Change CFP), see Figure 6.11. However, underlyingly there are only two changes: a change in the proportion of mood, which can be interpreted as a change between syntactic and semantic mood selection (Change M), and a change in the proportion of C-drop, which it was argued may reflect that the language started taking more independent clauses, and which indirectly increased the amount of the indicative mood (Change C). That is, Change MFP is the result of the addition of two underlying changes: Change C + Change M.

The next section explores whether this proposal is numerically sound.

6.1.5.2 Adding the effects

The idea we aim to prove is that the speed of change in mood in first person constructions (Change MFP) is similar to that obtained by combining the speed in mood for Construction O (Change MO), and the speed of complementizer drop in first person constructions (Change CFP). In order to do that, first we need to understand how multiple effects could be encoded in our equations.

In dynamical systems, for example models aiming to calculate the rate of change of a given population $N(t)$, the r parameter (the equivalent of the s we have been using thus far) can be affected by one or several factors. For example, the rate of growth of a given population (r) could be solely determined by the birth rate of that population ($r = b$), see (21), but it could also be affected by multiple factors (e.g., $r = \text{birth rate}(b) - \text{death rate}(d) + \text{migrations}(m)...$) (Vandermeer & Goldberg 2013). The important idea here is that when multiple factors affect the rate of change they appear in exponential models in the exponent as additive factors, see (22) (Vandermeer & Goldberg 2013).

$$(21) \quad N(t) = N_0 \cdot e^{r \cdot t} = N_0 \cdot e^{b \cdot t}$$

$$(22) \quad N(t) = N_0 \cdot e^{b \cdot t - d \cdot t + m \cdot t} = N_0 \cdot e^{(b-d+m) \cdot t}$$

Moreover, given that the Laplace function is the result of taking the negative absolute value of the x variable in the exponential, we obtain the formula in (23) to calculate the behavior of a given population when several factors affect the rate of change of the population.

$$(23) \quad N(t) = N_0 \cdot e^{-(r_1|t-t_0|+r_2|t-t_0|)}$$

Given that for the model fit we use $r = 1/s'$ (so that the speed of change is given in years and hence is more intuitive to understand), N_0 was represented as m , and t_0 as k , we obtain the general equation in (24).

$$(24) \quad M(t) = m \cdot e^{-\left(\frac{|t-k|}{s'_1} + \frac{|t-k|}{s'_2}\right)}$$

To test if the proposal is correct, we need to show that condition (25) holds.

$$(25) \quad e^{-\left(\frac{|t-k_{MFP}|}{s'_{MFP}}\right)} \approx e^{-\left(\frac{|t-k_{MO}|}{s'_{MO}} + \frac{|t-k_{CFP}|}{s'_{CFP}}\right)}$$

In order to show this, we first need to calculate the resulting rate of change of the right hand side of the equation. Making the assumption that $|t - k_{MO}| \approx |t - k_{CFP}|$,⁹ we obtain equation (26).

$$(26) \quad e^{-\left(\frac{1}{s'_{MO}} + \frac{1}{s'_{CFP}}\right)|t-k_{MC}|};$$

where k_{MC} is the mean between k_{MO} and k_{CFP} . By the general rules of rational functions we obtain (27):

$$(27) \quad e^{-\left(\frac{s'_{CFP}+s'_{MO}}{s'_{CFP}s'_{MO}}\right)|t-k_{MC}|}.$$

Then we just have to calculate the values of s'_{MO} , s'_{CFP} and k_{MC} . The first value has already been calculated, see Table 6.3 above. In order to calculate the values for the change in complementizers (Change CFP), a Laplace function was fitted to the data using the nonlinear least squares function in R. The values obtained for this as well as the other two changes appear in Table 6.5 and are plotted in Figure 6.13.

Table 6.5: Parameters of the fitted models with the confidence intervals.

Parameters	Model 1 (Change MFP)	Model 2 (ChangeMO)	Model 3 (Change CFP)
m	1.00 ± 0.08	0.68 ± 0.12	0.45 ± 0.11
k	1542 ± 8	1551 ± 42	1584 ± 28
s'	142 ± 14	398 ± 154	174 ± 98

⁹The confidence intervals for k overlap for Change MO and CFP in the nls regression.

6 Explaining Polarity Mood: The first person effect

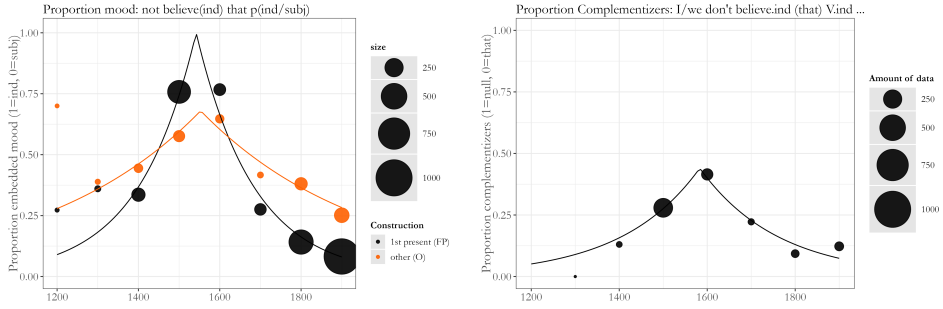


Figure 6.13: Fitting of the Laplace function to the change in mood and complementizer-drop.

Using these values in equation (27) ($s'_{MO} = 398$; $s'_{CFP} = 174$ and $k_{MC} = 1568$) the obtained result is that the predicted speed of change s' if a change had been affected by two cumulative changes is 121, see (28).

$$(28) \quad e^{-\left(\frac{174+398}{174 \cdot 398}\right)|t-1568|} = e^{-(0.0083)|t-1568|} = e^{-\frac{1}{121}|t-1568|}.$$

This is very close to the one obtained independently for Change MFP based on the corpus data: 142; and if one takes into account the error in the measurements, both the s' and k values overlap: $k_{MFP} = 1542 \pm 8$; $k_{addition} = 1568 \pm 35$; $s'_{MFP} = 142 \pm 14$; $s'_{addition} = 121 \pm 62$; This shows that the initial condition (29) holds. This is, perhaps, more intuitively shown by plotting the two resulting curves next to each other, see Figure 6.14.

$$(29) \quad e^{-\left(\frac{|t-k_{MFP}|}{s'_{MFP}}\right)} \approx e^{-\left(\frac{|t-k_{MO}|}{s'_{MO}} + \frac{|t-k_{CFP}|}{s'_{CFP}}\right)} = e^{-\left(\frac{|t-1542 \pm 8|}{142 \pm 14}\right)} \approx e^{-\frac{|t-1568 \pm 35|}{121 \pm 62}}$$

In conclusion, the rates of change are very close, and the hypothesis that the different speed across construction FP and O can be explained by means of a further linguistic change affecting Construction FP seems to be supported by the empirical data.

6.1.6 Summary of results

So far it has been shown that the speed of change across Construction FP and O are the same in the case of the verb *say* but different in the case of the verb *believe*. It was then proven that the different speed with the verb *believe* might be a result of another linguistic change indirectly affecting the proportions of mood: the location where the clause is merged in the structure. Given that semantic selection

6.2 The effect of first person with non-factive verbs

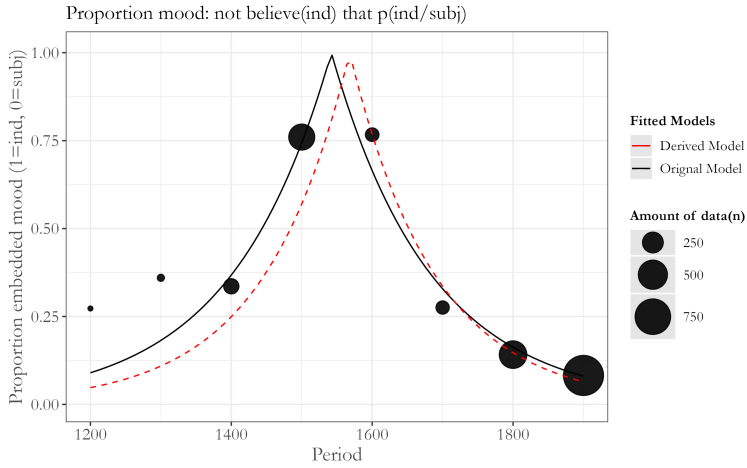


Figure 6.14: Cumulative effects. Comparison of adding the effect of Change MO and Change CFP vs. the effect of Change MFP.

but not syntactic mood selection can be long distance, in the case that the clause is adjoined as a parenthetical (or as a relative clause), syntactic selection will not be able to reach the clause and the only available option will be the indicative. Thus, an increase in the amount of independent like complement clauses would have as a result an increase in the amount of the indicative mood. Moreover, it has been proven that this proposed account is numerically sound, and predicts the correct behavior.

These results thus provide support for the account proposed in the previous section whereby the loss of the indicative in Construction FP and O with non-factive verbs is given a unified account: grammar G_2 wins over the other competing grammars.

The first person effect with semi-factive verbs is explained then as the emergence of speaker commitment (which leads to a contradiction with semi-factive verbs in FP contexts but not elsewhere). However, the proposal has left unexplained what causes the first person effect in non-factive verbs. That is, what explains the 25% difference between FP and O constructions with a non-factive verb? The next section aims to answer this question.

6.2 The effect of first person with non-factive verbs

In Old Spanish the difference in the frequencies of the indicative across constructions was already present in the language, even with non-factive verbs. In Fig-

ure 6.15 the results for non-factive verbs are displayed again. In the plot on the left is the raw data, and in the plot on the right appear the differences across constructions predicted by the GAM model calculated in Chapter 5. As can be seen, already by 1300 and 1400 there was a difference of about 25% more of indicative between Construction O and FP and this difference was statistically significant (as indicated by the red area in the plot on the right). The difference was reduced during 1500 and by 1600 it became non-significant, but then it changed back to its original 25% difference. This dip in 1600, I argued in the previous section, may have been due to an additional linguistic change – the merge position of the CP in the structure, which indirectly affected the proportions of mood – only affecting FP Constructions (first person and present tense). The question that still remains open is: what caused the initial 25% difference across constructions in the earlier stages of the language?

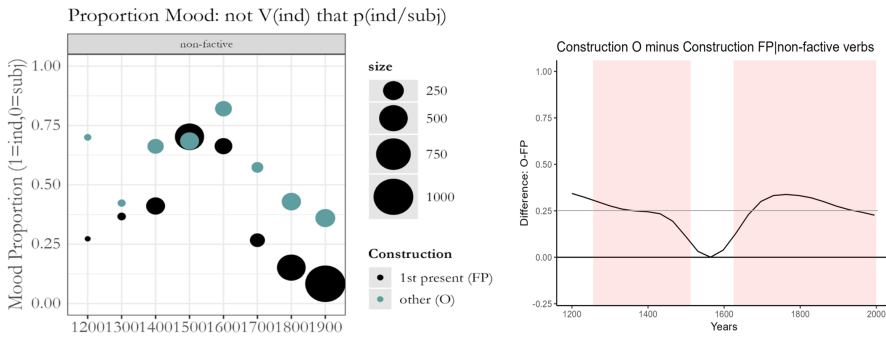


Figure 6.15: Summary data and GAM model of non-factive verbs.

In order to explore this question, we first need to propose a grammar which allows the indicative and the subjunctive mood to occur after negated non-factive predicates. Grammar G_2 (syntactic mood as a grammaticalized case marker) predicts the subjunctive in these contexts as it becomes similar to a marker of negation. Therefore, a grammar is missing for the initial stage in which both the indicative and the subjunctive could be used in these environments. Note that one cannot fully explain the initial stage as merely a combination of G_1 and G_2 because in the case of semi-factive verbs with a first person construction both grammars would predict the subjunctive mood, but there is mood variation even in this environment, see (30). Thus, a grammar that explains mood variation at the initial state with both non-factive and semi-factive negated verbs would be needed.

- (30) a. No sé que las sabéis vos → G??
 not know that them know.IND you
 ‘I don’t know if you know them.’
 (*Letras*, 1470–1485, Hernando del Pulgar, CDH)
- b. Yo no se que yo sia otra cosa sino vn animal. → G₁ and G₂.
 I not know that I be.SBJV other thing but an animal
 ‘I don’t know if I’m anything else but an animal.’
 (*Libro del Tesoro*, 1400–1425, CDH)

Drawing inspiration from the behavior of mood in Bulgarian, this section explores the possibility that this initial grammar, which will be referred to as G₀, is one in which the indicative indicates strong commitment and the subjunctive weak commitment. That is, the subjunctive is used as a sort of hedging device. This would explain the differences across constructions, as there is experimental evidence showing that hedging devices are more frequent with first person and present tense constructions (Koev et al. 2021). The first person effect observed in the present-day language with non-factive verbs is, I will argue, an artifact of the initial difference created by grammar G₀.

Section 6.2.1 shows how the proposed grammar predicts a small difference between Construction O and FP across verb types, and Section 6.2.2 discusses how this additional grammar can fit in the proposed analysis of the previous chapters, exploring which predictions the three-grammar interaction makes. Section 6.2.3 investigates these predictions more directly by showing the results of several simulations. The chapter then concludes with a summary of the key ideas of the chapter.

6.2.1 Explaining first person effect with non-factives

As was proposed in Chapter 4, the Old Spanish mood system resembled in several aspects the mood in Bulgarian. In that chapter it was explored the idea of applying Smirnova (2011)’s analysis – in which mood is used to express the degree of commitment – to Old Spanish. The key idea this section explores is that the ≈ 25% difference with negated predicates could be a result of this original use of the subjunctive as a hedging device. Koev et al. (2021) argue that hedging devices are expected to be more natural with a first person and present tense than other persons and tenses because they are more closely tied to the utterance context. The authors ran an experiment to test this prediction using a 2×2×2 factorial design, with person (first vs. third), tense (present vs. past) and position (main vs. subordinated) as the influencing factors. Participants had to rate on a

scale from 0 to 100 how acceptable they considered each sentence (see (31) and (32) for examples of target sentences).

(31) Main clause: I believe it is raining, but I am not sure

(32) Embedded clause: Suppose I believe it's raining, but I am not sure

Important for our purposes are the results for main clauses, which can be seen in Figure 6.16. The first bar shows the results of first person present tense constructions, which are the equivalent of what has been called Construction FP in this book. The next three bars show the results for non-first person present tense sentences and third and first person past tense sentences. These are a subset of the sentences which in this book I have called Construction O (note that they are not exactly the same environments but rather a subset, as what has been called Construction O also included second person and other tenses such as future and conditional). In accordance with the prediction, first person present tense constructions were rated more acceptable than non-first person present tense constructions. These results lend support to the claim that hedging constructions tend to be more acceptable with a first person subject and a matrix present tense.

If the subjunctive in Old Spanish worked as a hedging device, as argued in the previously, then one would have a natural explanation as to why first person constructions took significantly more subjunctive. Hedging sounds more natural in this context and hence it is expected to be more frequent.

In conclusion, this section has explored the possibility that the difference in mood frequencies between Construction FP and Construction O which was present since as early as 1300 could have been due to the subjunctive conveying weak commitment at the earlier stages of the language (G_0). This initial difference was increased with semi-factive verbs as a result of another competing grammar, G_1 , winning out, which made sentences with the indicative ungrammatical altogether in Construction FP. With non-factive verbs the difference was reduced at around the 1500s due to an additional change (the place of merging of complement clauses) which affected the proportions and the speed of change across constructions. Once this external change stopped having an impact, the system returned back to its initial state. The difference across constructions with non-factive verbs need not be explained then as a result of non-factive verbs leading to a semantic contradiction, but could be better understood as an artifact of previous stages of the language where the difference between the indicative and the subjunctive indicated strength of commitment. The next section extends the two competing grammars proposal from the previous chapter to also include this third grammar.

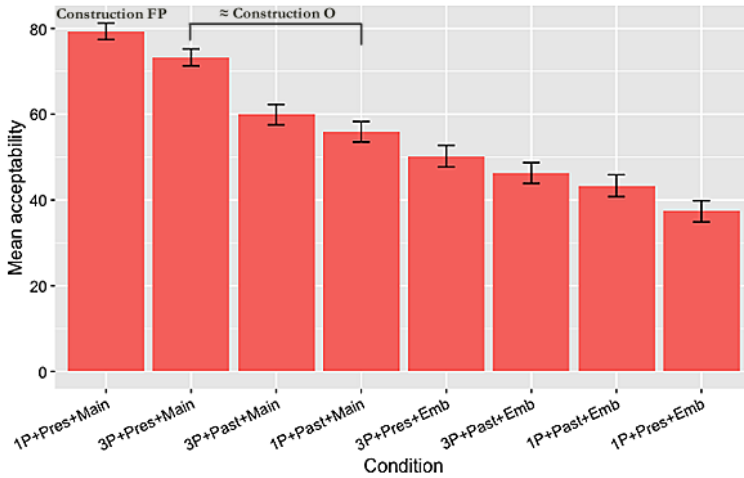


Figure 6.16: Experimental results from Koev et al. (2021) on the acceptability of hedging in different contexts. The image was modified to include the labels *Construction FP* and $\approx$ *Construction O* to indicate that the environment corresponded to what in this book has been called Construction FP (first person and present tense), and Construction O (all other contexts). The $\approx$ symbol indicates that the authors did not investigate second person constructions so it is not exactly the same environment.

6.2.2 Three-grammar competition

So far this chapter has argued that the two competing grammars account, although capable of explaining some interesting aspects of the phenomenon, could not explain the differences in mood proportion across constructions in the early stages of the languages. In particular, the account could not straightforwardly derive the use of both the indicative and the subjunctive after semi-factive verbs in Construction FP, and could not derive the $\approx 25\%$ mood proportion difference across constructions (irrespective of verb type). It was then proposed that another grammar G_0 , which was similar to the system in present-day Bulgarian, might have been in place at earlier stages. In this grammar the subjunctive, rather than lacking meaning, could convey weak epistemic commitment. This section explores to what extent the addition of this initial grammar makes the right predictions regarding which grammar will win out in diachrony.

Starting first with semi-factive verbs, G_0 will be able to parse a sentence whenever the indicative is used with strong commitment and the subjunctive with weak commitment. G_1 will be able to parse sentences with a non-first person subject (Construction O) and an indicative verb (indicating that the speaker pre-

supposes p) and with a subjunctive verb (marking lack of speaker presupposition). With a first person construction, as it was argued in Chapter 5, the only possibility would be to take the subjunctive. Finally, G_2 will be able to parse a sentence whenever the verb appears in the subjunctive (irrespective of commitment or construction). This is summarized in Table 6.6.

Table 6.6: Sentence type each grammar will be able to parse in negative sentences.

Semi-factives					
Construction	Commitment	Mood	G_0	G_1	G_2
FP	Strong	Ind	✓	x	x
FP	Weak	Ind	x	x	x
O	Strong	Ind	✓	✓	x
O	Weak	Ind	x	✓	x
FP	Strong	Sbjv	x	✓	✓
FP	Weak	Sbjv	✓	✓	✓
O	Strong	Sbjv	x	✓	✓
O	Weak	Sbjv	✓	✓	✓

In this case, G_2 is a subset of G_1 and as was discussed in Chapter 4 the prediction then is that grammar G_1 will win over G_2 . By introducing the competition with G_0 the pattern becomes more complicated as this grammar is neither a subset nor is it disjoint to either of the two other grammars; see Figure 6.17 for a graphical representation, where the set of sentences each grammar can parse is represented by a square and the advantages of each grammar are marked with an a . Following Kauhanen (2019)’s notation a_{ji} will stand for the advantage of grammar G_i over G_j . For instance, a_{12} represents the advantage of grammar G_2 over G_1 .

In order to better understand the system, with these advantages we can then create an advantage matrix A , as in (33) (Kauhanen 2019).

$$(33) \quad A = \begin{bmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ a_{20} & a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 0 & a_{01} & a_{02} \\ a_{10} & 0 & 0 \\ a_{20} & a_{21} & 0 \end{bmatrix}$$

In this matrix, the diagonal will be equal to zero as no grammar can have an advantage over itself ($a_{ii} = 0$). Moreover, given that G_2 is a subset of G_1 , a_{12} will

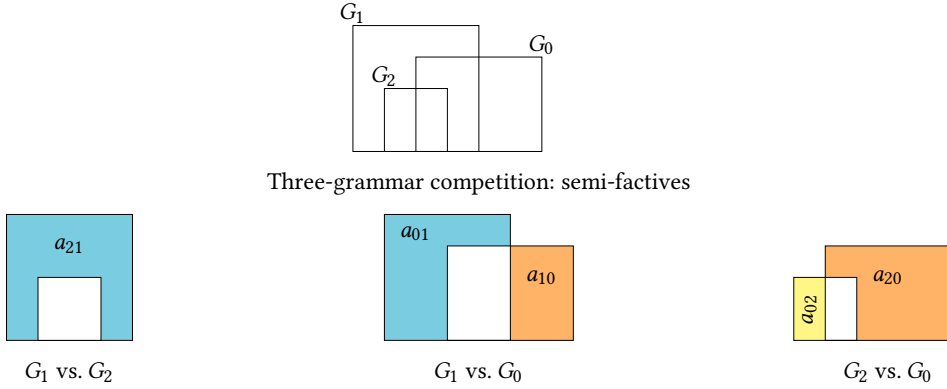


Figure 6.17: Graphical representation of the sentences each grammar can parse only taking into account contexts with a negated semi-factive verb.

also be equal to 0. This entails that G_1 has overall an advantage over G_2 as $a_{21} > 0$. The values of the other parameters are harder to determine, as one would have to calculate them based on corpus data from other languages (e.g., Bulgarian to obtain the value of G_0). One can, however, impose certain conditions on these parameters which, if satisfied, will result in the observed diachronic patterns for semi-factive verbs. Specifically, the argument put forth thus far is that G_1 has won over the other variants when used with a semi-factive verb; that is, G_1 is globally asymptotically dominant (GAD), see (34) (Kauhanen 2024).

- (34) Globally Asymptotically Dominant: A grammar G_k is defined to be globally asymptotically dominant if the state in which only G_k is used (its probability is 1) is globally asymptotically stable (Kauhanen 2024).

One case scenario in which a grammar would be GAD would be if condition (35) is satisfied. Notice, however, that this is a sufficient but not a necessary condition. If the system satisfies (35), then G_k will be GAD; but if it doesn't satisfy (35), the system may or may not be GAD (Kauhanen 2024).

- (35) GAD Sufficient Condition: G_k will be globally asymptotically dominant if:

$$\sum_{j \neq k} a_{kj} < \frac{1}{\sum_{j \neq k} \frac{1}{a_{jk}}}$$

(Kauhanen 2024)

In our given case, this means that in all those circumstances in which (36) is satisfied, the system will behave as expected, and grammar G_1 will win the other variants (see the next section for the simulations illustrating that this is the case).

$$(36) \quad a_{12} + a_{10} < \frac{1}{\frac{1}{a_{21}} + \frac{1}{a_{01}}}$$

Although this does not prove that the proposed three-grammar competition analysis presented in this chapter is correct, it does show that from a theoretical point of view the proposal is at least feasible. Moreover, it allows us to have testable predictions (we now know the range of values in which the three-grammar proposal would be correct).

Looking now at the set of sentences each grammar can parse when the verb is non-factive, we have that G_0 (hedging mood) will be able to parse a sentence whenever the indicative is used in a context which conveys strong commitment and the subjunctive weak commitment (this is so irrespective of the construction), see the fourth column in Table 6.7. G_1 (referential mood) will be able to parse all sentences with an indicative embedded verb, irrespective of construction and context, see the fifth column in Table 6.7. Finally, G_2 (grammaticalized mood) will be able to parse all sentences with a subjunctive verb in the embedded clause, see the sixth column in Table 6.7.

Table 6.7: Sentence type each grammar will be able to parse in negative declarative sentences.

Non-factives					
Construction	Commitment	Mood	G_0	G_1	G_2
FP	Strong	Ind	✓	✓	x
FP	Weak	Ind	x	✓	x
O	Strong	Ind	✓	✓	x
O	Weak	Ind	x	✓	x
FP	Strong	Sbjv	x	x	✓
FP	Weak	Sbjv	✓	x	✓
O	Strong	Sbjv	x	x	✓
O	Weak	Sbjv	✓	x	✓

Grammars G_1 and G_2 stand in a disjoint relationship, and hence the prediction is that the competition between these two grammars will be stable. When looking at G_0 , one observes that this grammar can parse some of the sentences the other two grammars could produce (whenever the indicative is used in a strong commitment context and the subjunctive in a weak commitment one), but there are also some sentences that this grammar will not be able to parse (namely, whenever the subjunctive is used to convey strong commitment or the indicative is used to convey weak commitment). The information is visually represented in Figure 6.18. As before, the set of sentences each grammar can parse is represented by a square and the notation a_{ji} will stand for the advantage of grammar G_i over G_j . We can then create an advantage matrix A' , which will enable us to understand the system better, see (37). In this case study, a'_{ii} is equal to 0 as no grammar has an advantage over itself. Moreover, given that G_1 and G_2 stand in a disjoint relationship a'_{21} will have the same value as a'_{12} . That is, both of them will be given an advantage of 1. The rest of the values cannot be straightforwardly calculated and hence the rest of the discussion will need to be more speculative in nature.

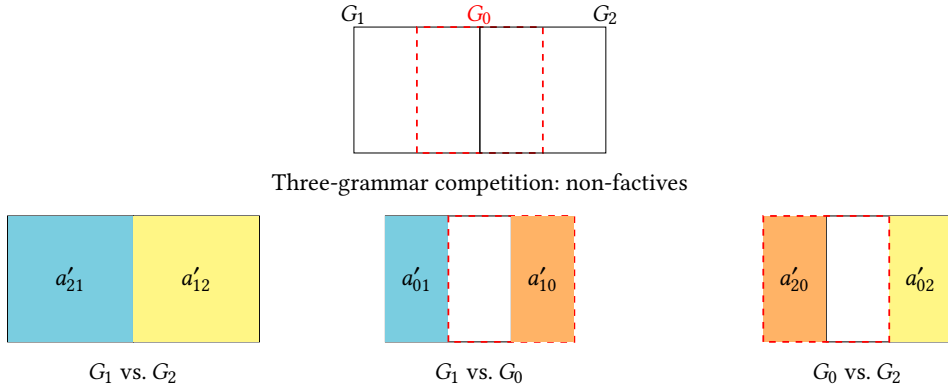


Figure 6.18: Non-factive verbs and the set of sentences each grammar can parse in negative declarative clauses.

$$(37) \quad A' = \begin{bmatrix} a'_{00} & a'_{01} & a'_{02} \\ a'_{10} & a'_{11} & a'_{12} \\ a'_{20} & a'_{21} & a'_{22} \end{bmatrix} = \begin{bmatrix} 0 & a'_{01} & a'_{02} \\ a'_{10} & 0 & 1 \\ a'_{20} & 1 & 0 \end{bmatrix}$$

A situation in which $a'_{10} = a'_{01}$ and $a'_{02} = a'_{20}$ will give rise to what has been termed as a Symmetric System (Kauhanen 2019). These systems, however, have been shown to give rise to stable variation and given the non-monotonic trajectory after non-factive predicates, this hypothesis will not predict the correct behavior. Interestingly, given that we are postulating that $a'_{12} = 1$ and $a'_{21} = 1$, it is hard to find a combination of values that would give the non-monotonic behavior observed for non-factive predicates (see next section for a full discussion of this issue). If G_1 and G_2 both have an advantage over G_0 , given that G_1 and G_2 don't have an advantage over each other, the prediction is that there will be stability. If G_1 has an advantage over G_0 , but G_0 has an advantage over G_2 the system will again stabilize (with a higher probability for G_1). Lastly, the inverse situation would emerge if G_2 has an advantage over G_0 , but G_0 has the advantage over G_1 ; the system will stabilize with a higher probability of being assigned for G_2 (see the simulations in the next section for evidence that this is indeed the case).

An alternative proposal, which was also discussed in Chapter 5, is that extra-linguistic factors (such as contact) may be at play. During contact with adult L2 speakers there might have been a preference/advantage for the semantic grammars, but once contact was lost the syntactic grammar had the advantage. If this were the case, this would derive the correct behavior for non-factive predicates, as the change in the advantages could derive a non-monotonic curve.

Summarizing, this section has shown that the proposed three-grammar competition approach is feasible on theoretical grounds: in the case of semi-factive verbs, if G_1 was globally asymptotically dominant, the expected behavior would be the one we encounter in the corpus data; in the case of non-factive predicates, it was argued that given the known parameters the non-monotonic behavior is difficult to obtain, but this could be solved if one takes into account the possible effect of contact during earlier stages of the language.

The underlying mathematical model used, however, assumes infinite populations. Since this is a non-realistic assumption, several agent-based simulations were run in order to test whether the predictions hold in cases of finite populations (see Section 6.2.3). An agent-based simulation is a way of recreating the behavior of a collection of discrete individuals, where the behavior of each individual is dictated by local rules, rather than global differential equations (see Winsberg 2013 for a discussion of how simulations are used in science and the different types of simulations). For example, a conversation between two individuals constitutes a local rule since it does not involve the whole population. The benefit of using these simulations is that they provide a controlled environment in which one can quantify the effect of each variable independently and in

combination with other factors, and provides a visual representation of how the different factors affect the mathematical model.

6.2.3 Simulations of the three-grammar competition

In order to test whether the same predictions hold in the case of finite populations, several computer simulations were run using the Julia programming language. The code used is heavily based on Kauhanen (2023)'s lecture notes, and it can be found in the supplementary materials of this book.

Firstly, a population of Variational Learners is created. Each of these learners has 4 attributes: the probability of G_0 (w_0), the probability of G_1 (w_1), the probability of G_2 (w_2), and a learning rate γ . These probabilities are then updated based on the linguistic input that each speaker hears. For simplicity, it was assumed that each generation of speakers learns the language from the previous generations (that is, the children would learn the language from the linguistic input of their parents, and the parents would have learned the language from the input of the grandparents, and so on). Following Kauhanen (2019), the linguistic input was determined by the advantage matrix and the probabilities of encountering speakers of either grammar in the previous generation. The variable c_0 stands for the probability of encountering a G_1 or G_2 speaker multiplied by the probability of those speakers uttering a sentence G_0 cannot parse; c_1 will stand for the probability of encountering a G_2 or G_0 speaker and those speakers uttering a sentence G_1 will not be able to parse, and c_2 will stand for the probability of encountering a G_1 or G_0 speaker multiplied by the probability of those speakers uttering a sentence G_2 cannot parse, see (38).

$$(38) \quad \begin{cases} c_0 = a_{01}w_1 + a_{02}w_2 \\ c_1 = a_{10}w_0 + a_{12}w_2 \\ c_2 = a_{20}w_0 + a_{21}w_1 \end{cases}$$

The way the probabilities are updated is based on the linear reward-penalty algorithm proposed, among others, by Bush & Mosteller (1955), Yang (2000), see (39).¹⁰ If a speaker hears a sentence its current grammar can parse, the probability of that grammar will be increased (indirectly reducing the probability of the other grammars); if they hear a sentence they cannot parse, the probability of that grammar will be reduced (indirectly increasing the probability of the other grammars).

¹⁰N stands for the number of competing grammars.

$$(39) \quad G_i \rightarrow s \begin{cases} w'_i = w_i + \gamma(1 - w_i) \\ w'_j = (1 - \gamma)w_j \text{ (if } j \neq i) \end{cases}$$

$$G_i \not\rightarrow s \begin{cases} w'_i = (1 - \gamma)w_i \\ w'_j = \frac{\gamma}{N - 1} + (1 - \gamma)w_j \text{ (if } j \neq i) \end{cases}$$

In order to model language change, each generation underwent a learning process of 10,000 steps. After that, it was assumed that the learners had converged into a combination of probabilities and the mean probability of each grammar for the end stage of each generation was calculated. The mean of each of these generations was then plotted to illustrate how the probabilities would change in time, see Figure 6.19 (see also the supplementary materials for more information).

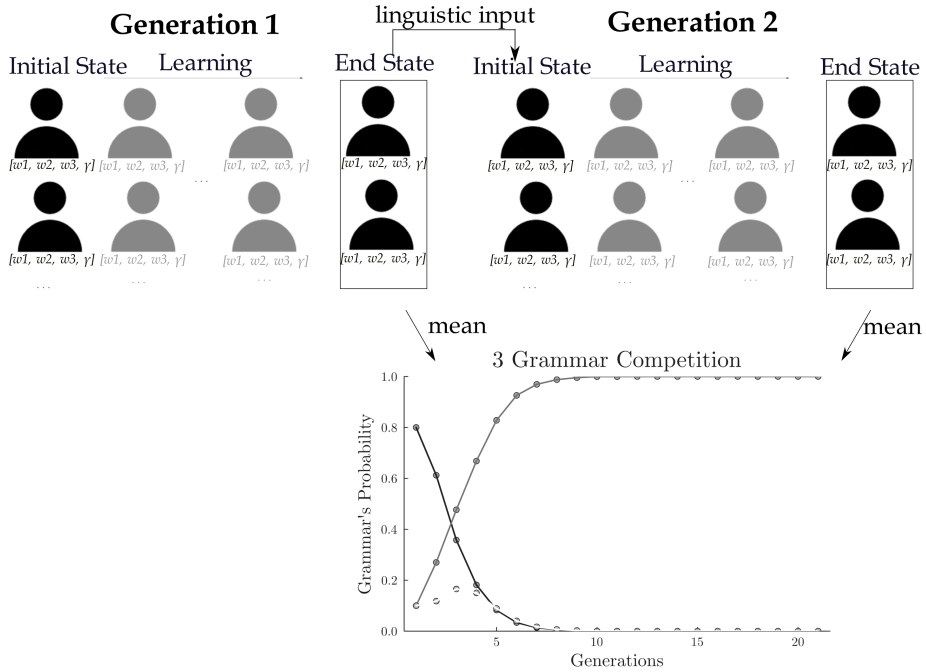


Figure 6.19: Simulation procedure.

It is important to highlight that to reduce computational costs, a number of simplifications (which do not necessarily hold in the real world, but which are common in the field (see Kauhanen 2023)) were made, to wit:

1. Generation t learns from the linguistic input of generation $t - 1$.
2. Speakers meet at random.
3. All learners have the same learning rate ($\gamma = 0.01$).
4. All learners have the same amount of input in the learning process (10,000 sentences per person).
5. All learners have the same starting probabilities (0.33 for each grammar) which are then updated.
6. Each generation has the same number of speakers ($n = 100$).¹¹ That is, Generation 1 has the same number of learners as subsequent generations.

The next subsection shows the results that were obtained.

6.2.3.1 Semi-factive verbs

In the case of semi-factives, it was argued in the previous section that the only value in the advantage matrix that we could estimate was a_{12} , which was equal to 0. It was further discussed that G_1 would win out over the other grammars (as was theoretically hypothesized) if the advantage matrix additionally satisfied condition (36), here repeated as (40).

$$(40) \quad a_{12} + a_{10} < \frac{1}{\frac{1}{a_{21}} + \frac{1}{a_{01}}}$$

These predictions seem to hold when looking at finite populations as well. The results of 4 different simulations can be found in Figure 6.20. On the x -axis appears the generation, and on the y -axis the mean probability of that generation for each of the grammars. The colors (line type) stand for G_0 (orange/dashed line), G_1 (blue/dashed-dotted line) and G_2 (yellow/continuous line). Simulation A (top-left) represents a situation in which GAD is satisfied, $a_{01} > a_{10}$ and $a_{02} < a_{20}$; simulation B (bottom-left) where GAD is satisfied, $a_{01} > a_{10}$ and $a_{02} > a_{20}$; simulation C (top-right) represents a situation when GAD is not satisfied, $a_{01} < a_{10}$ and $a_{02} > a_{20}$; and simulation D (bottom-right) represents a situation in which GAD is not satisfied, $a_{01} > a_{10}$, and $a_{02} < a_{20}$.

¹¹The number of people of each generation was kept low (only 100), because the numbers did not affect the diachronic patterns. An increase in population size would only result in smoother changing curves.

6 Explaining Polarity Mood: The first person effect

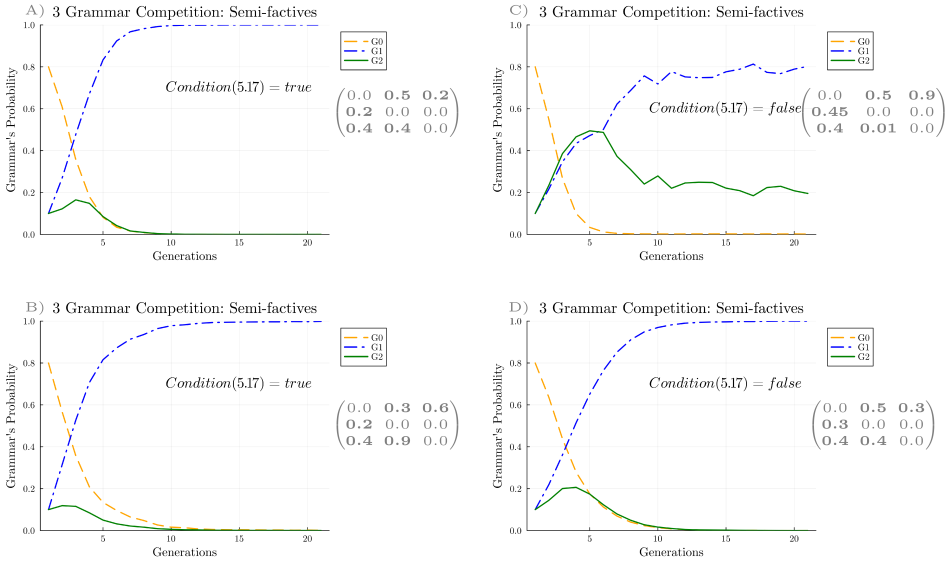


Figure 6.20: Three-grammar competition simulations for different advantage matrices. Bold values in the matrix indicate that an exact value could not be calculated/estimated.

Simulations A and B show that as long as condition (40) is satisfied, G_1 will win out over the other grammars. It does not matter how great the advantage between a_{01} and a_{10} is, nor whether the advantage of grammar G_2 is greater than that of G_0 or the other way around (i.e., G_1 will win when $a_{20} > a_{02}$ and also when $a_{20} < a_{02}$, the relation between these two grammars won't affect the end result). Simulation C shows that the important condition is (40) and not whether G_1 has an advantage over the other grammars ($a_{01} > a_{10}$ and $a_{01} > a_{10}$). In the simulation even if grammar G_1 has an advantage over the other two grammars it does not reach 100%, but rather fluctuates at 80% of usage. Lastly, simulation D shows that (40) is a sufficient condition but not a necessary one. If GAD is satisfied G_1 will out-compete the other variants (i.e. it will asymptotically move towards 1), but if it is not satisfied G_1 may or may not out-compete them. These simulations, thus, show that as long as condition (40) is satisfied the system will behave as expected even in the case of finite populations.

6.2.3.2 Non-factive verbs

Looking at non-factive verbs, the non-monotonic pattern observed in the data is a bit harder to explain. It was argued that, given that G_2 and G_1 stand in a

6.2 The effect of first person with non-factive verbs

disjoint relationship, their respective advantages (a_{12} and a_{21}) should be equal to one. However, it is not easy to find a combination of values with these restrictions which would give a non-monotonic curve (of all the attempted trials, no combination with such behavior was found). Interestingly, one very frequent outcome with such a restriction is stability across the three grammars. This is illustrated in Figure 6.21, where the outcome of 4 different simulations is illustrated. In simulation A $a_{01} > a_{10}$ and $a_{02} > a_{20}$; in simulation B $a_{01} < a_{10}$ and $a_{02} > a_{20}$; in simulation C $a_{01} > a_{10}$ and $a_{02} < a_{20}$; and lastly in simulation D $a_{01} = a_{10} = a_{02} = a_{20}$. Although the outcomes of the simulations differ in terms of how the system stabilizes, there is no grammar moving to 0 or 1; rather they all stabilize at intermediate values.

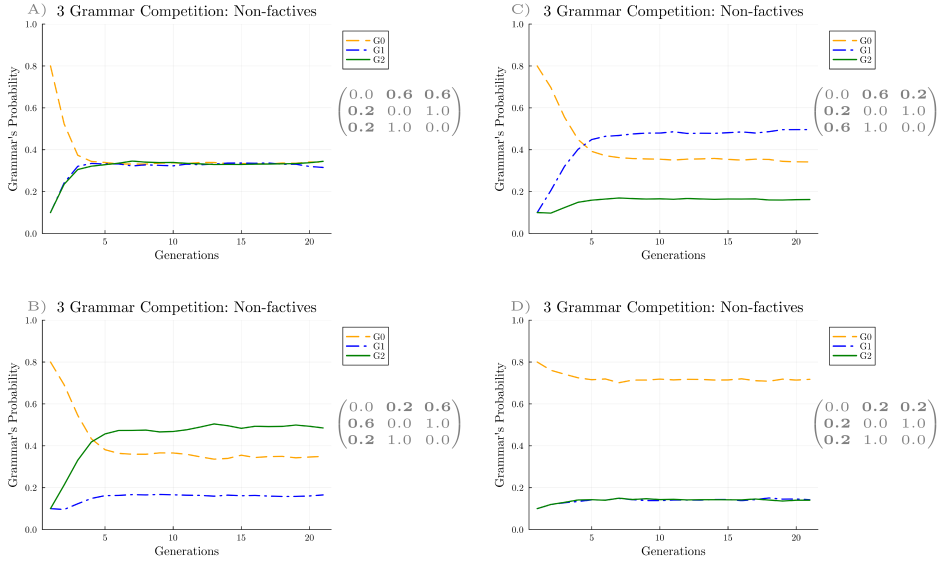


Figure 6.21: Three-grammar competition simulations for different advantage matrices. Bold values in the matrix indicate that an exact value could not be calculated/estimated.

In order to account for this non-monotonic behavior, a different explanation would be needed. One hypothesis that was mentioned in the previous section and in Chapter 5 is that of language contact/extra-linguistic factors. The idea is that although from a purely linguistic point of view the two grammars may not have any advantage over the other, specific cognitive biases may give an advantage to one or the other grammar. If these advantages are different depending on the type of learner (L2 adults vs. L1 children), then a change at the population level

regarding the proportions of each type of learner could have as a consequence a change in the diachronic development of a given linguistic phenomenon. Looking in particular at the case of subjunctive morphology, it has been reported that Spanish L1 children start using the subjunctive by the age of 2 (López Ornat et al. 1994). However, the acquisition literature also reports that children start realizing that other's beliefs may differ from their own at the age of 4–6 (Perner et al. 2015). That is, the necessary underlying concepts for acquiring grammar G_1 (that the world of evaluation lies in some accessible CS and that that CS can be different from one's own) and G_0 (the degree of commitment by someone toward a proposition) are acquired at a later stage than the first uses of the subjunctive morphology. This may give a preference to the syntactic system (at least at the earliest years of acquisition), as the key semantic concepts required to acquire G_1 and G_0 are not acquired yet.

On the other hand, adult learners would already understand the required semantic concepts to learn G_1 and G_0 (semantic selection), but may struggle with a purely syntactic system, as uninterpretable features have been argued to be hard for L2 adult learners (Tsimpili & Hulk 2013). Given that the analysis proposed thus far for G_2 is a purely syntactic mechanism, which is triggered by the need to delete uninterpretable features, the expectation is that L2 learners would struggle to learn this system.¹²

The hypothesis then would be that L2 learners would introduce an advantage for G_1 and G_0 , and L1 children would introduce an advantage for G_2 . For simplicity, this effect was modeled as a change in the advantage matrix.¹³ During contact with adult L2 learners (first 10 generations in the simulation), the advantage matrix needs to satisfy the following conditions: $a_{01} > a_{a10}$, $a_{02} < a_{20}$ and $a_{12} < a_{21}$ (G_1 has an advantage over G_0 and G_2 ; G_0 has an advantage over G_2). After the contact is lost and there are mostly L1 speakers (the following next 10 generations in the simulation), the following conditions to create the advantage matrix were followed: $a_{01} > a_{a10}$, $a_{02} > a_{20}$ and $a_{12} > a_{21}$ (G_2 wins over the other two grammars and G_1 wins over G_0). The exact values used can be found in (41), and the result is plotted in Figure 6.22.

¹²The language acquisition literature shows that L2 learners of Spanish tend to overgenerate the indicative in these environments (Borgonovo 2003), providing further support for this claim.

¹³Strictly speaking the advantage matrices are constant, and what changes is an extra parameter which would influence the value of the penalty probabilities (see Kauhanen 2019). In practical terms, however, a change in the advantage matrices would have a similar outcome, and has the advantage of being easier to code.

$$(41) \quad \text{Contact} = \begin{pmatrix} 0.0 & 0.7 & 0.2 \\ 0.3 & 0.0 & 0.5 \\ 0.7 & 1.0 & 0.0 \end{pmatrix}; \text{No contact} = \begin{pmatrix} 0.0 & 0.7 & 0.7 \\ 0.2 & 0.0 & 1.0 \\ 0.2 & 0.5 & 0.0 \end{pmatrix}$$

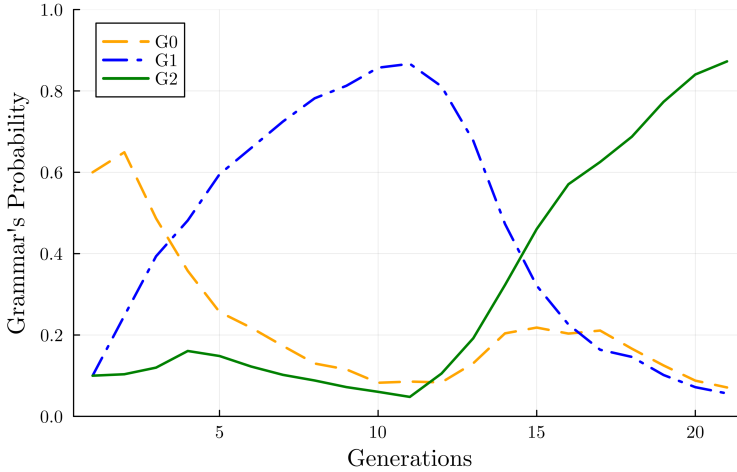


Figure 6.22: Three-grammar competition simulations with changing advantages with non-factive verbs.

As can be seen, by introducing a change in the advantage matrices we can obtain a non-monotonic behavior, whereby G_1 starts to win over the other grammars but after contact is interrupted G_2 starts to win over. This analysis, however, will only explain the corpus data if indeed during 1200–1600 it can be proven that there was a higher rate of L2 learners than in the periods afterward. During 1300–1600 Spanish was in contact with a variety of L2 speakers from other languages (Arabic (711–1610) and Portuguese (1580–1640) to name a few) (Ruhstaller & Gordón Peral 2017, Döhla 2014), but determining their numbers and whether the sociolinguistic situation was adequate for the type of effect proposed in this section will need further future work.

6.3 Conclusion

This chapter has proposed that the effect of first person in present-day Spanish can be better understood by differentiating two different causes for the effect. In the case of semi-factive verbs, the first person effect emerges due to a semantic contradiction when the indicative is used with a first person subject and present

tense verb. In the case of non-factive verbs, however, the effect seems to be an artifact of an earlier grammar in which the subjunctive was used as a hedging device. This created an asymmetry across constructions as hedging devices may be more frequent with first person constructions. The observed ungrammaticality of the indicative with a non-factive verb in Construction FP is then a result of this previous grammar coupled with a grammaticalization of the mood system (Grammar G_2 winning over G_0 and G_1). The proposed system was shown to be feasible when looking at the speed of change across constructions.

Overall, the proposed system not only allows us to derive in a principled way the first person effect but allows us to also derive the differences across predicate types observed in Chapter 5. That is, the proposal so far is capable of explaining two effects which none of the previous accounts could explain at the same time, see Table 6.8.

Table 6.8: Summary of the effects and approaches of Polarity Mood.

Approaches/effect	1st Person	NPI	Contrast	Verb Class	Clause type
Presuppositional	✓	✓	–	–	–
Neg-raising	–	✓	–	–	–
Information status	–	–	✓	–	–
Competing Grammars	✓	Ch. 6	Ch. 6	✓	Ch. 6

The next chapter will briefly cover how the proposed analysis could also account for the other effects, to wit: the behavior of strong NPIs, the effect of clause type and the influence of information status.

7 Explaining Polarity Mood: Other effects

The previous chapters explored the effect of verb class (Chapter 5) and first person (Chapter 6) on mood selection in polarity contexts. This chapter explores three other phenomena which have been argued to affect mood selection, to wit: clause type, information status and strong NPIs. Given the breadth of the domain of investigation the aim of this chapter is not to provide a fully-fledged theoretical explanation of all three effects, but rather it aims to provide a discursive basis on which a more formal theory can be constructed in the future.

7.1 Clause type

Although there is mood variation after negated predicates in declarative clauses (with a slight preference for the subjunctive), see (1), when the negated predicate appears in an imperative sentence, the only available option in present-day Spanish is the use of the indicative in the embedded clause (2).¹

(1) Negative Declarative:

No crees que el dinero ?crece/crezca de los árboles.
Not believe.IND that the money grow.PRS.IND/PRS.SBJV from the trees
'You don't believe that money grows on trees.'

(2) Negative Imperative:

No creas que el dinero crece/*crezca de los árboles.
not believe.SBJV that the money grow.PRS.IND/PRS.SBJV from the trees
'≈Don't you believe that money grows on trees!'

This phenomenon, first noticed in Borgonovo (2003), poses an interesting puzzle for current theories of the Polarity Subjunctive, as none of them are capable of explaining this effect (see Chapter 2 for an in-depth discussion of this). For

¹It must be noted here that negative imperative clauses in Spanish carry the subjunctive rather than the imperative (Real Academia Española 2010: 797).

instance, the Neg-raising/focus account (see Reider 1990) postulates that the indicative is used when the matrix verb is the focus of negation, see (3), whereas the subjunctive marks that the focus is the embedded clause, thus triggering a Neg-raising inference, see (4).

- (3) No [crees] que Juan es culpable.
not believe that John is.IND guilty
'You don't believe John is guilty.'
- (4) No crees [que Juan sea culpable].
not believe that John be.PRS.SBJV guilty
'You don't believe John is guilty.'
Neg-raising Inference: You believe John is not guilty.

However, imperative clauses like declarative ones can also give rise to neg-raising inferences, see (5). Thus, the fact that declarative clauses allow both moods but imperative sentences only select indicative embedded verbs is left unexplained.

- (5) No esperes que sepa una palabra de francés.
not expect that know.PRS.SBJV a word of French
'Don't expect they will know a word of French.'
Neg-raising Inference: (you should) expect she/he won't know a word of French.

The presuppositional account (see Quer 1998, among others) postulates that when the indicative is used, the sentence carries a presupposition that the speaker is committed to the embedded proposition *p*. However, in negative imperatives the speaker is, in most situations, not committed to *p*, see (6), and thus the prediction would be that these sentences should take the subjunctive, contrary to fact.

- (6) No creas que te voy a ayudar ✗ I am going to help you.
Not believe that you go.IND to help
'Don't think I will help you.'

Finally, the information status approach (Faulkner 2021) proposes that when the indicative is used, the information is new/contrastive, but when the subjunctive is used, it is old. However, given that imperative clauses can also present old information, see (7), the prediction is that in these environments there would also be mood alternation, contrary to fact, see (2) above.

- (7) Ya te lo he dicho muchas veces. No creas que el dinero
 already you it have told many times not believe that the money
 crece/*crezca de los árboles.
 grow.PRS.IND/PRS.SBJV from the trees
 ‘I have already told you many times. Don’t you think money grows from
 trees.’

Thus, none of the current approaches to the Polarity Subjunctive can explain the effect of imperative clauses on mood selection, as they all predict mood variation in these environments, rather than the indicative mood. The next section explains how the competing grammars approach could explain the effect of clause type.

7.1.1 Explaining the effect of clause type

The competing grammars approach developed in Chapters 4, 5 and 6 proposes that with non-factive predicates (e.g., *believe*), grammar G_2 (syntactic mood) started to win out over the other grammars, and that in the present-day language, G_2 has nearly won. Under this account, then, whatever is causing the effect of clause type, it must be the result of a clash at the syntactic level between the subjunctive mood and imperative clauses.

In particular, this section explores the hypothesis that similar to other dependent cases (e.g., accusative in Finnish), the subjunctive syntactic mood is a dependent mood.

7.1.1.1 The interaction between Case and clause type

It has been observed that in certain languages, for example Finnish (Poole 2023), when the object of a transitive verb appears in an imperative clause it takes the default nominative case rather than the expected accusative case. This is illustrated in (8) where the object *book* appears in the accusative case in a matrix declarative clause, but changes to the nominative when used in an imperative sentence.

- (8) Finnish: (Poole 2023: 360)
- a. Pekka osti kirja-n.
 Pekka.NOM bought book-ACC
 ‘Pekka bought the/a book.’
 - b. Osta kirja!
 buy.IMP book.NOM
 ‘Buy the/a book!’

A similar phenomenon is attested in Estonian where an object that would normally be marked as genitive gets assigned nominative case in imperative clauses (Miljan & Cann 2013). For example, in sentence (9a) the object *ice-cream* is assigned genitive case, but that same object would be assigned nominative case in an imperative clause, see (9b).

- (9) a. Poiss sõi (ühe) jäätise.
 boy.NOM eat one.GEN ice-cream.GEN
 ‘The boy ate an ice-cream.’ (Miljan & Cann 2013: 335)
- b. Söö üks jäättis!
 eat.IMP one.NOM ice-cream.NOM
 ‘Eat an ice-cream!’ (Miljan & Cann 2013: 335)

The effect of clause type on case is also attested in negative sentences. For example, the Genitive of Negation is unacceptable in imperatives in Russian, whereas in declarative clauses both accusative and genitive are possible (Timberlake 1975).²

- (10) Genitive of Negation in imperatives in Russian (Timberlake 1975: 129)
- a. Smotri ne poterjaj očki/*očkov!
 see.IMP not lose glasses.ACC/*GEN
 ‘Watch out you don’t lose your glasses!’

These data show that case marking can be affected by the type of clause. As such, if the subjunctive mood is also analyzed as the equivalent of dependent case, it would not be unexpected that it also is affected by the clause type. The question that arises is: how can one formalize this idea?

7.1.1.2 Dependent Case theory

Within the Generative tradition there are two main theories of Case assignment: Agreement-based theories (also known as functional-head case theory),

²In Lithuanian, however, the genitive is still compulsory in negative imperatives (p.c. Milena Šereikaitė). The effect of imperative clauses seems to be language-specific.

- (i) a. Nepamesk šio pieštuko.
 not.lose this.GEN pencil.GEN
- b. *Nepamesk šį pieštuką.
 not-lose this.ACC pencil.ACC
 ‘Don’t lose this pencil!’

and dependent case theories (see Poole 2024 for a recent overview). According to agreement-based theories structural cases result from the relation between a functional head and a DP.³ For example, the T head assigns nominative case to the DP subject with which it agrees in ϕ -features and the v head assigns accusative case to the DP object with which it also agrees (even if not overtly), see (11) (Chomsky 2001).

- (11) **Agreement-based Case:** a functional head X assigns case to a DP iff it enters into an Agree relationship with the DP (Chomsky 2001)

[X_{[u ϕ],[uCase]} ... [DP_{[i ϕ],[uCase]}]]

According to dependent case theories structural cases result from the relation between two DPs which are unvalued for case and which stand in a c-command relationship (Baker 2015, Marantz 1991, McFadden 2004: among others).

- (12) **Dependent Case:** If there are two distinct NPs in the same spell-out domain/clause⁴ such that NP1 c-commands NP2, then value the case feature of NP2 as accusative unless NP1 has already been marked for case. (Baker 2015: 48)

[DP_[Case:]... [DP_[Case:acc]]]

There is much debate in the literature regarding which is the better formalization to capture the data (Levin & Preminger 2015), and whether the two systems might be at play in different languages/constructions (Baker 2015).

Important for our purposes is that in the case of Finnish described in the previous section, Poole (2023) has proposed that the data could be explained in terms of dependent case. In a plain declarative clause, the subject DP and the object DP will enter the derivation with unvalued features, see Step 1 in (14). The subject would license dependent case on the object (because it is c-commanding the object DP and has an unvalued Case feature), see Step 2 in (14). Given there is no other DP c-commanding the subject, it would remain unvalued for case throughout the derivation. The subject would then be realized as having unmarked (nominative) case at PF (Phonological Form), see Step 3 in (14).⁵

³Here I am using NP as equivalent to DP as the literature uses different terminology to refer to the same thing.

⁴The original proposal in Marantz (1991) used a clause as domain, but this has been modified to the notion of phases in more current versions (Baker & Vinokurova 2010).

⁵In the Minimalist Program there are two interface layers: Phonological Form (PF) and Logical Form (LF). The derivation would start with a Numeration (set of lexical items), to which the operations Select, Merge and Move are applied. Once these operations are in place the derivation is sent to Spell-out, and it is split into PF and LF (Hornstein et al. 2005: p. 80).

- | | |
|---|--|
| <p>(13) Pekka osti kirja-n
 Pekka.NOM bought book.ACC
 ‘Pekka bought the/a book.’
 (Poole 2023: 360)</p> | <p>(14) Step 1. [DP_[Case:]...[DP_[Case:]]]
 Step 2.
 [DP_[Case:]...[DP_[Case:acc]]]
 Step 3.
 [DP_[Case:nom]...[DP_[Case:acc]]]</p> |
|---|--|

In imperative sentences, however, see (15), no other DP c-commands the object (even if the subject is overtly pronounced, Poole 2023 argues that it would appear in a parenthetical, not in the canonical subject position). Given the absence of another DP with an unvalued case feature, no dependent case can be established, see Step 2 in (16). Hence the object is assigned the unmarked nominative case at the end of the derivation, see Step 3 in (16).

- | | |
|---|---|
| <p>(15) Osta kirja!
 buy.IMP book
 ‘Buy a/the book!’
 (Poole 2023: p. 360)</p> | <p>(16) Step 1. [[DP_[Case:]]]
 Step 2. [[DP_[Case:]]]
 Step 3. [[DP_[Case:nom]]]</p> |
|---|---|

Finally, in a sentence such as (17) the subject DP would receive lexical genitive case, see Step 1 in (18). As such, there is no other DP than the object DP with an unvalued case and no dependent case could be established with the object DP, see Step 2 in (18). As a last step, at PF the object would be assigned nominative unmarked case, see Step 3 in (18).

- | | |
|---|--|
| <p>(17) Minu-n täytyy osta-a
 I-GEN need buy-INF
 kirja.
 book.NOM
 ‘I have to buy the/a book.’
 (Poole 2023: 360)</p> | <p>(18) Step 1.
 [DP_[Case:gen]...[DP_[Case:]]]
 Step 2.
 [DP_[Case:gen]...[DP_[Case:]]]
 Step 3.
 [DP_[Case:gen]...[DP_[Case:nom]]]</p> |
|---|--|

Note that although Baker (2015)’s formulation of dependent case is phase-bound,⁶ there is much debate in the literature regarding this last point. Looking at Finnish, for example, accusative dependent case can occur across vP phases,

⁶The idea that there is cyclic transfer to the interfaces (PF and LF) has a long history in the syntactic literature (Bobaljik & Wurmbrand 2013). That is, rather than sending a derivation to PF and LF once, the derivation is sent at multiple stages to the Interfaces. There is much disagreement regarding what constitutes a phase, although the general agreement is that vPs and CPs are phases.

see (19)–(20). Based on this data, Poole (2023) argues that dependent case assignment is not subject to the phase impenetrability condition at the *vP*-level.

- (19) **Hän** kävi [_{vP} avaa-ma-ssa **ove-n**]
 s/he.NOM went open-INF door-ACC
 ‘S/he went to open the door.’

- (20) **Käy** [_{vP} avaa-ma-ssa ovi].
 go.IMP open-IMP door.NOM
 ‘Go open the door!’

The impact of different phases, thus, seems to be language-specific and more research is needed in this respect.

Leaving aside the issue about domains, the account proposed in Poole (2023) is capable of deriving the correct case assignment pattern in Finnish. The next section attempts to extend this machinery to account for the effect of imperative clauses on mood selection in Spanish.

7.1.1.3 Dependent mood theory

Chapters 3 and 4 explored the idea that the subjunctive might be becoming a grammaticalized structural case (similar to the Lithuanian genitive case). An agreement-based account of case was extended to explain the Polarity Subjunctive. This section explores the possibility of instead interpreting it as a ‘dependent’ mood.

Starting first with negative declarative clauses, as in (21), one could postulate that the two verbs carry an unvalued mood feature, see Step 1 in (22).⁷ Given that both verbs have an unvalued mood feature and the first one *c*-commands the second one, they can enter into a dependent mood relation whereby the embedded verb is assigned the subjunctive,⁸ see Step 2 in (22). Because there is no other verb that *c*-commands the matrix verb, this verb remains unvalued for mood throughout the derivation, and is labeled as the unmarked indicative mood at PF.

⁷Note that here we are not saying that VP carries the uninterpretable feature, as it is usually only the verb that is marked for mood and other elements in the Verbal phrase (objects, adverbs, etc.) do not carry a mood feature. This contrasts with DPs which have a case feature in all the elements inside the DP (i.e., determiners, adjectives, etc.).

⁸Assignment might be affected by the type of NegP. If positive the value would be the indicative and if negative the subjunctive.

(21) Negative Declarative Clause

No crees que sea mentira.
not believe.IND that be.PRS.SBJV lie
'You don't believe it's a lie.'

- (22) **Step 1.** [V_[Mood:] [V_[Mood:]]]
Step 2. [V_[Mood:] [V_[Mood:sbjv]]]
Step 3. [V_[Mood:ind] [V_[Mood:sbjv]]]

When dealing with an imperative mood clause, however, the matrix verb would have already been assigned lexical subjunctive mood (similar to the case in Finnish where the subject was assigned genitive lexical case), see Step 1 in (24). Only phrases with unvalued mood features factor into the calculus of dependent mood, and as such no dependency is established between the two verbs, see Step 2 in (24). The verb remains unvalued throughout the derivation and is assigned the unmarked indicative mood at PF, step 3 in (24).

(23) Negative Imperative Clause

No creas que es mentira.
not believe.SUBJ that be.IND lie
'Don't believe it's a lie!'

- (24) **Step 1.** [V_[Mood:sbjv] [V_[Mood:]]]
Step 2. [V_[Mood:sbjv] [V_[Mood:]]]
Step 3. [V_[Mood:sbjv] [V_[Mood:ind]]]

The proposed analysis relies on the assumption that *v*Ps are not phases for dependent mood (which is also argued for in the case of dependent case in Poole 2023), and importantly that the CP in this case is also not a phase for dependent mood. Given that the status of a given phase seems to be language-specific – the *v*P has been argued to be a phase in Sakha (Baker & Vinokurova 2010), but not in Finnish (Poole 2023) – it would not be too far-fetched to argue that in Spanish the CP would not count as a phase for this phenomenon. Other authors have also argued that subjunctive clauses might not be phases in Spanish. For example, Torrego (2018) argues that given that phase heads cannot carry unvalued features, subjunctive clauses under negation might be non-phasal because they carry an unvalued negative feature:

a plausible conjecture is that subjunctive clauses under negation have a defective C with respect to negative features, making the subjunctive C dependent on the matrix for valuation. The present suggestion recalls proposals

by Progovac (1994) and Laka Mugarza (1990), who posit negative complementizers as well as proposals concerning the postulation of negative features in the CP-domain. (Torrego 2018: 104)

Evidence that these may not be phases in Spanish is provided, for example, by the fact that n-words are allowed in these structures. The n-word *nada* in postverbal position needs the appearance of a negative element preceding it, (25). This negative element, however, needs to c-command the n-word and it has to be within the same phase as the n-word (see Giannakidou & Mari 2017 for a recent overview of current analyses). In sentence (26), however, the n-word *nada* is licensed across a supposedly CP phase. In order to account for these data the literature has proposed that CPs with a subjunctive embedded verb are not phases in Spanish.

- (25) *Sabe **nada**.
 knows nothing
 Intended: ‘He/she knows nothing.’
- (26) No creo [CP que sepa **nada**].
 not believe that know.PRS.SBJV nothing
 ‘I don’t believe that they(sg.) know a thing.’

Assuming that negative clauses with an unvalued negative C feature are not phases, the proposal would then derive the correct effect of clause type on mood. Given that these negated CPs are not phases, they would allow long distance dependent mood.

Notice that the analysis predicts that not only imperative clauses but also any other case in which the matrix mood is lexically selected should not be able to license the subjunctive dependent mood in the embedded clause. This prediction seems to be corroborated by the empirical data. In sentence (27), taken from Borgonovo (2003), the verb *want* lexically selects for the subjunctive mood in the verb *believe*. Given that the verb *believe* has been lexically assigned the subjunctive mood, this verb, despite being preceded by negation, is not capable of triggering dependent subjunctive mood in the most deeply embedded clause (as only verbs unvalued for mood can trigger dependent mood). As a consequence, the only available option is the use of the indicative in the most deeply embedded verb.

- (27) La prensa *quiere* que la gente no *crea* que el
 The media want.IND that the people not believe.PRS.SBJV that the
 ministro *es*/**sea* culpable.
 minister be.PRS.IND/PRS.SBJV guilty
 ‘The media wants people not to believe that the minister is guilty.’

This idea, thus, would be reminiscent of the idea that the subjunctive is a marker of subordination discussed in Chapter 3, but with some important differences. Firstly, not all subordinated clauses would be marked with the subjunctive, but rather only those that have an indicative (not lexically selected) matrix verb. That is, in the dependent case theory the proposal is that case assignment follows the pattern: lexical case > dependent case > unmarked case. Following the same hierarchy for mood would mean that lexical mood is assigned first, then dependent mood and finally unmarked mood. The subjunctive dependent mood would only appear in the absence of a lexically specified mood.

The next section of this chapter will have a look at whether the proposed analysis in combination with the 3-grammar competition model proposed in Chapter 4 fits with the diachronic picture as well.

7.1.2 The effect of imperative clauses in diachrony

Chapter 5 presented the data for negative clauses with a matrix verb in the indicative. The way this data was gathered and annotated appears schematized in Figure 7.1.

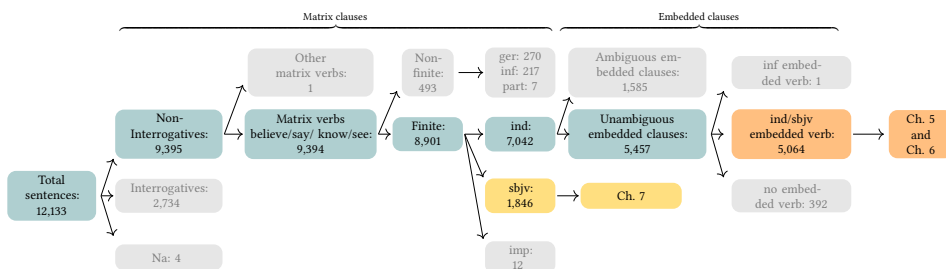


Figure 7.1: Data Extraction Process from Chapter 5

In this chapter we look at the behavior of mood when the matrix verb is in the subjunctive mood (1,846 sentences in total, see yellow box in the schematic in Figure 7.1). The data were then further annotated according to the type of matrix

clause and the mood of the embedded verb. Regarding matrix clauses they were divided into imperative ones, as in (28), and ‘other’. This last category included conditional sentences (29), purpose sentences (30), adversative sentences (31) and embedded clauses (32), among others. All of these allow the matrix verb to appear in the subjunctive morphology, but are not imperative clauses.

- (28) No creas que esto es de voluntad!
 not believe.SBJV that this is of will
 ‘Don’t believe this is voluntary!’
- (29) si no creyera que otras ocupaciones más nobles os sirven de
 if not believe.SBJV that other occupations more noble you serve.IND of
 entretenimiento, no me quedaría aquí
 entertainment, not CLT stay here
 ‘If I didn’t think that other more noble occupations serve you as
 entertainment, I wouldn’t stay here.’
- (30) para que no creyesses que a mi palabra faltava
 so that not believe.SBJV that to my word miss.IND
 ‘So that you wouldn’t believe that I went back on my word...’
- (31) Aunque no creyeras que había dios...
 although not believe.SBJV that have.IND God
 ‘Although you didn’t believe that there was a God...’
- (32) No es maravilla que no crea que hay ...
 not is wonder that not believe.SBJV that have.IND
 ‘It is no wonder that they do not believe that there is...’

Regarding the embedded verb, the sentences were divided into *indicative*, *subjunctive* and *NA* (not applicable). This last category was created to include cases in which there was no embedded verb, as in (33).

- (33) No digas que no.
 not say.SBJV that no
 ‘Don’t say no!’

The amount of data assigned to each category can be seen in Figure 7.2. Of special importance for this section of the book are imperative clauses with an embedded verb either in the indicative or the subjunctive (1,032). The other clauses,

although interesting as well, do not have enough data to draw any clear generalizations: there are only 761 tokens and these include a mixture of conditionals, purpose clauses, adversatives, etc. This means that for each sentence type there was an even smaller number of tokens. The data will nonetheless be presented in the following sections for completeness as the approach presented in the previous section predicts that these sentences (given that the matrix verb is not in the indicative) would mostly select the indicative as well.

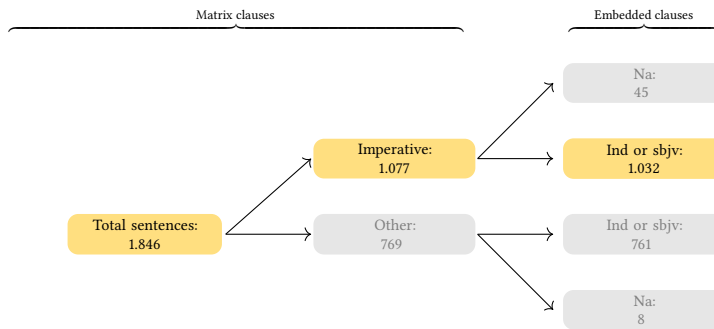


Figure 7.2: Data Extraction process for subjunctive clauses.

The results are plotted in Figure 7.3. On the top row are the results of both negative imperative – yellow dots – and other constructions (conditional, adversatives, etc.) – gray dots – with a subjunctive matrix verb. On the bottom row are the results already presented in Chapter 5 for negative clauses with a matrix indicative verb, which have been included here for comparison purposes. As in previous graphs, the x -axis represents the time period, and the y -axis the proportion of embedded mood: 1 indicates that all instances were in the indicative and 0 that all were in the subjunctive.

In addition, the proposal from Chapter 6 regarding which grammar is hypothesized to be winning is superimposed on the graph. For semi-factive verbs the prediction was that G_1 would win out over the other two grammars and that by 1600–1700 that change was almost complete. For non-factive verbs the proposal was that G_1 and G_2 both won out over G_0 , but the two were themselves in a disjoint relationship and hence neither was predicted to win out over the other. It was then hypothesized that due to extra-linguistic factors one grammar might have been preferred to the other grammar: between 1400 and 1600 G_1 was winning out, but after 1700 G_2 gained an advantage. This state of affairs explained the non-monotonic behavior of non-factive predicates.

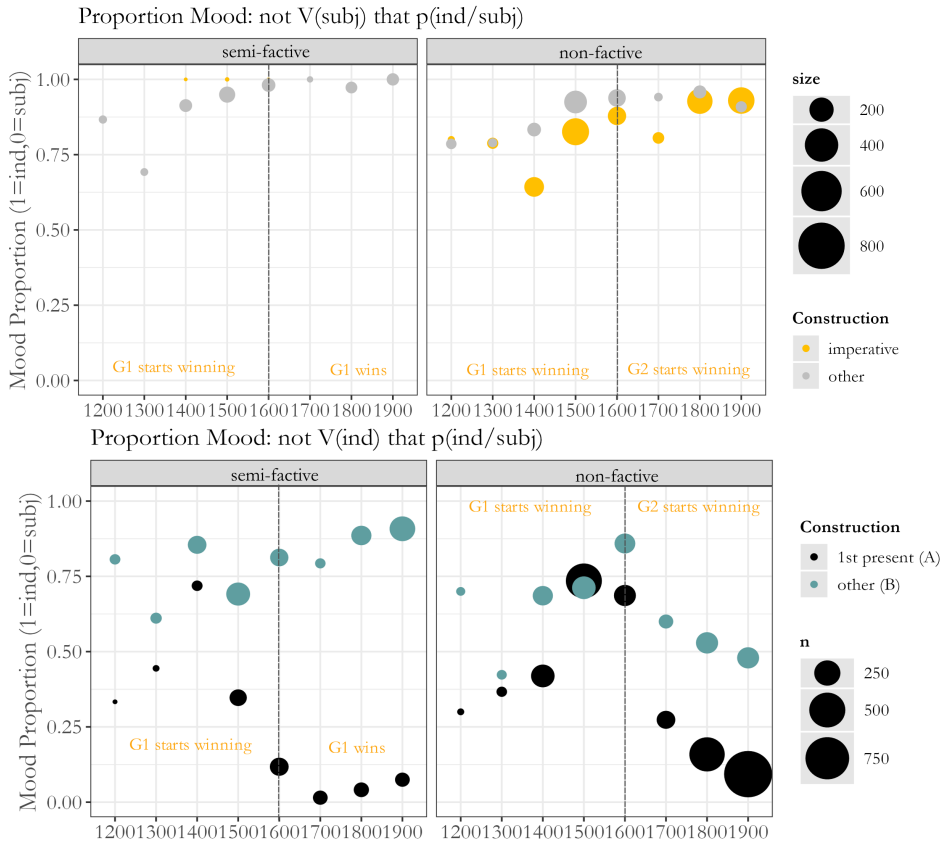


Figure 7.3: Comparison of the development of mood selection with a matrix verb in the subjunctive (top row) and the matrix verb in the indicative (bottom row); per verb class and construction. The construction ‘other’ includes conditional, purpose and adversative clauses. Construction FP stands for first person and present tense and Construction O for all other tenses and subjects (see Chapter 5).

Looking at the behavior of negative imperative clauses (the focus of this chapter⁹) one observes that by 1400 the proportion of the indicative was about 65% and after that period the construction started to take more instances of the indicative. By 1800 almost all tokens are in the indicative (92%). These results are interesting in light of the three-grammar competition developed in Chapter 4. The last section has argued that G₂ (syntactic mood) could only select the indicative mood after negative imperatives, given its analysis as a form of dependent

⁹Mood in conditionals and purpose clauses has been included in the graph for completeness.

mood marking. Similarly, G_1 (semantic selection à la Schlenker 2005) only predicts the indicative mood after negated non-factives, irrespective of clause type (see Section 5.2.1).

The prediction is that starting in 1400, given that grammar G_1 will start to win over G_0 in non-factive contexts, there should be more indicative being used. This is confirmed by the graph, where an increase in the indicative mood from 1400 to 1600 is observed. After 1600 grammar G_2 was argued to start winning out over G_1 in non-factive contexts. Given that G_2 also selects the indicative mood in imperative contexts the prediction is that there won't be a change in the proportions of mood after 1700. This prediction is again corroborated by the data. A summary of what each of the grammars predicts in the different contexts can be found in Table 7.1. Only the predictions of non-factive predicates are included as these are the only ones that can take a negative imperative clause.¹⁰

Table 7.1: Mood selection prediction of each grammar in negative declarative and imperative clauses.

Grammar	Non-factive	
	Negative Declarative	Negative Imperative
G_0	ind/sbjv	ind/sbjv
G_1	ind	ind
G_2	sbjv	ind

In summary, the proposed analysis could derive the observed effect of imperative clauses in the present-day language, while also accounting for its development in diachrony. The next section of this chapter looks at the effect of information status.

¹⁰Semi-factive verbs for the most part cannot appear in negative imperatives irrespective of mood.

(i) *No sepas que Ana va/vaya a venir!
 not know.IMP that Ana go.IND/SBJV to come
 *‘Don’t know that Ana is coming.’

7.2 Information status

7.2.1 Mood and information status

The idea that information status can affect mood selection has been explored at least since Lunn (1989)'s work on *Spanish mood and the prototype of assertability* and has been further developed by Mejías-Bikandi (1998), among others. One influential hypothesis that has been put forth is that the subjunctive mood is used whenever the information is old/given (Mejías-Bikandi 1998). As such, sentence (34) can only be uttered in a context in which the topic of going on a strike has been discussed previously.

- (34) Pocos trabajadores creen que haya que ir a la huelga.
 few workers believe that have.SBJV that to.go to the strike
 'Few workers believe we should go on strike.' (Mejías-Bikandi 1998: 945)

More recently, however, Faulkner (2021) has proposed that the indicative is used to mark new/contrastive information and the subjunctive is the default/elsewhere case. In other words, the only mood that is restricted in its distribution by information status is the indicative and the subjunctive will be possible in contexts where the information is both new or old. In order to prove this, Faulkner (2021) carried out an experiment and found that while the subjunctive was rated as acceptable across contexts – 3.44 out of 4 on the Likert Scale, see Table 7.2 – the indicative obtained statistically higher ratings when used in environments in which the information was new – 2.54 vs. 2.15, respectively – (see Chapter 2 and Faulkner 2021 for more information on this).

Table 7.2: Acceptability ratings results from Faulkner (2021: 76–78) for negated epistemics.

Mood	Uninformative	Informative	Context-Free
Ind	2.15(2 median)	2.54(3 median)	2.55(4 median)
Sbjv	3.44 (4 median)	3.44 (4 median)	3.53 (4 median)

These results provide empirical support for analyzing the subjunctive as a default mood, as it was equally acceptable across constructions, but raises questions regarding how to analyze the indicative. Although there was a significant raise in the acceptability of the indicative across environments, with a slight preference for the indicative in contexts in which the information was new, the effect was

rather small (less than 0.5 difference in means), and the indicative never reached the same level of acceptance as the subjunctive, irrespective of context (2.5 vs 3.5).

A semantic analysis in which the indicative is used to mark that the information is new would leave unexplained why the indicative is always rated lower than the subjunctive (about 1 point difference). This is so because the analysis predicts that at least in those contexts where the information is new the indicative should be as grammatical as the subjunctive.¹¹

Under the competing grammars approach presented in this book, however, the fact that the subjunctive is always rated as more acceptable after negated verbs like *believe* is expected given that in these environments it has been argued that G_2 (syntactic selection) is winning, and this grammar only allows the subjunctive in negative declarative clauses. The (small) effect of information status on the indicative could potentially be understood as a consequence of the competing grammars trying to specialize for a given meaning in diachrony, but not being successful. This behavior also seems to be characteristic of other cases of competing grammars, in which the competition undergoes an intermediate stage in which information status plays a relevant role for some time, but ends up disappearing due to one of the grammars taking over the other. This last idea is explored in more detail in the following subsection.

7.2.2 Competing grammars and information status

Recent work investigating cases of Grammar Competition has shown that for several linguistic phenomena, information status (IS) seems to play a (transient) role during the competition process. This has been argued to be at least the case for Jespersen's cycle, head directionality (e.g., OV vs VO) and the emergence of Differential Object Marking (DOM). The following sections explain these phenomena and the role of information status in more detail.

7.2.2.1 Jespersen's cycle and information status

The term Jespersen's Cycle is used to describe a series of common diachronic changes in the expression of negation (Jespersen 1917, Dahl 1979). In STAGE I of the cycle negation is expressed via a pre-verbal particle, in STAGE II both a

¹¹If one follows the principle of Maximize Presupposition! (Heim 1991), the account would actually predict that the indicative should be preferred (and thus be more grammatical than the subjunctive) in such contexts. Importantly the account never predicts the subjunctive to have a higher acceptance than the indicative in informative contexts.

preverbal and a posverbal element become obligatory to mark negation and in STAGE III the postverbal element is the only one that is obligatory, see (35). The most frequently cited example to illustrate this change is French. In Old French the pre-verbal particle *ne* was used to mark negation (STAGE I). Then, the presence of post-verbal *pas* became obligatory (STAGE II), and in modern colloquial French pre-verbal *ne* is no longer obligatory, *pas* being capable of expressing negation on its own, see (36) (Dahl 1979). Similarly in the history of English the pre-verbal particle *ne* was used to mark negation (1150–1300). It then became obligatory to have both *ne* and *not* (1150–1400), and in the last stage *not* could be used alone (1300–1500), see (37) (Wallage 2008).

- (35) a. STAGE I. NEG V (Jespersen's Cycle)
 b. STAGE II. NEG V NEG
 c. STAGE III. V NEG
- (36) a. Jeo **ne** di (Cycle in French)
 b. Je **ne** di **pas**
 c. je dis **pas**
- (37) a. we **ne** mugen þat don (Cycle in English)
 b. I **ne** may **nat** denye it
 c. I know **nat** the cause

These changes have been frequently analyzed as a case in which three grammars (one for each stage in the cycle) are competing with each other (Wallage 2008, Walkden & Breitbarth 2019). Important for our purposes is that this competition between grammars has been shown to be affected by information status (see for example Schwenter 2006). For example, in the case of Medieval French the change from STAGE I to STAGE II has been argued to be characterized by the loss of a pragmatic distinction between plain negation and reinforced negation (Hansen & Visconti 2009). That is, plain negation (*ne*) was used in contexts where the information was new, and emphatic negation (*ne ... pas*) when the information was old, but this distinction became blurred leading to STAGE II in which emphatic negation was used across contexts.¹²

In the case of Early English Wallage (2013) shows that although *ne... not* is favored when the propositions present old information and *ne* is favored when

¹²There is to my knowledge no study exploring whether information status also affects the last stage in the cycle in French. That is whether the spread of *pas* as a marker of negation is also affected by whether the information is old or new.

the information is discourse- new, there does not seem to be any statistically significant pragmatic bleaching (at least during STAGE I to STAGE II). This, however, does not completely rule out the effect of pragmatic factors as the author acknowledges:

the transition from stage two *ne...not* to stage three *not* does not simply involve loss of the morpheme *ne*, but also a change in the discourse-functional constraints on *not* when it begins to appear independently of *ne*. [...] If this is right, more detailed research is required to provide a more precise characterisation of the role of pragmatic change or loss of pragmatic marking in the transition from stage two to stage three of the Jespersen Cycle. However, the remainder of this paper will focus particularly on the role of pragmatic factors in the transition from stage one *ne* to stage two *ne...not*. (Wallage 2013: 13)

In any case, what this work seems to propose is not that pragmatic factors are not involved in the cycle (there is a pragmatic component in the sense that one variant is preferred in one pragmatic context over the other), but rather that pragmatic bleaching might not be the key internal trigger of the cycle.

From these two case studies it can thus be concluded that pragmatic factors play a relevant role in the cycle – although there is more controversy regarding whether the pragmatic effect is the trigger of the change or is a consequence of an underlying syntactic change. The next subsection will present another linguistic phenomenon in which pragmatic factors have also been argued to play a relevant role in the competition between the variants.

7.2.2.2 Head directionality and information status

Another common example of language change in which it has been proposed that the change could be understood in terms of competing grammars is that of word order, and more specifically changes in the ordering of a given head (V, T, etc.) and its complement. For instance, looking at English, the language changed from allowing both OV and VO orders during the OE period (38), to only allowing VO order in the present-day language, see (39) (Taylor 2005, Struik & van Kemenade 2020).

- (38) a. wel bid þæm þe mot | æfter deaddæge / **drihten** *secean*
 well is that who may after death.day Lord seek
 ‘well is it with him who after his death-day may seek the Lord’
 (cobeowulf, 8.183.150 in Taylor 2008: 127)

- b. þeah þe he ne meahte / on mere *drifan* | *hringedstefnan*
 though that he NEG might on sea drive ring-prowed.ship
 ‘though he might not steer on the sea, the ring prowed ship’
 (cobeowulf, 36.1129.928 in Taylor 2008: 128)

- (39) a. * I have **the book** *seen* (OV)
 b. I have *seen* **the book** (VO)

Proposals aiming to explain the possibility of both orders range from underlying OV order with rightward movement in OE, underlying VO order with leftward movement, or competition in underlying base order OV and VO (see Struik & van Kemenade 2020 for a recent overview of the different hypotheses).

(40) OV + rightward movement

(41) VO + leftward movement

(42) OV and VO

Whether the alternation is derived via movement or through a difference in the base position of the objects, the general consensus is that IS played a defining factor in the position of the object: OV in general was more likely to involve an object with a discourse-old referent and VO with a discourse-new referent (Taylor & Pintzuk 2012).¹³ That is, while the two grammars competed, IS had an effect on whether one grammar or the other was used. The IS effect is not restricted to English but seems to have played a major role in other Germanic languages such as Old Saxon, Old High German and Dutch (Struik 2022, Walkden 2014).

Within the Romance languages, similar observations have been made for the loss of OV orders (Ledgeway 2012, Lahousse & Lamiroy 2012, Zaring 2010, Eide 2014, Danckaert 2017, Pujol i Campeny 2019, Wolfe 2021). For example, in Old French SOV orders were possible to a certain extent, see (43). In the transition from this system to the modern SVO one, IS has been suggested to play a role (Zaring 2010).

- (43) Li quens Rollant Gualter de l’Hum apelet. (Old French)
 the count Rolland Walter de l’Hum calls
 ‘Count Roland calls over Walter de l’Hum.’
 (Roland 803; M-N, 75, in Zaring 2010: 4)

¹³Struik & van Kemenade (2020) argue that the situation is more complex and although the OV order tends to be new, VO orders show a mixture of both new and old information.

7.2.2.3 Differential object marking and information status

The last linguistic phenomenon that will be described in this section is Differential Object Marking (DOM). This is a phenomenon whereby only a subset of objects require additional grammatical marking. For example, sentence (44) shows that in Spanish the object *la niña* ‘the girl’ needs to be accompanied by *a*, whereas the object *la casa* ‘the house’ cannot carry such a marker. Whether an object is marked is determined by semantic factors such as animacy, definiteness, etc. (Bossong 1991, von Heusinger 2008, Hill & Mardale 2021).¹⁴

- (44) a. Veo *a la niña*. (Spanish)
 see DOM the girl
 ‘I see the girl.’
 b. Veo *la casa*. (Spanish)
 see the house
 ‘I see the house.’

DOM is also well attested across the Romance languages and can be found in Asturian, many Italo-Romance dialects, and even some Gallo-Romance varieties (Gerards 2023). An example from Central Asturian is provided in (45a).

- (45) a. Quiero mucho *a esti* home. (Central Asturian)
 want much DOM this man
 ‘I love this man very much.’ (Corral Esteban 2020: 103)

While some of these systems have been argued to be affected by semantic factors, others have been claimed to rely mostly on information structure (Dalrymple & Nikolaeva 2011.) In some cases, the emergence of these systems has been argued to be preceded by an intermediate stage in which the presence of the additional marker is determined by information status. For instance, Iemmolo (2010: 259) shows that in Sicilian DOM was originally used to mark IS and it was only later that more semantic notions started to determine the distribution of the phenomenon.¹⁵

¹⁴There is a vast literature on DOM in Romance, to which this short overview cannot do justice. See Kabatek et al. (2021), Irimia & Pineda (2019), Ledgeway (2022) Mardale & Irimia (2023), among many others, on the diachronic and cross-linguistic variation of this phenomenon.

¹⁵See Tal et al. (2022) for an experimental exploration of why DOM may emerge first as systems to mark IS.

When the DOM system begins to loosen its link with information structure, the informational status of the component loses importance as the grammaticalization process goes on, and DOM becomes mainly regulated by merely semantic factors such as animacy and definiteness (this happens because topical objects are usually definite) (Iemmolo 2010: 259)

Further evidence that DOM systems may first start to be used as markers of IS comes from Romance languages which usually do not display DOM, but do when the object has been dislocated or fronted (D'Alessandro 2021). This is exemplified in the case of Italian in examples (46).

- (46) a. A me non mi ha invitato.
 DOM me NEG me has invited
 ‘As for me, s/he hasn’t invited me.’
 b. * Me non mi ha invitato.
 me NEG me has invited
 (D'Alessandro 2021: 320)

7.2.3 Discussion

In conclusion, this section has explored several linguistic changes which had an intermediate stage in which the phenomenon was constrained by IS, namely: Jespersen’s cycle, head directionality, and DOM.¹⁶ The claim is not that all linguistic changes need to undergo a stage in which IS plays a central role in the distribution of the phenomenon, but rather that it might be a frequent stage due to strategies used by learners to cope with a system with several grammatical options. More specifically, the claim is that due to the Principle of Contrast (Clark 1987) – whereby during the acquisition process children assign contrasting forms different meanings – when two grammatical possibilities emerge, the learner will try to assign a different usage to each of these. In the absence of any other salient linguistic feature, a common strategy could be to assign one grammar as marking that the information is discourse-old and the other as discourse-new. This difference may not be salient enough or the community may not converge on the direction of specialization¹⁷ and in the long term it would tend to disappear, leading to the extinction of one of the grammars. This would explain why in

¹⁶Other diachronic changes also seem to follow this general pattern. For example, the loss of V2 in French and Venetan (Wolfe 2024).

¹⁷Given the continuous nature of information status another possibility would be that full specialization cannot occur, as it would require a categorical dimension (p.c. George Walkden).

many cases of grammar competition there seems to be a stage in which IS has an effect.

If the explanation given to the previous cases could be extrapolated more generally to any case of grammar competition, then one would have a natural way of explaining the effect of IS (and its small size) on mood that was reported in Faulkner (2021). Given that the competition between grammars is ongoing in present-day Spanish the effect of IS is still visible in the language. However, because the competition is at an advanced stage, the effect of IS is losing its force, thus explaining why the effect is small in size.

7.3 Strong negative polarity items

7.3.1 Strong negative polarity items and mood

As was described in Chapter 2, the two main analyses of the Polarity Subjunctive were designed to account for the distribution of strong NPIs in Spanish.¹⁸ A strong NPI is a negative element which is licensed only in anti-additive environments such as matrix negation (see Zwarts 1998 among others).¹⁹

In Spanish examples of strong NPIs are words such as *en años* ‘in years’, *hasta las 5* ‘until 5’, or *una gota de vino* ‘a single drop of wine’, see examples (47)-(48).

- (47) * El tren va a llegar **hasta las diez**.
the train goes to arrive until the ten

- (48) El tren **no** va a llegar **hasta las diez**.
the train not goes to arrive until the ten
‘The train won’t arrive until ten.’

The key observation in the literature is that when the indicative is selected in the embedded clause the presence of a strong NPI is degraded, see (49).

¹⁸The distribution of n-words is also affected by mood, but this topic will not be covered in this chapter.

¹⁹An anti-additive environment is one where the following entailments are satisfied:

1. $f(X) \vee f(Y) \implies f(X \wedge Y)$
2. $f(X \vee Y) \implies f(X) \wedge f(Y)$
3. $f(X) \wedge f(Y) \implies f(X \vee Y)$

(49) Spanish strong NPIs: (Rivero 1971: 310)

- a. *No creen que probaste **gota de vino**. $\leadsto$ p is true.
not believe that try.IND drop of wine
'They don't believe you tried a single drop of wine.'
- b. No creen que probaras **gota de vino**.
not believe that try.SBJV drop of wine
'They don't believe you tried a single drop of wine.'

The Neg-raising account argues that when the indicative is used, negation is interpreted high (non-neg-raising); whereas if the subjunctive is used, negation is interpreted low (neg-raising). If this characterization of the data is correct, one would have a direct way of explaining the ban on NPIs with the indicative mood. A non-neg-raising environment is not anti-additive (Gajewski 2005), and hence it would not constitute the right environment to license strong NPIs.

However, as was shown in Chapter 2, this characterization of the phenomenon has some problems. One of them is that corpus data show that both the indicative and the subjunctive can be used with a neg-raising inference (i.e., negation is interpreted low). This point is also corroborated by recent experimental data which found that the neg-raising inference can indeed be licensed with both indicative and subjunctive embedded verbs (Montero et al. 2025). As part of the experiment, participants were asked whether they could get a Neg-raising interpretation in three different constructions: non-neg-raising (NNR), neg-raising (NR) and neg-raising with a strong NPI (NR + NPI), each environment with the verb in the indicative and the subjunctive. The results can be found in Figure 7.4. The *x*-axis indicates the different constructions, the *y*-axis the proportion of Neg-raising inferences obtained and the color indicates the mood in the embedded verb (blue for the subjunctive and yellow for the indicative).

As can be seen, the non-neg-raising predicates had a very low proportion of Neg-raising inferences, indicating that participants understood the task. Neg-raising constructions (with and without the NPI) had a higher proportion of Neg-raising inferences (about 75–80% of the time). Importantly, the statistical analysis found a main effect of sentence type ($\chi^2 = 249.51$, $p < 0.001$), but no main effect of mood ($\chi^2 = 0.41$, $p = 0.52$) nor an interaction between sentence type and mood ($\chi^2 = 2.46$, $p = 0.29$) in the proportion of Neg-raising inferences. Thus, the correlation proposed in the literature (ind $\rightarrow$ NNR; sbjv $\rightarrow$ NR) could not be replicated in an experimental setting, which calls then for a different explanation for the interaction between mood and NPIs.

Another option explored in the literature by Homer (2008) for Italian and by Quer (1998) for Catalan is that certain presuppositions can disrupt the licensing of

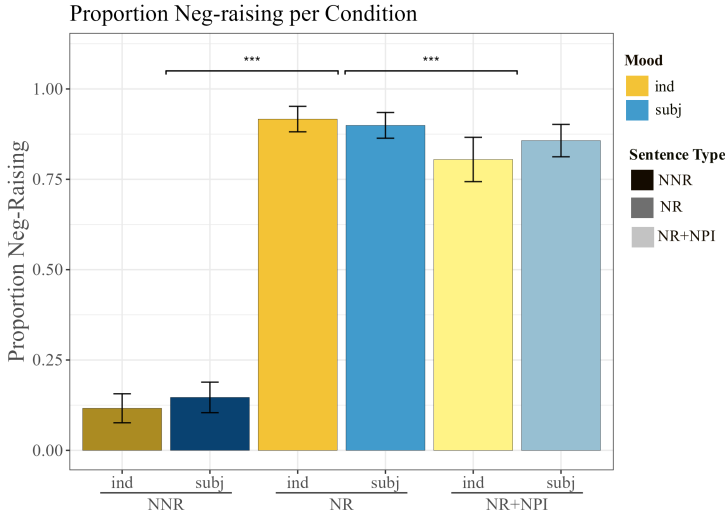


Figure 7.4: Experimental Results from Montero et al. (2025) regarding the proportion of Neg-raising inference across three constructions: non-neg-raising (NNR), neg-raising (NR) and neg-raising with a strong NPI (NR + NPI); and two moods: indicative vs. subjunctive

strong NPIs. For example, in Italian Homer (2008) has argued that the indicative carries a presupposition whereby the speaker is committed to the truth of the embedded proposition (even in non-factive contexts), see (50).

- (50) a. Gianni non pensa che Maria è andata a Parigi.
 Gianni NEG thinks that Maria be.IND gone to Paris
 ‘Gianni doesn’t think that Mary has been to Paris.’
 $\leadsto$ The speaker believes that Maria has been to Paris
- b. Gianni non pensa che Maria sia andata a Parigi.
 Gianni NEG thinks that Maria be.SBJV gone to Paris
 ‘Gianni doesn’t think that Mary has been to Paris.’

This presupposition would block the licensing of (strong) NPIs in Italian, as the context would be no longer anti-additive, see (51). If one takes into account non-truth conditional content (NT), (51a) does not entail (51b).

- (51) Anti-additive: $F(x \vee y) \Leftrightarrow F(x) \wedge F(y)$
- a. $F(x \vee y)$: Anna didn’t think that Adam or Miles would come.IND
 NT: Anna thought that neither Adam nor Miles would come $\wedge$ the speaker believes that Adam or Miles would come

- b. $F(x) \wedge F(y)$: Anna didn't think that Adam would come and Anna didn't think that Miles would come
 NT: Anna didn't think Adam would come $\wedge$ the speaker believes Adam would come $\wedge$ Anna didn't think that Miles would come $\wedge$ the speaker believes Miles would come.

Extending this analysis to the Spanish data, however, is not straightforward given that experimental evidence shows that with non-factive predicates there are possible interpretations without the presupposition of speaker commitment, even when the verb is in the indicative (Montero & Romero 2023). An alternative possibility is that there is also another presupposition which blocks the licensing of NPIs. This is the option explored in Montero et al. (2025). This work proposes that the indicative may carry the presupposition that the proposition p has been discussed in the previous discourse ($p \in CPS$)²⁰ and that this presupposition is the one that blocks the licensing of NPIs. Similar to the previous case, the use of the indicative would create an environment which is no longer anti-additive as (52b) $\neq$ (52a).

- (52) Anti-additive: $F(x \vee y) \Leftrightarrow F(x) \wedge F(y)$
 a. $F(x \vee y)$: Anna didn't think that Adam or Miles would come.IND
 NT: Anna thought that neither Adam nor Miles would come $\wedge$ it was mentioned in the previous discourse that Adam or Miles would come.
 b. $F(x) \wedge F(y)$: Anna didn't think that Adam would come and Anna didn't think that Miles would come
 NT: Anna didn't think Adam would come $\wedge$ it was mentioned in the previous discourse that Adam would come $\wedge$ Anna didn't think that Miles would come $\wedge$ it was mentioned in the previous discourse that Miles would come.

The next section briefly covers to what extent this proposal is compatible with the competing grammars approach.

7.3.2 Strong negative polarity items and the competing grammars approach to mood

The competing grammars approach developed thus far has proposed that there might have been three grammars in competition: G_0 (hedging semantic mood), G_1 (referential semantic mood) and G_2 (syntactic mood). In the case of non-factive

²⁰CPS stands for what Portner (2009) calls the Common Propositional Space.

contexts it was argued that G_0 selected for the indicative and the subjunctive (with certain semantic constraints), G_1 selected for the indicative and G_2 selected for the subjunctive mood. The interaction between the indicative mood and NPIs must then necessarily be caused by a grammar in which mood is semantic, as only G_1 and G_0 select for the indicative mood to begin with. This is already a welcome result given that there is agreement in the literature that the distribution of strong NPIs is constrained semantically rather than syntactically (Gianollo 2016).²¹

In the previous chapters it was further argued that G_0 lost the competition with the other grammars by 1400 and G_1 and G_2 were competing with one another during the subsequent periods. This reduces the source of the ungrammaticality to the G_1 grammar, which is based on the referential mood system proposed by Schlenker (2005). The proposal developed in Montero et al. (2025) whereby the indicative carries a presupposition which blocks the strong NPIs is compatible with Schlenker (2005)'s account insofar as in this system, subjunctive mood is a default and the indicative carries a presupposition.²² Thus, the competing grammars approach is compatible with the proposal described in the previous section and could provide a possible avenue to explain the ungrammaticality of NPIs with embedded indicative mood.

7.4 Conclusion

This chapter has explored three other effects which interact with Polarity Mood in Spanish: the effect of imperative clauses, the effect of information status and the distribution of NPIs. The effect of imperative clauses can be characterized as a result of polarity mood being a dependent mood (similar to dependent accusative case in Finnish which is also affected by imperative clauses). The effect of information status reported in Faulkner (2021) as well as its size might be a result of general strategies used by language learners when acquiring multiple grammars. That is, the effect could be explained as a result of speakers attempting to specialize across one linguistic domain, but not being successful (leading to the extinction of G_0 and G_1 in non-factive contexts). Lastly, the ban on having an NPI with an embedded indicative verb might be a remnant from grammar G_1 : in this grammar the indicative used to carry a presupposition that blocked the licensing of NPIs, as proposed in Homer (2008) or Montero et al. (2025). Although

²¹This is not the case, for example, for n-words.

²²The account, however, raises the question of why there might be two presuppositions attached to the indicative ($p \in CS$ vs. $p \in CPS$).

the theories behind these explanations need further work and refinements, the intention of the chapter was to show possible lines of argumentation in which the effects could be explained under the competing grammars approach.

8 Conclusion and future directions

Goals and research questions. The aim of this book was to investigate the phenomenon of Polarity Mood (PM)– mood selection triggered by a negative operator, see (1)– in Spanish, and how it has changed in diachrony.

- (1) Auri **no** dijo que el gato **iba/fuera** a volver.
Auri not said that the cat GO.PST.IND/PST.SBJV to return
‘Auri didn’t say that the cat was going to come back.’

In particular, it aimed to investigate the interaction of PM with five other linguistic phenomena, to wit:

1. Verb class (see Chapter 5): if the matrix verb is semi-factive, the mood alternation conveys a presuppositional difference with regard to speaker commitment; but if the verb is non-factive such a presuppositional difference is not as clear cut – it could be triggered, but is not as salient as with semi-factive verbs – (Montero & Romero 2023).
2. The First Person Effect (see Chapter 6): if the matrix subject is first person and the matrix verb is in present tense the use of the indicative in the embedded clause is degraded (Quer 1998).
3. Clause type (see Chapter 7, Section 7.1): if the matrix clause is an imperative clause, the use of the subjunctive in the embedded clause is degraded (Borgonovo 2003).
4. Information status (see Chapter 7, Section 7.2): the indicative is perceived as more grammatical if the information presented in the embedded clause is discourse-new (Faulkner 2021).
5. Strong NPIs (see Chapter 7, Section 7.3): if the embedded clause contains a strong NPI, the use of the indicative is degraded (Rivero 1971).

New Empirical Observations. In order to better understand these effects and when they emerged in the Spanish language, a corpus study was conducted using the CDH corpus (Real Academia Española 2013).

8 Conclusion and future directions

With regard to the effect of verb class, it was shown that the two verb classes behaved differently also in diachrony, see Figure 8.1. With semi-factive verbs, there was a decrease in the use of the indicative in Construction FP (with a first person subject and a present tense matrix verb) starting in 1400, but the proportion of the indicative mood remained stable at about 75% in Construction O (all other environments). With non-factive verbs, there was an up-and-then-down behavior in the use of indicative with a peak in 1600. Starting in 1700, the use of the indicative across both constructions started to decline.

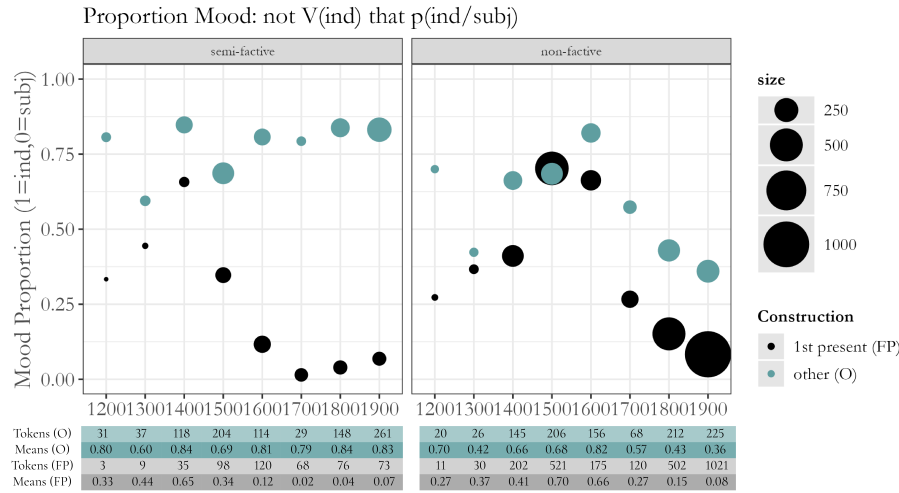


Figure 8.1: Mood selection with a matrix indicative verb.

With regard to the First Person Effect, it was shown that as early as 1400 there was a difference across constructions: Construction FP took significantly more subjunctive (about 25% more) than Construction O; but this difference changed depending on the verb type, see Figure 8.2. With semi-factive verbs, the difference across constructions increased as time passed by. With non-factive verbs, the difference was reduced during 1500–1600, but then the initial difference was restored.

Lastly, with regard to the effect of clause type, it was shown that even if imperative clauses could take both the indicative and the subjunctive in the past, as time passed by the use of the subjunctive decreased and by 1700 most tokens were already in the indicative, see yellow dots in Figure 8.3.

New Theoretical Proposal. In order to account for the different effects as well as their emergence in the history of Spanish, a new proposal for mood selection was

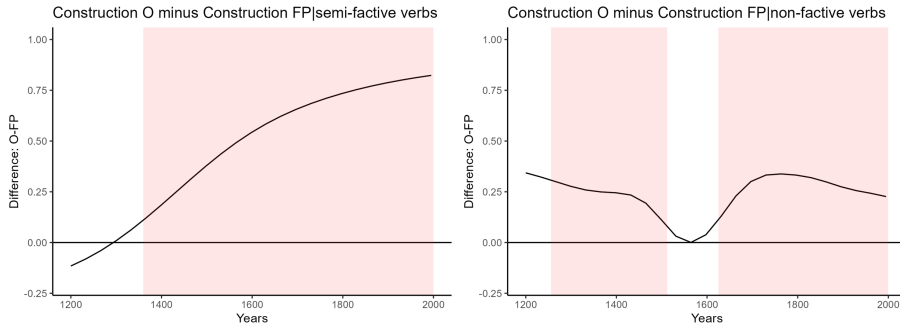


Figure 8.2: Difference curve between Construction FP and Construction O.

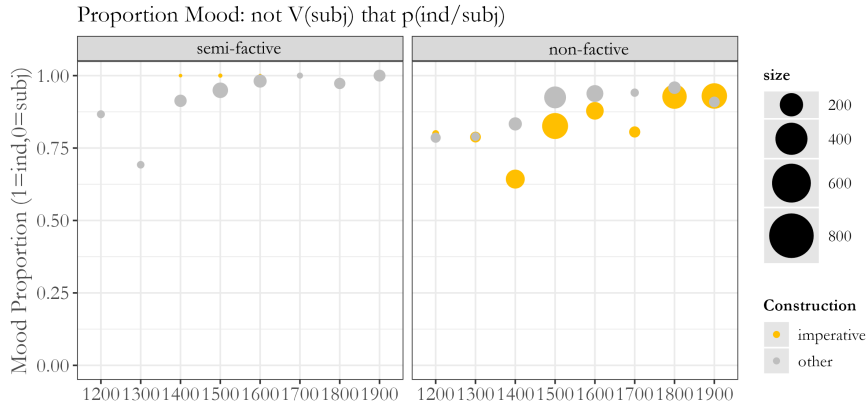


Figure 8.3: Mood selection with a matrix subjunctive verb.

put forth. Drawing inspiration from the competing grammars framework (Kroch 1989, Yang 2000) and the theory of Variational Specialization (Wallenberg 2019), it was proposed that there were three different selectional mechanisms in the history of Spanish: G_0 in which the subjunctive mood behaves as a hedging device and indicates the degree of commitment – similar to mood in present-day Bulgarian (Smirnova 2011); G_1 in which the indicative is used to express that the world of evaluation lies in some accessible Context Set (G_1), see Schlenker (2005); and G_2 in which mood is analyzed as a purely syntactic marker (similar to Genitive case in Lithuanian). These three grammars were in competition, but rather than tracking their probabilities in the language as a whole, the probabilities of the grammars were tracked differently across contexts. This led to a situation in which grammar G_1 won when the matrix verb was semi-factive, see the im-

age on the right in Figure 8.4, and grammar G_2 when the verb was non-factive. The competition in non-factive contexts, however, took longer to be resolved, see the image on the left in Figure 8.4, and as a consequence, in the present-day language there might still be a mixture of grammars G_1 and G_2 , when the verb is non-factive.

Under this account, the different behavior across verb classes can be explained as a result of different grammars winning in different contexts. With semi-factive verbs, G_1 won out (and hence the alternation is semantically driven). With non-factive verbs, grammar G_2 (syntactic mood) seems to be winning out (which would result in the loss of a semantic distinction after these verbs). The First Person Effect is caused by two different triggers: in the case of non-factive verbs, the difference is a result of the initial grammar G_0 showing a preference for the subjunctive mood if the matrix predicate was first person and present tense, as hedging devices are might be more frequent in these constructions (Koev et al. 2021). Given that the speed of change is the same across constructions –disregarding the additional change in complementizer drop – the initial difference was preserved in the system. In the case of semi-factive verbs, the initial difference was increased as a result of grammar G_1 winning out, as in this grammar having an embedded indicative leads to a contradiction in Construction FP but not in Construction O. The effect of imperative clauses can be derived if one assumes that grammar G_2 , syntactic mood, behaves similarly to other dependent cases, as in Marantz (1991). That is, Polarity Subjunctive can only be licensed if there is another unvalued verbal element with which to agree. If the matrix verbs is not unvalued (as in imperative clauses and other clauses in which the matrix verb carries the subjunctive morphology), the embedded verb wouldn't have an unvalued feature with which to enter into a dependent mood relation, and the prediction is that Polarity dependent mood won't be licensed. The effect of information status can be understood as a result of the competing grammars trying to specialize but failing to do so: either because at the population level there were opposing tendencies or because the system tried to specialize along a non-categorical dimension, thus precluding full specialization from happening (Wallenberg 2019). Lastly, the effect of NPIs can be seen as a remnant of the G_1 grammar, as in this grammar the indicative carries a presupposition which blocks the licensing of NPIs, as proposed in Montero et al. (2025).

It was, then, shown that the three-grammar competition proposal is feasible on theoretical grounds, and an estimate of which values would make the proposal correct was discussed. With semi-factive verbs, G_1 would have won out over all other grammars if condition (2) was satisfied (Kauhanen 2024).¹

¹This is only a sufficient condition and G_1 may have won even if this condition was not satisfied.

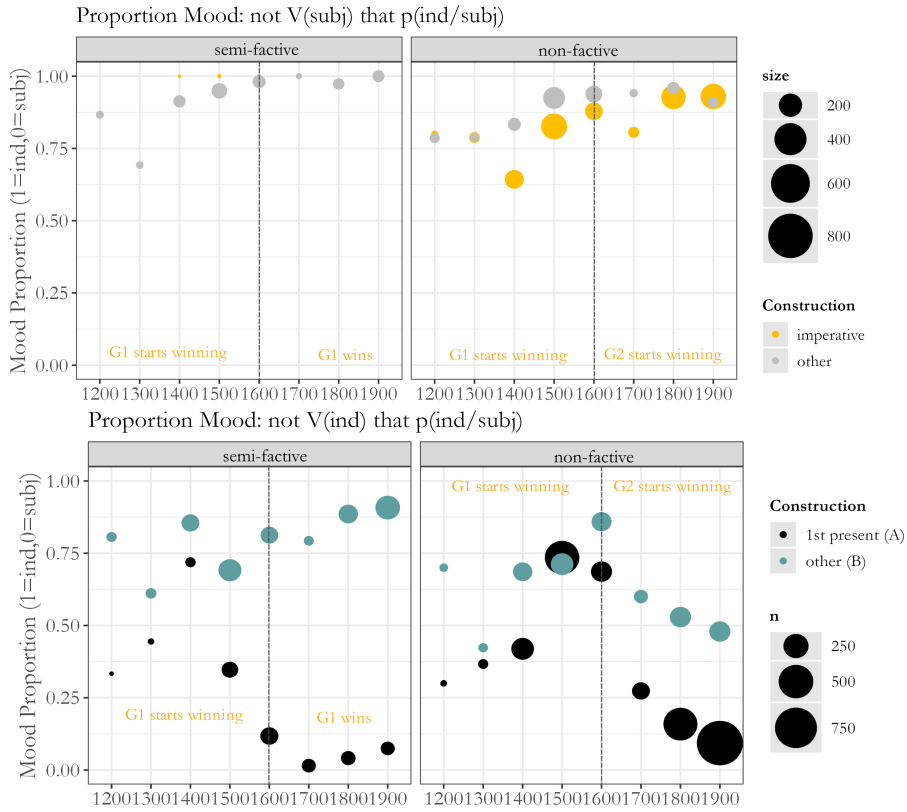


Figure 8.4: Three-grammar competition: proposal for the negated verbs studied in this book.

$$(2) \quad a_{12} + a_{10} < \frac{1}{\frac{1}{a_{21}} + \frac{1}{a_{01}}}$$

With non-factive verbs, G_1 and G_2 are disjoint and as there is no advantage of one grammar over the other, the model predicts stability between the two grammars. However, if one takes into account the contact situation present in the Peninsula at the time, the non-monotonic behavior of these predicates could be explained. During 1200–1500 there was contact in the Peninsula which gave an advantage to G_1 (semantic system), but after contact G_2 (syntactic system) gained an advantage (as children may struggle less to learn a purely syntactic system).

8 Conclusion and future directions

In conclusion, the proposed analysis is capable of explaining five effects which have been argued to affect PM, see Table 8.1.

Table 8.1: Summary of the effects and approaches of Polarity Mood.

Approaches/effect	1st Person	NPI	Contrast	Verb Class	Clause Type
Presuppositional	✓	✓	–	–	–
Neg-raising	–	✓	–	–	–
Information status	–	–	✓	–	–
Competing Grammars	✓	✓ /?	✓	✓	✓

Future work. Although the proposed theory is capable of explaining a wide range of effects, there remain many important aspects of Polarity Mood that could be further investigated.

First, the three-grammar competition proposal, although feasible in theoretical grounds, has not been proven. In order to do that, one would have to calculate the advantages of each of the grammars, by carrying corpus work in other languages (e.g., using corpus data from Bulgarian to calculate the advantage of grammar G_0); and the contact situation would have to be studied in more depth in order to corroborate the idea that this is what caused the non-monotonic behavior of non-factive predicates.

Second, PM is a very intricate phenomenon and there are several other effects which have an impact on it, but which had to be left out of the book. For example, interrogative embedded clauses behave differently than declarative embedded clauses insofar as they only allow for the indicative mood in the embedded clause, see (3).

(3) Present-day Peninsular Spanish

Auri no sabía qué pensaba/*pensase el misterioso gato.

Auri not knew what think.PST.IND/PST.SBJV the mysterious cat

‘Auri didn’t know what the mysterious cat thought.’

The effect of indirect questions is also present in other languages like French (Amsili 2014), see (4), but it was not present in Old Spanish (Nieuwenhuijsen 2016), see (5), nor is it present in some dialectal varieties of present-day Spanish (Alonso Pascua 2023). Moreover, other Romance languages, such as Italian, don’t show this constraint either (Squartini 2010), see (6). The constraint, thus, emerged in French and (some dialects of) Spanish, but not in Italian.

(4) **French:** (Amsili 2014: 12)

Je me demande s' il est/*soit malade.
I me wonder if he be.PRS.IND/PRS.SBJV sick
'I wonder if he's sick.'

(5) **Old Spanish:**

De mí non sé qué sea.
of me not know what be.PRS.SBJV
'I don't know what will be of me.'
(1398, *Poesías (Cancionero de Baena)*, *Arcediano de Toro*, CDH)

(6) **Italian:** (Squartini 2010: 246)

- a. Chiedono se sia partito.
ask if be.PRS.SBJV left
'They asked whether he has left.'
- b. Non sapevo cosa significasse.
not knew what mean.PST.SBJV
'I didn't know what it meant.'

There also seems to be an interaction with matrix interrogative clauses. As was discussed in Chapter 5, there are some interrogative clauses in which the alternation is not licensed and only the indicative is allowed, see (7).

- (7) ¿No creías que ya no *estuvieras/estabas enamorada de Dani?
Not believe that no.longer not bePST.SBJV/PST.IND in.love of Dani
'Didn't you think you were no longer in love with Dani?'

Furthermore, in the book the behavior of strong NPIs could not be studied from a diachronic perspective, because these elements are too infrequent in the corpus. However, apart from strong NPIs, n-words (which are more frequent in corpora) have also been argued to interact with mood (Bosque 2012). For instance, the n-word *ningún* in postverbal position is only licensed in sentence (8) if the embedded verb has subjunctive morphology. This constraint wasn't at play at earlier stages of the language, as a sentence (9) taken from the CDH corpus demonstrates. In this sentence the n-word *ninguno* is licensed despite the fact that the embedded verb is in the indicative. Moreover, there seems to be some dialectal variability, as examples with an n-word and the indicative are encountered as late as the 1950s in the corpus. For example, in the work of Pío de Baroja (who was born in the Basque Country) there are many instances where this occurs, see (10).

- (8) No sabíamos que *estaba/estuviera metido en **ningún** lío.
not knew that be.PST.IND/PST.SBJV tucked in any trouble
'We didn't know that s/he was in any trouble.' (Bosque 2012: 15)
- (9) aunque ahora no creo que ay **ninguno** en la tierra
although now not believe that is.IND any in the earth
'Although now I don't believe there is any on Earth'
(1566, Pedro Arias de Benavides, *Secretos*)
- (10) no creo que había por entonces **ningún** cuadro en el Louvre
not believe that have.IND by then any painting in the Louvre
'I don't believe there was at that time any painting in the Louvre'
(1944–1949, Pío de Baroja, *Desde la última vuelta del camino. Memorias*, CDH)

Moreover, the present book has focused on the interaction of mood with negation but mood selection has also been shown to interact with tense. For example, in Icelandic the verb *know* selects the indicative mood in present tense but allows for both the indicative and the subjunctive in the embedded clause when the verb appears in past tense. In Italian, negated semi-factives allow for mood alternation in the embedded clause only if the embedded verb is in past tense (Giorgi & Pianesi 1997).

- (11) **Italian:** (Giorgi & Pianesi 1997: 225)
- a. Mario non sa che Gianni è/*sia arrivato.
Mario not know that Gianni be.PRS.IND/PRS.SBJV arrived
'Mario doesn't know that Gianni has arrived'
- b. Mario non sapeva che Gianni partiva/partisse l'indomani.
Mario not knew that Gianni leave.PST.IND/PST.SBJV the.day.after
'Mario didn't know that Gianni had left the day after.'

Lastly, the book has concentrated on Spanish, but it would be fruitful to explore the phenomenon and its evolution in other languages, and determine whether the observed differences could be explained in terms of the three-grammar competition approach.

For example, in Romanian negation does not affect mood selection when the matrix verb is a predicate of certainty, such as *be sure* (Farkas 1992). In affirmative and negative clauses this predicate selects for the indicative mood, see (12). In Spanish, in contrast, this predicate takes the indicative in affirmative clauses,

but selects for the subjunctive when negated, see (13). A possible explanation would be that while in Spanish G_2 (syntactic mood) has won with non-factive verbs (as suggested in this book), in Romanian G_1 (semantic mood à la Schlenker 2005) might have won the competition in these environments, hence the reason why the indicative is selected in both affirmative and negative clauses. Future research could investigate whether other cases of cross-linguistic variability could be explained via the three-grammar competition approach.

(12) **Romanian:** (Farkas 1992: 72)

- a. E sigur că a venit Ana.
is sure that(IND) has come Ana
'It sure that Ana has come.'
- b. No e sigur că a venit Ana/ *să fi venit Ana.
not is sure that(IND) has come Ana/ SBJV PAST come Ana
'It's not sure that Ana has come/came.'

(13) **Spanish:**

- a. Es seguro que va a venir.
is seguro that go.IND to come
'It's sure that they(sg) will come.'
- b. No es seguro que *va/vaya a venir.
not is sure that go.IND/SBJV to come
'It's not sure that they(sg) will come.'

In conclusion, the proposed competing grammars approach to mood provides new insights into the phenomenon of PM and its development in the history of Spanish, paving the way for new analyses in the domain of cross-linguistic variability.

Appendix A: (Un)ambiguous sentences

In the annotation of the corpus, as explained in Section 5.1.2, embedded clauses were divided into ambiguous and unambiguous declarative clauses. The argument for this is that if indeed in some cases the ambiguous ones are instances of interrogative clauses, then we would not be looking at true embedded declarative clauses, which behave differently from interrogative ones.

A comparison of including ambiguous sentences in the results or not is illustrated in Figure A.1. As can be seen, this choice did not have a big effect on the overall results, with the possible exception of first person constructions with a semi-factive verb. For all other constructions the difference between including ambiguous sentences is below 10%. For first person constructions the difference is of 10–20%. Importantly, the overall diachronic tendencies seem to be maintained, and the only construction where we see a different pattern (1st present with semi-factive verbs), we know from the present-day language that these constructions do not take the indicative (hence an artifact from interrogative clauses).

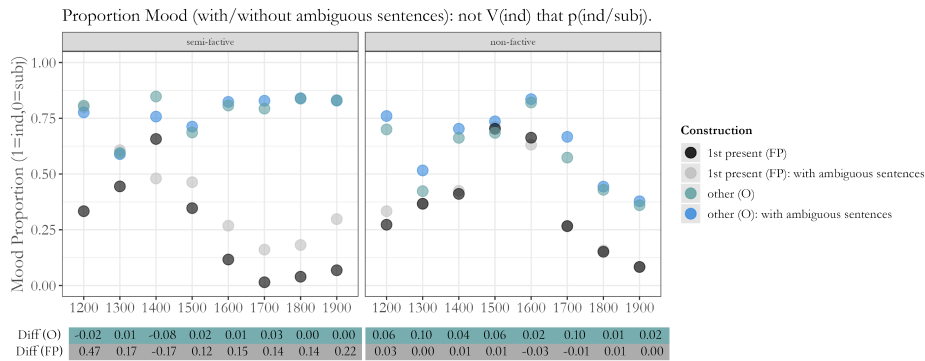


Figure A.1: Comparison of results with and without ambiguous sentences included (sentences which are ambiguous between being a complement that-clause or an interrogative what-clause without accent.)

In some cases the difference is negative, which means that there was subjunctive being used when we included ambiguous sentences. This result is not unexpected given that in present-day Spanish interrogative *what*-clauses must appear

A (Un)ambiguous sentences

in the indicative. This was not the case at previous stages of the language where both the indicative or the subjunctive could be used, see (1) (Jensen & Lathrop 1973, Nieuwenhuijsen 2001, 2016).

- (1) a. de mí non sé qué sea
 of me not know what be.PRS.SBJV
 ‘I don’t know what will be of me.’
 (1398, *Poesías (Cancionero de Baena)*, Arcediano de Toro)
- b. non sé qué fagas
 not know that do.SBJV
 ‘I don’t know what would you do.’ (1400, *Cancionero castellano del s. XV de la Biblioteca Estense de Módena*)
- c. no sabe qué se haga
 not know what CLT do.SBJV
 ‘He/she doesn’t know what would they do to themselves’
 (1508, *Cancionero*, Fray Ambrosio Montesino)

A table with all the tokens and proportions is provided on the next page.

Semi-factive			1200	1300	1400	1500	1600	1700	1800	1900
Const. FP	w/ ambiguous	Tokens	10	28	125	218	231	118	121	131
		Mean	0.80	0.61	0.48	0.46	0.27	0.16	0.18	0.29
Const. FP	w/o ambiguous	Tokens	3	9	35	98	120	68	76	73
		Mean	0.33	0.44	0.65	0.34	0.12	0.02	0.04	0.07
		Mean Diff.	0.47	0.17	-0.17	0.12	0.15	0.14	0.14	0.22
Const. O	w/ ambiguous	Tokens	76	73	235	320	187	35	181	327
		Mean	0.78	0.59	0.76	0.71	0.82	0.82	0.84	0.83
Const. O	w/o ambiguous	Tokens	31	37	118	204	114	29	148	261
		Mean	0.80	0.60	0.84	0.69	0.81	0.79	0.84	0.83
		Mean Diff.	-0.02	-0.01	-0.08	0.02	0.01	0.03	0.00	0.00
Non-factive			1200	1300	1400	1500	1600	1700	1800	1900
Const. FP	w/ ambiguous	Tokens	12	30	212	572	198	121	516	1051
		Mean	0.33	0.37	0.43	0.71	0.63	0.26	0.16	0.08
Const. FP	w/o ambiguous	Tokens	11	30	202	521	175	120	502	1021
		Mean	0.27	0.37	0.41	0.70	0.66	0.27	0.15	0.08
		Mean Diff.	0.0.3	0.00	0.01	0.01	-0.03	-0.01	0.01	0.00
Const. O	w/ ambiguous	Tokens	25	31	175	273	201	93	230	246
		Mean	0.76	0.52	0.70	0.74	0.84	0.67	0.44	0.38
Const. O	w/o ambiguous	Tokens	20	26	145	206	156	68	212	225
		Mean	0.70	0.42	0.66	0.68	0.82	0.57	0.43	0.36
		Mean Diff.	0.06	0.10	0.04	0.06	0.02	0.10	0.01	0.02

Appendix B: The two subjunctive forms

As discussed in detail in Chapter 5 Section 5.1.2, there are two distinct imperfective subjunctive forms in Spanish: the *-ra* form and the *-se* form. The *-ra* form however came from a Latin indicative form and as such was ambiguous between an indicative and a subjunctive form in earlier stages of the language. The literature estimates that the form was ambiguous until the 17th century, a moment after which the form is considered to be mostly a subjunctive marker (Jensen & Lathrop 1973).

Given that the form was ambiguous, but only during certain periods (Jensen & Lathrop 1973), and that the distribution is uneven across different dialects in the Peninsula (Lara Bermejo 2019) the overall results presented in Chapter 4 included the *-ra* form. This methodological decision, however, did not have an effect on the overall results. Figure B.1 shows the difference of including the *-ra* and not including it.

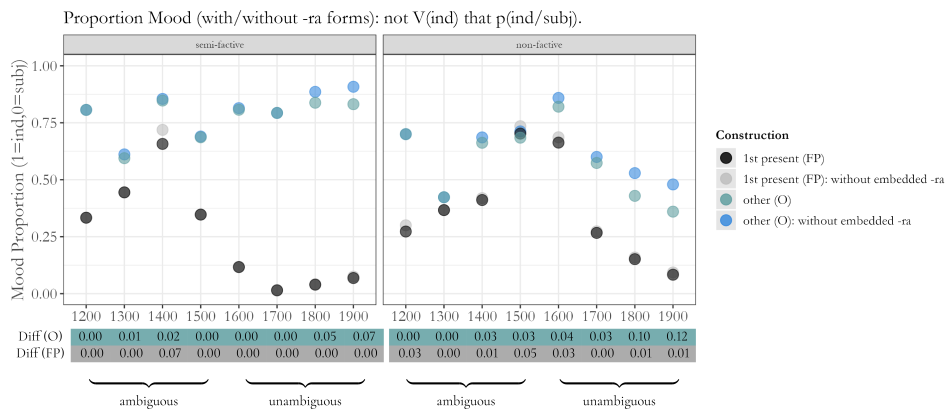


Figure B.1: Comparative of including or excluding imperfective subjunctive *-ra* forms in the overall results.

As can be seen, the difference is very small across all constructions and periods (always below a 13% difference in the frequencies), and the period with the highest impact is when the form is no longer ambiguous. Most importantly, the diachronic patterns are left unaffected, and the key observation from Chapter 4

B The two subjunctive forms

still holds: with the exception of the first person construction with a semi-factive verb, the rest of the constructions show a decrease in the use of the indicative.

For the interested reader, a full table with all the tokens and proportions in different contexts is provided below:

Semi-factive			1200	1300	1400	1500	1600	1700	1800	1900
Const. FP	w/ -ra	Tokens	3	9	35	98	120	68	76	73
		Mean	0.33	0.44	0.65	0.34	0.12	0.02	0.04	0.07
Const. FP	w/o -ra	Tokens	3	9	32	98	119	67	73	67
		Mean	0.33	0.44	0.72	0.34	0.12	0.02	0.04	0.07
		Mean Diff.	0.00	0.00	0.07	0.00	0.00	0.00	0.00	0.00
Const. O	w/ -ra	Tokens	31	37	118	204	114	29	148	261
		Mean	0.80	0.60	0.84	0.69	0.81	0.79	0.84	0.83
Const. O	w/o -ra	Tokens	31	36	117	203	113	29	140	239
		Mean	0.80	0.61	0.86	0.69	0.81	0.79	0.89	0.90
		Mean Diff.	0.00	0.01	0.02	0.00	0.00	0.00	0.05	0.07
non-factive			1200	1300	1400	1500	1600	1700	1800	1900
Const. FP	w/ -ra	Tokens	11	30	202	521	175	120	502	1021
		Mean	0.27	0.37	0.41	0.70	0.66	0.27	0.15	0.08
Const. FP	w/o -ra	Tokens	10	30	198	498	169	117	479	913
		Mean	0.30	0.37	0.42	0.75	0.69	0.27	0.16	0.09
		Mean Diff.	0.003	0.00	0.01	0.05	0.03	0.00	0.01	0.01
Const. O	w/ -ra	Tokens	20	26	145	206	156	68	212	225
		Mean	0.70	0.42	0.66	0.68	0.82	0.57	0.43	0.36
Const. O	w/o -ra	Tokens	20	26	140	198	149	65	172	169
		Mean	0.70	0.42	0.69	0.71	0.86	0.60	0.53	0.48
		Mean Diff.	0.00	0.00	0.03	0.03	0.04	0.03	0.10	0.12

Appendix C: Model fit of the verb *say* (Hubbert curve)

Following the same methodology as in Chapter 6, the data of the verb *say* was fitted in R using the `nls` function, but this time to the Hubbert curve. The Hubbert curve as well as the parameters obtained for it appear in equation (1) and Table C.1, respectively. The model fit has been plotted in Figure C.1.

(1)
$$M(t) = \frac{e^{-\frac{t-k}{m}}}{(1 + e^{-\frac{t-j}{r}})^2}$$

Table C.1: Values of the best fitting Hubbert curve (the verb *say*)

Parameter	Model 1 (Construction FP)	Model 2 (Construction 0)
k	1506 ± 1126	1144 ± 246
j	1666 ± 820	1335 ± 822
m	-361 ± 1532	-178 ± 384
r	-158 ± 106	-237 ± 216

C Model fit of the verb say (Hubbert curve)

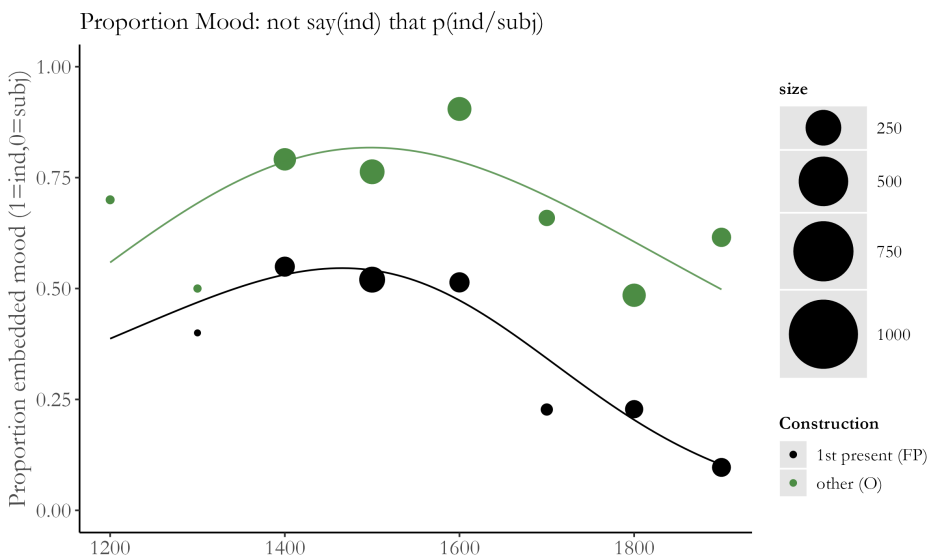


Figure C.1: Fitting of the Hubbert curve to the data from the verb say.

References

- Abney, Steven Paul. 1987. *The English noun phrase in its sentential aspect*. Cambridge: Massachusetts Institute of Technology. (Doctoral dissertation).
- Abrusán, Márta. 2019. Semantic anomaly, pragmatic infelicity, and ungrammaticality. *Annual Review of Linguistics* 5. 329–351. DOI: 10.1146/annurev-linguistics-011718-011938.
- Abusch, Dorit. 2002. Lexical alternatives as a source of pragmatic presuppositions. In Brendan Jackson (ed.), *Semantics and linguistic theory*, 1–19. DOI: 10.3765/salt.v12i0.2867.
- Achard, Michel. 2002. The meaning and distribution of French mood inflections. In René Dirven, Ronald W. Langacker & John R. Taylor (eds.), *Grounding: The epistemic footing of deixis and reference*, 197–250. Berlin: De Gruyter. DOI: 10.1515/9783110899801.197.
- Adger, David. 2006. Combinatorial variability. *Journal of Linguistics* 42(3). 503–530. DOI: 10.1017/S002222670600418X.
- Akkuş, Faruk. 2017. Agreement and anaphora in the context of imposters. In Aaron Kaplan, Abby Kaplan, Miranda K. McCarvel & Edward J. Rubin (eds.), *Proceedings of the 34th West Coast Conference on Formal Linguistics*, 24–31. Somerville: Cascadia Proceedings Project.
- Albelda Marco, Marta & Marlies Jansegers. 2019. From visual perception to evidentiality: A functional empirical approach to *se ve que* in Spanish. *Lingua* 220. 76–97. DOI: 10.1016/j.lingua.2019.01.005.
- Alcoba, Santiago. 1999. La flexión verbal. In Violeta Demonte Barreto & Ignacio Bosque Muñoz (eds.), *Gramática descriptiva de la lengua española*, vol. 3, 4915–4991. Madrid: Espasa Calpe.
- Alonso Pascua, Borja. 2023. Desplazamientos del subjuntivo en español europeo: Una aproximación dialectal. *Moenia* 29. 1–30. DOI: 10.15304/moenia.id9156.
- Álvarez, Rosario & Xosé Xove. 2002. *Gramática da lingua galega*, vol. 5. Santiago de Compostela: Editorial Galaxia.
- Ambrazas, Vytautas, Emma Geniušienė, Aleksas Girdenis, Nijolė Sližienė, Dalija Tekorienė, Adelė Valeckienė & Elena Valiulytė. 1997. *Lithuanian grammar*. Vilnius: Baltos Lankos.

References

- Amsili, Pascal. 2014. *Towards a multifactorial analysis of French subjunctive*. Paper presented at the Universidade de São Paulo.
- Andrés-Suárez, Irene. 1994. *El verbo español: Sistemas medievales y sistema clásico*. Madrid: Gredos.
- Arias Bedoya, Franklin Yessid. 2021. El subjuntivo en español e italiano: Una descripción contrastiva. *Cuadernos de Lingüística Hispánica* 37. 1–26. DOI: 10.19053/0121053X.n37.2021.12798.
- Arkadiev, Peter. 2016. Long-distance genitive of negation in Lithuanian. In Axel Holvoet & Nicole Nau (eds.), *Argument realization in Baltic*, 37–81. John Benjamins. DOI: 10.1075/vargreb.3.01ark.
- Azofra Sierra, María Elena. 2010. *Morfosintaxis histórica del español: De la teoría a la práctica*. Madrid: Editorial UNED.
- Azofra Sierra, María Elena. 2019. Morfosintaxis diacrónica. In Emilio Ridruejo (ed.), *Manual de lingüística española*, 96–132. Berlin, Boston: De Gruyter. DOI: 10.1515/9783110362084-005.
- B-Violette, Laurence. 2019. *The development of the polarity subjunctive in Romance languages*. Harvard University. (Doctoral dissertation).
- Baayen, Harald, Fabian Tomaschek, Susanne Gahl & Michael Ramscar. 2017. The Ecclesiastes principle in language change. In Marianne Hundt, Sandra Mollin & Simone E. Pfenninger (eds.), *The changing English language: Psycholinguistic perspectives*, 21–48. Cambridge: Cambridge University Press. DOI: 10.1017/9781316091746.002.
- Badia i Margarit, Antoni M. 1994. *Gramàtica de la llengua catalana: Descriptiva, normativa, diatòpica, diastràtica*. Barcelona: Enciclopèdia Catalana.
- Bailey, Charles-James. 1973. *Variation and linguistic theory*. Washington: Center for Applied Linguistics.
- Bailyn, John F. 1997. Genitive of negation is obligatory. In Wayles Browne, Ewa Dornish, Natasha Kondrashova & Draga Zec (eds.), *Formal approaches to Slavic linguistics: The Cornell meeting*, vol. 4, 84–114. Ann Arbor: Michigan Slavic Publications.
- Baker, Mark C. 2015. *Case*. Cambridge: Cambridge University Press. DOI: 10.1017/CBO9781107295186.
- Baker, Mark C. & Nadya Vinokurova. 2010. Two modalities of case assignment: Case in Sakha. *Natural Language & Linguistic Theory* 28. 593–642. DOI: 10.1007/s11049-010-9105-1.
- Barrie, Michael Jonathan Mathew, Audrey Li, Martina Wiltschko & Jong Un Park. 2021. In defense of DP (or KP). *Linguistic Research* 38(2). 207–238.
- Baunaz, Lena & Genoveva Puskás. 2022. *A cross-linguistic approach to the syntax of subjunctive mood*. Cham: Springer. DOI: 10.1007/978-3-031-04540-0.

- Becerra, Ariel R. 2006. *Gramática Rusa*. Manuscript, Pamplona.
- Becker, Martin G. 2010a. Mood in Portuguese. In Björn Rothstein & Rolf Thieroff (eds.), *Mood in the languages of Europe*, 179–197. Amsterdam: John Benjamins. DOI: 10.1075/slcs.120.10bec.
- Becker, Martin G. 2010b. Mood in Rumanian. In Björn Rothstein & Rolf Thieroff (eds.), *Mood in the languages of Europe*, 251–270. Amsterdam: John Benjamins. DOI: 10.1075/slcs.120.13squ.
- Bello, Andrés. 1847. *Gramática de la lengua castellana destinada al uso de los americanos*. Madrid: EDAF.
- Bezinska, Yanka, Iva Novakova & Jean-Pierre Chevrot. 2018. Acquisition de la causativité chez des enfants français et bulgares âgés de 3 à 6 ans: Complexité syntaxique ou complexité sémantique? *CogniTextes: Revue de l'Association Française de Linguistique Cognitive* 17. 1–34. DOI: 10.4000/cognitextes.938.
- Bianchi, Valentina & Mara Frascarelli. 2010. Is topic a root phenomenon? *IBERIA: An International Journal of Theoretical Linguistics* 2(1). 43–88.
- Bjarnadóttir, Valgerður. 2014. *Non-canonical case-marking on core arguments in Lithuanian: A historical and contrastive perspective*. Stockholm: Stockholm University. (Doctoral dissertation).
- Blake, Robert J. & Carlos Sánchez Lancis. 2018. La alternancia de modo en las subordinadas sustantivas: El caso de *esperar* en la historia del español. In María Luisa Arnal Purroy, Rosa María Castañer Martín, José M.^a Enguita Utrilla, Vicente Lagüéns Gracia & María Antonia Martín Zorraquino (eds.), *Actas del X Congreso Internacional de Historia de la Lengua Española*, vol. 1, 513–530. Zaragoza: Editorial Cometa, S. A.
- Blas Arroyo, José Luis. 2023. Individual variation and frequency change in Early Modern Spanish: Alignment and intra-speaker (in)stability in a corpus of 18th century ego-documents. *Journal of Historical Linguistics* 15(1). 1–43. DOI: 10.1075/jhl.22059.bla.
- Błaszczak, Joanna. 2001. *Covert movement and the genitive of negation in Polish*. Potsdam: Universitätsbibliothek Potsdam.
- Bobaljik, Jonathan David & Susi Wurmbrand. 2013. Suspension across domains. In Ora Matushansky & Alec P. Marantz (eds.), *Distributed morphology today: Morphemes for Morris Halle*, chap. 11, 185–198. Cambridge: MIT Press. DOI: 10.7551/mitpress/9701.003.0013.
- Bonamente, Massimiliano. 2017. Theory of probability. In Massimiliano Bonamente (ed.), *Statistics and analysis of scientific data* (Graduate Texts in Physics), 1–15. New York: Springer. DOI: 10.1007/978-1-4614-7984-0_1.

References

- Borgonovo, Claudia. 2003. Mood and focus. In Josep Quer, Jan Schroten, Mauro Scorretti, Petra Sleeman & Els Verheugd-Daatzelaar (eds.), *Romance languages and linguistic theory 2001: Selected papers from 'Going Romance' Amsterdam, 6-8 December 2001*, 17–30. Amsterdam: John Benjamins. DOI: 10.1075/cilt.245.
- Bosque, Ignacio. 2012. Mood: Indicative vs. subjunctive. In José Ignacio Hualde, Antxon Olarrea & Erin O'Rourke (eds.), *The handbook of Hispanic linguistics*, 373–394. Chichester: John Wiley. DOI: 10.1002/9781118228098.ch19.
- Bosson, Georg. 1991. Differential object marking in Romance and beyond. In Dieter Wanner & Douglas A. Kibbee (eds.), *New analyses in Romance linguistics*, vol. 69, 143–170.
- Bove, Kathryn Paige. 2018. *Mood selection and doxastic predicates in Yucatec Spanish*. Athens: University of Georgia. (Doctoral dissertation).
- Burnett, Heather. 2016. Variation as a testing ground for grammatical theory: Variable negative concord in Montréal French. *Linguistic Variation* 16(2). 267–299.
- Busch, Hans-Jörg. 2017. *A complete guide to the Spanish subjunctive*. London: Routledge.
- Bush, Robert R. & Frederick Mosteller. 1955. *Stochastic models for learning*. New York: John Wiley.
- Butt, Miriam. 2006. The dative-ergative connection. In Olivier Bonami & Patricia Cabredo Hofherr (eds.), *Empirical issues in syntax and semantics*, vol. 6, 69–92. Paris: Centre national de la recherche scientifique.
- Bybee, Joan & Suzanne Fleischman. 1995. Modality in grammar and discourse: An introductory essay. In Joan Bybee & Suzanne Fleischman (eds.), *Modality in grammar and discourse*, 1–14. Amsterdam: John Benjamins.
- Camacho-Taboada, Victoria & Ángel L. Jiménez-Fernández. 2013. Focus fronting and root phenomena in Spanish and English. In Joseph Emonds & Markéta Janebová (eds.), *Language use and linguistic structure. Proceedings of the Olo-mouc Linguistics Symposium*, vol. 3, 47–60. DOI: 10.5507/ff.14.24440606.03.
- Caplan, Spencer & Kajsa Djärv. 2019. What usage can tell us about grammar: Embedded verb second in Scandinavian. *Glossa: A Journal of General Linguistics* 4(1). 1–37. DOI: 10.5334/gjgl.867.
- Carstens, Vicki. 2010. Implications of grammatical gender for the theory of uninterpretable features. In Michael T. Putnam (ed.), *Exploring crash-proof grammars*, vol. 3 (Language Faculty and Beyond), 31–58. John Benjamins. DOI: 10.1075/lfab.3.03car.
- Carstens, Vicki & Michael Diercks. 2013. Agreeing how? Implications for theories of agreement and locality. *Linguistic Inquiry* 44(2). 179–237. DOI: 10.1162/LING_a_00125.

- Chierchia, Gennaro & Sally McConnell-Ginet. 1990. *Meaning and grammar: An introduction to semantics*. Cambridge: MIT Press.
- Chomsky, Noam. 2001. Derivation by phase. In Michael Kenstowicz (ed.), *Ken Hale: A life in language* (Current Studies in Linguistics 36), 1–52. Cambridge: MIT Press. DOI: 10.7551/mitpress/4056.003.0004.
- Clark, Eve V. 1987. The principle of contrast: A constraint on language acquisition. In Eve V. Clark (ed.), *Mechanisms of language acquisition*, 1–33. Hove: Psychology Press.
- Company Company, Concepción. 2006. *Sintaxis histórica de la lengua española. Primera parte: La frase verbal*. México: UNAM/FCE.
- Coniglio, Marco. 2008. Modal particles in Italian. *University of Venice Working Papers in Linguistics* 18. 91–129.
- Coniglio, Marco & Roland Hinterhölzl. 2020. Discourse conditions on relative clauses: A crosslinguistic and diachronic study on the interaction between mood, verb position and information structure. In Anke Holler, Katja Suckow & Israel de la Fuente (eds.), *Information structuring in discourse*, vol. 40 (Current research in the semantics / pragmatics interface), 201–233. Leiden: Brill. DOI: 10.1163/9789004436725_009.
- Coretta, Stefano. 2023. *tidymv: Tidy Model Visualisation for Generalised Additive Models*. <https://rdocumentation.org/packages/tidymv/versions/3.4.2>.
- Corral Esteban, Avelino. 2020. A study of DOM in Asturian ('Dialectu Vaqueiru'). *Journal of Language Contact* 13(1). 96–140.
- D'Alessandro, Roberta. 2021. Syntactic change in contact: Romance. *Annual Review of Linguistics* 7(1). 309–328.
- Dahl, Östen. 1971. The genitive and the subjunctive in Russian. *Scando-Slavica* 17(1). 135–139. DOI: 10.1080/00806767108600577.
- Dahl, Östen. 1979. Typology of sentence negation. *Linguistics* 17. 79–116. DOI: 10.1515/ling.1979.17.1-2.79.
- Dalrymple, Mary & Irina Nikolaeva. 2011. *Objects and information structure*. Cambridge: Cambridge University Press.
- Danckaert, Lieven. 2017. The loss of Latin OV: Steps towards an analysis. In Enoch Aboh, Eric Haeberli, Genoveva Puskás & Manuela Schönenberger (eds.), *Elements of comparative syntax: Theory and description*, 401–444. Berlin: De Gruyter. DOI: 10.1515/9781501504037-015.
- Davies, Mark. 2009. *Corpus del español*. <<https://www.corpusdelespanol.org/hist-gen/>>.
- de Haan, Ferdinand. 2006. Typological approaches to modality. In Wolfgang Klein & Stephen Levinson (eds.), *The expression of modality*, 27–69. Berlin: De Gruyter.

References

- De Mulder, Walter. 2010. Mood in French. In Björn Rothstein & Rolf Thieroff (eds.), *Mood in the languages of Europe*, 157–178. Amsterdam: John Benjamins. DOI: 10.1075/slcs.120.09dem.
- Deal, Amy Rose. 2025. Current models of Agree. *Berkeley Papers in Formal Linguistics* 4(1). 1–35. DOI: 10.5070/BF2.48662.
- Delbecque, Nicole & Béatrice Lamiroy. 1999. La subordinación sustantiva: Las subordinadas enunciativas en los complementos verbales. In Violeta Demonte Barreto & Ignacio Bosque Muñoz (eds.), *Gramática descriptiva de la lengua española*, vol. 2, 1965–2082. Madrid: Espasa Calpe.
- Despić, Miloje. 2017. Investigations on mixed agreement: Polite plurals, hybrid nouns and coordinate structures. *Morphology* 27. 253–310. DOI: 10.1007/s11525-017-9301-3.
- Dindelegan, Gabriela Pană & Martin Maiden. 2013. *The grammar of Romanian*. Oxford: Oxford University Press.
- Djäv, Kajsa. 2019. The syntax of assertion and presupposition. In Katherine Blake, Forrest Davis, Kaelyn Lamp & Joseph Rhyne (eds.), *Proceedings of SALT 29*, vol. 29, 392–419. Linguistic Society of America. DOI: 10.3765/salt.v29i0.4626.
- Djäv, Kajsa, Jérémy Zehr & Florian Schwarz. 2018. Cognitive vs. emotive factives: An experimental differentiation. In Robert Truswell, Chris Cummins, Caroline Heycock, Brian Rabern & Hannah Rohde (eds.), *Proceedings of Sinn und Bedeutung*, vol. 21, 367–386.
- Döhla, Hans-Jörg. 2014. Diachronic convergence and divergence in differential object marking between Spanish and Portuguese. In Kurt Braunmüller, Karoline Kühl & Steffen Höder (eds.), *Stability and divergence in language contact. Factors and mechanisms*. 265–289. Amsterdam: John Benjamins.
- Dudley, Rachel. 2014. Spanish imposters and verbal agreement. In Chris Collins (ed.), *Cross-linguistic studies of imposters and pronominal agreement*, 42–70. New York: Oxford University Press. DOI: 10.1093/acprof:oso/9780199336852.003.0003.
- Eberenz, Rolf. 1991. Castellano antiguo y español moderno: Reflexiones sobre la periodización en la historia de la lengua. *Revista de Filología Española* 71(1/2). 79–106. DOI: 10.3989/rfe.1991.v71.i1/2.652.
- Eberenz, Rolf. 2023. Periodización de la historia de la lengua española. In Steven N. Dworkin, Gloria Clavería Nadal & Álvaro S. Octavio de Toledo y Huerta (eds.), *Lingüística histórica del español/The Routledge Handbook of Spanish Historical Linguistics*, 317–327. London: Routledge.

- Eide, Kristine Gunn. 2014. Studying word order differences in a historical parallel corpus: An example from Old Spanish and Old Portuguese. *Oslo Studies in Language. Corpus-based studies in contrastive linguistics* 6(1). 181–199. DOI: 10.5617/osla.693.
- Emiliano, Paulo C., Mário J. Vivanco & Fortunato S. De Menezes. 2014. Information criteria: How do they behave in different models? *Computational Statistics & Data Analysis* 69. 141–153.
- Enciclopèdia Catalana. 1998. *Gran diccionari de la llengua catalana*. Online Dictionary. <https://www.diccionari.cat/>.
- Enrique-Arias, Andrés (ed.). 2013. *Diacronía de las lenguas iberorrománicas: Nuevas aportaciones desde la lingüística de corpus*. Madrid: Iberoamericana Editorial Vervuert.
- Enrique-Arias, Andrés & Javier Pueyo Mena. 2008–2026. *Biblias Medievales [en línea]*. <http://www.bibliamedieval.es>.
- Etxepare, Ricardo. 1996. On null complementizers in Spanish. *Anuario del Seminario de Filología Vasca* 30(2). 469–496.
- Fábregas, Antonio. 2014. A guide to subjunctive and modals in Spanish: Questions and analyses. *Borealis – An International Journal of Hispanic Linguistics* 3(2). 1–94. DOI: 10.7557/1.3.2.3064.
- Farkas, Donka. 1992. On the semantics of subjunctive complements. In Paul Hirschbühler & Ernst Frideryk Konrad Koerner (eds.), *Romance languages and modern linguistic theory*, 69–104. Berlin: De Gruyter. DOI: 10.1075/cilt.91.07far.
- Farkas, Donka F. 1981. *Intensionality and Romance subjunctive relatives*. The University of Chicago. (Doctoral dissertation).
- Faulkner, Tris. 2021. *A systematic investigation of the Spanish subjunctive: Mood variation in subjunctive clauses*. Georgetown University. (Doctoral dissertation).
- Faulkner, Tris. 2022. The two Spanish subjunctives: The required and default subjunctives. *Borealis – An International Journal of Hispanic Linguistics* 11(1). 70–100. DOI: 10.7557/1.11.1.6334.
- Fernández, Olga. 2000. Force and finiteness in the Spanish complementizer system. *Probus* 21. 23–49. DOI: 0.1515/prbs.2009.002.
- Frías Conde, Francisco Xavier. 2015. Notas acerca do verbo galego. *Revista de Linguas y Literaturas Catalana, Gallega y Vasca* 20. 7–51. DOI: 10.5944/rllcgv.vol.20.2015.15920.
- Fukushima, Noritaka. 2015. Pasado, presente y futuro del subjuntivo en español. *Centro virtual Cervantes. Kobe*. 45–61.

References

- Furuya, Kaori. 2019. A cross-linguistic study of person agreement in imposter constructions. *Theory and Practice in Language Studies* 9(1). 1–10. DOI: 10.17507/tpls.0901.01.
- Gajewski, Jon Robert. 2005. *Neg-raising: Polarity and presupposition*. Cambridge: Massachusetts Institute of Technology. (Doctoral dissertation).
- Gajewski, Jon Robert. 2011. Licensing strong NPIs. *Natural Language Semantics* 19. 109–148. DOI: 10.1007/s11050-010-9067-1.
- Gallego, Muriel & Emilia Alonso-Marks. 2014. Degrees of subjunctive vitality among monolingual speakers of Peninsular and Argentinian Spanish. *Borealis – An International Journal of Hispanic Linguistics* 3(2). 95–104. DOI: 10.7557/1.3.2.3129.
- Gardiner, Shayna. 2015. Taking possession of the constant rate hypothesis: Variation and change in Ancient Egyptian possessive constructions. *University of Pennsylvania Working Papers in Linguistics* 21(2). 69–78.
- Gärtner, Hans-Martin & Þórhallur Eypórssón. 2020. Varieties of dependent v2 and verbal mood: A view from Icelandic. In Rebecca Woods & Sam Wolfe (eds.), *Rethinking verb second*, 1–34. Oxford: Oxford University Press. DOI: 10.1093/oso/9780198844303.003.0009.
- Geeraerts, Dirk. 1999. Diachronic prototype semantics. A digest. In Andreas Blank & Peter Koch (eds.), *Historical semantics and cognition*, 91–108. Berlin: De Gruyter. DOI: 10.1515/9783110804195.91.
- Gerards, David Paul. 2023. *Differential object marking in the Romance languages*. Oxford. <https://oxfordre.com/linguistics/view/10.1093/acrefore/9780199384655.001.0001/acrefore-9780199384655-e-648>.
- Giannakidou, Anastasia. 2009. The dependency of the subjunctive revisited: Temporal semantics and polarity. *Lingua* 119(12). 1883–1908. DOI: 10.1016/j.lingua.2008.11.007.
- Giannakidou, Anastasia & Alda Mari. 2017. Biased modality and epistemic weakness with the future and MUST: Non-veridicality, partial knowledge. In Joanna Blaszczak, Anastasia Giannakidou, Dorota Klimek-Jankowska & Krzysztof Migdalski (eds.), *Tense, mood, and modality: New perspectives on old questions*, 1–27. Chicago: University of Chicago Press.
- Giannakidou, Anastasia & Alda Mari. 2021. *Truth and veridicality in grammar and thought: Mood, modality, and propositional attitudes*. Chicago: University of Chicago Press.
- Gianollo, Chiara. 2016. The Latin system of negation at the syntax-semantics interface. *Rivista di Grammatica Generativa* 38. 115–135.
- Giorgi, Alessandra. 2009. Toward a syntax of the subjunctive mood. *Lingua* 119(12). 1837–1858. DOI: 10.1016/j.lingua.2008.11.008.

- Giorgi, Alessandra & Fabio Pianesi. 1997. *Tense and aspect: From semantics to morphosyntax*. New York: Oxford University Press. DOI: 10.1093/oso/9780195091922.001.0001.
- Giorgi, Alessandra & Fabio Pianesi. 2001. Imperfect dreams: The temporal dependencies of fictional predicates. *Probus: International Journal of Latin & Romance Linguistics* 13(1). 31–68. DOI: 10.1515/prbs.13.1.31.
- GITHE, Grupo de Investigación Textos para la Historia del Español. 2022. *CODEA+ 2022 (Corpus de Documentos Españoles Anteriores a 1900)*. DOI: 10.37536/CODEA.2015.
- Gras, Pedro & María Sol Sansiñena. 2015. An interactional account of discourse-connective *que*-constructions in Spanish. *Text & Talk* 35(4). 505–529. DOI: 10.1515/text-2015-0010.
- Hahm, Hyun-Jong. 2010. *A cross-linguistic study of syntactic and semantic agreement: Polite plural pronouns and other issues*. Austin: The University of Texas at Austin. (Doctoral dissertation).
- Hansen, Maj-Britt Mosegaard & Jacqueline Visconti. 2009. On the diachrony of “reinforced” negation in French and Italian. In Corinne Rossari, Claudia Ricci & Adriana Spiridon (eds.), *Grammaticalization and pragmatics: Facts, approaches, theoretical issues*, 137–171. Brill. DOI: 10.1163/9789004253193_009.
- Hanssen, Federico. 1913. *Gramática histórica de la lengua castellana*. Breinigsvill: Nabu.
- Heim, Irene. 1991. Artikel und Definitheit. In Arnim von Stechow & Dieter Wunderlich (eds.), *Semantik: Ein internationales Handbuch der zeitgenössischen Forschung*, 487–535. Berlin: De Gruyter.
- Heim, Irene. 1992. Presupposition projection and the semantics of attitude verbs. *Journal of Semantics* 9(3). 183–221. DOI: 10.1093/jos/9.3.183.
- Heim, Irene & Angelika Kratzer. 1998. *Semantics in generative grammar*. Massachusetts: Blackwell.
- Hill, Virginia & Alexandru Mardale. 2021. *The diachrony of differential object marking in Romanian*. Oxford: Oxford University Press. DOI: 10.1093/oso/9780192898791.001.0001.
- Hintikka, Jaakko. 1961. Modality and quantification. *Theoria* 27(3). 119–128. DOI: 10.1111/j.1755-2567.1961.tb00020.x.
- Hiraiwa, Ken. 2005. *Dimensions of symmetry in syntax: Agreement and clausal architecture*. Cambridge: Massachusetts Institute of Technology. (Doctoral dissertation).
- Homer, Vincent. 2008. Disruption of NPI licensing: The case of presuppositions. In Tova Friedman & Satoshi Ito (eds.), *Proceedings of SALT 18*, 429–446. DOI: 10.3765/salt.v18i0.2483.

References

- Hooper, Joan B. & Sandra A. Thompson. 1973. On the applicability of root transformations. *Linguistic Inquiry* 4(4). 465–497.
- Horn, Laurence R. 1978. Remarks on neg-raising. In Peter Cole (ed.), *Pragmatics*, vol. 9 (Syntax and Semantics), 129–220. Leiden: Brill. DOI: 10.1163/9789004368873_007.
- Hornstein, Norbert, Jairo Nunes & Kleanthes K. Grohmann. 2005. *Understanding minimalism*. Cambridge: Cambridge University Press. DOI: 10.1017/CBO9780511840678.
- Iemmolo, Giorgio. 2010. Topicality and differential object marking: Evidence from Romance and beyond. *Studies in Language* 34(2). 239–272. DOI: 10.1075/sl.34.2.01iem.
- Ingason, Anton Karl, Julie Anne Legate & Charles Yang. 2013. The evolutionary trajectory of the Icelandic new passive. *University of Pennsylvania Working Papers in Linguistics* 19(2). 91–100.
- Ingham, Richard. 2014. Old French negation, the Tobler/Mussafia law, and V2. *Lingua* 147. 25–39. DOI: 10.1016/j.lingua.2013.11.005.
- Institut d'Estudis Catalans. 2016. *Gramàtica de la llengua catalana*. Barcelona: Institut d'Estudis Catalans.
- Irimia, Monica Alexandrina & Anna Pineda. 2019. Differential object marking and scales: Insights from diachrony. *Proceedings of the Linguistic Society of America* 4. 57–1. DOI: 10.3765/plsa.v4i1.4561.
- Jardón, Natalia & Sigríður Mjöll Björnsdóttir. 2026. Penso que poida existir: Aspectos semánticos do uso do subxuntivo en galego e islandés. *Cumieira das Letras* 4. 1–10.
- Jensen, Frede & Thomas A. Lathrop. 1973. *The syntax of the Old Spanish subjunctive*. Berlin: De Gruyter.
- Jespersen, Otto. 1917. *Negation in English and other languages*. Copenhagen: Bianco Lunos Bogtrykkeri.
- Kabatek, Johannes, Philipp Obrist & Albert Wall. 2021. *Differential object marking in Romance: The third wave*. Berlin: De Gruyter.
- Kagan, Olga. 2007. Property-denoting NPs and non-canonical genitive case. In Tova Friedman & Masayuki Gibson (eds.), *Proceedings of semantics and linguistic theory*, vol. 17, 148–165. Linguistic Society of America. DOI: 10.3765/salt.v17i0.3090.
- Kagan, Olga. 2012. *Semantics of genitive objects in Russian: A study of genitive of negation and intensional genitive case*. Dordrecht: Springer.
- Kaufmann, Magdalena & Sanae Tamura. 2020. Possibility and necessity in Japanese: Prioritizing, epistemic, and dynamic modality. In Wesley Jacobsen &

- Yukinori Takubo (eds.), *Handbook of Japanese semantics and pragmatics*, 537–586. Berlin: De Gruyter. DOI: 10.1515/9781614512073-011.
- Kauhanen, Henri. 2019. Stable variation in multidimensional competition. In Anne Breitbarth, Miriam Bouzouita, Lieven Danckaert & Melissa Farasyn (eds.), *The determinants of diachronic stability*, 263–290. Amsterdam: Benjamins.
- Kauhanen, Henri. 2021. A bifurcation threshold for contact-induced language change. *Glossa: A Journal of General Linguistics* 7(1). 1–32. DOI: 10.16995/glossa.8211.
- Kauhanen, Henri. 2023. *Language dynamics*. (Lecture notes, University of Konstanz).
- Kauhanen, Henri. 2024. *Variational learning: Anatomy of an algorithm*. Talk presented at the workshop Understanding grammar change: Digital resources and evolutionary modelling, Edinburgh, 4–5 June 2024.
- Kauhanen, Henri & George Walkden. 2018. Deriving the constant rate effect. *Natural Language & Linguistic Theory* 36. 483–521. DOI: 10.1007/s11049-017-9380-1.
- Kempchinsky, Paula. 2009. What can the subjunctive disjoint reference effect tell us about the subjunctive? *Lingua* 119(12). 1788–1810. DOI: 10.1016/j.lingua.2008.11.009.
- Kiparsky, Paul. 1998. Partitive case and aspect. In Miriam Butt & Wilhelm Geuder (eds.), *The projection of arguments: Lexical and compositional factors* (CSLI Lecture Notes 83), 265–307. Stanford: CSLI Publications.
- Kocher, Anna. 2023. *Complementizers on edge: On the boundaries between syntax and pragmatics in Ibero-Romance* (Open Romance Linguistics 1). Berlin: Language Science Press. DOI: 10.5281/zenodo.6907848.
- Koev, Todor, Cory Bill & Maryam Mohammadi. 2021. *Believe is strong but subjective: Experimental evidence from hedging*. In Patrick Grosz, Luisa Martí, Hazel Pearson, Yasutada Sudo & Sarah Zobel (eds.), *Proceedings of Sinn und Bedeutung* 25, 480–497. University of Konstanz. DOI: 10.18148/sub/2021.v25i0.950.
- Kratzer, Angelika. 1981. The notional category of modality. In Hans J. Eikmeyer & Hannes Rieser (eds.), *Words, worlds, and contexts: New approaches in word semantics* (Research in Text Theory 6), 38–74. Berlin: De Gruyter. DOI: 10.1515/9783110842524-004.
- Kroch, Anthony S. 1989. Reflexes of grammar in patterns of language change. *Language Variation and Change* 1(3). 199–244. DOI: 10.1017/S0954394500000168.
- Kroch, Anthony S. 1994. Morphosyntactic variation. In Katharine Beals, Jeanette Denton, Robert Knippen, Lynette Melnar, Hisami Suzuki & Erica Zeinfeld (eds.), *Papers from the 30th regional meeting of the Chicago Linguistics So-*

References

- ciety: *Parasession on variation and linguistic theory*, vol. 2, 180–201. Chicago: Chicago Linguistic Society.
- Labov, William. 1972. *Sociolinguistic patterns*. Philadelphia: University of Pennsylvania Press.
- Lahousse, Karen & Béatrice Lamiroy. 2012. Word order in French, Spanish and Italian: A grammaticalization account. *Folia Linguistica: Acta Societatis Linguisticae Europaeae* 46(2). 387–415.
- Laka Mugarza, Miren Itziar. 1990. *Negation in syntax: On the nature of functional categories and projections*. Massachusetts Institute of Technology. (Doctoral dissertation).
- Lakoff, George. 1973. Hedges: A study in meaning criteria and the logic of fuzzy concepts. *Journal of Philosophical Logic* 2(4). 458–508.
- Landau, Idan. 2016. DP-internal semantic agreement: A configurational analysis. *Natural Language & Linguistic Theory* 34. 975–1020. DOI: 10.1007/s11049-015-9319-3.
- Lapesa, Rafael. 1968. *Historia de la lengua española*. Madrid: Escelicer.
- Lara Bermejo, Víctor. 2019. El pretérito imperfecto de subjuntivo en la Península Ibérica del siglo XX. *Verba: Anuario Galego de Filoloxía* 46. 313–338. DOI: 10.15304/verba.46.5047.
- Larm, Lars. 2014. Modality packaging in Japanese: The encoding of modal meanings and subjectivity. *Studies in Pragmatics* 16. 20–46.
- Lavandera, Beatriz. 1983. Shifting moods in Spanish discourse. In Flora Klein-Andreu (ed.), *Discourse perspectives on syntax*, 209–236. New York: Academic Press.
- Ledgeway, Adam. 2012. *From Latin to Romance: Morphosyntactic typology and change*. Oxford: Oxford University Press.
- Ledgeway, Adam. 2022. Disentangling parameters: Romance differential object marking and the distribution of head and edge features. In Gabriela Alboiu, Daniela Isac, Alexandru Nicolae, Mihaela Tănase-Dogaru & Alina Tigău (eds.), *A life in linguistics. A Festschrift for Alexandra Cornilescu on her 75th birthday*, 439–458. Bucharest: Bucharest University Press.
- Levin, Theodore & Omer Preminger. 2015. Case in Sakha: Are two modalities really necessary? *Natural Language & Linguistic Theory* 33. 231–250.
- Lewis, David. 1973. *Counterfactuals*. Oxford: Blackwell.
- Lindschouw, Jan. 2017. Tense, aspect, mood. In Andreas Dufter & Elisabeth Stark (eds.), *Manual of Romance morphosyntax and syntax*, vol. 17, 397–448. Germany: De Gruyter. DOI: 10.1515/9783110377088-011.

- López Fonseca, Antonio. 2009. Modo y modalidad. In Olga Álvarez Huerta & José Miguel Baños Baños (eds.), *Sintaxis del latín clásico*, 441–466. Madrid: Liceus E-Excellence.
- López Ornat, Susana, Almudena Fernández, Pilar Gallo & Sonia Mariscal. 1994. *La adquisición de la lengua española*. Madrid: Siglo XXI de España.
- Lunn, Patricia V. 1989. Spanish mood and the prototype of assertability. *Linguistics* 27(4). 687–702. DOI: 10.1515/ling.1989.27.4.687.
- Magni, Elisabetta. 2010. Mood and modality. In Pierluigi Cuzzolin & Philip Baldi (eds.), *New perspectives on historical Latin syntax. Constituent syntax: Adverbial phrases, adverbs, mood, tense*, vol. 2, 193–275. De Gruyter. DOI: 10.1515/9783110215458.193.
- Maiden, Martin. 2018. *The Romance verb: Morphomic structure and diachrony*. Oxford: Oxford University Press.
- Manzini, Maria Rita. 2008. Sentential complementation: The subjunctive. In Peter Coopmans, Martin B. Everaert & Jane Grimshaw (eds.), *Lexical specification and insertion*, 241–268. Amsterdam: John Benjamins. DOI: 10.1075/cilt.197.12man.
- Marantz, Alec. 1991. Case and licensing. In Germán F. Westphal, Benjamin Ao & Hee-Rahk Chae (eds.), *Proceedings of the Eastern States Conference on Linguistics*, 234–253. Ohio State University.
- Mardale, Alexandru & Monica Alexandrina Irimia (eds.). 2023. *Differential object marking in Romance: Towards microvariation*. Amsterdam: John Benjamins.
- Mari, Alda & Paul Portner. 2021. Mood variation with belief predicates: Modal comparison and the raisability of questions. *Glossa: A Journal of General Linguistics* 40(1). 132. DOI: 10.16995/glossa.5726.
- Marques, Rui. 2010. Modality, context change potential and mood selection in European Portuguese. In Martin G. Becker & Eva-Maria Remberge (eds.), *Modality and mood in romance: Modal interpretation, mood selection, and mood alternation*. Berlin: De Gruyter. DOI: 10.1515/9783110234343.2.133.
- Marques, Rui. 2024. A unitary account of indicative/subjunctive mood choice. *Probus* 36(1). 113–142. DOI: 10.1515/probus-2024-2007.
- Martins, Ana Maria. 2008. On the origin of the Portuguese inflected infinitive: A new perspective on an enduring debate. In Laurel J. Brinton (ed.), *Historical linguistics 1999: Selected papers from the 14th International Conference on Historical Linguistics*, 207–222. Amsterdam: John Benjamins. DOI: 10.1075/cilt.215.15mar.
- Mazzola, Giulia, Bert Cornillie & Malte Rosemeyer. 2022. Asyndetic complementation and referential integration in Spanish: A diachronic probabilistic grammar account. *Journal of Historical Linguistics* 12(2). 194–240. DOI: 10.1075/jhl.20031.maz.

References

- McCloskey, James. 1991. *There, it, and agreement*. *Linguistic Inquiry* 22(3). 563–567.
- McFadden, Thomas. 2004. *The position of morphological *case in the derivation: A study on the syntax-morphology interface*. University of Pennsylvania. (Doctoral dissertation).
- Meinunger, André. 2017. Desires, wishes and hopes – desiderative predicates and presuppositions. *STUF-Language Typology and Universals* 70(4). 611–635. DOI: 10.1515/stuf-2017-0026.
- Mejías-Bikandi, Errapel. 1998. Pragmatic presupposition and old information in the use of the subjunctive mood in Spanish. *Hispania* 81(4). 941–948. DOI: 10.2307/345807.
- Menéndez Pidal, Ramón. 2005. *Historia de la lengua española*. Madrid: Fundación Ramón Menéndez Pidal.
- Menzel, Christopher. 2016. Possible worlds. In Edward N. Zalta & Uri Nodelman (eds.), *The Stanford Encyclopedia of Philosophy*. Stanford: Stanford University. <https://plato.stanford.edu/archives/sum2024/entries/possible-worlds/>.
- Menzies, Peter & Helen Beebee. 2001. Counterfactual theories of causation. In Edward N. Zalta & Uri Nodelman (eds.), *The stanford encyclopedia of philosophy*. Stanford: Stanford University. <https://plato.stanford.edu/entries/causation-counterfactual>.
- Messick, Troy. 2020. *3/4 agreement patterns beyond hybrid nouns*. Manuscript Rutgers University.
- Miljan, Merilin & Ronnie Cann. 2013. Rethinking case marking and case alternation in Estonian. *Nordic Journal of Linguistics* 36(3). 333–379. DOI: 10.1017/S0332586513000309.
- Mitkovska, Liljana & Eleni Bužarovska. 2015. Variation in clausal complementation: Macedonian and Bulgarian predicates ‘hope’ and ‘believe’. *Dimenzije Značenja. Zagrebačka slavistička škola*. 189–242.
- Montero, Raquel, Leah Doroski & Maribel Romero. 2025. Spanish neg-raising: Mood effects on NPI licensing. In Federica Longo & Daniele Panizza (eds.), *Proceedings of Sinn und Bedeutung* 29, 1036–1053. DOI: 10.18148/sub/2024.v29.1261.
- Montero, Raquel & Maribel Romero. 2023. Examining the meaning of polarity subjunctive. In Suet-Ying Lam & Satoru Ozaki (eds.), *NELS53: Proceedings of the fifty-third annual meeting of the North East Linguistic Society*, vol. 2, 219–232. Graduate Linguistics Student Association.
- Moreno Bernal, Jesús & Bautista Horcajada. 1997. Sobre *no* y *non* en español medieval. *Revista de Filología Románica* I(14). 345–361.

- Morosi, Paolo. 2025. *(In)definiteness and (non-)maximality: Italian as compared to other Romance languages*. Barcelona: Universitat Autònoma de Barcelona. (Doctoral dissertation).
- Murray, Sarah E. 2016. Two imperatives in Cheyenne: Some preliminary distinctions. In Monica Macaulay, Margaret Noodin & Randolph Valentine (eds.), *Papers of the Forty-Fourth Algonquian Conference*, 242–256. New York: SUNY Press.
- Mutet, Sylvie. 2022. *Manuel des modes et modalités*, vol. 29. Berlin: De Gruyter.
- Nauze, Fabrice. 2008. *Modality in typological perspective*. Amsterdam: University of Amsterdam. (Doctoral dissertation).
- Neidle, Carol. 1988. *The role of case in Russian syntax*. Dordrecht: Kluwer Academic Publishers.
- Nieuwenhuijsen, Dorien. 2001. Modo verbal en las oraciones interrogativas indirectas. *Nueva Revista de Filología Hispánica* 49(2). 339–362.
- Nieuwenhuijsen, Dorien. 2016. Notas sobre la aportación del análisis estadístico a la lingüística de corpus. In Johannes Kabatek & Carlota de Benito Moreno (eds.), *Lingüística de corpus y lingüística histórica iberorrománica*, 215–237. Berlin: De Gruyter.
- Núñez, Salvador. 1991. *Semántica de la modalidad en Latín*. Granada: Universidad de Granada.
- O'Donnell, John & Cordelia Hall. 2006. Set theory. In Cordelia Hall & John O'Donnell (eds.), *Discrete mathematics using a computer*, 189–206. London: Springer. DOI: 10.1007/978-1-4471-3657-6_4.
- Olivier, Marc. 2022a. *A corpus study of clitic placement with infinitives in the diachrony of French*. Ulster: Ulster University. (Doctoral dissertation).
- Olivier, Marc. 2022b. Diachronie de la proclise et de l'enclise avec l'infinitif en français médiéval (12e-15e siècles). *Studia Linguistica Romanica* 8. 10–35.
- Palmer, Frank Robert. 2001. *Mood and modality*. Cambridge: Cambridge University Press.
- Partee, Barbara H. 1995. Quantificational structures and compositionality. In Emmon Bach, Eloise Jelinek, Angelika Kratzer & Barbara H. Partee (eds.), *Quantification in natural languages* (Studies in Linguistics and Philosophy 54), 541–601. Dordrecht: Springer. DOI: 10.1007/978-94-017-2817-1_17.
- Penny, Ralph J. 2006. *Gramática histórica del español*. 2nd. Barcelona: Ariel.
- Perner, Josef, Michael Huemer & Brian Leahy. 2015. Mental files and belief: A cognitive theory of how children represent belief and its intensionality. *Cognition* 145. 77–88. DOI: 10.1016/j.cognition.2015.08.006.
- Perry, John R. 1998. Semantics, possible worlds. *Routledge Encyclopedia of Philosophy* 39. DOI: 10.4324/9780415249126-U039-1.

References

- Pesetsky, David & Esther Torrego. 2004. Tense, case, and the nature of syntactic categories. In Jacqueline Guéron & Jacqueline Lecarme (eds.), *The syntax of time*, 495–537. Cambridge: MIT Press. DOI: 10.7551/mitpress/6598.003.0021.
- Pesetsky, David & Esther Torrego. 2008. The syntax of valuation and the interpretability of features. In Simin Karimi, Vida Samiian & Vida Samiian (eds.), *Phrasal and clausal architecture: Syntactic derivation and interpretation. In honor of Joseph E. Emonds*, 262–294. Amsterdam: John Benjamins. DOI: 10.1075/la.101.14pes.
- Pinkster, Harm. 1995. *Sintaxis y semántica del latín*. Madrid: Ediciones Clásicas.
- Pinkster, Harm. 2021. *The Oxford Latin syntax. Volume II: The complex sentence and discourse*. Oxford: Oxford University Press.
- Pintzuk, Susan. 2003. Variationist approaches to syntactic change. In Brian D. Joseph & Richard D. Janda (eds.), *The handbook of historical linguistics*, 509–528. Blackwell Publishing Ltd. DOI: 10.1002/9780470756393.ch15.
- Polinsky, Maria. 1996. *American Russian: An endangered language*. Manuscript. USC-UCSD.
- Pons Rodríguez, Lola. 2008. Las construcciones imitativas del Accusativus cum infinitivo: Modelos latinos y consecuencias romances. *Revista de Historia de la Lengua Española* 3. 117. DOI: 10.54166/rhle.2008.03.04.
- Pons Rodríguez, Lola. 2010. *La lengua de ayer. Manual práctico de Historia del Español*. Madrid: Arco/Libros.
- Poole, Ethan. 2023. Improper case. *Natural Language & Linguistic Theory* 41(1). 347–397. DOI: 10.1007/s11049-022-09541-6.
- Poole, Ethan. 2024. Dependent-case assignment could be Agree. *Glossa: A Journal of General Linguistics* 9(1). DOI: 10.16995/glossa.9894.
- Portner, Paul. 1997. The semantics of mood, complementation, and conversational force. *Natural Language Semantics* 5(2). 167–212. DOI: 10.1023/A:1008280630142.
- Portner, Paul. 2009. *Modality*. Oxford: Oxford University Press.
- Portner, Paul. 2018. *Mood*. Oxford: Oxford University Press.
- Portner, Paul & Aynat Rubinstein. 2012. Mood and contextual commitment. In Anca Chereches (ed.), *Proceedings of SALT 22*, 461–487. Linguistic Society of America. DOI: 10.3765/salt.v22i0.2642.
- Portner, Paul & Aynat Rubinstein. 2020. Desire, belief, and semantic composition: Variation in mood selection with desire predicates. *Natural Language Semantics* 28(4). 343–393. DOI: 10.1007/s11050-020-09167-7.
- Postma, Gertjan. 2010. The impact of failed changes. In Anne Breitbarth, Christopher Lucas, Sheila Watts & David Willis (eds.), *Continuity and change in grammar*, 269–302. Amsterdam: John Benjamins.

- Progovac, Ljiljana. 1994. *Negative and positive polarity*. Cambridge: Cambridge University Press.
- Pujol i Campeny, Afra. 2019. *Word order in Old Catalan*. Cambridge: University of Cambridge. (Doctoral dissertation).
- Pujol i Campeny, Afra. 2025. Fronting in Old Catalan: Asymmetries between narration and reported speech. *Transactions of the Philological Society* 123(1). 1–28. DOI: 10.1111/1467-968X.12299.
- Quer, Josep. 1998. *Mood at the interface*. Utrecht: Universiteit Utrecht. (Doctoral dissertation).
- Quer, Josep. 2010. On the (un)stability of mood distribution in Romance. In Martin G. Becker & Eva-Maria Remberger (eds.), *Modality and mood in romance: Modal interpretation, mood selection, and mood alternation*, 163–180. Berlin: De Gruyter. DOI: 10.1515/9783110234343.2.163.
- Quer, Josep. 2017. Subjunctives. In de Martin Everaert & Henk Van Riemsdijk (eds.), *The Blackwell companion to syntax*, 660–684. Chichester: John Wiley. DOI: 10.1002/9781118358733.wbsyncom113.
- Ramonienė, Meilutė & Ian Press. 2015. *Colloquial Lithuanian: The complete course for beginners*. London: Routledge.
- Real Academia Española. 2008. *Banco de datos (CREA) [en línea]*. *Corpus de referencia del español actual*. <<http://www.rae.es>> [Accessed: 01/07/2023].
- Real Academia Española. 2010. *Nueva gramática de la lengua española manual*. Madrid: Espasa Calpe.
- Real Academia Española. 2013. *Corpus del diccionario histórico de la lengua española (CDHLA) [en línea]*. <https://apps.rae.es/CNDHE> (1 May, 2023).
- Real Academia Española. 2024. *Diccionario de la lengua española*. <https://dle.rae.es/>.
- Real-Puigdollers, Cristina. 2021. A minimalist approach to the syntax of *p*: A Romance perspective. *Linguistic Variation* 21(1). 90–134. DOI: 10.1075/lv.00034.rea.
- Reider, Michael. 1990. Neg-transportation, neg-trace, and the choice of mood in Spanish. *Hispania* 73(1). 212–222.
- Ridruejo Alonso, Emilio. 1999. Modo y modalidad. El modo en las subordinadas sustantivas. In Violeta Demonte Barreto & Ignacio Bosque Muñoz (eds.), *Gramática descriptiva de la lengua española*, vol. 2, 3209–3252. Madrid: Espasa Calpe.
- Ritter, Elizabeth. 1991. Two functional categories in noun phrases: Evidence from Modern Hebrew. In Susan Rothstein (ed.), *Perspectives on phrase structure: Heads and licensing* (Syntax and Semantics), 37–62. Leiden: Brill. DOI: 10.1163/9789004373198_004.

References

- Rivero, María-Luisa. 1971. Mood and presupposition in Spanish. *Foundations of Language* 7(3). 305–336.
- Rodina, Yulia. 2008. *Semantics and morphology: The acquisition of grammatical gender in Russian*. Universitetet i Tromsø. (Doctoral dissertation).
- Romoli, Jacopo. 2015. The presuppositions of soft triggers are obligatory scalar implicatures. *Journal of Semantics* 32(2). 173–219. DOI: 10.1093/jos/fft017.
- Rothstein, Björn & Rolf Thieroff. 2010. *Mood in the languages of Europe*. Amsterdam: John Benjamins.
- Ruhstaller, Stefan & María Dolores Gordón Peral. 2017. The lexical impact of language contact with Arabic on Spanish and Catalan. *Lexicographica* 33(2017). 277–296. DOI: 10.1515/lexi-2017-0014.
- San Martín, Itziar. 2007. Beyond the infinitive vs. subjunctive rivalry. In Luis Eguren & Olga Fernández Soriano (eds.), *Coreference, modality, and focus: Studies on the syntax-semantics interface*, vol. 111, 171–190. Amsterdam: John Benjamins. DOI: 10.1075/la.111.09mar.
- Santorini, Beatrice. 1992. Variation and change in Yiddish subordinate clause word order. *Natural Language & Linguistic Theory* 10(4). 595–640. DOI: 10.1007/BF00133331.
- Santorini, Beatrice. 1993. The rate of phrase structure change in the history of Yiddish. *Language Variation & Change* 5(3). 257–283. DOI: 10.1017/S0954394500001502.
- Schiegg, Markus & Judith Huber. 2023. *Intra-writer variation in historical sociolinguistics*. Bern: Peter Lang. DOI: 10.3726/b19157.
- Schlenker, Philippe. 2005. The lazy Frenchman's approach to the subjunctive. In Twan Geerts, Ivo Ginneken & Haike Jacobs (eds.), *Romance languages and linguistic theory 2003*, 269–309. Amsterdam: John Benjamins. DOI: 10.1075/cilt.270.15sch.
- Schwenter, Scott A. 2006. Fine-tuning Jespersen's cycle. In Gregory Ward & Betty J Birner (eds.), *Drawing the boundaries of meaning: Neo-Gricean studies in pragmatics and semantics in honor of Laurence R. Horn*, 327–344. Amsterdam: John Benjamins. DOI: 10.1075/slcs.80.19sch.
- Siegel, Laura. 2009. Mood selection in Romance and Balkan. *Lingua* 119(12). 1859–1882. DOI: 10.1016/j.lingua.2008.11.006.
- Sigurðsson, Einar Freyr & Milena Šereikaitė. 2023. The dual face of structural object case: On Lithuanian genitive of negation. *Journal of Linguistics* 60(1). 161–212. DOI: 10.1017/S0022226723000130.
- Silk, Alex. 2018. Commitment and states of mind with mood and modality. *Natural Language Semantics* 26. 125–166. DOI: 10.1007/s11050-018-9144-4.

- Simonenko, Alexandra & Anne Carlier. 2022. The evolution of possessive phrases and the rise of DP in French, Spanish, and Portuguese. *Glossa: A Journal of General Linguistics* 7(1). 1–35. DOI: 10.16995/glossa.5703.
- Simonenko, Alexandra, Benoît Crabbé & Sophie Prévost. 2019. Agreement syncretization and the loss of null subjects: Quantificational models for Medieval French. *Language Variation and Change* 31(3). 275–301. DOI: 10.1017/S0954394519000188.
- Sitaridou, Ioanna. 2017. Objects. In Elisabeth Stark & Andreas Dufter (eds.), *Manual of Romance morphosyntax and syntax*, 89–153. Berlin: De Gruyter. DOI: 10.1515/9783110377088-003.
- Smirnova, Anastasia. 2011. *Evidentiality and mood: Grammatical expressions of epistemic modality in Bulgarian*. Columbus: The Ohio State University. (Doctoral dissertation).
- Smith, Peter W. 2017a. *Possible and impossible agreement mismatches*. Manuscript Goethe-Universität, Frankfurt.
- Smith, Peter W. 2017b. The syntax of semantic agreement in English. *Journal of Linguistics* 53(4). 823–863. DOI: 10.1017/S0022226716000360.
- Sorace, Antonella. 2011. Pinning down the concept of “interface” in bilingualism. *Linguistic Approaches to Bilingualism* 1(1). 1–33. DOI: 10.1075/lab.1.1.01sor.
- Soutet, Olivier. 2000. *Le subjonctif en français*. Paris: Editions Ophrys.
- Squartini, Mario. 2010. Mood in Italian. In Björn Rothstein & Rolf Thieroff (eds.), *Mood in the languages of Europe*, 237–250. Amsterdam: John Benjamins. DOI: 10.1075/slcs.120.13squ.
- Stone, Matthew. 1997. The anaphoric parallel between modality and tense. *IRCS Technical Reports* 97-06. 1–45.
- Struik, Tara. 2022. *Information structure triggers for word order variation and change: The OV/VO alternation in the West-Germanic languages*. Utrecht: Landelijke Onderzoekschool Taalwetenschap.
- Struik, Tara & Ans van Kemenade. 2020. On the givenness of OV word order: A (re)examination of OV/VO variation in Old English. *English Language & Linguistics* 24(1). 1–22. DOI: 10.1017/S1360674318000187.
- Svenonius, Peter. 2001. Case and event structure. In Niina Zhang (ed.), *Proceedings of the workshop on syntax of predication* 26, 197–217. Berlin: ZAS Papers in Linguistics. DOI: 10.21248/zaspil.26.2001.144.
- Szabolcsi, Anna. 1983. The possessor that ran away from home. *The Linguistic Review* 3. 89–102. DOI: 10.1515/tlir.1983.3.1.89.
- Tal, Shira, Kenny Smith, Jennifer Culbertson, Eitan Grossman & Inbal Arnon. 2022. The impact of information structure on the emergence of differential

References

- object marking: An experimental study. *Cognitive Science* 46(3). 1–31. DOI: 10.1111/cogs.13119.
- Taylor, Ann. 1994. Variation in past tense formation in the history of English. *University of Pennsylvania Working Papers in Linguistics* 1(1). 1–20.
- Taylor, Ann. 2005. Prosodic evidence for incipient VO order in Old English. *English Language & Linguistics* 9(1). 139–156. DOI: 10.1017/S1360674305001504.
- Taylor, Ann. 2008. A note on the position of the verb in Old English rhythmic prose. *York Papers in Linguistics* 9. 124–141.
- Taylor, Ann & Susan Pintzuk. 2012. Verb order, object position and information status in Old English. *York Papers in Linguistics* 2(12a). 29–52.
- Thieroff, Rolf. 2010. Mood in German. In Björn Rothstein & Rolf Thieroff (eds.), *Mood in the languages of Europe*, vol. 120 (Studies in Language Companion Series), 133–157. Amsterdam: John Benjamins. DOI: 10.1075/slcs.120.08thi.
- Timberlake, Alan. 1975. Hierarchies in the genitive of negation. *Slavic and East European Journal* 19(2). 123–138. DOI: 10.2307/306765.
- Torrego, Esther. 2018. Phasehood and Romance adverbial *because*-clauses. In Ángel J. Gallego & Roger Martin (eds.), *Language, syntax, and the natural sciences*, 99–113. Cambridge: Cambridge University Press. DOI: 10.1017/9781316591529.006.
- Torrego, Esther & Juan Uriagereka. 1992. *Indicative dependents*. Manuscript, University of Massachusetts and University of Maryland.
- Travis, Lisa & Greg Lamontagne. 1992. The case filter and licensing of empty k. *Canadian Journal of Linguistics/Revue canadienne de linguistique* 37(2). 157–174. DOI: 10.1017/S0008413100021939.
- Truckenbrodt, Hubert. 2006. On the semantic motivation of syntactic verb movement to C in German. *Theoretical Linguistics* 32(3). 257–306. DOI: 10.1515/TL.2006.018.
- Truckenbrodt, Hubert. 2019. Co-anchoring with the matrix clause: French verbal mood and German V2. In Maria Teresa Espinal, Elena Castroviejo, Manuel Leonetti, Louise McNally & Cristina Real-Puigdollers (eds.), *Proceedings of Sinn und Bedeutung* 23, 391–408. DOI: 10.18148/sub/2019.v23i2.620.
- Tsimpli, Ianthi Maria & Aafke Hulk. 2013. Grammatical gender and the notion of default: Insights from language acquisition. *Lingua* 137. 128–144. DOI: 10.1016/j.lingua.2013.09.001.
- Uriagereka, Juan. 1988. *On government*. Mansfield: University of Connecticut. (Doctoral dissertation).
- Vandermeer, John H. & Deborah E. Goldberg. 2013. *Population ecology: First principles*. Princeton: Princeton University Press. DOI: 10.1515/9781400848737.

- Veiga, Alexandre. 2006. Las formas verbales subjuntivas. Su reorganización modo-temporal. In Concepción Company Company (ed.), *Sintaxis histórica de la lengua española*, vol. 1, 95–242. Mexico City: FCE & UNAM.
- Vicente, Ángeles. 2020. Andalusi Arabic. In Christopher Lucas & Stefano Manfredi (eds.), *Arabic and contact-induced change*, 225–244. Berlin: Language Science Press. DOI: 10.5281/zenodo.3744521.
- Villalta, Elisabeth. 2008. Mood and gradability: An investigation of the subjunctive mood in Spanish. *Linguistics and Philosophy* 31. 467–522. DOI: 10.1007/s10988-008-9046-x.
- Vivanco, Margot. 2019. Haciendo encajar las piezas en las causativas analíticas con ‘hacer’. *Verba: Anuario Galego De Filoloxía* 46. 69–108. DOI: 10.15304/verba.46.4225.
- von Fintel, Kai. 2006. Modality and language. In Donald M. Borchert (ed.), *Encyclopedia of philosophy*, 2nd edn., vol. 10, 1–16. Detroit: MacMillan.
- von Fintel, Kai & Irene Heim. 2011. *Intensional semantics*. Unpublished lecture notes.
- von Heusinger, Klaus. 2008. Verbal semantics and the diachronic development of DOM in Spanish. *Probus* 20. 1–31. DOI: 10.1515/PROBUS.2008.001.
- Wagner, Laura. 1996. The transition from *haver* to *ter* in Portuguese. *University of Pennsylvania Working Papers in Linguistics* 3(2). 113–145.
- Walkden, George. 2014. Object position and Heavy NP Shift in Old Saxon and beyond. In Kristin Bech & Kristine Gunn Eide (eds.), *Information structure and syntactic change in Germanic and Romance languages*, 313–340. Amsterdam: John Benjamins. DOI: 10.1075/la.213.12wal.
- Walkden, George & Anne Breitbarth. 2019. Complexity as L2-difficulty: Implications for syntactic change. *Theoretical Linguistics* 45(3-4). 183–209. DOI: 10.1515/tl-2019-0012.
- Wallage, Phillip. 2008. Jespersen’s Cycle in Middle English: Parametric variation and grammatical competition. *Lingua* 118(5). 643–674. DOI: 10.1016/j.lingua.2007.09.001.
- Wallage, Phillip. 2013. Functional differentiation and grammatical competition in the English Jespersen Cycle. *Journal of Historical Syntax* 2(1). 1–25. DOI: 10.18148/hs/2013.v2i1.4.
- Wallenberg, Joel C. 2019. A variational theory of specialization in acquisition and diachrony. In Anne Breitbarth, Melissa Farasyn, Lieven Danckaert & Miriam Bouzouita (eds.), *The determinants of diachronic stability*, 245–262. Amsterdam: John Benjamins.

References

- Wallenberg, Joel C. & Josef Fruehwald. 2013. Optionality is stable variation is competing grammars. In *The 25th Scandinavian Conference of Linguistics, Formal Ways of Analyzing Variation (FWAV) Workshop 25*, 1–31.
- Wallenberg, Joel C., Henri Kauhanen, George Walkden & Caroline Heycock. 2026. Grammar competition and variational learning. In Adam Ledgeway, Edith Aldridge, Anne Breitbarth, Katalin E. Kiss, Joseph Salmons, Alexandra Simonenko, Martin Everaert & Henk Van Riemsdijk (eds.), *The Wiley-Blackwell Companion to Diachronic Linguistics*. Oxford: Wiley-Blackwell.
- Wechsler, Stephen & Larisa Zlatić. 2000. A theory of agreement and its application to Serbo-Croatian. *Language* 76(4). 799–832. DOI: 10.2307/417200.
- Westergaard, Marit. 2007. English as a mixed V2 grammar: Synchronic word order inconsistencies from the perspective of first language acquisition. *Poznań Studies in Contemporary Linguistics* 43(2). 107–131. DOI: 10.2478/v10010-007-0015-2.
- Wickham, Hadley & Sievert Carson. 2016. *Ggplot2: Elegant graphics for data analysis*. New York: Springer.
- Wieling, Martijn. 2018. Analyzing dynamic phonetic data using generalized additive mixed modeling: A tutorial focusing on articulatory differences between L1 and L2 speakers of English. *Journal of Phonetics* 70. 86–116. DOI: 10.1016/j.wocn.2018.03.002.
- Wieling, Martijn, John Nerbonne & R. Harald Baayen. 2011. Quantitative social dialectology: Explaining linguistic variation geographically and socially. *PLoS ONE* 6(9). e23613. DOI: 10.1371/journal.pone.0023613.
- Wiemer, Björn. 2021. A general template of clausal complementation and its application to South Slavic: Theoretical premises, typological background, empirical issues. In Björn Wiemer & Barbara Sonnenhauser (eds.), *Clausal complementation in South Slavic*, 29–159. Berlin: De Gruyter. DOI: 10.1515/9783110725858-002.
- Winsberg, Eric. 2013. *Computer simulations in science*. Stanford Encyclopedia of Philosophy. <https://plato.stanford.edu/cite.html>.
- Witkoś, Jacek. 2008. Genitive of negation in Polish and single-cycle derivations. *Journal of Slavic Linguistics* 16(2). 247–287. DOI: <https://dx.doi.org/10.1353/jsl.0.0002>.
- Wolfe, Sam. 2021. *Syntactic change in French* (Oxford Studies in Diachronic and Historical Linguistics 47). Oxford University Press.
- Wolfe, Sam. 2024. A conspiracy theory for the loss of V2 in Romance. *Journal of Historical Syntax* 8(5). 1–42. DOI: 10.18148/hs/2024.v8i5.194.

- Wood, Jim & Einar Freyr Sigurðsson. 2014. Icelandic verbal agreement and pronoun-antecedent relations. In Chris Collins (ed.), *Cross-linguistic studies of imposters and pronominal agreement*, 196–237. New York: Oxford Academic. DOI: 10.1093/acprof:oso/9780199336852.003.0009.
- Wood, Simon N. 2017a. *Generalized additive models: An introduction with R*. Chapman & Hall/CRC. DOI: 10.1201/9781315370279.
- Wood, Simon N. 2017b. *Mgcv-package Mixed GAM Computation Vehicle with GCV/AIC/REML smoothness estimation and GAMMs by REML/PQL*. published on the Comprehensive R Archive Network, CRAN).
- Yang, Charles. 2000. Internal and external forces in language change. *Language Variation and Change* 12(3). 231–250. DOI: 10.1017/S0954394500123014.
- Yang, Charles. 2002. Grammar competition and language change. In David W. Lightfoot (ed.), *Syntactic effects of morphological change*, 367–380. Cambridge University Press. DOI: 10.1093/acprof:oso/9780199250691.003.0021.
- Yang, Charles. 2010. Three factors in language variation. *Lingua* 120(5). 1160–1177.
- Yang, Charles & Silvina Montrul. 2017. Learning datives: The Tolerance Principle in monolingual and bilingual acquisition. *Second Language Research* 33(1). 119–144. DOI: 10.1177/02676583166673.
- Zahner-Ritter, Katharina, Tianyi Zhao, Marieke Einfeldt & Bettina Braun. 2022. How experience with tone in the native language affects the L2 acquisition of pitch accents. *Frontiers in Psychology* 13. 903879. DOI: 10.3389/fpsyg.2022.903879.
- Zamorano Aguilar, Alfonso. 2001. *El subjuntivo en la historia de la gramática española, 1771–1973*. Madrid: Arco Libros.
- Zaring, Laurie. 2010. Changing from OV to VO: More evidence from Old French. *Ianua. Revista Philologica Romanica* (10). 1–18.
- Zimmermann, Richard. 2023. An improved test of the constant rate hypothesis: Late Modern American English possessive *have*. *Corpus Linguistics and Linguistic Theory* 19(3). 323–352. DOI: 10.1515/cllt-2021-0038.
- Zwarts, Frans. 1998. Three types of polarity. In Fritz Hamm & Erhard Hinrichs (eds.), *Plurality and quantification* (Studies in Linguistics and Philosophy 69), 177–238. Dordrecht: Springer. DOI: 10.1007/978-94-017-2706-8_5.

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Mood alternations

This book investigates Polarity Mood (PM) in Peninsular Spanish and how the phenomenon has evolved diachronically. In particular, it explores from a theoretical and empirical perspective the interaction of PM with five other linguistic phenomena: (i) the verb class of the matrix predicate, (ii) the presence of a first-person subject; (iii) the type of matrix clause, (iv) information structure, and (v) the presence of Negative Polarity Items. Drawing inspiration from the competing grammars framework, the book starts by proposing an account of mood morphology in terms of three competing systems: one in which mood morphology is used as a hedging device to express the degree of commitment; another in which mood is analyzed as a pronoun which indicates that the world of evaluation lies in some accessible Context Set; and another in which mood is used as a purely grammaticalized syntactic marker. The second part of the book shows how the historical competition between these systems explains the diachronic development observed in corpus data, as well as the present-day mood selection patterns.